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Easy Piping User Manual

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Easy Piping



User Manual Version 4.7.3



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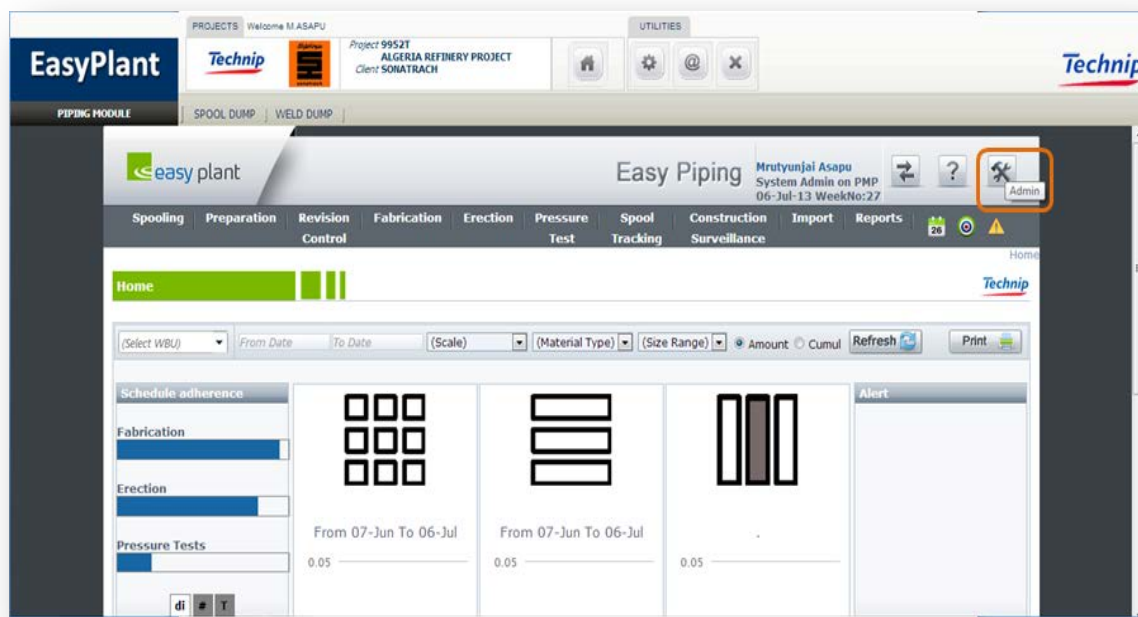
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[*Easy Piping Part 1: Set up*

ACCESS TO THE PROJECT SETUP

The user can access the setup screen from the home page “Admin Module” of Easy Piping (Below Image) this screen is accessible only for the System admin, Project admin and site admin defined by the project. This screen allows the user to perform various activities like defining a new project, defining the System referential, project referential, define access rights and import settings.

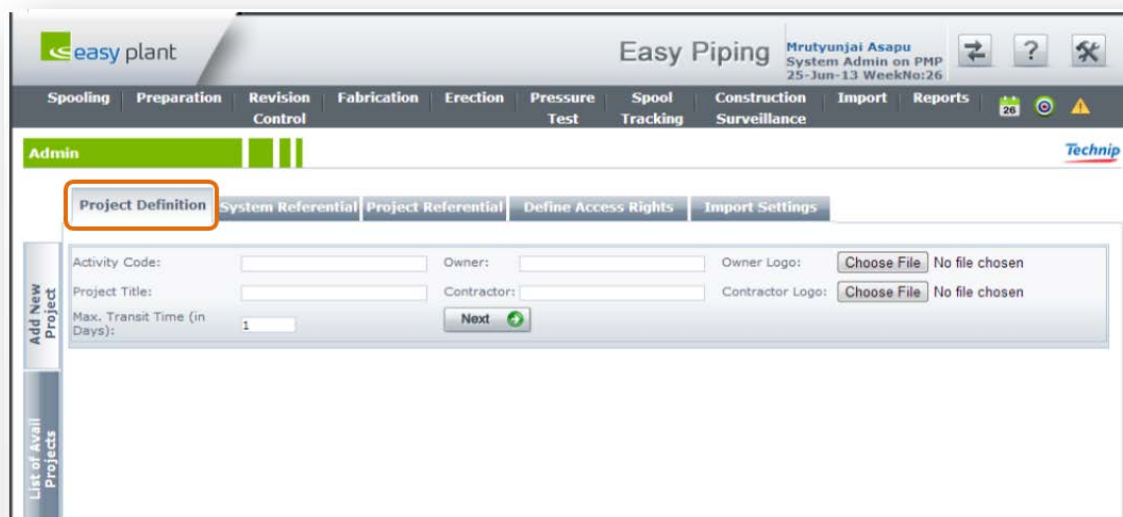


The Below Matrix will help the user to know the sequence of operations during setup

PROCESS	ACTIVITY NO.	PROJECT REFERENTIAL ACTIVITIES	STEP 1	STEP 2	PREDECESSOR	COMMENTS
GENERAL PROJECT DEFINITION	Act 3.1	Sub-contractor list	X			
SPOOLING	Act 3.7	Service class/Material type list	X			
SPOOLING	Act 3.8	Weld Type List	X			Should define before weld progress entry
SPOOLING	Act 3.9	INDE Matrix		X	Act 3.7, Act 3.8, Act 3.11	
SPOOLING	Act 3.10	Rework Code	X			
SPOOLING	Act 3.11	Thickness	X			
FAB & ERECTION	Act 3.12	Progress Weight Factor	X			
FAB & ERECTION	Act 3.3	Area Classification	X			
FAB & ERECTION	Act 3.4	PDS Area/Sub-contractor list		X	Act 3.1, Act 3.3	
FAB & ERECTION	Act 3.5	WPS List	X			
FAB & ERECTION	Act 3.6	Welders Qualification		X	Act 3.1, Act 3.5	Should define before weld progress entry
FAB & ERECTION	Act 3.12	Project Piping Material List	X			Should define before weld progress entry
TESTPACK	Act 3.13	Joint Category Definition	X			Should define before Bolting progress entry
TESTPACK	Act 3.14	Unit of Time Reference	X			Should define before Bolting progress entry
TESTPACK	Act 3.15	Jointer List	X			Should define before Bolting progress entry
TESTPACK	Act 3.16	Blinding Team	X			Should define before Testpack progress entry
TESTPACK	Act 3.17	Finishing Team	X			Should define before Testpack progress entry
TESTPACK	Act 3.18	Reinstatement Team	X			Should define before Testpack progress entry
TESTPACK	Act 3.19	System	X			Should define before testpack assignment
TESTPACK	Act 3.20	Sub System		X	Act 3.19	
TESTPACK	Act 3.21	Line Checker Team	X			Should define before Testpack progress entry
TESTPACK	Act 3.26	Pressure Unit	X			Should define before Testpack preparation
TESTPACK	Act 3.27	Line Service	X			Should define before Testpack preparation
SPOOL TRACKING	Act 3.22	Location Category	X			
SPOOL TRACKING	Act 3.23	Locations		X	Act 3.22	
SPOOL TRACKING	Act 3.24	Devices	X			
SPOOL TRACKING	Act 3.25	PDA Users	X			

1. PROJECT DEFINITION

The project definition module helps the user to setup a new project, define the custom columns and defines the project parameters like project activity no; client and client's logo, contractor and contractor's logo, sub-contractor and its logo etc., the user can modify or delete the existing projects



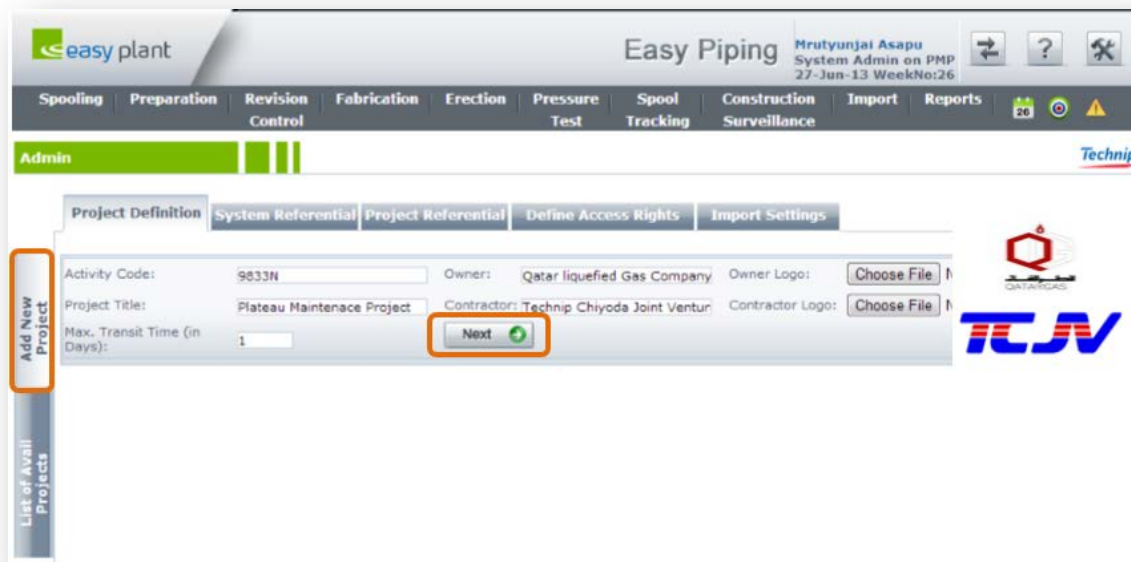
Under this module there are two tabs namely “Add a New project” and “List of Available Projects” which are described here below

1.1 Add New Project

The user need to follow the below parameters while creating a new project

- Activity Code: The alphanumeric string defined for the project Ex: 9833N
- Project Title: Project Name Ex: Plateau Maintenance Project
- Owner: The name of the client company
- Contractor: The name of the contractor company
- Owner Logo (System will accept only the following file types ".bmp" ".jpg" and size should be less than 200 KB)
- Contractor Logo (System will accept only the following file types ".bmp" ".jpg" and size should be less than 200 KB)
- Maximum Transit Time (default value 1) this value is used in the 'Spool Tracking' module to calculate the no. of days a spool has been in transit from a location. The last tracking date of a spool will be compared with the current date, and if the difference exceeds the 'Maximum Transit Time', the spool will be listed in the 'Transit out' section.

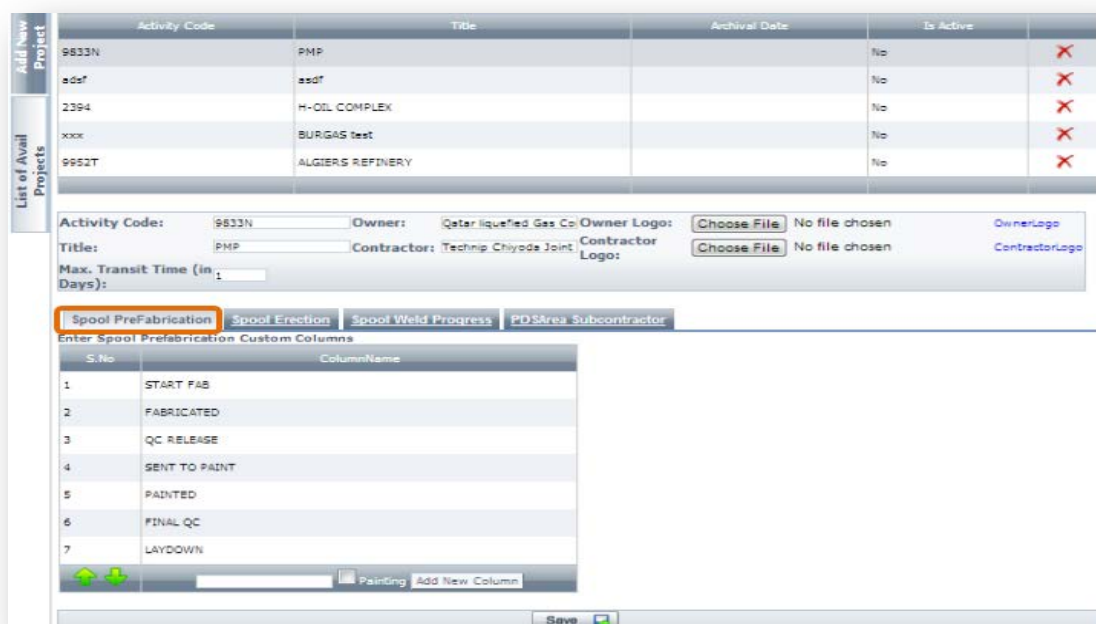
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The screenshot shows the 'Easy Piping' software interface. The 'Project Definition' tab is active. The 'Next' button is highlighted with a red box. The 'Add New Project' button is also visible on the left sidebar.

On clicking the icon  , the user will see a grid with 4 tabs:

- Spool Prefabrication
- Spool Erection
- Spool Weld Progress
- PDS Area Subcontractor



The screenshot shows the 'Easy Piping' software interface with the 'Spool Prefabrication' tab selected. The 'Spool Prefabrication' tab is highlighted with a red box. The 'Add New Project' button is also visible on the left sidebar.

Activity Code	Title	Archival Date	Is Active
9833N	PMP		No
adsf	asdt		No
2394	H-OIL COMPLEX		No
xxxx	BURIGAS test		No
9952T	ALGIERS REFINERY		No

Activity Code: 9833N Owner: Qatar liquefied Gas Co. Owner Logo: Choose File No file chosen
 Title: PMP Contractor: Technip Chiyoda Joint Contractor Logo: Choose File No file chosen
 Max. Transit Time (in Days): 1

Spool Prefabrication Spool Erection Spool Weld Progress PDS Area Subcontractor

Enter Spool Prefabrication Custom Columns

S.No	ColumnName
1	START FAB
2	FABRICATED
3	QC RELEASE
4	SENT TO PAINT
5	PAINTED
6	FINAL QC
7	LAYDOWN

Printing Add New Column

Save

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Prefabrication, Erection and Weld tabs show the default columns that are available to the user to enter the progress information in the 'Progress' module.

In the Prefabrication tab, the following columns are the default columns that will appear under Spool Fabrication:

- Start Fab
- Fabrication
- QC Release

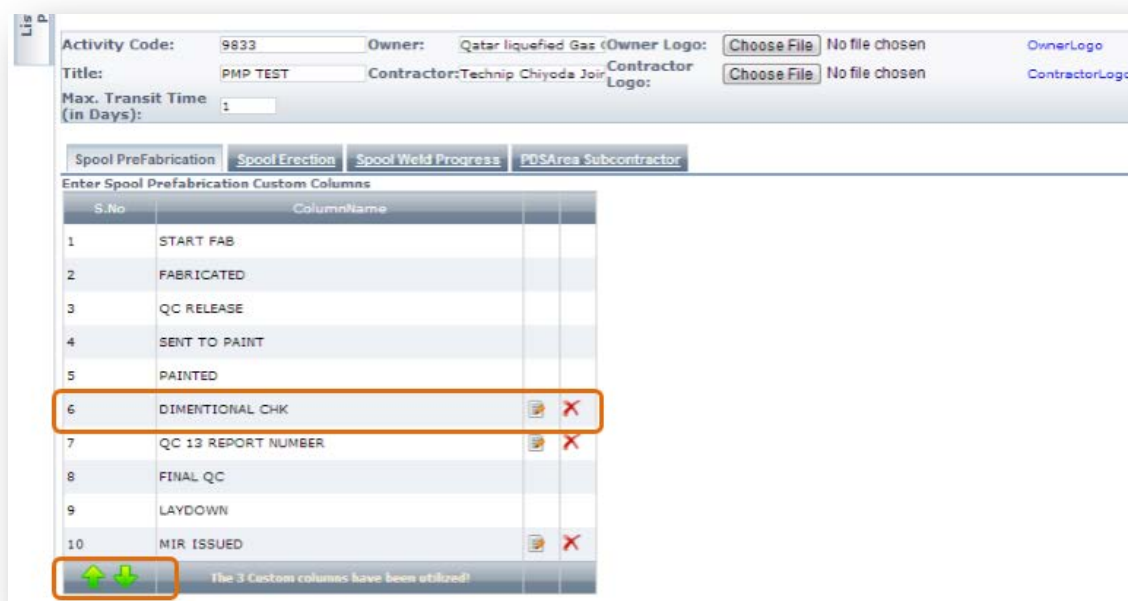
The following columns are the default columns that will appear under Painting:

- Sent to Paint
- Painted
- Final QC
- Laydown

Prefabrication & Erection : User can add up to 3 custom columns to the 'Prefabrication' and 'Erection' tabs. The user has to specify whether the newly added column should appear under 'Painting' or 'Fabrication' by checking or clearing the 'Painting' checkbox.

All columns under this tab are used to record progress in the respective modules and therefore they are expected to be 'date' values. These columns can be positioned in any order by clicking on the 'up' and 'down' arrows. These custom columns will appear in the Progress module in the order defined.

The custom columns added by the user can be edited and deleted at any time, even if the project has some values in front of these columns, in such case these values will be lost.



S.No	ColumnName		
1	START FAB		
2	FABRICATED		
3	QC RELEASE		
4	SENT TO PAINT		
5	PAINTED		
6	DIMENSIONAL CHK		
7	QC 13 REPORT NUMBER		
8	FINAL QC		
9	LAYDOWN		
10	MIR ISSUED		

the PDS Area/ Subcontractor: The user can add 3 custom columns to the PDS Area/ Subcontractor tab. The Use of this option is to define custom attributes at PDS Area level. The Columns which are defined in the following screen

These custom columns will appear in the Project referential of PDS Area/ Subcontractor . It will allows the user to link the information to the subcontractor by design area.(Refer to Para 3.4)

They will also appear as filters in "Summary report", "Weekly fabrication" and "Progress Report-Fabrication" reports.

List of Avail Projects

14002v	ASABUG	No	
9833	PMP TEST	No	

Activity Code: Owner: Owner Logo: No file chosen [OwnerLogo](#)

Title: Contractor: Contractor Logo: No file chosen [ContractorLogo](#)

Max. Transit Time (in Days):

Spool Prefabrication

Spool Erection

Spool Weld Progress

PDSArea Subcontractor

Enter PDSArea Subcontractor Custom Columns

S.No	ColumnName		
1	NEW		
			Add New Column

Spool Category Definition: Spool Category Definition options allows user to define custom attributes to all spools. In this Option user can define 3 custom attributes at Spool Area level. The Columns which are defined in the following screen

There defined attributes can be updated against each spool through import module.(Refer to Para 3.4)

They will also appear as filters in “Fabrication”, “Weekly fabrication”, “Erection” and “Weekly Erection Reports”.

Spool PreFabrication	Spool Erection	Spool Weld Progress	PDSArea Subcontractor	Spool Category Definition
--------------------------------------	--------------------------------	-------------------------------------	---------------------------------------	--

Enter Spool Category Definition Custom Columns

S.No	ColumnName		
1	PAINT_SUBCON		
2	FAB_SUBCON		
3	ERECTION_SUBCON		
  The 3 Custom columns have been utilized!			



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1.2 List of available projects

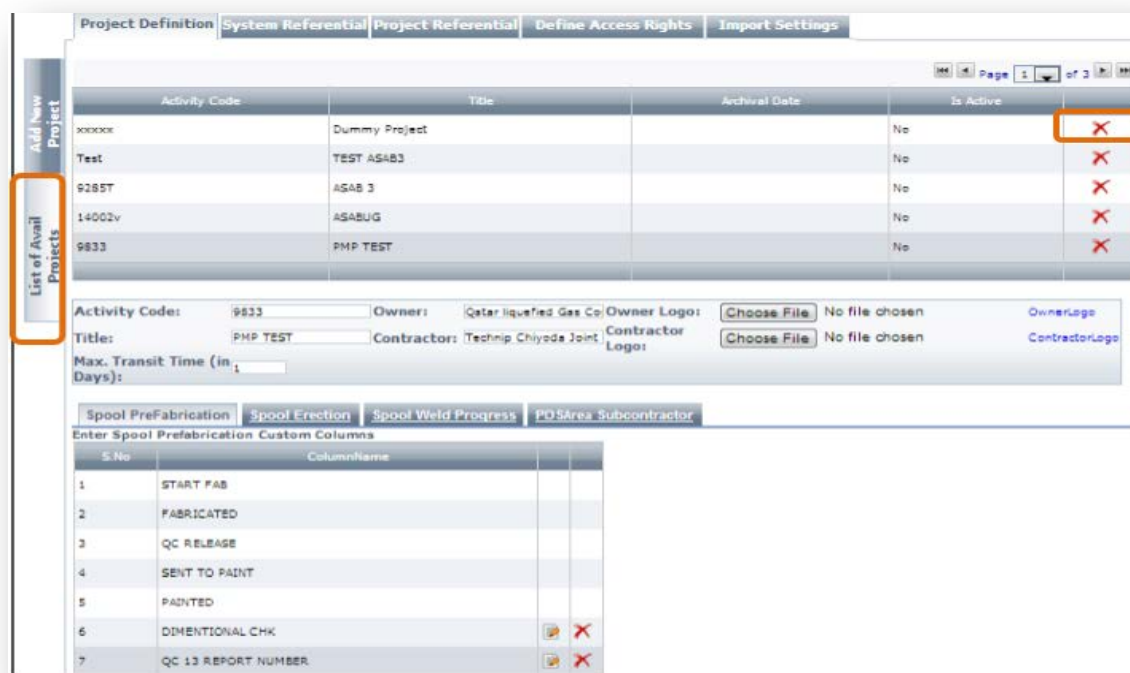
The tab allows the user to view project's information and edit some of the associated data. The following parameters can be modified when modifying a project in SYSTEM:

- Activity Code
- Project Title
- Owner
- Contractor
- Owner Logo
- Contractor Logo
- Maximum Transit Time

The custom columns can be added at a later stage as well. These custom columns can also be deleted at any stage in the project. However, all the data belonging to these columns will also be deleted.

The System Administrator can delete a project. However, he must be logged into another project to delete a particular project. All project specific information like project referential, isometrics, spools, joints, batches, progress information, and examination information will be deleted.

The user can choose the icons () & () to delete and edit the details respectively.



Activity Code	Title	Archival Date	Is Active
xxxxxx	Dummy Project		No 
Test	TEST ASAB3		No 
9285T	ASAB 3		No 
14002v	ASABUG		No 
9833	PMP TEST		No 

Activity Code: 9833 Owner: Qatar Liquefied Gas Co Owner Logo:  No file chosen OwnerLogo: 

Title: PMP TEST Contractor: Technip Chiyoda Joint Contractor Logo:  No file chosen ContractorLogo: 

Max. Transit Time (in Days):

Spool PreFabrication Spool Erection Spool Weld Progress PDS Area Subcontractor

Enter Spool Prefabrication Custom Columns

S.No	ColumnName
1	START FAB
2	FABRICATED
3	QC RELEASE
4	SENT TO PAINT
5	PAINTED
6	DIMENSIONAL CHK 
7	QC 13 REPORT NUMBER 

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2. SYSTEM REFERENTIAL

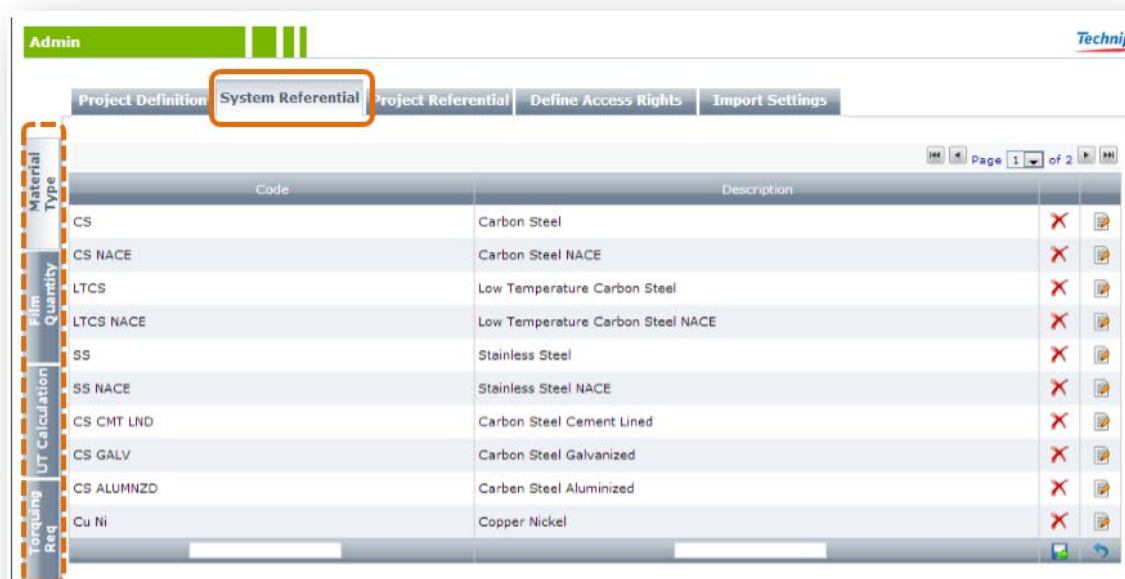
The System referential module will allow the System admin to define all the lists that the application might refer to through the course of the application, irrespective of the project selected. These lists include the following:

- Material Type List
- Film Quantity per Diameter
- UT Calculation
- Torquing Requirement

The above referential is common to all projects.

Once these lists are defined, they will appear as a dropdown list whenever needed in the various screens of the application so that the user can select values out of these lists. The objective of this approach is to maintain the consistency of data entry across the application, which will result in better and accurate reports. Moreover, it will ease up the data entry process.

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.



2.1 Material Type

In the below picture all the lists are shown as explained earlier, as the screen shows the first sub tab material list. This screen will allow the user to define all the different types of materials. The user needs to define the Code and Description both the fields are mandatory. Once it is defined, the user is not allowed to delete the material type if it is assigned for any service class but can be modified at any time. The same will display in all kind of reports

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

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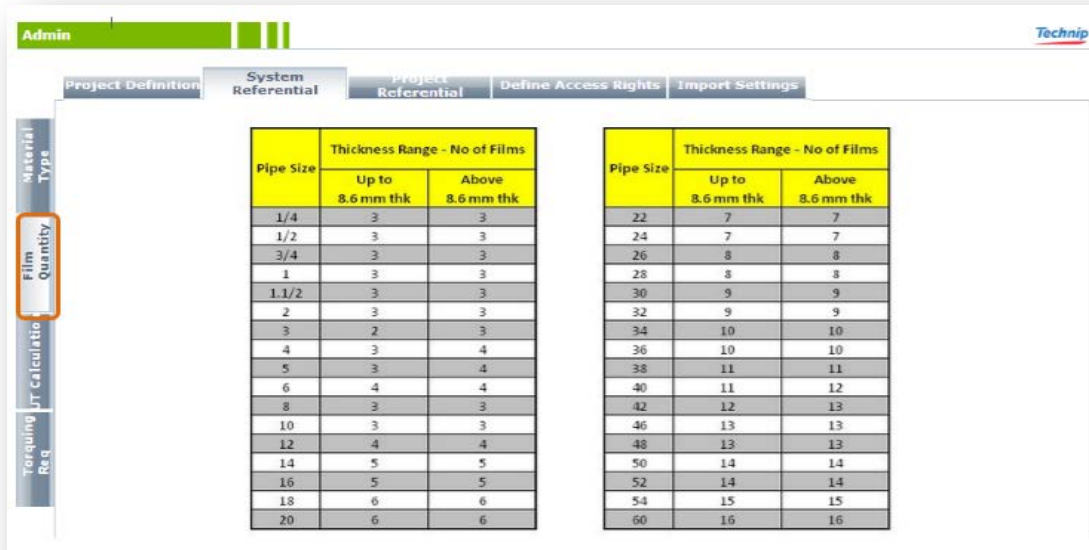
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Code	Description		
CS	Carbon Steel	X	
CS NACE	Carbon Steel NACE	X	
LTCS	Low Temperature Carbon Steel	X	
LTCS NACE	Low Temperature Carbon Steel NACE	X	
SS	Stainless Steel	X	
SS NACE	Stainless Steel NACE	X	
CS CMT LND	Carbon Steel Cement Lined	X	
CS GALV	Carbon Steel Galvanized	X	
CS ALUMINZD	Carbon Steel Aluminized	X	
Cu Ni	Copper Nickel	X	

2.2 Film Quantity per Diameter

The screen displays to the user the film quantities based on pipe diameter and thickness. This number will be displayed to the user in the relevant Progress screen. This screen is static and user will not be allowed to edit or delete any information. However the user can edit the total number of films in the RT progress screen.



Pipe Size	Thickness Range - No of Films	
	Up to 8.6 mm thk	Above 8.6 mm thk
1/4	3	3
1/2	3	3
3/4	3	3
1	3	3
1.1/2	3	3
2	3	3
3	2	3
4	3	4
5	3	4
6	4	4
8	3	3
10	3	3
12	4	4
14	5	5
16	5	5
18	6	6
20	6	6

Pipe Size	Thickness Range - No of Films	
	Up to 8.6 mm thk	Above 8.6 mm thk
22	7	7
24	7	7
26	8	8
28	8	8
30	9	9
32	9	9
34	10	10
36	10	10
38	11	11
40	11	12
42	12	13
46	13	13
48	13	13
50	14	14
52	14	14
54	15	15
60	16	16

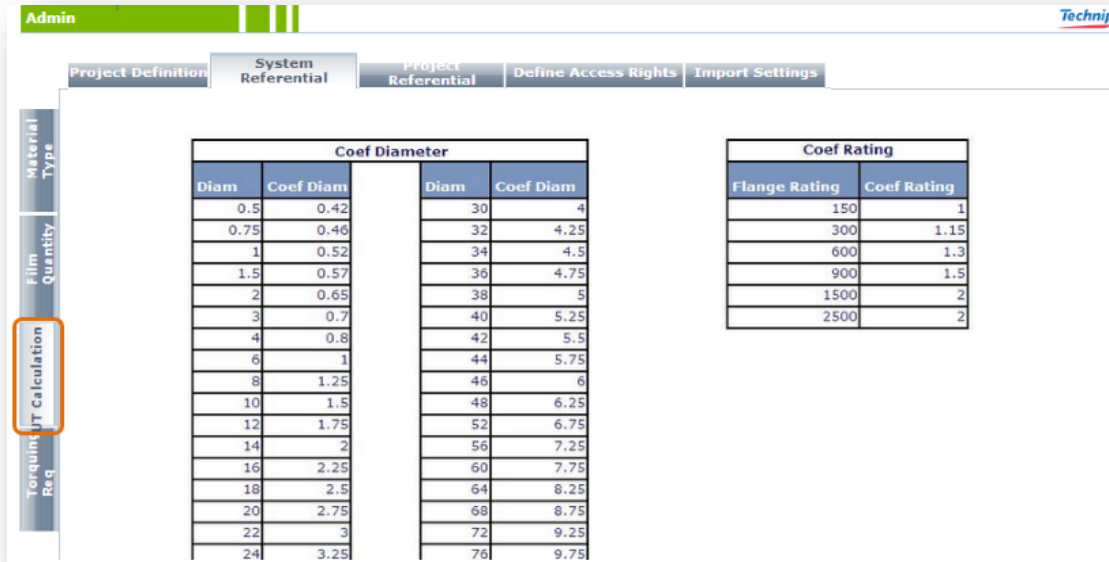
2.3 UT Calculation

The screen displays to the user the values of referential “Coefficient Diameter” and “Coefficient Rating”. This screen is also static and user cannot edit or modify at any time. The values are based on Technip standards.

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These values are used for UT calculations based on the diameters and it will export the values in the “Flange joint progress”.



Coef Diameter			
Diam	Coef Diam	Diam	Coef Diam
0.5	0.42	30	4
0.75	0.46	32	4.25
1	0.52	34	4.5
1.5	0.57	36	4.75
2	0.65	38	5
3	0.7	40	5.25
4	0.8	42	5.5
6	1	44	5.75
8	1.25	46	6
10	1.5	48	6.25
12	1.75	52	6.75
14	2	56	7.25
16	2.25	60	7.75
18	2.5	64	8.25
20	2.75	68	8.75
22	3	72	9.25
24	3.25	76	9.75

Coef Rating	
Flange Rating	Coef Rating
150	1
300	1.15
600	1.3
900	1.5
1500	2
2500	2

2.4 Torquing requirement

The user can view the methods of bolting. This screen is a static screen, the torquing methods are defined in this referential none of the values can be edit or modify.

- Manual
- Torquing
- Tensioning



SNo.	Methods
1	Manual
2	Torquing
3	Tensioning

3. PROJECT REFERENTIAL

The project referential module will be used to define all the lists that the application might refer to through the course of the application. These lists include the following:

- Sub-contractor list
- Progress Weight Factor
- Area Classification
- PDS Area/Sub-contractor list
- WPS List
- Welders Qualification
- Service class/Material type list
- Weld Type List
- NDE Matrix
- Rework Code
- Thickness
- Project Piping Material List
- Joint Category Definition
- Unit of Time Reference
- Jointer List
- Blinding Team
- Finishing Team
- Reinstatement Team
- System
- Sub System
- Line Checker Team
- Location Category
- Locations
- Devices
- PDA Users
- Pressure Unit
- Line Service

Once the above lists are defined, they will appear as a dropdown list whenever needed in the various screens of the application so that the user can select values out of these lists.

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3.1 Subcontractor List

The user can define all the subcontractors specified by the project, the user need to define the following details

- Subcontractor Code
- Subcontractor Description

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

Once it is defined, the System will not allow the user to delete if the subcontractor is already assigned to any PDS Area. However the user can edit or modify the code and description at any time

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Subcontractor	Code	Description		
Progress Weight Factor	DESCON	Descon Engineering Qatar LLC	X	
Area Classification	DEBAJ	Deutsche Babcock Al Jaber LLC	X	
PDS Area/Subcontractor	ELETECH	ELETECH	X	
WPS List	KETTANEH	KETTANEH Constructions	X	
Welder Qualification	OCT	Consolidated Contractors	X	
Quality Inspector				
Service Class/Matrl.Type				
Weld Type				
NDE Matrix				
Rework Code				
Thickness				
Project Piping Material				
Joint Category Definition				

3.2 Progress Weight Factor

The user need to define the progress weight factors, these weight factors are used to calculate the completion of the following activities.

- Spool Prefabrication
- Spool Painting
- Spool Erection

Each activity in each phase will have its own progress factor; the total of progress factors for all the activities should be one hundred. In the reports module, the progress will be reported either by phase (prefabrication, painting, erection), or as an overall progress. The weight values can be modified at any time, but if modified, it will have a direct impact on the weight calculation of the generated reports. It should be noted that there will be discrepancies between reports before modifying the weight and after modifying the weight.

After every change please press () to update the details.

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3.3 Area Classification

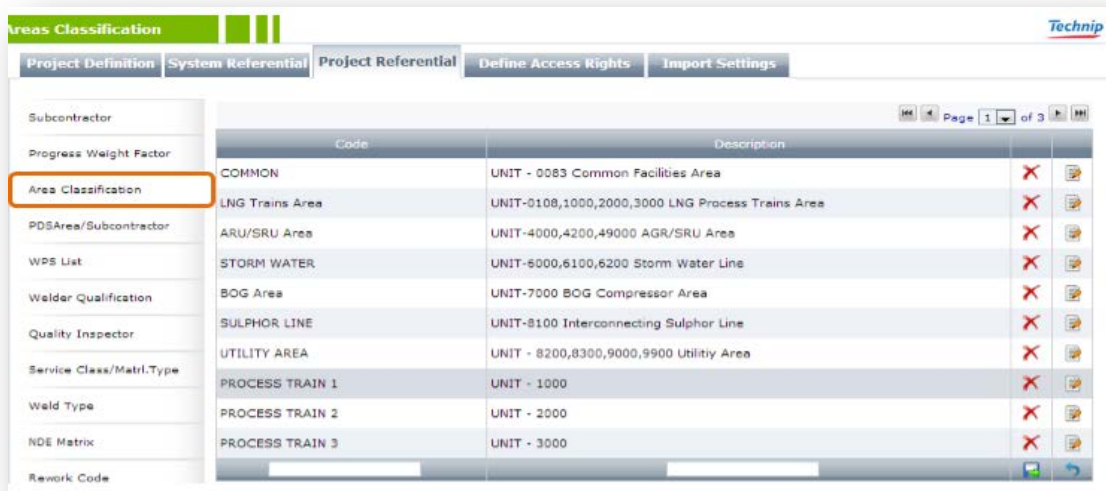
This screen helps the user to define the Area classification of the project, the user need to define the following details

- Area Classification code
- Area Classification description

The same details will be used while linking certain PDS areas in another referential ([Refer to Para 3.4](#)) this codes are used to generate reports by different area classification. The user can see all the areas in the dropdown while generating various reports.

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

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Code	Description		
COMMON	UNIT - 0083 Common Facilities Area	X	
LNG Trains Area	UNIT-0108,1000,2000,3000 LNG Process Trains Area	X	
ARU/SRU Area	UNIT-4000,4200,49000 AGR/SRU Area	X	
STORM WATER	UNIT-6000,6100,6200 Storm Water Line	X	
BOG Area	UNIT-7000 BOG Compressor Area	X	
SULPHOR LINE	UNIT-8100 Interconnecting Sulphor Line	X	
UTILITY AREA	UNIT - 8200,8300,9000,9900 Utility Area	X	
PROCESS TRAIN 1	UNIT - 1000	X	
PROCESS TRAIN 2	UNIT - 2000	X	
PROCESS TRAIN 3	UNIT - 3000	X	

3.4 PDS Area/Subcontractor

This screen is used to define the referential based on the PDS (Design area) and assign to the following details

- Shop Sub-contractor
- Field Sub-contractor
- Area Classification
- AG/UG

In addition of above, the user can specifies the area is unit or rack and any custom column if any, which are predefined while defining project ([Refer to Para 1.2](#))

Note: The subcontractor can be assigned only by Design area, if the user wants to change the subcontractor of a particular Isometric, It can be done manually in the browse latest module.

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

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PDSArea	Shop Subcontractor	Field Subcontractor	Area Classification	UNIT/RACK/SLPR	AG/UG
1801R	DESCON	DESCON	PROCESS TRAIN 1	RACK	AG
1801RT	DESCON	DESCON	PROCESS TRAIN 1	RACK	AG
1802A	DESCON	DESCON	PROCESS TRAIN 1	UNIT	AG
1803A	DESCON	DESCON	PROCESS TRAIN 1	UNIT	AG
1804A	DESCON	DESCON	PROCESS TRAIN 1	UNIT	AG
1812R	DESCON	DESCON	PROCESS TRAIN 1	RACK	AG
2801R	DESCON	DESCON	PROCESS TRAIN 2	RACK	AG
2802A	DESCON	DESCON	PROCESS TRAIN 2	UNIT	AG
2803A	DESCON	DESCON	PROCESS TRAIN 2	UNIT	AG
2804A	DESCON	DESCON	PROCESS TRAIN 2	UNIT	AG

3.5 WPS List

This screen is used to define the WPS approved by the Quality team. The WPS defined in this screen is used in the Welder Qualification module in order to define the WPS for a particular qualified welder. The user defines each WPS with material type, Dia, thickness and link to a subcontractor.

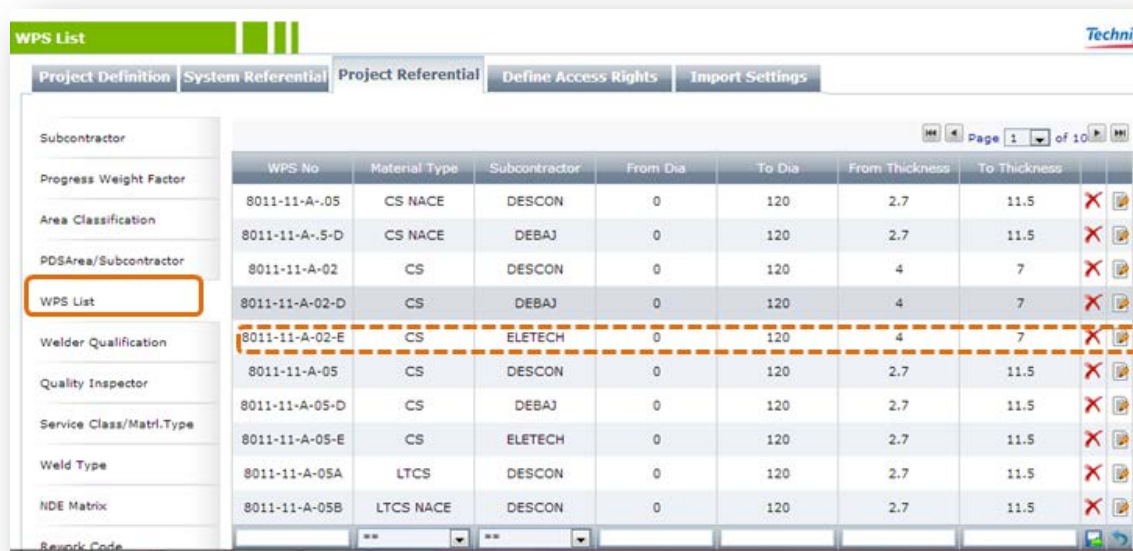
Validations

- All fields are mandatory.
- Diameter and Thickness allow only numeric values.
- To Dia should be greater than or equal to From Dia.
- To Thickness should be greater than or equal to From Thickness.

During data import from spooling, Easy Piping will automatically check that a WPS exists in the WPS list to cover new joints. If WPS (based on joint dia, thickness and material type) does not exist, SYSTEM will alert the user but import can proceed.

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

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WPS No	Material Type	Subcontractor	From Dia	To Dia	From Thickness	To Thickness	
8011-11-A-.05	CS NACE	DESCON	0	120	2.7	11.5	
8011-11-A-.5-D	CS NACE	DEBAJ	0	120	2.7	11.5	
8011-11-A-02	CS	DESCON	0	120	4	7	
8011-11-A-02-D	CS	DEBAJ	0	120	4	7	
8011-11-A-02-E	CS	ELETECH	0	120	4	7	
8011-11-A-05	CS	DESCON	0	120	2.7	11.5	
8011-11-A-05-D	CS	DEBAJ	0	120	2.7	11.5	
8011-11-A-05-E	CS	ELETECH	0	120	2.7	11.5	
8011-11-A-05A	LTCS	DESCON	0	120	2.7	11.5	
8011-11-A-05B	LTCS NACE	DESCON	0	120	2.7	11.5	

3.6 Welder Qualification

This screen is used to define the welder code and name of welders qualified for the project. The user needs to link the welder to a subcontractor and WPS. This screen allows the user to add, delete and edit the welder code until and unless there is no records for the related welder. The same welders will be seen in a dropdown list in the relevant progress module. In order to qualify the welders, the user needs to create the WPS List. ([Refer above Para 3.5](#))

All the fields are mandatory

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

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Code	First Name	Last Name	Subcontractor	Qualification
W-RTW-050	Mao	nan	ELETECH	WPS-PMP-01, WPS-PMP-02, WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-06	Qiu	Yongli	ELETECH	WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-07	Ma	Zhenxi	ELETECH	WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-08	Qian	Itjun	ELETECH	WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-09	Zhao	Jianling	ELETECH	WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-10	Hu	Jiayong	ELETECH	WPS-PMP-01, WPS-PMP-02, WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-11	Li	Zhongkui	ELETECH	WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-12	Zhang	Genming	ELETECH	WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06
W-RTW-12-E	Zhang	Genming	DESCON	WPS-PMP-03-E, WPS-PMP-05-D
W-RTW-13	Han	Yangyang	ELETECH	WPS-PMP-03, WPS-PMP-04, WPS-PMP-05, WPS-PMP-06

3.7 Service class/Material type

This screen allows the user to define all the service class available for the project and link to the material type. In order to link, the user have to define the material type ([Refer to the Para 2.1](#)). All the service classes are corresponding codes uploaded from the spooling data. The service class cannot be deleted but it can be edit or modify at any time.

The Material type is used while defining the WPS ([Refer to Para 3.5](#)) and for generation of the reports as well.

Service Class	Material Type
1A0JLX	SS
1C0JLX	SS
1C0KLX	SS
1C0KLNX	SS NACE
1C1KLX	SS
1C3AS1X	CS NACE
1C3AS2X	CS NACE
1C3AS3X	CS NACE
1J0MX	CS PTFE
1N4ASX	CS

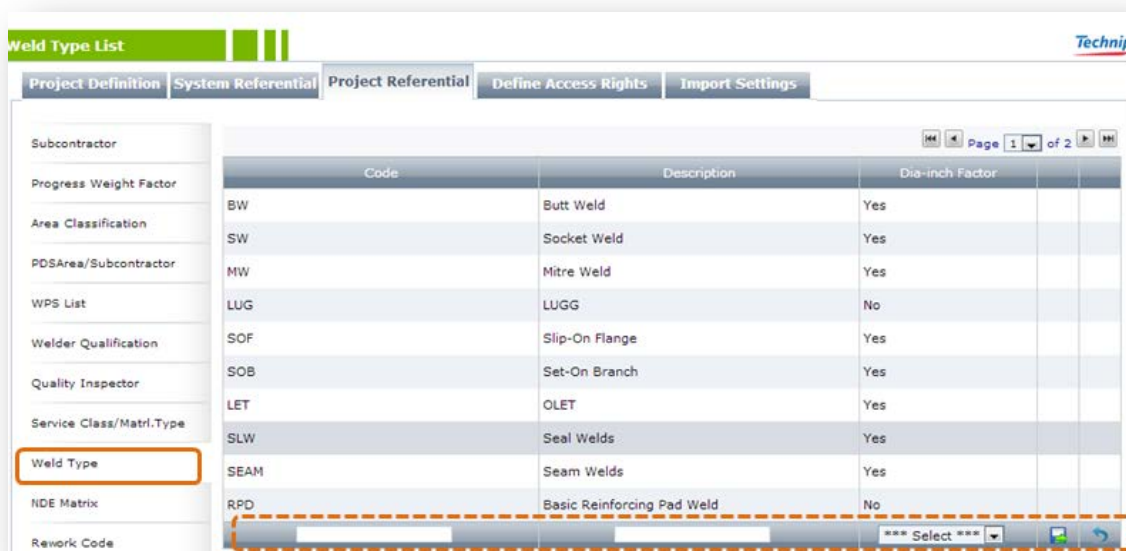
3.8 Weld type List

This screen contains references of all the different types of welds that will be uploaded automatically from the spooling data. In this screen the user can only add new weld types. The list contains the following

- Weld type code (This type code is from spoolgen data)
- Weld type description
- Dia inch factor – User to define Yes or No

Note: If the user defines “Yes” in Dia-inch Factor for a weld type code, the joint will be counted in the Dia inch calculation in the reports.

If the user defines “No” In Dia-Inch factor for a weld type code, the joint will not be counted in the Dia inch calculations in the reports



Code	Description	Dia-inch Factor	
BW	Butt Weld	Yes	
SW	Socket Weld	Yes	
MW	Mitre Weld	Yes	
LUG	LUGG	No	
SOF	Slip-On Flange	Yes	
SOB	Set-On Branch	Yes	
LET	OLET	Yes	
SLW	Seal Welds	Yes	
SEAM	Seam Welds	Yes	
RPD	Basic Reinforcing Pad Weld	No	

3.9 NDE Matrix

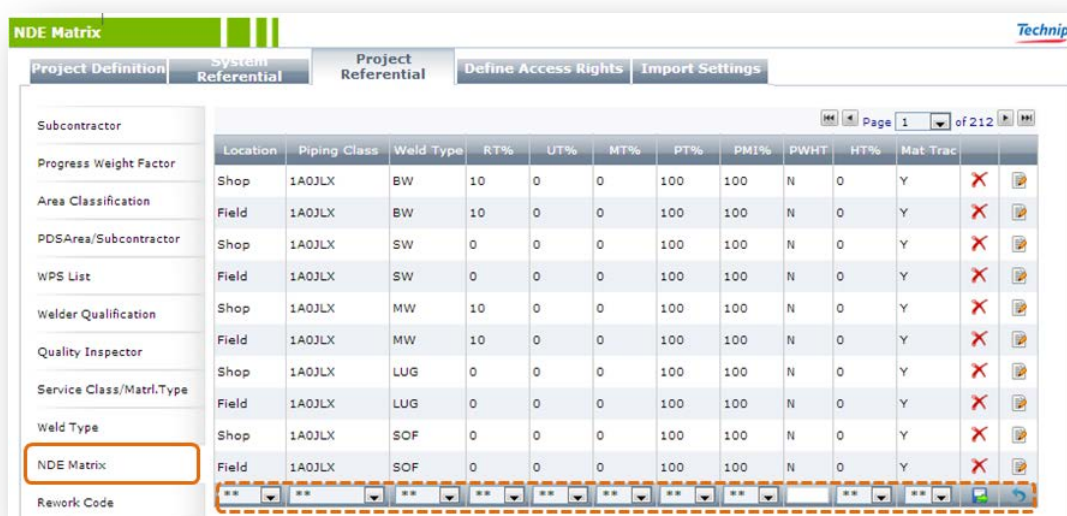
This screen helps the user to define the NDE (Non Destructive Examination) % required for the different types of weld categories.

- User needs to define the various types of testing parameters like RT, MT, PT and UT etc. based on the piping class, weld type and weld location.
- All the fields in this screen will be available as a dropdown list where the user can select the required information.
- The columns RT, UT, MT, PT, PMI, and HT are dropdown lists having the values for different percentages ranging from 0 to 100.
- The columns PWHT, Material Traceability is a dropdown list that will allow “Y” or “N” and the user can also define the thickness which required PWHT.
- The matrix can be filled in an excel format and upload in the System through the import module.
- During import of spools, the System will validate whether the imported welds have the combination of piping class, location and weld type is already defined in the NDE matrix. If not, the System will alert the user and import is not allowed.
- During a weld Progress entry, the System will automatically check the NDE % defined in the NDE matrix and allocate a batch based on the examination category to the joint. This case is particularly for 5% or 10% categories.

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- In the case of NDE 100 %, the system will automatically allocate the joints to the relevant NDE category during the spooling data import.

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.



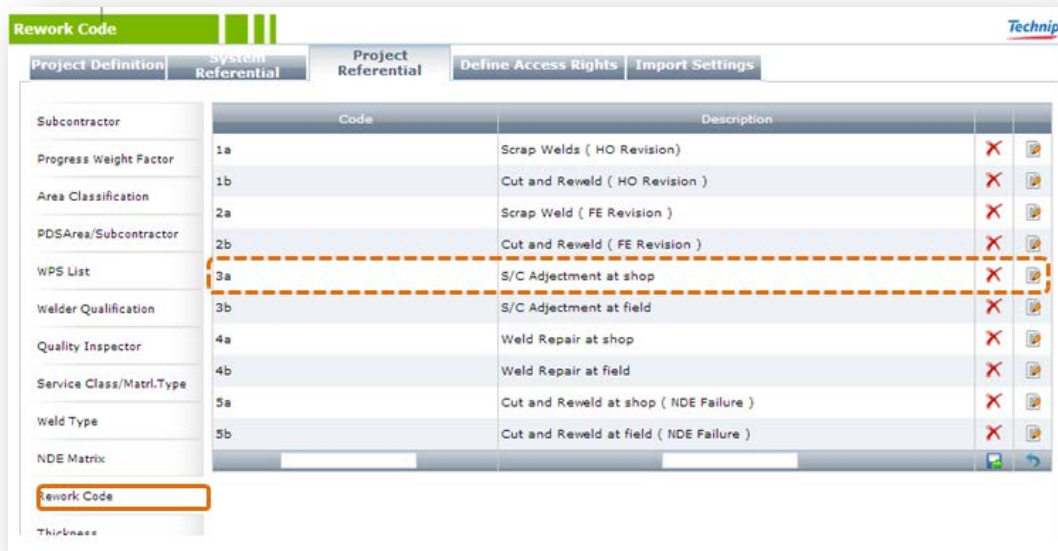
Location	Piping Class	Weld Type	RT%	UT%	MT%	PT%	PM1%	PWHT	HT%	Mat Trac		
Shop	1A0JLX	BW	10	0	0	100	100	N	0	Y		
Field	1A0JLX	BW	10	0	0	100	100	N	0	Y		
Shop	1A0JLX	SW	0	0	0	100	100	N	0	Y		
Field	1A0JLX	SW	0	0	0	100	100	N	0	Y		
Shop	1A0JLX	MW	10	0	0	100	100	N	0	Y		
Field	1A0JLX	MW	10	0	0	100	100	N	0	Y		
Shop	1A0JLX	LUG	0	0	0	100	100	N	0	Y		
Field	1A0JLX	LUG	0	0	0	100	100	N	0	Y		
Shop	1A0JLX	SOF	0	0	0	100	100	N	0	Y		
Field	1A0JLX	SOF	0	0	0	100	100	N	0	Y		

3.10 Rework Code

This screen helps the user to define the different rework codes and their description.

- Rework Code and Description are textboxes
- Both fields are mandatory.
- The user can Add, Edit, Delete at any time during the project phase.
- This code will be displayed to the user in a dropdown list in the weld progress screen.
- There is no limit in number of codes.

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Code	Description
1a	Scrap Welds (HO Revision)
1b	Cut and Reweld (HO Revision)
2a	Scrap Weld (FE Revision)
2b	Cut and Reweld (FE Revision)
3a	S/C Adjectment at shop
3b	S/C Adjectment at field
4a	Weld Repair at shop
4b	Weld Repair at field
5a	Cut and Reweld at shop (NDE Failure)
5b	Cut and Reweld at field (NDE Failure)

3.11 Thickness

The screen helps the user to define criteria to automatically allocate the Thickness for the welds. The criteria are based on Piping class and Diameter of the joint. The thickness information will not be reflected in Isometric. All the fields are mandatory

User has to enter values for the following columns:

- Service Class
- Dia Inch
- Thickness
- Flange Rating

During Spooling import, System will automatically check that a weld has a corresponding thickness defined in the System by checking its piping class and dia-inch. If it does not exist, System will alert the user and import will be stopped. The same service class and diameter should not have two different thicknesses. It should be always unique.

This screen also helps the user to define flange ratings. During bolting import, System will automatically check that a joint has a corresponding flange rating defined in the System by checking diameter. If it does not exist, System will alert the user and import will be stopped.

- Duplicate values for thickness for same piping class and dia-inch are not allowed. Validation will occur
- There should be only 1 Flange Rating entry for a particular combination of piping class, dia inch and thickness.

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ServiceClass	Diainch	Thickness	Flange Rating
15C0KLX	0.5	3.73	1500
15C0KLX	0.75	3.91	1500
15C0KLX	1	4.55	1500
15C0KLX	1.5	5.08	1500
15C0KLX	2	8.74	1500
15C0KLX	3	5.49	1500
15C0KLX	20	12.7	1500
1A0JLX	0.75	2.87	150
1A0JLX	10	4.19	150
1A1A	0.5	3.73	150

3.12 Project Piping Material Class

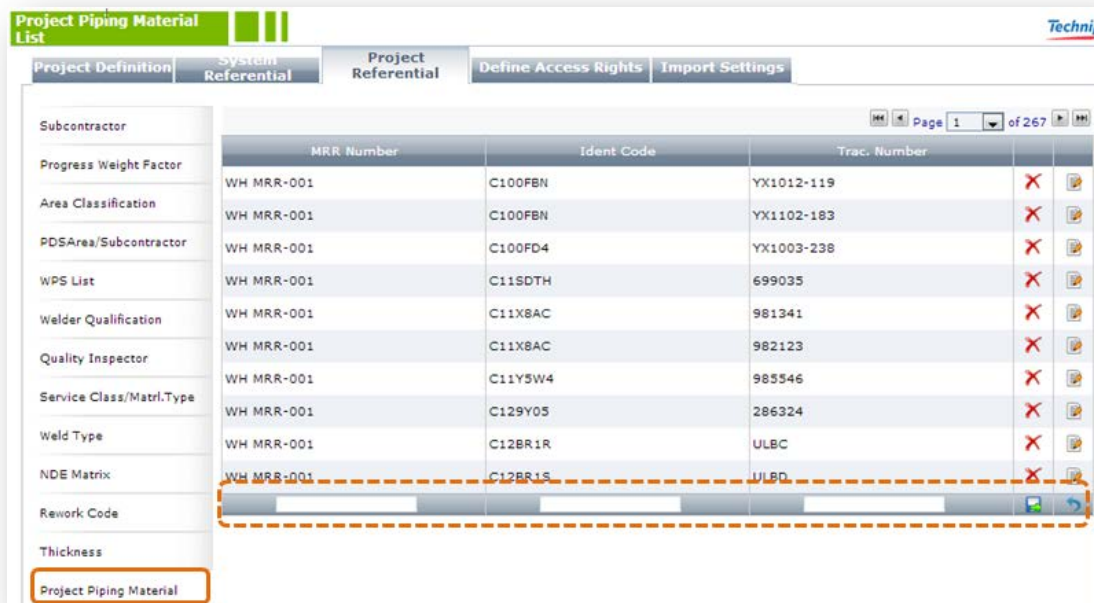
The user need to define the define the following details

- MRR No(Material reception request)
- Ident code
- Trace number (Heat number or File No)

This referential is used in the material traceability screen in progress module. The above details can be imported by using Excel format in the import settings module. It allows the system to validate the codes of trace number vs. Ident code.

Note: The file can be exported from SP Mat

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MRR Number	Ident Code	Trac. Number	Status
WH MRR-001	C100FBN	YX1012-119	X
WH MRR-001	C100FBN	YX1102-183	X
WH MRR-001	C100FD4	YX1003-238	X
WH MRR-001	C11SDTH	699035	X
WH MRR-001	C11X8AC	981341	X
WH MRR-001	C11X8AC	982123	X
WH MRR-001	C11Y5W4	985546	X
WH MRR-001	C129Y05	286324	X
WH MRR-001	C12BR1R	ULBC	X
WH MRR-001	C12BR1S	ULBD	X

3.13 Joint Category Definition

This referential is used for Flange management and Test pack management. The flange joints will be assigned for the below mentioned parameters to manage the Bolt torquing activities. User cannot delete once it is defined for any flange joint.

Each flange joint is associated to a category corresponding to the timing of the joint execution. These categories are directly linked to the test-pack limits and the precommissioning activities.

User has to define the details for the following columns:

- Joint Definition
- Timing
- Category
- Reason
- Coefficient category

Joint definition is the subject to define for a particular joint when it has to be performed. Below are the following example cases

- Joint to be done before Pressure Test
- Joint to be done after Pressure test but before Precomm
- Joint to be done after precomm

Timing is the subject to define the period (Before Pressure test, before Precomm, After Precomm)

Category is the subject to categorize the activities by defining with codes like (X, Y, Z) or (A, B, C) and it varies from project.

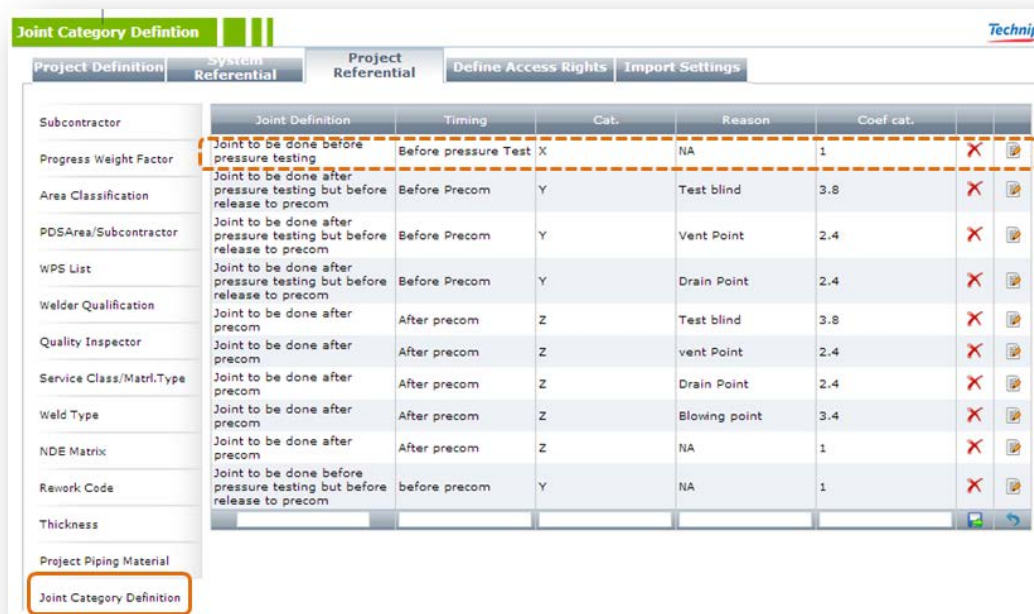
Reason is the subject to define the bolting point like vent point, Test blind point, drain point, blowing point etc.

The coefficient category is the value which is defined based on the punch category and reason. There are some standard coefficient values based on above criteria

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Punch	Reason	Coef punch	content
A	NA	1	1 Torquing
B or B1	Test blind	3.8	3 torquing +2 breaking joint
B or B1	Vent point	2.4	2 torquing +1 breaking joint
B or B1	Drain point	2.4	2 torquing +1 breaking joint
B1	Blowing point	3.4	3 torquing +1 breaking joint

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.



Joint Definition	Timing	Cat.	Reason	Coef cat.		
Joint to be done before pressure testing	Before pressure Test	X	NA	1		
Joint to be done after pressure testing but before release to precom	Before Precom	Y	Test blind	3.8		
Joint to be done after pressure testing but before release to precom	Before Precom	Y	Vent Point	2.4		
Joint to be done after pressure testing but before release to precom	Before Precom	Y	Drain Point	2.4		
Joint to be done after precom	After precom	Z	Test blind	3.8		
Joint to be done after precom	After precom	Z	vent Point	2.4		
Joint to be done after precom	After precom	Z	Drain Point	2.4		
Joint to be done after precom	After precom	Z	Blowing point	3.4		
Joint to be done after precom	After precom	Z	NA	1		
Joint to be done before pressure testing but before release to precom	before precom	Y	NA	1		

3.14 Unit of time reference

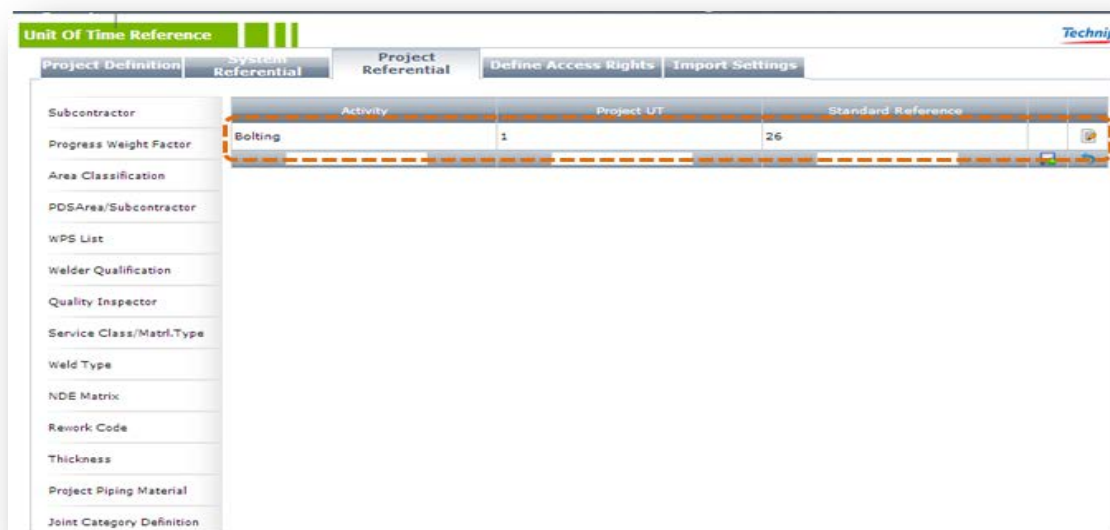
This reference is used to calculate the work volume (Unit of time required to execute the jointing). This referential is used in 'Generate Flange Joint Progress' template. The 'UT Calculation' column is calculated based on the 'Project UT' values.

The user have to define the following details

- Activity
- Project UT (Unit of Time)
- Standard Reference

The equivalent quantities are calculated from the quantity of each bolting operation by using the adequate multiplier factors for Piping Size, Flange Rating and punch category

UT = Reference point quantity x coef Diam x Coef Rating x Coef category



Activity	Project UT	Standard Reference
Bolting	1	26

3.15 Jointer List

User has to define the following columns:

- Jointer Code
- Jointer Description

This referential will be used in the flange management. If the user tries to delete a Jointer which is already assigned to a Flange Joint, the System won't allow the deletion. However, the System will allow modifying the Jointer code and description even if it is assigned to a Flange joint. Duplicate values are not allowed to be entered in Jointer code.

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3.16 Blinding Team

This screen allows the user to define the following details

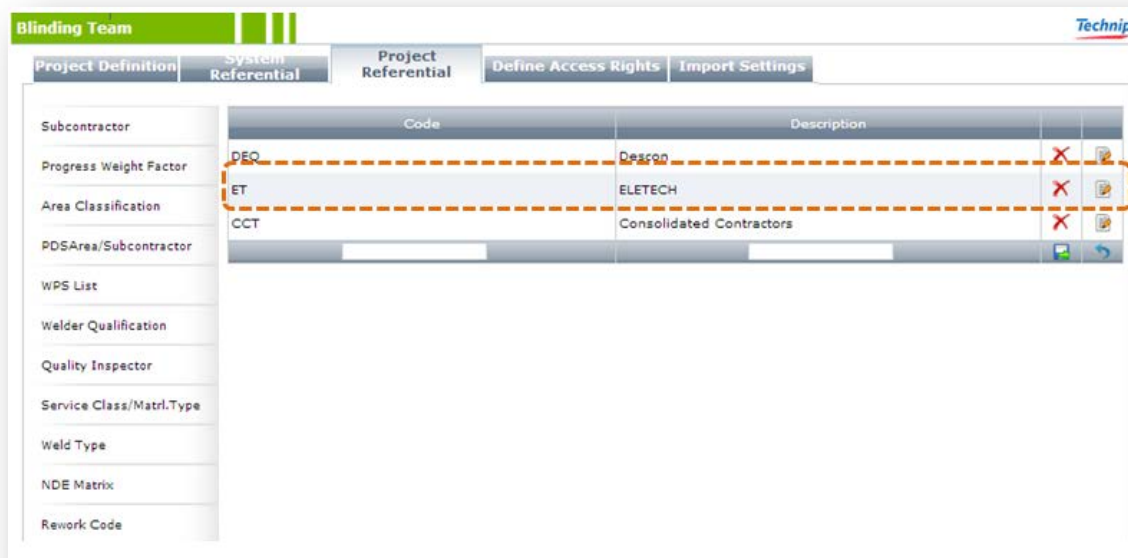
- Code (Blinding team)
- Description

This referential is used in the pressure test module. If the user tries to delete a Blinding Team which is already used in blinding module, the System won't allow the deletion. However, the System will allow modifying the code and description even if it is assigned to blinding module.

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

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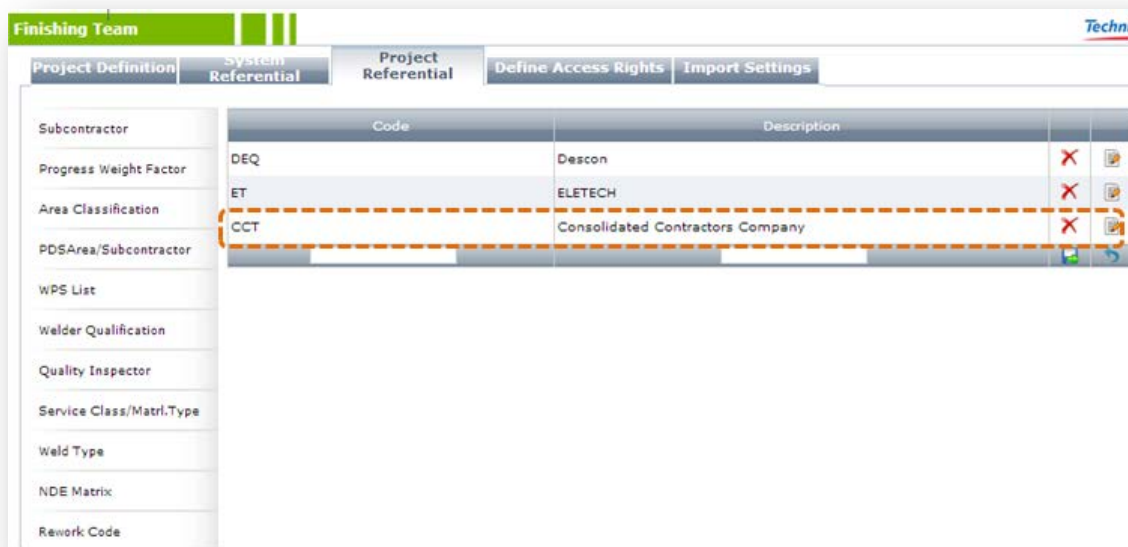
Code	Description	Delete	Edit
DEQ	Descon	X	
ET	ELETECH	X	
CCT	Consolidated Contractors	X	

3.17 Finishing Team

This screen allows the user has to define the following columns:

- Code
- Description

This referential is used for Pressure Test (Item Clearance Module). If the user tries to delete a Finishing Team which is already used in Item clearance module, the System won't allow the deletion. However, the System will allow modifying the code and description even if it is assigned to Item clearance module.



Code	Description	Delete	Edit
DEQ	Descon	X	
ET	ELETECH	X	
CCT	Consolidated Contractors Company	X	

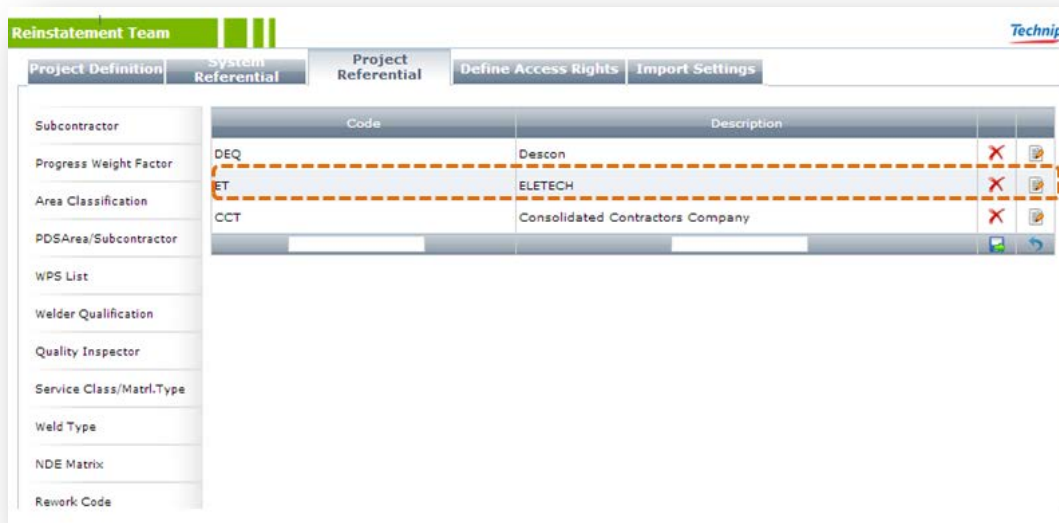
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3.18 Reinstatement Team

User needs to enter values for the following columns:

- Code
- Description

This referential is used for Pressure Test (Reinstatement Module).



Code	Description
DEQ	Descon
ET	ELETECH
CCT	Consolidated Contractors Company

3.19 System

User needs to define the following columns:

- Code
- Description

This referential is used for Test Pack module. The test packs will be assigned to the System which is defined by the project. Both fields are mandatory.

Note: This referential will be integrated with easy plant.

After every change press () to update the details and user can use the icons () & () to delete and edit the details respectively.

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Code	Description	System
00-94-24	Service water distribution	
01-01-01	Other Revamping items Train 1	
01-08-01	Fuel Gas Compressor 01- K0802	
01-82-01	Sea water systems Train 1	
01-84-01	Dry Gas Flare Train 1	
01-92-01	Low Pressure Steam Train 1	
01-93-01	End Flash gas Train 1	
01-94-01	Fresh cooling water supply Train 1	
01-94-03	Fresh cooling water return Train 1	
01-95-02	SA Distribution Train 1	

3.20 Sub System

This screen allows the user to define the sub System which is linked to the System. This referential is used in the test pack module. The user needs to define the following details

- Code
- Description
- System

The column System will be auto drop down.



Code	Description	System
07-83-01	Fixed Spray System	07-83-01
01-94-01	Fresh Cooling Water Supply Train 1	01-94-01
02-92-02	Low Pressure Steam Train 02	02-92-02
02-93-04	FG Mixing Drum Train 2	02-93-04
03-94-07	Fresh Cooling Water Supply Train 3	03-94-07
00-94-24	Service water distribution	00-94-24
01-01-01	Other Revamping Trains 1	01-01-01
01-08-01	Fuel Gas Compressor 01-K0802	01-08-01
01-82-01	Sea Water Systems Train 1	01-82-01
01-84-01	Dry Gas Flare Train 1	01-84-01

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3.21 3.21 Line Checker Team

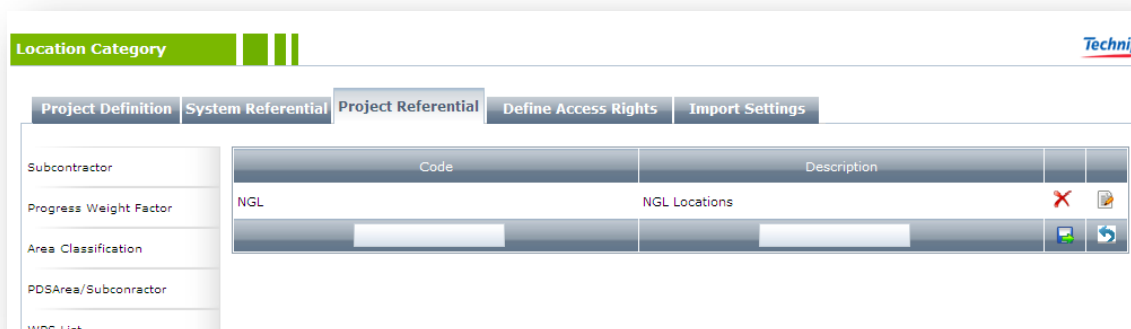
This screen helps the user to define the line checking teams for a specific project. The test packs or Isometrics will be assigned for the line checker team. This referential is used in the test pack module. The user needs to enter the code and description.



Subcontractor	Code	Description		
Progress Weight Factor	AK	Anil Kumar	X	
Area Classification	GS	Govinda Samy	X	
PDSArea/Subcontractor	JC	Jose Coria	X	
WPS List	DH	DongtaoHou	X	
Welder Qualification	FB	Freddie Belmes	X	
Quality Inspector	MA	Mohammed Ansari	X	
Service Class/Matrl.Type	UK	Unnikrishna	X	
Weld Type	JF	Jianguang Feng	X	
NDE Matrix	LM	Lihua Ma	X	
Rework Code	EDA	Efren D. Abba	X	
Thickness				

3.22 Location Category

This referential is used in the spool tracking module. This screen helps the user to define the code and descriptions for the different areas of a project.

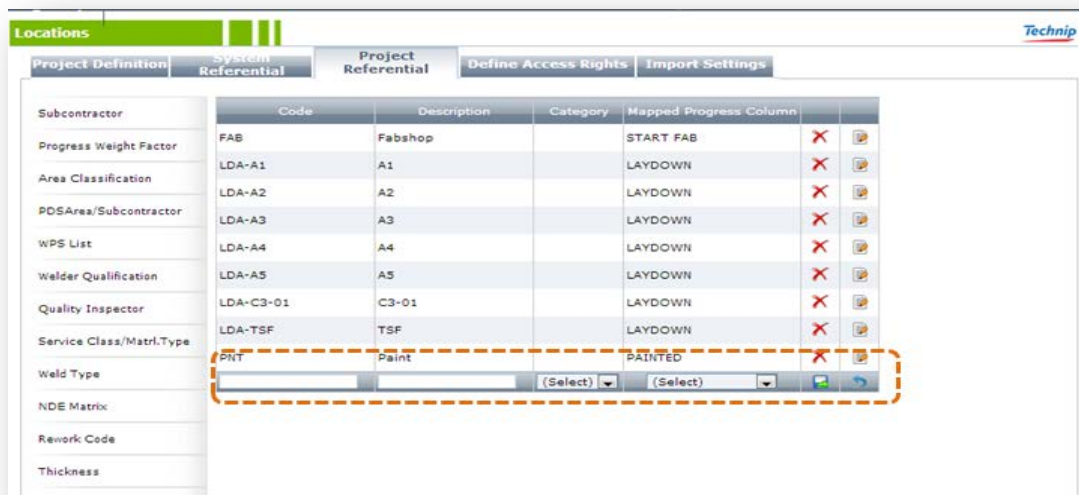


Subcontractor	Code	Description		
Progress Weight Factor	NGL	NGL Locations	X	
Area Classification				
PDSArea/Subcontractor				
WPS List				

3.23 Location

This referential is used in the spool tracking module. This screen helps the user to define the Code, Description, Category, Mapped progress columns. It establishes a relation between spool location and spool status where system will provide an analysis of consistencies.

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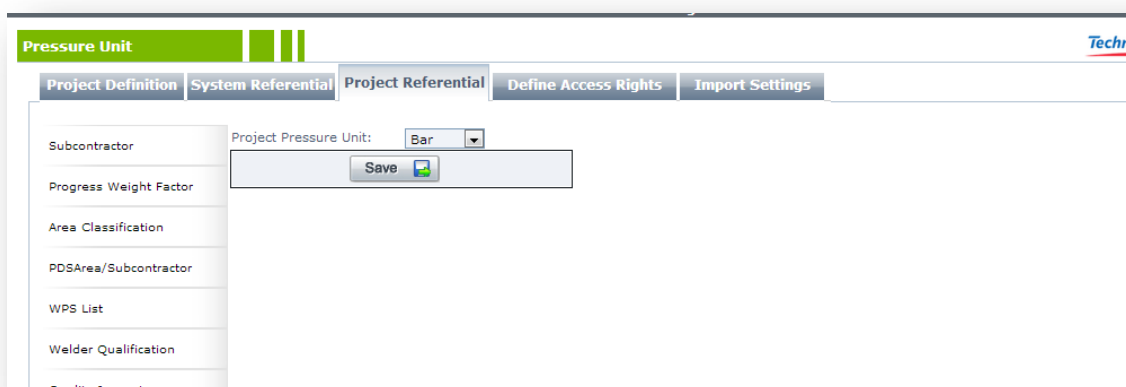


Code	Description	Category	Mapped Progress Column	Status
FAB	Fabshop		START FAB	X
LDA-A1	A1		LAYDOWN	X
LDA-A2	A2		LAYDOWN	X
LDA-A3	A3		LAYDOWN	X
LDA-A4	A4		LAYDOWN	X
LDA-A5	A5		LAYDOWN	X
LDA-C3-01	C3-01		LAYDOWN	X
LDA-TSF	TSF		LAYDOWN	X
PNT	Paint		PAINTED	X

3.24 Pressure Unit

This is a referential for the assigned Pressure unit of the Project. Only one Pressure unit will be assigned to a project and this will be used to in Test Pack Builder Module when adding new Test Packs. Normally the following will be used on any project

- Bar
- Psi



Project Pressure Unit: Bar

Save

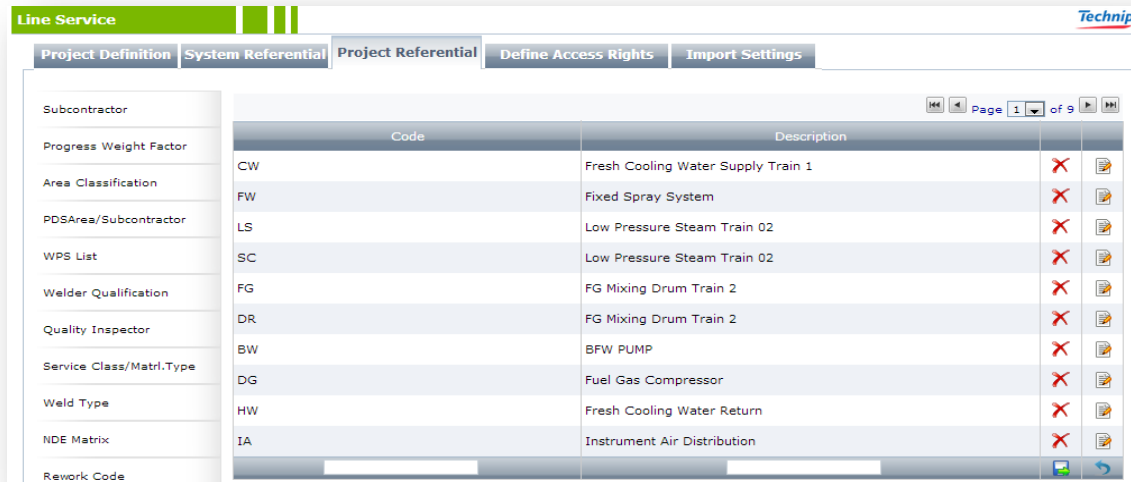
3.25 Line Service

This screen helps the user to define the line service code and description. This is used in the testpack builder module to define the line service of the specific testpack. These codes will be the auto dropdown while creating a test pack.

Note: The Codes must be related to the line service code included in the spooled Iso's data

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Code	Description		
CW	Fresh Cooling Water Supply Train 1	X	
FW	Fixed Spray System	X	
LS	Low Pressure Steam Train 02	X	
SC	Low Pressure Steam Train 02	X	
FG	FG Mixing Drum Train 2	X	
DR	FG Mixing Drum Train 2	X	
BW	BFW PUMP	X	
DG	Fuel Gas Compressor	X	
HW	Fresh Cooling Water Return	X	
IA	Instrument Air Distribution	X	

3.26 Unit Classification

Unit Classification is another level of Hierarchy which stays above Area Classification. This screen helps user to define Units of the project. These Units are shall appear in “Area Classification” definition as drop down while defining Areas.

This Units shall appear in Report Filters as selection criteria while generating reports.



Code	Description		
1	1	X	

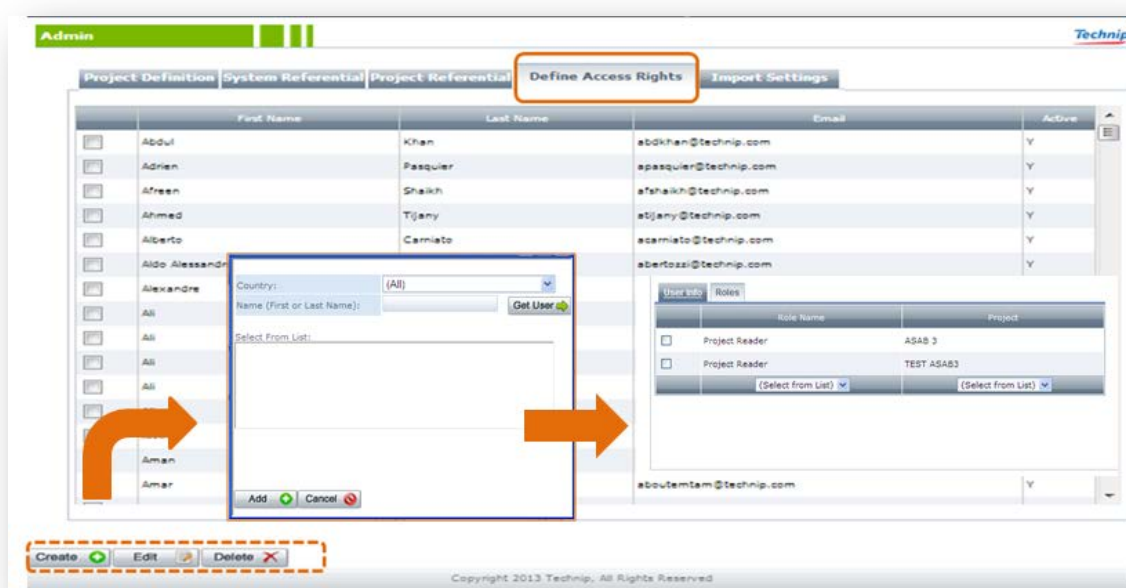
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4. DEFINE ACCESS RIGHTS

This module allows the administrator to create a new user and assign a role as per the project specific. The roles can be edit and deleted at any time. The role is based on project based, user can access to several projects with different rights.

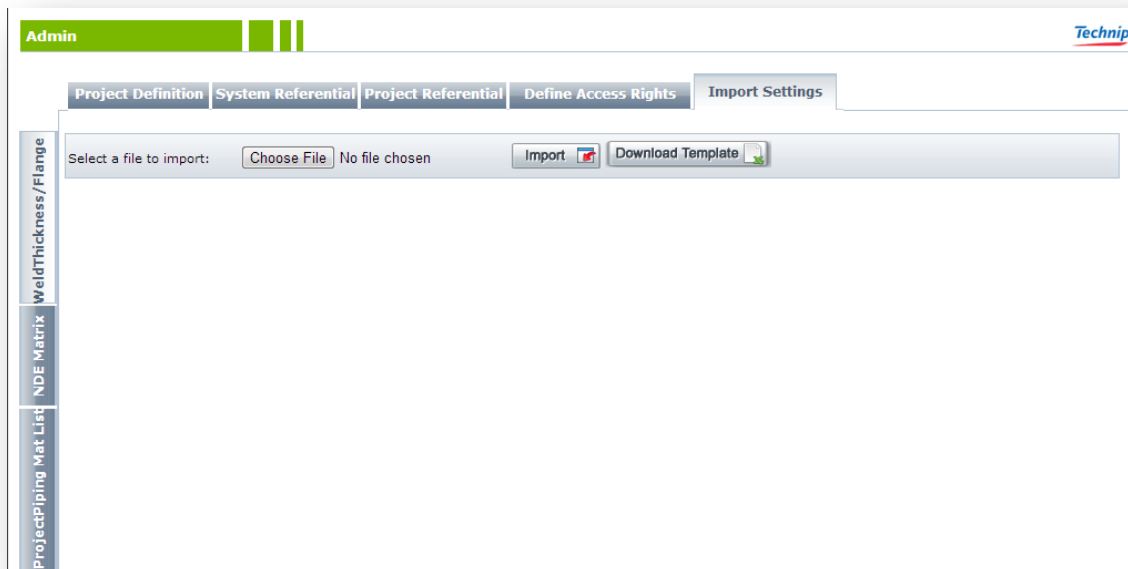


- When assigning a new user a “subcontractor” role, the System will display the list of subcontractors available for this project (from the “subcontractors” referential)
- Subcontractor can view only data related to his PDS area. This is maintained by forcing all the “Subcontractor” dropdown lists in all the screens to be disabled and set the selected value as logged-in subcontractor.

5. IMPORT SETTINGS

This module helps the user to generate an empty excel template in which the user can enter the relevant information and import into the System. There are several sub modules where the user can perform the imports for different kind of information which will be used throughout the process. These are the following sub modules available in this module.

- Weld thickness/Flange
- NDE Matrix
- Project Piping Material List
- Spooling Images
- Spooling Material
- Spooling Class Material



5.1 Weld Thickness/Flange

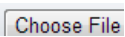
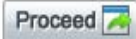
This module helps the user to generate an empty excel template in which the user can enter the weld thickness details and import to the System. The user needs to define the following details in the template

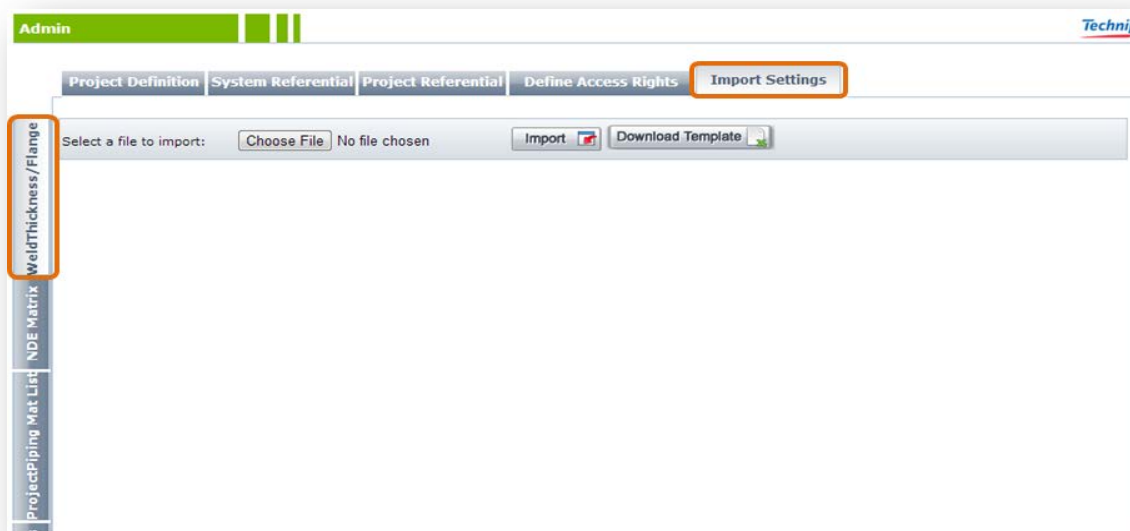
- Piping Class
- Dia Inch
- Thickness
- Flange Rating

The user should choose the option  to generate the empty template.

During the import procedure the System will validate the information provided by the user. In this case the user needs to follow certain rules

- Piping class, Dia-inch, Thickness and Flange Rating are mandatory fields.
- There should be no duplicate entries, i.e., there should be only 1 thickness entry for a particular combination of piping class, dia inch and Flange Rating.
- Piping class entered in the excel file should exist in the project referential.
- Numeric values are only allowed for Dia-inch, Thickness and Flange Rating.

The user needs to click the  option to select the file to import. After this action the user has to click on the  option to import the file. By choosing the option  the data will be saved successfully in the database.



5.2 Import NDE Matrix

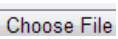
This module helps the user to generate an empty excel template in which the user can enter the NDE Matrix details and import to the System. The user needs to define the following details in the template

- Location
- Piping class
- Weld Type
- RT%
- UT%
- MT%
- PT%
- PMI%
- PWHT
- HT%
- Material Traceability

The user should choose the option  to generate the empty template.

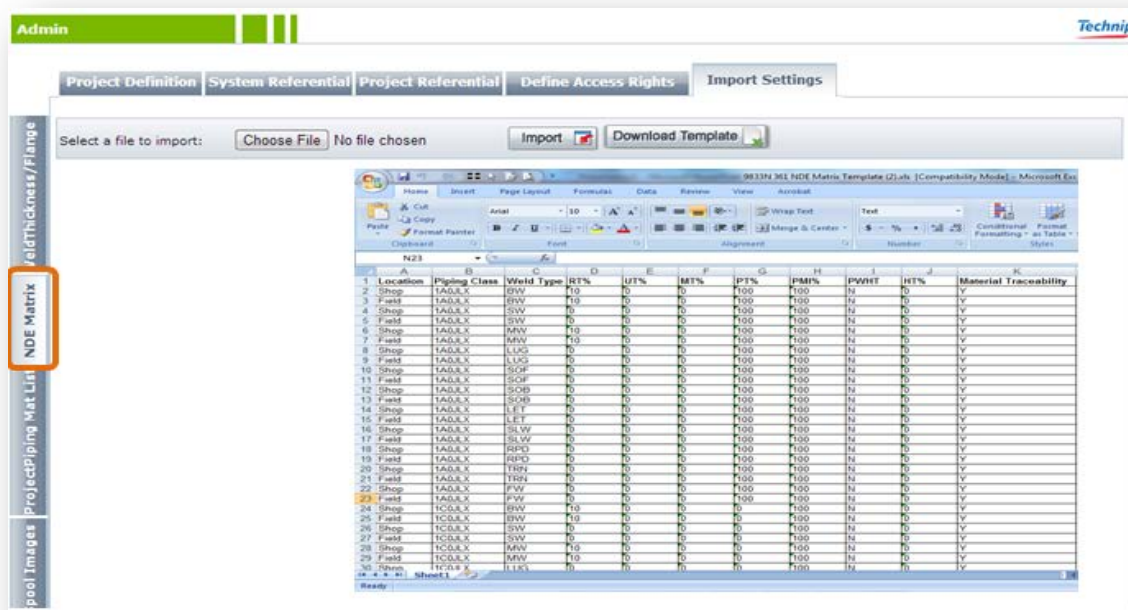
During the import procedure the System will validate the information provided by the user. In this case the user needs to follow certain rules

- Piping class entered in the excel file should exist in the project referential.
- Weld type entered in the excel file should exist in the project referential.
- 'Location' column should only have the values 'Shop' or 'Field'.
- RT, UT, MT, PT, PMI, HT should have values 0, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100.
- PWHT and 'Material Traceability' should only have values 'Y' or 'N'.

The user needs to click the  option to select the file to import. After this action the user has to click on the  option to import the file. By choosing the option

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the data will be saved successfully in the database.



5.3 Import Project Piping Material List

This module helps the user to generate an empty excel template in which the user can enter the Project material details and import to the System. The user needs to define the following details in the template

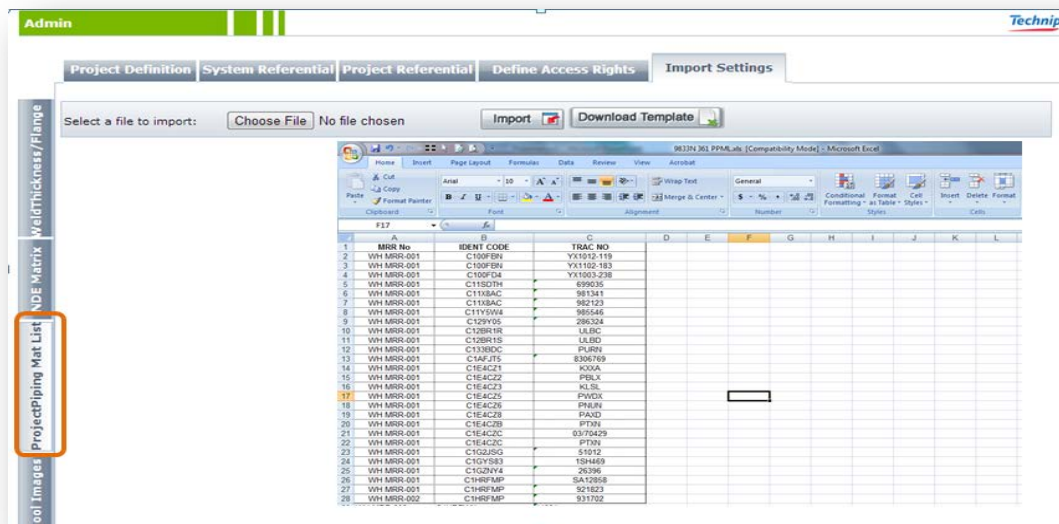
- MRR No
- Ident Code
- Heat No.

The user should choose the option to generate the empty template.

The user needs to click the option to select the file to import. After this action the user has to click on the option to import the file. By choosing the option

the data will be saved successfully in the database.

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The screenshot shows the Technip software interface with the 'ProjectPiping Mat List' window open. The window displays a table with the following data:

MRR No	IDENT CODE	TRAC NO
WH MRR-001	C100FBN	YX1012-119
WH MRR-001	C100FBN	YX1102-183
WH MRR-001	C100FBN	YX1103-178
WH MRR-001	C115DTH	699035
WH MRR-001	C115BAC	981341
WH MRR-001	C115BAC	982123
WH MRR-001	C115VWJ	985546
WH MRR-001	C129F05	286324
WH MRR-001	C129R1R	ULBC
WH MRR-001	C129R1S	ULBD
WH MRR-001	C133BDC	PURN
WH MRR-001	C1AFJ15	8306769
WH MRR-001	C1EAC21	KDAA
WH MRR-001	C1EAC22	PELX
WH MRR-001	C1EAC23	KL SL
WH MRR-001	C1EAC25	PWDX
WH MRR-001	C1EAC26	PELX
WH MRR-001	C1EAC28	PAVD
WH MRR-001	C1EAC2B	PTXN
WH MRR-001	C1EAC2C	0370429
WH MRR-001	C1EAC2C	PTXN
WH MRR-001	C1G2J5G	51912
WH MRR-001	C1GY583	184469
WH MRR-001	C1G2Y14	26396
WH MRR-001	C1HRFMP	SA12058
WH MRR-001	C1HRFMP	921823
WH MRR-002	C1HRFMP	911702

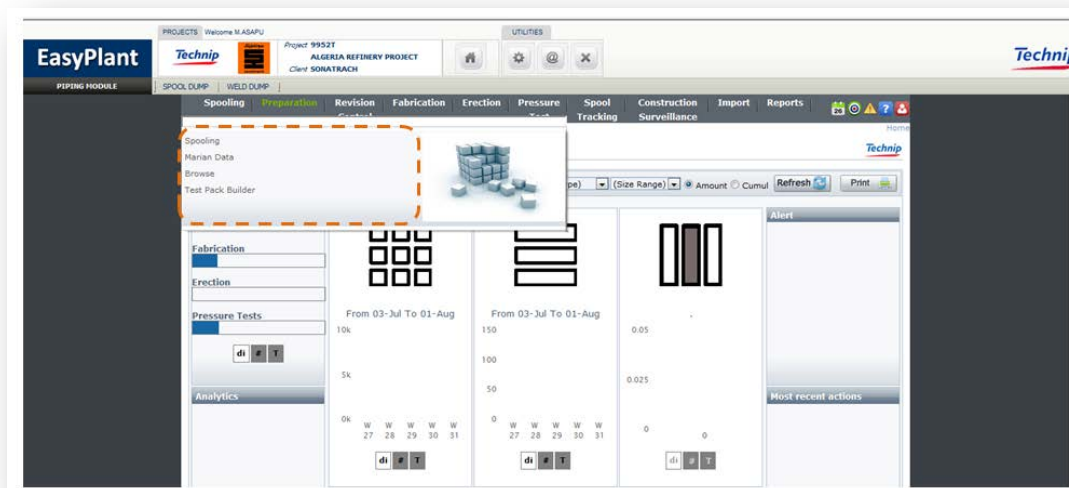
[*Easy Piping Part 2: Preparation*]

PREPARATION MODULE

The Preparation module is organized in 4 different sections. This module allows the user to import all the information's like spooling data, material status data which will be used during the operation phase. This module also helps navigate the status of latest and history of each isometrics. The details of each section are explained below in detail.

- Spooling
- Marian Data
- Browse
- Test pack Builder

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6. SPOOLING

This module is interrelated with the spooling outputs. This module helps the user to import the spooling information (Iso NO, Spool NO, Weld No, Size Etc.) into Easy Piping. Both the New spooled Isometrics and revised spooled Isometrics are able to import in this screen. This information's are exported from the spooling in a predefined .txt or .csv format. This module is having 3 tabs which are namely "Spooling", "Ident Code", "Bolting report" and works with the similar function.

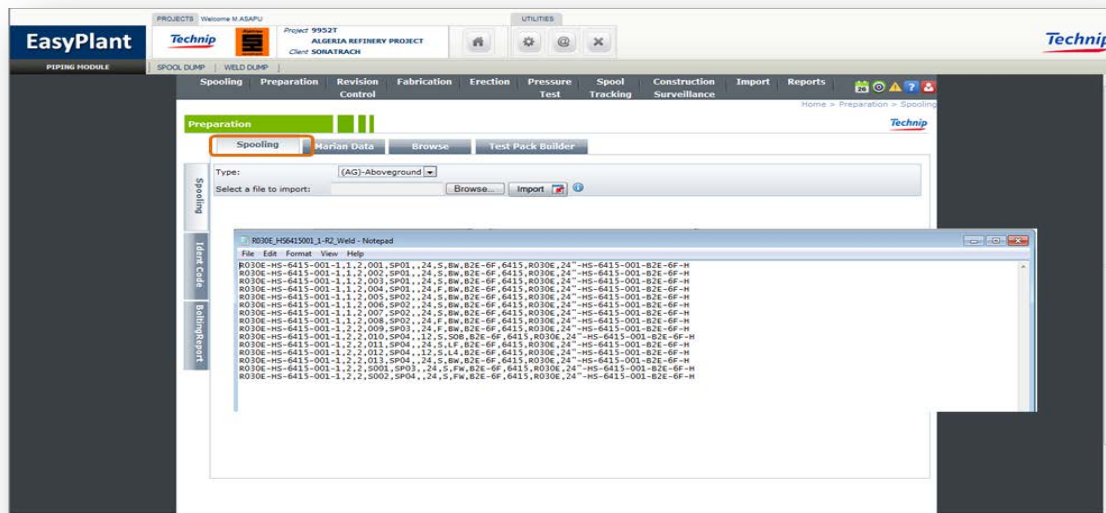
ISOMETRIC NO	SHT	REV	WELD NO	SPL No	Field Pipe	SIZE	WELD CAT	WELD TYPE	SERVICE CLASS	WBU	PDS Area	PIPELINE NO
R030E-HS-6415-001-1	1	2	1	SP01		24 S	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	1	2	2	SP01		24 S	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	1	2	3	SP01		24 S	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	1	2	4	SP01		24 F	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	1	2	5	SP02		24 S	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	1	2	6	SP02		24 S	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	1	2	7	SP02		24 S	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	1	2	8	SP02		24 F	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	2	2	9	SP03		24 F	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	2	2	10	SP04		12 S	SOB	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	2	2	11	SP04		24 S	LF	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	2	2	12	SP04		12 S	L4	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	2	2	13	SP04		24 S	BW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	2	2	S001	SP03		24 S	FW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	
R030E-HS-6415-001-1	2	2	S002	SP04		24 S	FW	B2E-6F		6415 R030E	24"-HS-6415-001-B2E-6F-H	

6.1 Spooling

In the spooling section. The user can use the Icon  and select the file and use the icon  to import all the weld data into Easy Piping. This information will be shown in the browse screens, fabrication and erection modules also.

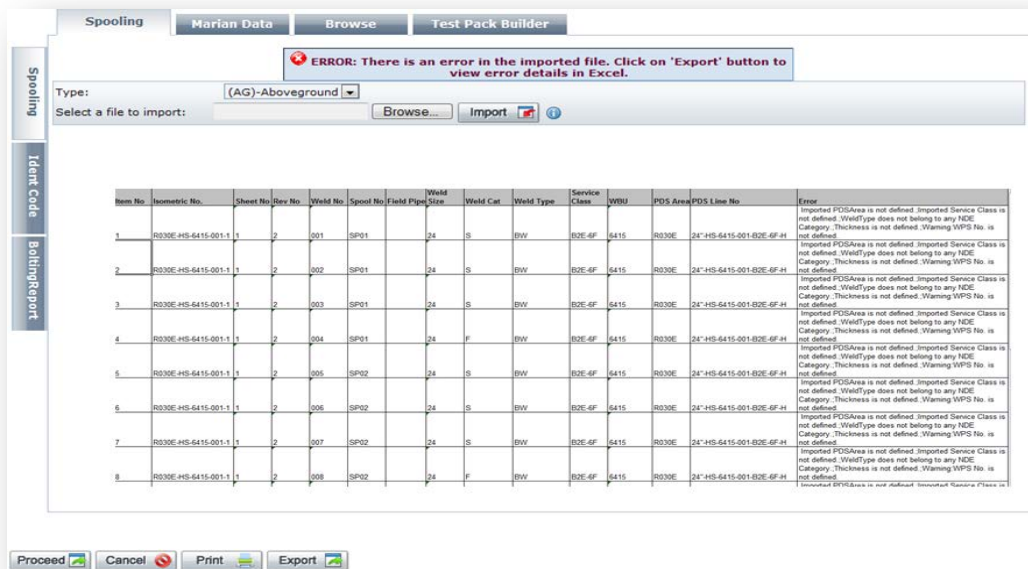
Note: The file size should not exceed 4MB

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Importing Data

Once the user imports the “weldsumm data” into Easy piping. The system will validate the information. If the file consist any errors the system will display an error message and user can able to export the error file. The user can choose the option  to export the file.



Note:

- PSMS will validate whether PDS Area, Service class and Weld Type data that is being imported exists in the corresponding referential. If not, the import process will stop and the user will be notified.
- PSMS will validate whether the ‘thickness’ for the imported welds exist in the ‘Thickness’ referential by checking the weld’s Piping class and dia-inch. If it exists, PSMS will assign the corresponding thickness to the weld, otherwise an error message will appear.
- PSMS will validate whether each imported weld has a corresponding NDE in the NDE matrix referential by checking the weld’s location, piping class and weld type. If it does not exist, an error message will appear against the particular weld.

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- PSMS will check that a WPS exists in the WPS list to cover new joints. If WPS (based on joint dia, thickness and material type) does not exist, PSMS will alert the user with a warning message but import can proceed.
- PSMS will check whether there are different entries for pipeline no. for a particular ISO. If so, the user will be alerted. An ISO should have the same pipeline no. for all spools and welds.
- PSMS will check whether there are different entries for service class for a particular ISO. If so, the user will be alerted. An ISO should have the same service class for all spools and welds.
- As long as the imported file is not validated by PSMS (contains errors), the import process will not be completed.

Once all the errors are rectified by the user the system will display the results.

Preparation						
Spooling Marian Data Browse Test Pack Builder						
Spooling	Summary of Imported ISO's					
	List of Isometrics					
	ISO Number	Revision No	Pipeline No	PDS Area	WBU	Status
	A060A-P-2002-015-1	0	16"-P-2002-015-I3-6F-T	A060A	2002	I3-6F Imported Successfully
Ident Code						
Boiling Report						

Revision Management

If the imported data have no previous revisions (For ex: Importing ISO R0 for the first time) associated to it in the database, then Easy Piping will simply import the data to the database.

In case the imported data is having a previous revision (For ex: Importing ISO R1 or R2 or R3, etc. when R0 already exists in the system) stored in Easy Piping, then the revisions management process will be triggered. The revisions management process will alert the user informing him that a previous revision of the imported data was found, the user can select to proceed to resolve the conflict and take action, or he can choose to cancel the entire import process.

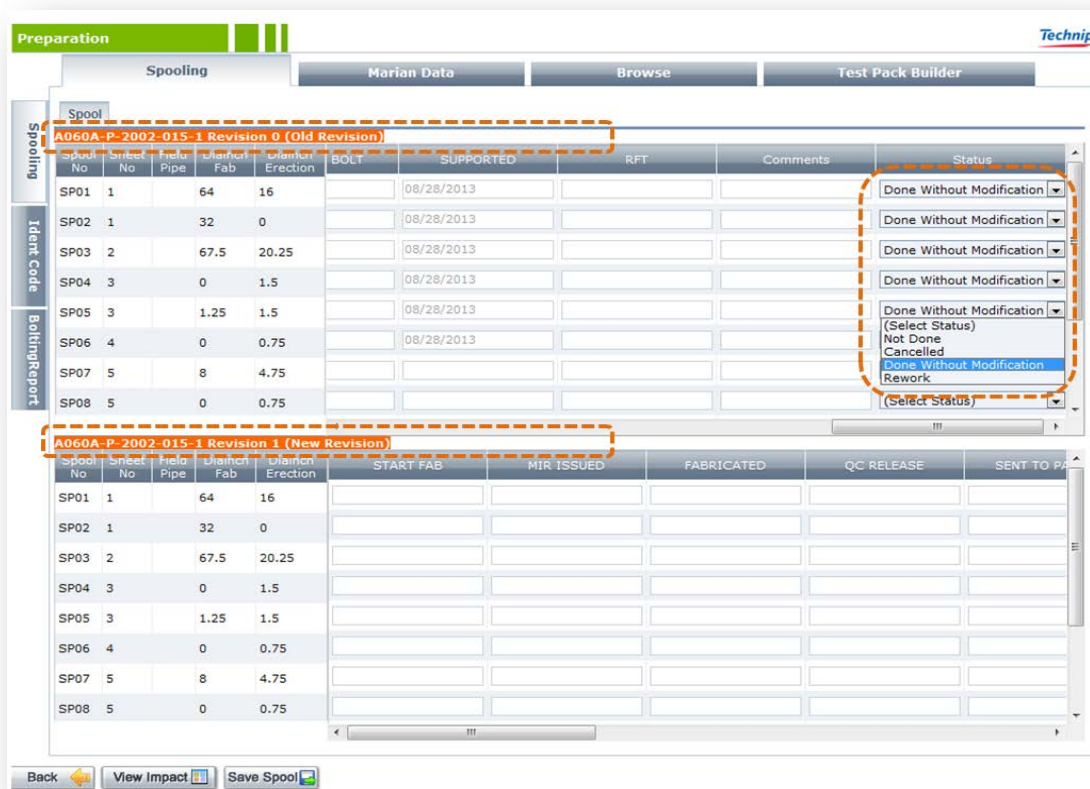
Preparation						
Spooling Marian Data Browse Test Pack Builder						
Spooling	List of Isometrics which has conflict from previous revision. Click on the ISO to resolve.					
	List of Isometrics					
	ISO Number	Revision No	Pipeline No	PDS Area	WBU	Status
	A060A-P-2002-015-1	1	16"-P-2002-015-I3-6F-T	A060A	2002	I3-6F Conflict
Ident Code						
Boiling Report						

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If the user chooses to proceed, Easy Piping will display a list of all the ISOs which are found to have a previous revision in the database, next to each ISO, there will be a “status” column to indicate whether the conflict of each ISO has been resolved or not. The user can click on each of the ISO’s one by one to resolve the conflict.

Each time the user selects an ISO to resolve the conflict, a wizard will be launched to allow the user to visually compare the definition and progress information related to the new version versus the previous version of each spool. The wizard starts by comparing the prefabrication and erection in the same screen, and if necessary the welds related to each ISO. In each step, the system will show the original progress data on the top, versus the new data in the bottom.

When comparing between a spool of an older revision and a spool of a newer revision, the older ones will appear on top with all progress information related to “Prefabrication” and “Erection” on the same grid, while in the bottom the revised spool will appear with blank progress information.



the progress information will either be copied to the newer revision of the spool or not, as explained in the table below:

	Status	Action	Check Weld?
	Not Done	The user will select this value if there is no progress information. No action to be taken in this case, the newer version of the spool will remain empty, same applies to the weld	No
	Cancelled	The user will select this value if the spool has been cancelled. No action to be taken in this case, the spool will not appear in the newer revision.	No

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	Done, but revised w/o modification	This value refers to Spools which are fabricated in the older revision, but the newer version of the spool hasn't changed, so the progress information of the older revision will be copied to the newer revision. Weld progress information of the older revision will also be copied to welds in the newer revision. The 'Spool Revision No.' will remain the same.	No.
	Rework	If the user selects this value, it means that the spool in the older revision has been fabricated (or fabrication has started) but it has changed in the newer revision, which will result in rework. In this case, the system will copy the "Fabrication Start Date", "Sent to Paint" and "Paint" from the older revision to the new revision. The system will force the user to go to the level of the weld, and validate welds one by one. The 'Spool Revision No.' will also be changed to the new ISO rev no.	Yes

Concerning the weld progress, the "status" column will also apply based on the following values (this is only considered if the spool is "Rework"):

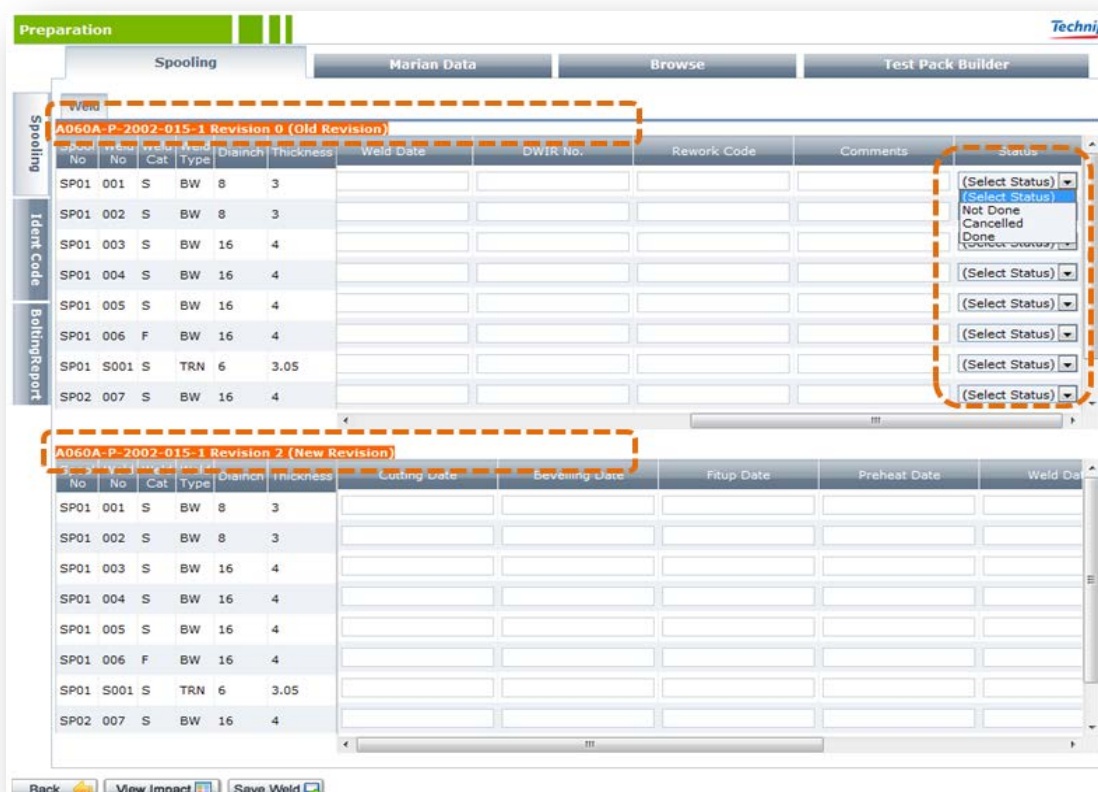
Status	Action
--------	--------

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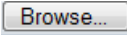
	Not Done	The user will select this value if there is no progress information. No action to be taken in this case, the newer version of the weld will remain empty.
	Cancelled	The user will select this value if the weld has been cancelled. No action to be taken in this case, the weld will disappear in the newer revision.
	Done, but revised w/o modification	This value refers to welds which are fabricated in the older revision, but the newer version of the weld hasn't changed, so the progress information of the older revision will be copied to the newer revision.

When the user has finished resolving the conflict of a specific ISO, Easy Piping will redirect the user back again into the first screen which displays the list of ISO's to be resolved, the status of the resolved ISO's will be updated to indicate that the conflict was resolved. This process will keep going until the user has finished resolving the conflict for all the ISO's. The user can then finish the process and complete the import. Once the import process is successfully finishes, the system will assign automatically all the new revision numbers to the spools impacted.



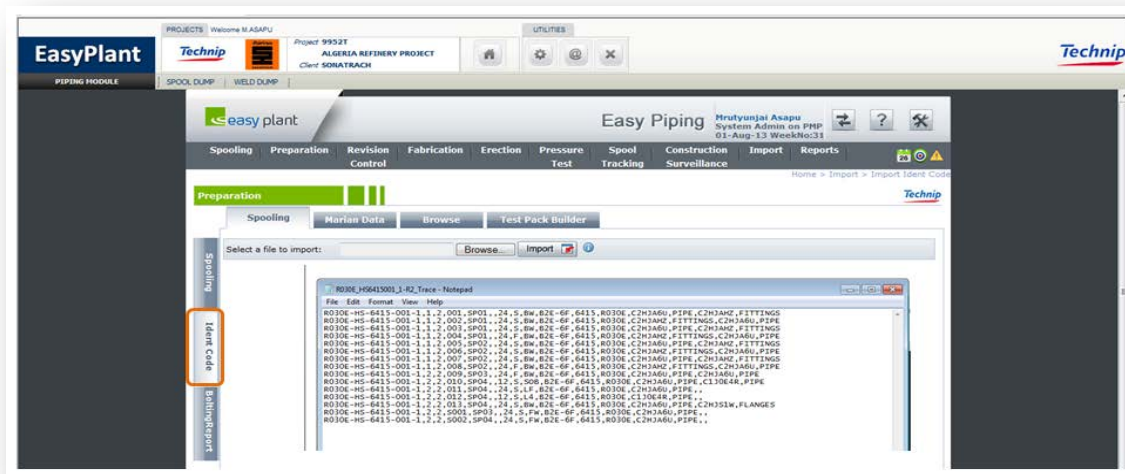
This revision management can be done manually without .txt file. The user can edit the Isometrics by using "Manual revision management" which is generally used by site spooling team. The detailed process is explained in the [Para 6.5.3](#)

6.2 Ident Code

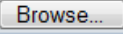
In the Ident code section, the user can use the Icon  and select the file and use the Icon  to import all the trace files from spooling output .txt files into Easy Piping. This Information will be shown in the material traceability screen. ([Refer to Para 7.2](#))

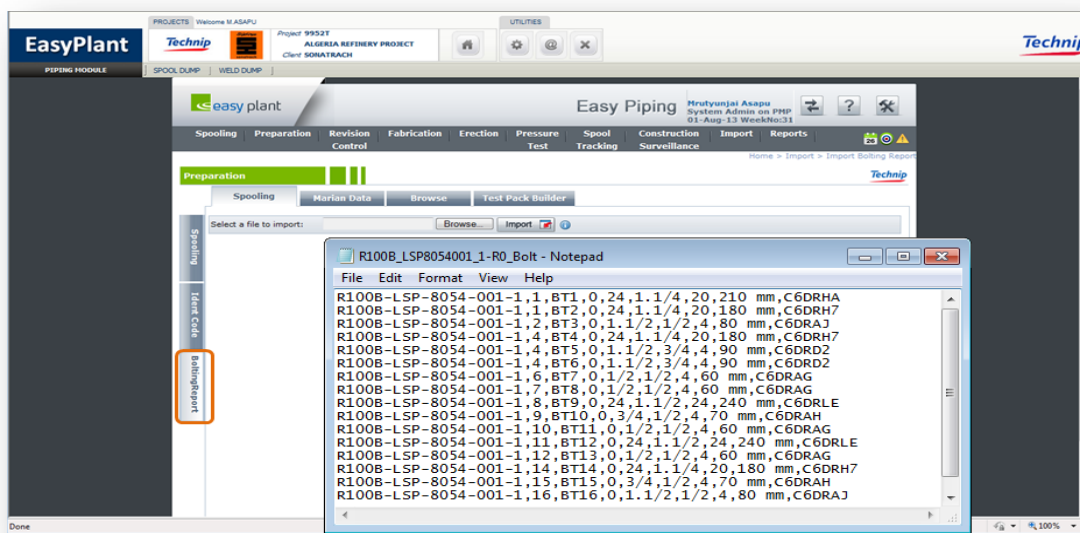
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Note: The file size should not exceed 4MB



6.3 Bolting Report

In the bolting report section, the user can use the Icon  and select the file and use the Icon  to import all the Bolting records from spooling outputs .txt files into Easy Piping. This information will be shown in the Erection progress screen.



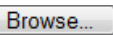
6.4 Marian Data

This module helps the user to import the material status “Marian Data” and this file is generated from SP-MAT (Marian). SP-MAT forecast runs and reservation runs are executed by spools. The export file from Marian will have the information of material status and weight of each spool. This should be an excel format with specified columns which is designed for Easy Piping Import.

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The user can use the Icon  and select the file and use the Icon  to import the Marian data into Easy Piping. This information will be seen in the browse screens, where the weight of each spool, forecast dates of material arrival and status of the spools are shown as

- **ALREADY COMPLETED:** This status states that all the materials required for a particular spool have been already issued or reserved before the forecast run. SP-MAT will consider that materials are reserved until they are on a MIR and the MIR is posted. Some of the reserved materials might therefore have been already given to contractor (MIR not yet posted) whereas some have not yet been given to contractor (not yet on a MIR). The date of availability for such spools will be determined based on the latest availability of its items: items already on MIR will be considered available at the issue date attached to the MIR whereas materials not yet on MIR will be considered available at the date of the Forecast run.

Filter											
(PDS Area)	Search Line No.	(AG)-Aboveground	Search Priority	(Serviceclass)							
S2P2-27-HCX-1003-01	(Subcontractor)	(Area Classification)	Search	Cancel							
ISO Number	Rev	Service Class	Type	WBU	PDS Area Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type		
S2P2-27-HCX-1003-01	0A	31036	AG	200	S2P2 27-HCX-1003-31036-44"-N	28440.8	ELECO	ELECO	CS NACE		
Spool No.	Spool Rev	ISO Sheet	Field Pipe	Shop Dia	Field Dia	LB/SB	Material Availability	ETA	Spool Weight	Priority	Progress Status
SP01	0A	1		88.75	44.75	LB	ALREADY COMPLETE	12/24/2011	6615.32	05/01/2011	RFT
SP02	0A	1		0.75	0	SB	ALREADY COMPLETE	12/24/2011	2.16	08/01/2011	RFT
SP03	0A	2		0	44	LB	ALREADY COMPLETE	12/24/2011	5455.83	05/01/2011	Supported
SP04	0A	2		0	44	LB	ALREADY COMPLETE	12/24/2011	5455.83	05/01/2011	Supported
SP05	0A	2		4	44	LB	ALREADY COMPLETE	12/24/2011	5455.83	05/01/2011	Supported

- **NOW COMPLETE:** This status states that all the materials of a required spool are available at the date of forecast run.

Filter											
(PDS Area)	Search Line No.	(AG)-Aboveground	Search Priority	(Serviceclass)							
S1S9-28-FW-9648-03	(Subcontractor)	(Area Classification)	Search	Cancel							
ISO Number	Rev	Service Class	Type	WBU	PDS Area Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type		
S1S9-28-FW-9648-03	0.	18100	AG	300	S1S9 28-FW-9648-18100-3"-N	418.75	ELECO	ELECO	CS		
Spool No.	Spool Rev	ISO Sheet	Field Pipe	Shop Dia	Field Dia	LB/SB	Material Availability	ETA	Spool Weight	Priority	Progress Status
SP01	0.	1		11	0	LB	NOW COMPLETE	12/24/2011	142.09	09/01/2011	RFT
SP02	0	4		13	0	LB	NOW COMPLETE	12/24/2011	47.65	09/01/2011	RFT
SP03	0	5		17	0	LB	NOW COMPLETE	12/24/2011	152.68	09/01/2011	RFT
SP04	0	7		14	0	LB	NOW COMPLETE	12/24/2011	76.33	09/01/2011	RFT
Spool No	ISO Sheet	Field Pipe	Weld No.	Weld Size	Thickness	Weld Category	Weld Type				
						S-Shop Weld	BW-Butt Weld				

- **TO COMPLETE:** This status states that the forecast run has identified all the materials but some materials are not available in the ware house and the forecast run has found all the materials in the Purchase Order(PO) with the last material date available.

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Spooling Marian Data Browse Test Pack Builder

Filter (PDS Area) Search Line No. (AG)-Aboveground Search Priority (Serviceclass)

SSCL-28-HC-8071-01 (Subcontractor) (Area Classification) Search Cancel

ISO Number	Rev	Service Class	Type	WBU	PDS Area Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type
SSCL-28-HC-8071-01	1	13460	AG	300	SSCL 28-HC-8071-13460-3"-N	197.62	ELECO	ELECO	SS NACE

Spool No.	Spool Rev	ISO Sheet	Field Pipe	Shop Dia	Field Dia	LB/SB	Material Availability	ETA	Spool Weight	Priority	Progress Status
SP01	1	1	23	9	LB	TO COMPLETE	11/19/2011	33.23	11/01/2011	RFT	
SP02	1	2	24	0	LB	TO COMPLETE	11/19/2011	66.13	11/01/2011	RFT	

Spool No. ISO Sheet Field Pipe Weld No. Weld Size Thickness Weld Category Weld Type

S-Shop Weld BW-Butt Weld

- INCOMPLETE: This status states that all the materials of a particular spool are not available and there is no date for full availability of material of that spool.

Spooling Marian Data Browse Test Pack Builder

Filter (PDS Area) Search Line No. (AG)-Aboveground Search Priority (Serviceclass)

SSCD-28-HCX-8167-01 (Subcontractor) (Area Classification) Search Cancel

ISO Number	Rev	Service Class	Type	WBU	PDS Area Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type
SSCD-28-HCX-8167-01	0	93460	AG	300	SSCD 28-HCX-8167-93460-2"-IS	332.94	ELECO	ELECO	SS NACE

Spool No.	Spool Rev	ISO Sheet	Field Pipe	Shop Dia	Field Dia	LB/SB	Material Availability	ETA	Spool Weight	Priority	Progress Status
SP01	0	1	15	5	LB	INCOMPLETE	28.27			RFT	
SP02	0	1	6.25	0	LB	INCOMPLETE	28.46			RFT	
SP03	0	2	8	2	LB	ALREADY COMPLETE	12/28/2011	12.44		RFT	
SP04	0	2	8	2	LB	INCOMPLETE	50.36			RFT	
SP05	0	2	8	2	LB	INCOMPLETE	39.48			RFT	

The Import file should not contain any inappropriate data. If the file is having any errors, the system will alert the user with the appropriate error message and by highlighting the error in red color. If any of the imported data will overwrite the data in the database, it will be highlighted in yellow to alert the user.

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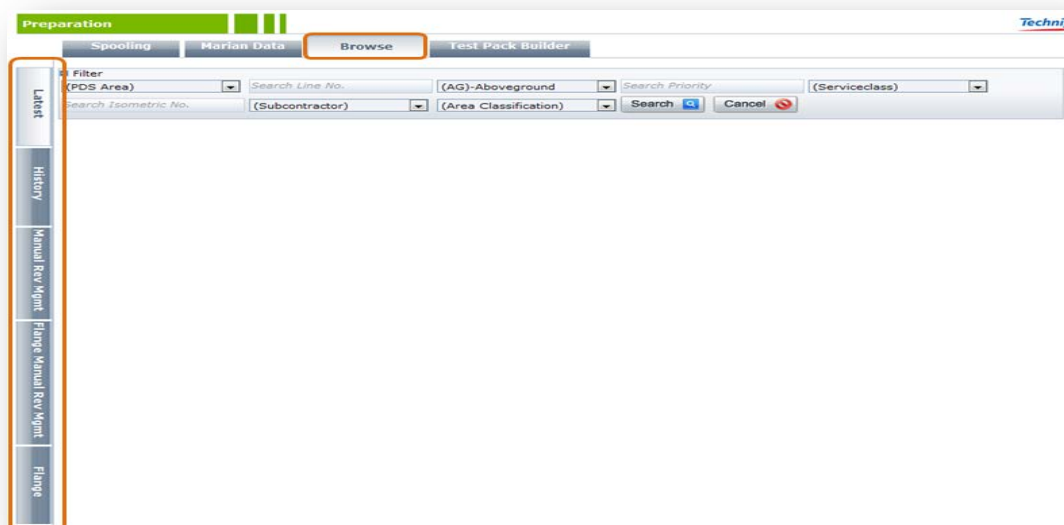
The screenshot shows the 'Easy Piping' software interface. The 'Preparation' tab is active, and the 'Browse' sub-tab is selected. A table is displayed with columns for FAH CODE, Run Number, Run Date, Unit, Area, Line, sheet, Iso No, Spool no, Issue Stat, Weight, Completion Status, and Completion Date. The table contains 26 rows of data, including run numbers, dates, units, areas, lines, sheets, ISO numbers, spool numbers, issue status, weight, completion status, and completion dates.

FAH CODE	Run Number	Run Date	Unit	Area	Line	sheet	Iso No	Spool no	Issue Stat	Weight	Completion Status	Completion Date
1	RUN027-9	1	7/3/2012	4900	4004R	L504090509L_01	01	4004R_L504090509L_01	SPO2	03	1.83 NOW COMPLETE	7/3/2012
2	RUN027-9	1	7/3/2012	4900	4004R	L504090509L_01	01	4004R_L504090509L_01	SPO3	03	3.92 NOW COMPLETE	7/3/2012
3	RUN027-9	1	7/3/2012	4900	4004R	L504090509L_01	01	4004R_L504090509L_01	SPO4	03	2.21 NOW COMPLETE	7/3/2012
4	RUN027-9	1	7/3/2012	4900	4004R	L504090509L_02	02	4004R_L504090509L_02	SPO2	03	2.21 NOW COMPLETE	7/3/2012
5	RUN027-9	1	7/3/2012	4900	4004R	L504090509L_02	02	4004R_L504090509L_02	SPO3	03	1.21 NOW COMPLETE	7/3/2012
6	RUN027-9	1	7/3/2012	4900	4004R	L504090509L_02	02	4004R_L504090509L_02	SPO4	03	2.13 NOW COMPLETE	7/3/2012
7	RUN027-9	1	7/3/2012	4900	4004R	L5040905320G_07	07	4004R_L5040905320G_07	SPO1	03	432.13 NOW COMPLETE	7/3/2012
8	RUN027-9	1	7/3/2012	4900	4004R	L5040905320G_07	07	4004R_L5040905320G_07	SPO2	03	4.15 NOW COMPLETE	7/3/2012
9	RUN027-9	1	7/3/2012	4900	4004R	L5040905320G_07	07	4004R_L5040905320G_07	SPO3	03	9.03 NOW COMPLETE	7/3/2012
10	RUN027-9	1	7/3/2012	4900	4004R	L5040905320G_07	07	4004R_L5040905320G_07	SPO4	03	2.89 NOW COMPLETE	7/3/2012
11	RUN027-9	1	7/3/2012	4900	4031R	L504090512L_29	29	4031R_L504090512L_29	SPO1	0301	2.64 NOW COMPLETE	7/3/2012
12	RUN027-9	1	7/3/2012	4900	4031R	L504090512L_29	29	4031R_L504090512L_29	SPO2	0301	1.09 NOW COMPLETE	7/3/2012
13	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_01	01	6001R_L508860543L_01	SPO1	03	80.98 NOW COMPLETE	7/3/2012
14	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_01	01	6001R_L508860543L_01	SPO2	03	2.22 NOW COMPLETE	7/3/2012
15	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_01	01	6001R_L508860543L_01	SPO3	03	2.5 NOW COMPLETE	7/3/2012
16	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_01	01	6001R_L508860543L_01	SPO4	03	3.42 NOW COMPLETE	7/3/2012
17	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_02	02	6001R_L508860543L_02	SPO1	03	99.28 NOW COMPLETE	7/3/2012
18	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_02	02	6001R_L508860543L_02	SPO2	03	2.22 NOW COMPLETE	7/3/2012
19	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_02	02	6001R_L508860543L_02	SPO3	03	1.09 NOW COMPLETE	7/3/2012
20	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_03	03	6001R_L508860543L_03	SPO1	03	266.22 NOW COMPLETE	7/3/2012
21	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_03	03	6001R_L508860543L_03	SPO2	03	1.09 NOW COMPLETE	7/3/2012
22	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_03	03	6001R_L508860543L_03	SPO3	03	1.83 NOW COMPLETE	7/3/2012
23	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_03	03	6001R_L508860543L_03	SPO4	03	4.67 NOW COMPLETE	7/3/2012
24	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_04	04	6001R_L508860543L_04	SPO1	03	119.66 NOW COMPLETE	7/3/2012
25	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_04	04	6001R_L508860543L_04	SPO2	03	2.22 NOW COMPLETE	7/3/2012
26	RUN027-9	1	7/3/2012	5100	6001R	L508860543L_04	04	6001R_L508860543L_04	SPO3	03	2.22 NOW COMPLETE	7/3/2012

6.5 Browse

This module helps the user to navigate in a hierarchical manner between the Isometrics, Spools and welds. It will allow the user to view all the information related to each category (Iso, Spool and weld). The user with proper rights like system admin or project admin can also edit some information in this screen. This module is organized in the following tabs as below mentioned.

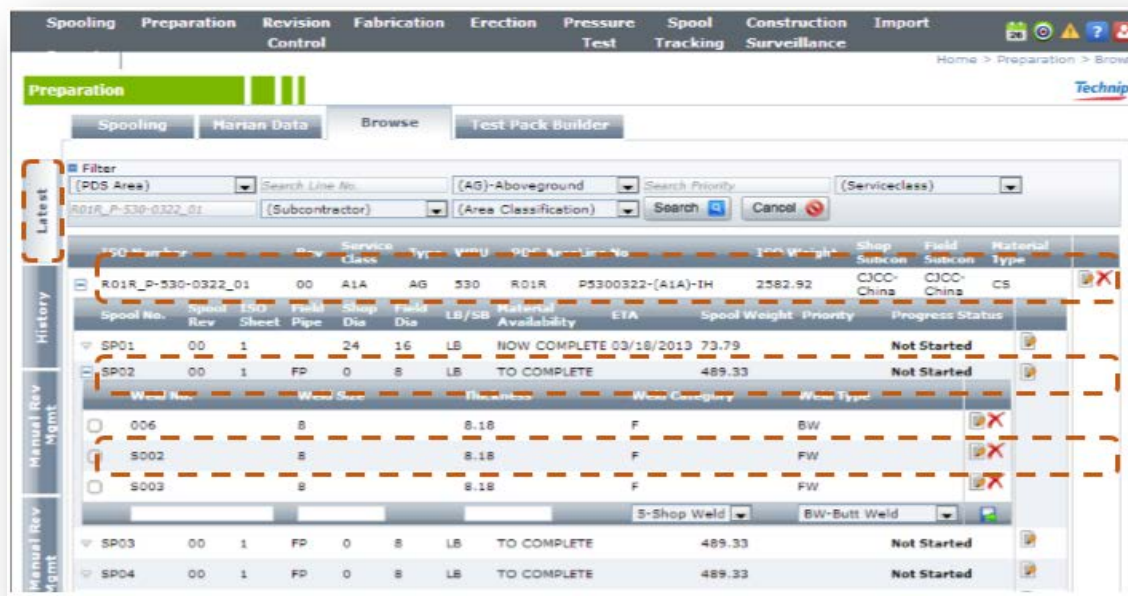
- Latest
- History
- Manual Revision Management



6.5.1 Browse Latest

This module helps the user to search different parameters from the filters like PDS area, Line No, Isometric No, Service class, Subcontractor, Area Classification etc.

In this menu, the search results will display the latest information in a hierarchical format starting from the ISO, the user can select to expand and explore a specific Iso and related spools also will appear, the user can roll down to the level of the weld to explore the welds belonging to a specific Spool.



This module helps the user also to edit some information like

- -Type
- -Shop subcontractor
- -Field subcontractor
- -Material type

The above fields can be edited at any time during the project phase. The shop and field subcontractor can be edited on the ISO level as mentioned in the [\(Refer to Para 3.4\)](#)

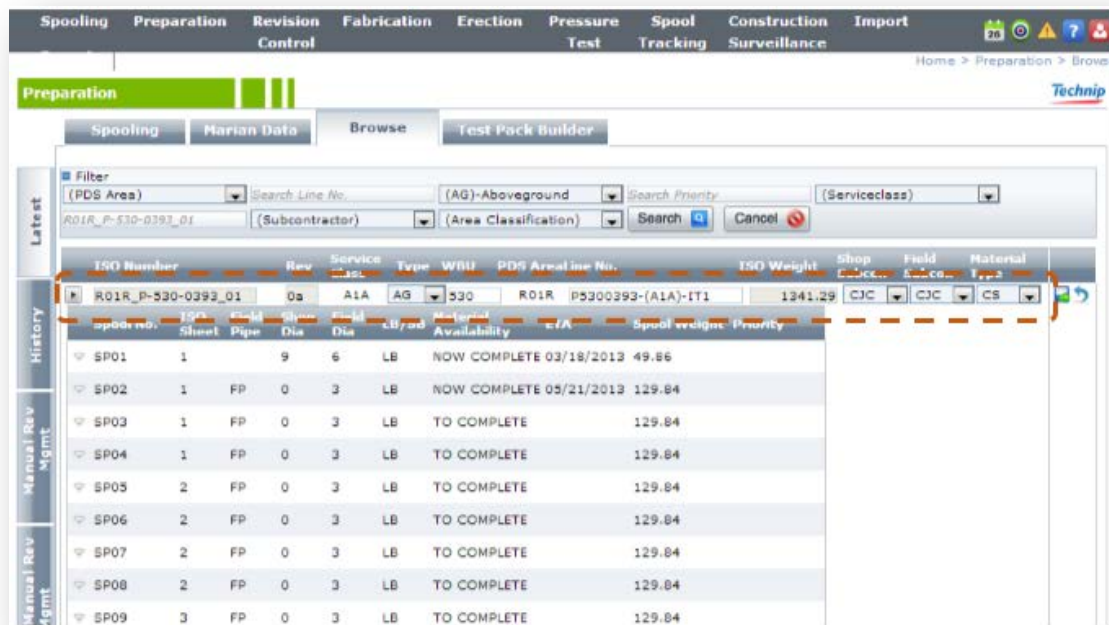
Note:

- All the welds in a given spool cannot be deleted. A spool should have at least 1 weld.
- When a weld is progressed, the user cannot delete the particular weld, spool or ISO, i.e., if an ISO or spool has even 1 weld that is progressed, then it cannot be deleted. Only welds that are not progressed can be deleted.
- When a weld is progressed, then that particular weld cannot be edited.
- When an ISO is deleted, PSMS asks the user whether it is a Soft Delete or Hard Delete.

Soft delete means that the deleted ISO will go to "Browse History" with a status="soft deleted" and comments="date/by (whom deleted)".

Hard delete means that the record (ISO, Spool, and Weld) will be deleted permanently from the database.

The user can use this  icon to edit the information and  to save the information.



6.5.2 Browse History

This module helps the user to view the history of an Isometric with his various revisions. In this module the user cannot edit any information. The ISO's that have been soft deleted from the "Browse Latest" will appear in the "Browse History". the search results will display all the information in a hierarchical format starting from the ISO, the user can select to expand and explore a specific Iso and related spools also will appear, the user can roll down to the level of the weld to explore the welds belonging to a specific Spool.

In terms of the information displayed in the screen, the same columns of "Browse Latest Revision" will appear in "Browse History", in addition to that, an extra field "Comments" will appear in the level of the spool and weld. The comments column can have different information depending on the following:

- In a revised ISO/Spool/Weld, the "comments" column will display information on the status of each spool and weld when revised (not done, done without modifications, rework, cancelled).
- In a soft-deleted ISO, the "comments" column will display information related to who/when the ISO was soft-deleted.

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Preparation

Spooling Marian Data Browse Test Pack Builder

Filter (PDS Area) Search Line No. (AG)-Aboveground Search Priority (Serviceclass)

4002A_MV04020298_01 (Subcontractor) (Area Classification) Search Cancel

ISO Number

4002A_MV04020298_01

Rev	Service Class Type	WBU	PDS Area	Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type
1F4	1C0KLX	AG	4200	4002A MV04020298-24"(1C0KLX)--	9224.41	DEBAJ	ELETECH	SS
1F3	1C0KLX	AG	4200	4002A MV04020298-24"(1C0KLX)--	9224.41	DEBAJ	ELETECH	SS
1F2	1C0KLX	AG	4200	4002A MV04020298-24"(1C0KLX)--	9224.41	DEBAJ	ELETECH	SS
1F1	1C0KLX	AG	4200	4002A MV04020298-24"(1C0KLX)--	9224.41	DEBAJ	ELETECH	SS
1	1C0KLX	AG	4200	4002A MV04020298-24"(1C0KLX)--	6754.31	DEBAJ	ELETECH	SS
0	1C0KLX	AG	4200	4002A MV04020298-24"(1C0KLX)-2	4476.52	DEBAJ	ELETECH	SS

ISO Number

4002A_AG04020354_01

Rev	Service Class Type	WBU	PDS Area	Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type
0F3	1C3AS1X	AG	4200	4002A AG040203540-6"(1C3AS1X)-A	657.91	DESCON	ELETECH	CS NACE

Spool No.	Spool Rev	ISO Sheet	Field Pipe	Shop Dia	Field Dia	LB/SB	Material Availability	ETA	Spool Weight	Priority	Status	Comments
SP01	0	1	18	24	LB	NOW COMPLETE	09/27/2011	269.33	08/01/2012			
SP02	0	1	30	24	LB	NOW COMPLETE	09/27/2011	138.27	08/01/2012			
SP03	0	1	18	6	LB	NOW COMPLETE	09/27/2011	176.04	08/01/2012			
SP04	0	2	28.25	2	LB	NOW COMPLETE	01/24/2012	65.06	08/01/2012			
SP05	0	2	6	4	LB	NOW COMPLETE	11/12/2011	9.21	08/01/2012			

Rev	Service Class Type	WBU	PDS Area	Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type
0F1	1C3AS1X	AG	4200	4002A AG040203540-6"(1C3AS1X)-A	657.91	DESCON	ELETECH	CS NACE

Spool No.	Spool Rev	ISO Sheet	Field Pipe	Shop Dia	Field Dia	LB/SB	Material Availability	ETA	Spool Weight	Priority	Status	Comments
SP01	0	1	18	12	LB	NOW COMPLETE	09/27/2011	269.33	08/01/2012		Done Without Modification	
SP02	0	1	30	18	LB	NOW COMPLETE	09/27/2011	138.27	08/01/2012		Done Without Modification	
SP03	0	1	18	6	LB	NOW COMPLETE	09/27/2011	176.04	08/01/2012		Done Without Modification	
SP04	0	2	28.25	2	LB	NOW COMPLETE	01/24/2012	65.06	08/01/2012		Done Without Modification	
SP05	0	2	6	4	LB	NOW COMPLETE	11/12/2011	9.21	08/01/2012		Done Without Modification	

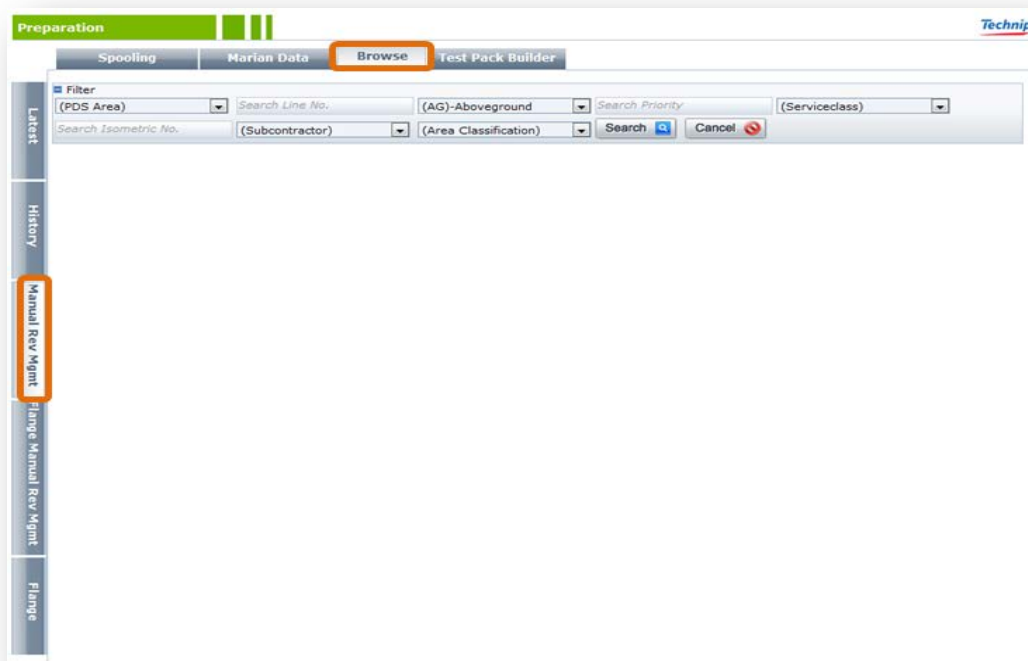
Rev	Service Class Type	WBU	PDS Area	Line No.	ISO Weight	Shop Subcon	Field Subcon	Material Type
0	1C3AS1X	AG	4200	4002A AG040203540-6"(1C3AS1X)-A	657.91	DESCON	ELETECH	CS NACE

Spool No.	Spool Rev	ISO Sheet	Field Pipe	Shop Dia	Field Dia	LB/SB	Material Availability	ETA	Spool Weight	Priority	Status	Comments
SP01	0	1	18	12	LB	NOW COMPLETE	09/27/2011	269.33	08/01/2012		Done Without Modification	
SP02	0	1	30	18	LB	NOW COMPLETE	09/27/2011	138.27	08/01/2012		Done Without Modification	
SP03	0	1	18	6	LB	NOW COMPLETE	09/27/2011	176.04	08/01/2012		Done Without Modification	
SP04	0	2	28.25	2	LB	NOW COMPLETE	01/24/2012	65.06	08/01/2012		Done Without Modification	
SP05	0	2	6	4	LB	NOW COMPLETE	11/12/2011	9.21	08/01/2012		Done Without Modification	

6.5.3 Manual revision Management

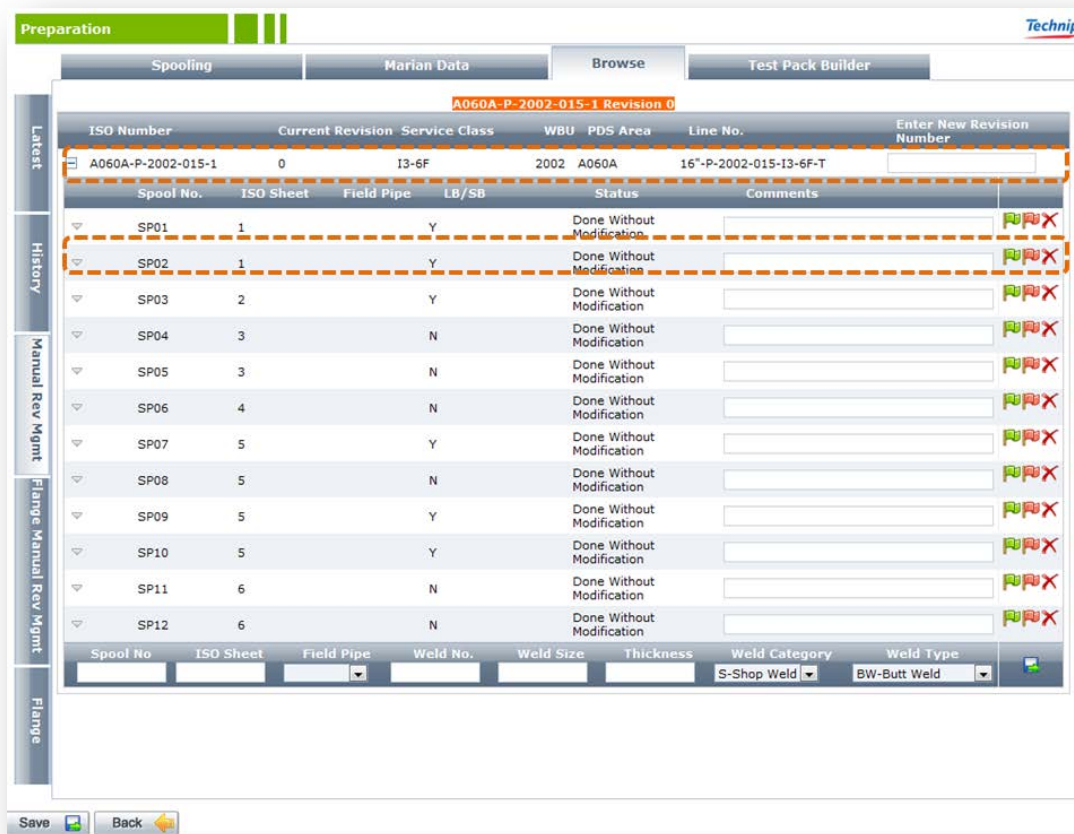
This module helps the user to do revision management manually. In this module the user can revise the isometric without the use of spooling .txt file. This module will be mostly used by the site spooling team to revise the field revision Isometrics.

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The user can search the isometrics from the fields as shown above, upon searching the isometrics the system will display the list of isometrics as a dropdown and the user can select the required isometric. After selection of Isometric, the system will take to the revision management screen where the user can revise the isometric as required. The system will show in a hierarchical manner with a default spool status "Done without Modification"

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In this screen, the user needs to put the revision number manually and add the comments wherever required during the revision process. All the previous revisions will go the browse history where user can view the changes made in that particular revision.

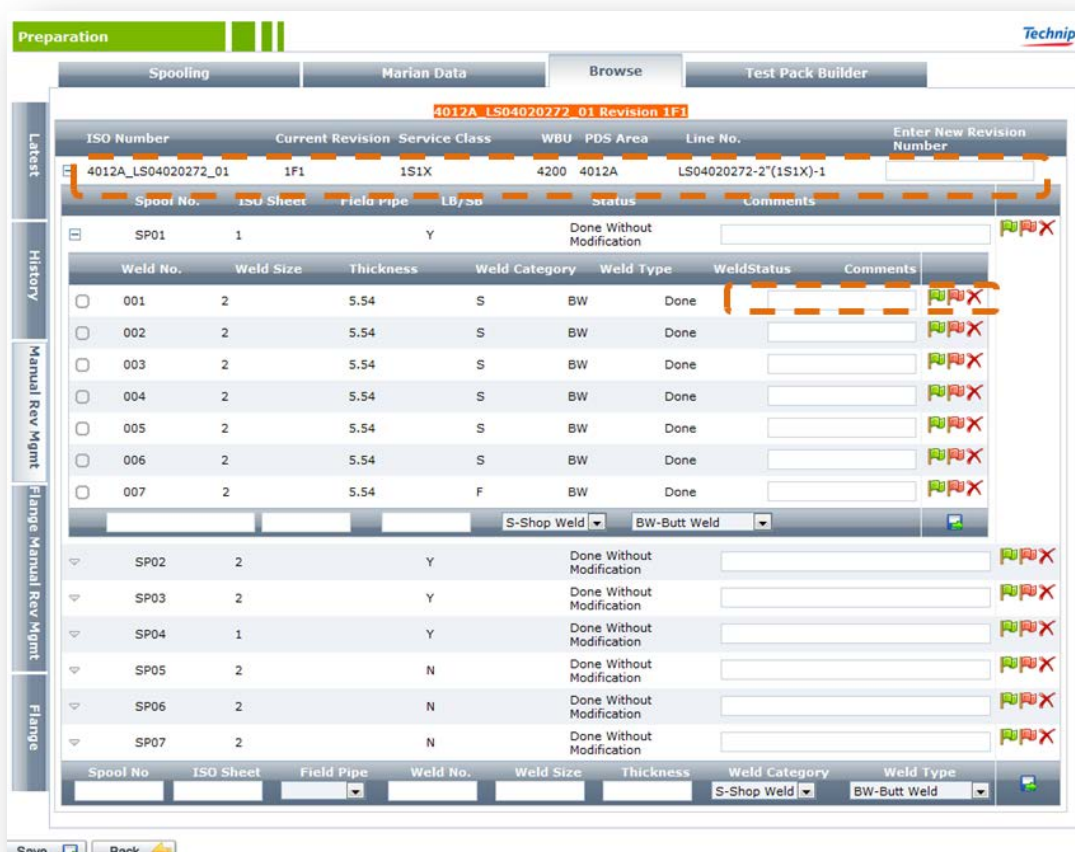
The progress information will either be copied to the newer revision of the spool or not, as explained in the table below:

	Status	Action	Check Weld?
	Not Done	The user will select this value if there is no progress information. No action to be taken in this case, the newer version of the spool will remain empty, same applies to the weld	No
	Cancelled	The user will select this value if the spool has been cancelled. No action to be taken in this case, the spool will not appear in the newer revision.	No
	Done, but revised w/o modification	This value refers to Spools which are fabricated in the older revision, but the newer version of the spool hasn't changed, so the progress information of the older revision will be copied to the newer revision. Weld progress information of the older revision will also be	No.

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		copied to welds in the newer revision. The 'Spool Revision No.' will remain the same.	
	Rework	If the user selects this value, it means that the spool in the older revision has been fabricated (or fabrication has started) but it has changed in the newer revision, which will result in rework. In this case, the system will copy the "Fabrication Start Date", "Sent to Paint" and "Paint" from the older revision to the new revision. The system will force the user to go to the level of the weld, and validate welds one by one. The 'Spool Revision No.' will also be changed to the new ISO rev no.	Yes

The user can click on the Icon  to "expand" to view on weld level.

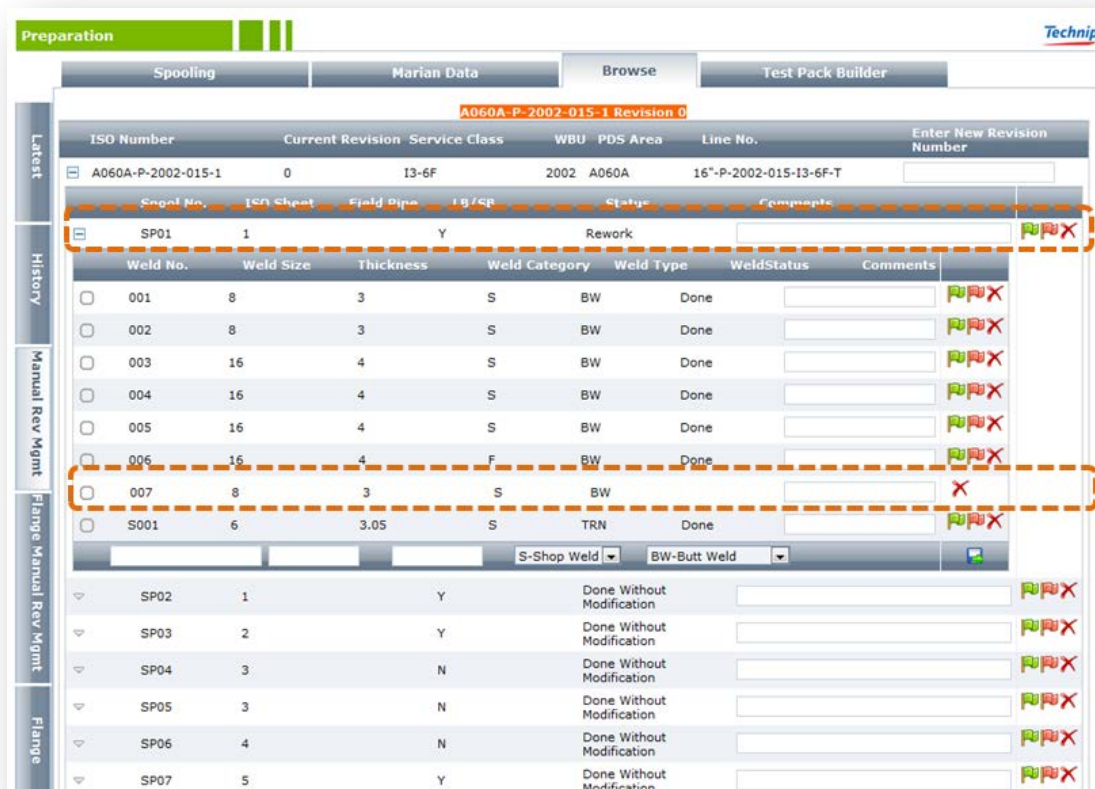


If the user goes to the weld level and did any actions then spool will consider as 'Rework'. For example, If user add new joint or made any modification to the existing joint then the spool status will changed to 'Rework'; automatically.

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Concerning the weld progress, the “status” column will also apply based on the following values (this is only considered if the spool is “Rework”):

	Status	Action
	Not Done	The user will select this value if there is no progress information. No action to be taken in this case, the newer version of the weld will remain empty.
	Cancelled	The user will select this value if the weld has been cancelled. No action to be taken in this case, the weld will disappear in the newer revision.
	Done, but revised w/o modification	This value refers to welds which are fabricated in the older revision, but the newer version of the weld hasn't changed, so the progress information of the older revision will be copied to the newer revision.



The user should choose the option () to save the manual revision process. Before save, user has to enter the new revision number of the specific ISO. If the entered ISO Revision number is already exists system will discard the revision number then user has to enter new revision number accordingly. Once the process is finished the system will display the below pop up

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Preparation

Spooling

Marian Data

Browse

Test Pack Builder

Latest

History

Manual Rev Mgmt

Flange Manu

A060A-P-2002-015-1 Revision 0

ISO Number	Revision Number	Line Number	PDS Area	WBU	Service Class	Status
A060A-P-2002-015-1	1	16"-P-2002-015-13-6F-T	A060A	2002	I3-6F	Imported Successfully (Conflict Resolved)

Message from webpage

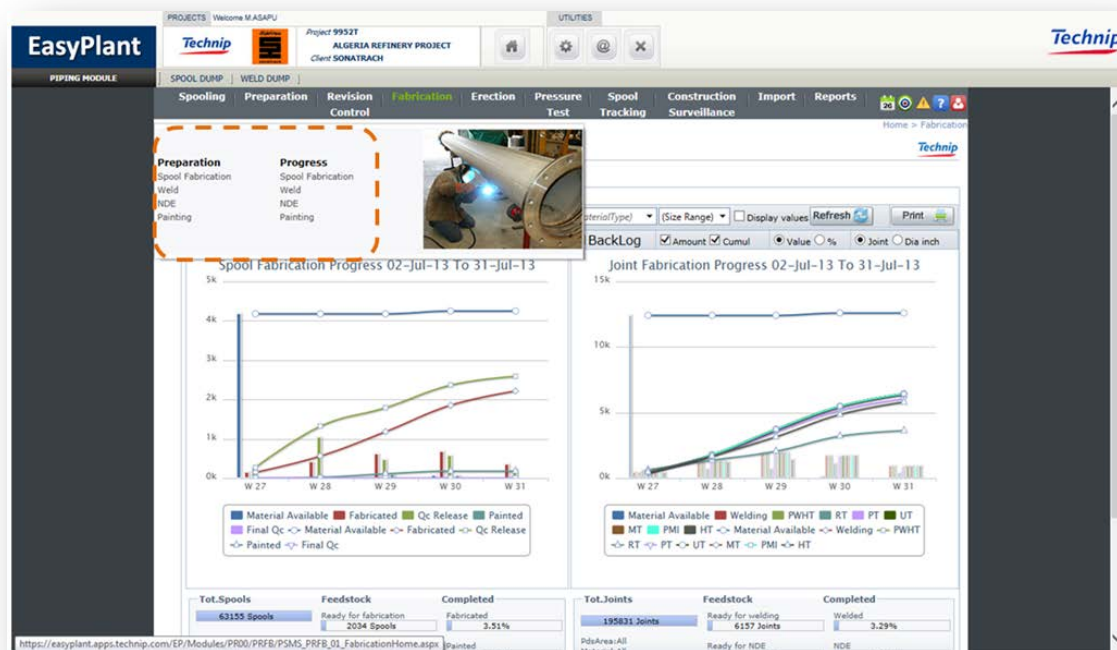
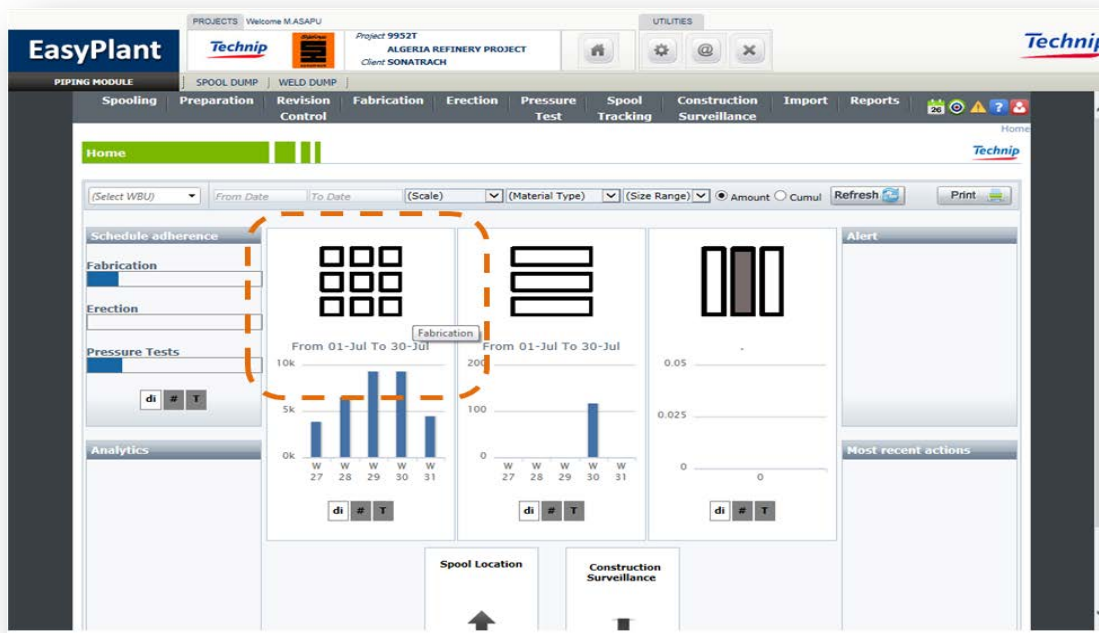

Manual Revision process was completed. Please review the summary of Revised ISO!

OK

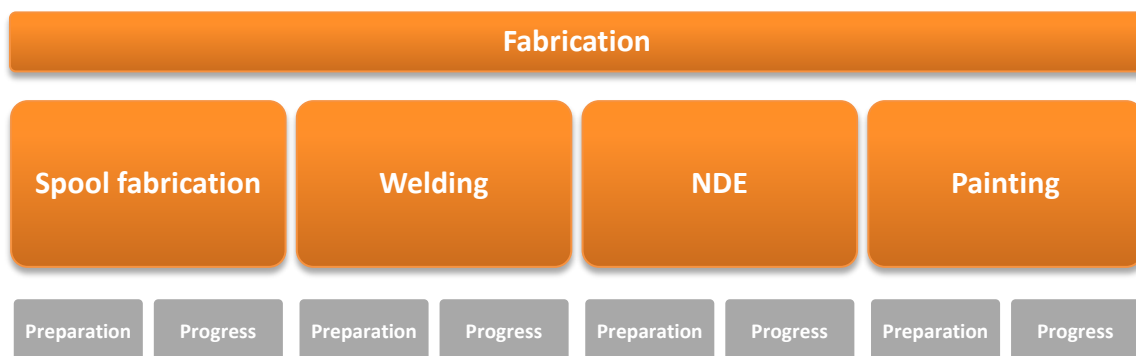
[*Easy Piping Part 3: Fabrication & Spool Tracking*

7. FABRICATION MODULE

The Fabrication module is organized in 4 different sections corresponding to the different site activities. These sections allow the user to plan & prepare the workload and record the Progress for each activity. The user can also generate the fabrication reports from this screen. All the sections and its uses are defined clearly in the below details.



DIFFERENT SECTIONS DURING FABRICATION PHASE



SPOOL FABRICATION

This screen contains two different sections which is preparation and progress. The preparation screen allows the user to plan and prepare the workload. The progress screen consists of several steps that user needs to follow. There can be some additional steps which shall be added if it is required for a project ([Refer to Para 1.1](#)). Definition of each step has to be agreed in the initial stage of the project

- Start fab
- Material Check
- Fabricated
- QC release
- Sent to paint
- Painted
- Final QC
- Laydown

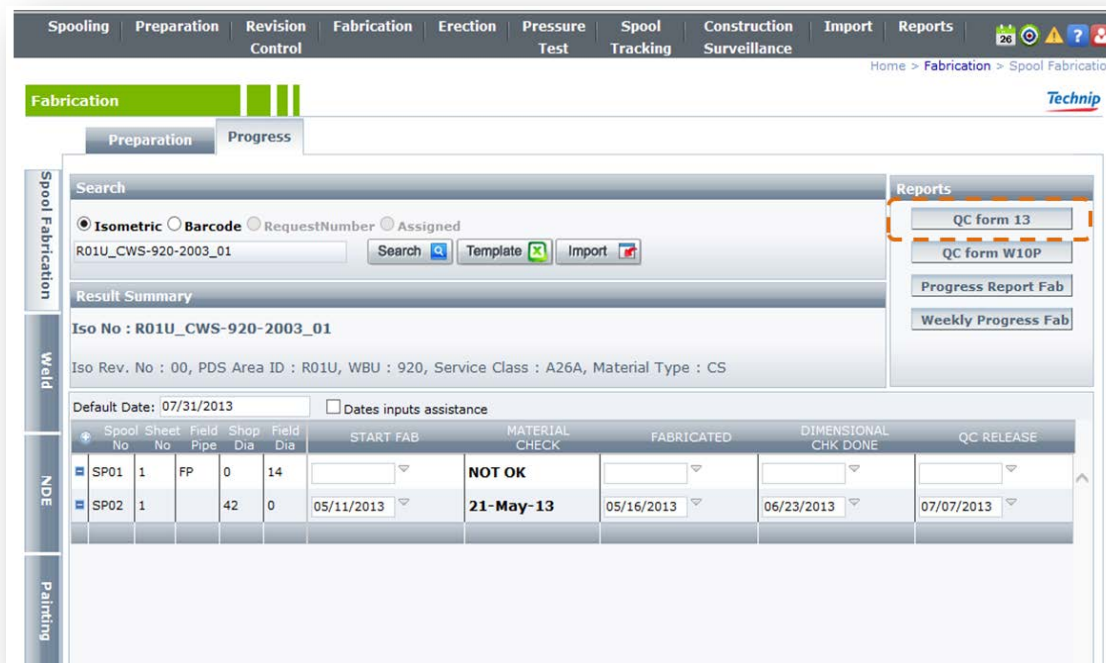
7.1 Start Fab

The user needs to update this step as soon as the spool has been started to fabricate, once this field is updated the user needs to generate the QC-13 Form

The user can generate the QC forms in two methods.

1. Using search method – User can search for single Iso and spool
2. Using Excel Import method – User can List upto Max 50 spools in excel and generate the forms
 - QC-13 form is the daily progress report form, at spool level. It should be generated right after “start fab” date has been recorded
- The QC Form should be duly filled by workers about the details of the shop joints for the selected spool
 - Material traceability
 - Welding

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Spool No	Sheet No	Field Pipe	Shop Dia	Field Dia	START FAB	MATERIAL CHECK	FABRICATED	DIMENSIONAL CHK DONE	QC RELEASE
SP01	1	FP	0	14		NOT OK			
SP02	1		42	0	05/11/2013	21-May-13	05/16/2013	06/23/2013	07/07/2013

7.2 Material Check

In the weld progress screen, the user can access the “material traceability” pop-up window. The user will enter the heat numbers filled in the QC – 13 forms by foreman. Easy piping will validate the entered heat numbers which are predefined in the project referential. ([Refer to Para 3.12](#))
When heat numbers are correctly recorded, the “material check” status of the corresponding spools will be automatically updated.

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Material Traceability

Filter (FDSName) Search Line No. (AG) Aboveground Search Isometric No. (Service/dest) (Subcontractor) Search priority (Area/classification) Y Search Cancel

Default Heat Number: Note: Please Save, Before going to next page!

ISO Number	Spool No	Heat No	Weld No	Part	Weld Type	Size	Group 1	Start Code 1	Heat Number 1	Group 2	Start Code 2	Heat Number 2
SAL-27-HCK-1014-01	SP01	1	001	S	BW	36	Flanges	CC74DV5	71R2	Pipe	CC70T3A	352959
SAL-27-HCK-1014-01	SP01	1	002	F	BW	36	Pipe	CC70T3A		Pipe	CC70T3A	
SAL-27-HCK-1014-01	SP02	1	003	F	BW	36	Pipe	CC70T3A		Pipe	CC70T3A	
SAL-27-HCK-1014-01	SP03	1	004	F	BW	36	Pipe	CC70T3A		Fittings	CBE2W05	
SAL-27-HCK-1014-01	SP04	1	005	S	BW	36	Fittings	CBE2W05	BN053	Pipe	CC70T3A	352959
SAL-27-HCK-1014-01	SP04	1	006	F	BW	36	Pipe	CC70T3A		Fittings	CBE2W05	
SAL-27-HCK-1014-01	SP05	1	007	S	BW	36	Fittings	CBE2W05	BN053	Pipe	CC70T3A	352959
SAL-27-HCK-1014-01	SP05	1	008	F	BW	36	Pipe	CC70T3A		(Select)		
SAL-27-HCK-8126-02	SP01	1	001	S	BW	2	Pipe	CD1736	62617	Fittings	CDPRT5	BC024
SAL-27-HCK-8126-02	SP01	1	002	S	BW	2	Fittings	CDPRT5	BC024	Pipe	CD1736	62617

ISOMETRIC NO: 52E6-25-IA-1006-01

SHO P JOINTS	TRACEABILITY			
	DESC	IDENT CODE	HEAT NO (1)	SIZE
009	Flanges	C2FBK07	39370	4
	Pipe	CAM0FR	318598	
010	Pipe	CAM0FR	318598	4
	Fittings	C9K6PK	BL338	
011	Fittings	C9K6PK	BL338	4

7.3 Weld Progress

In the fabrication module, the weld progress screen will show only shop joints only. Welding progress is filled in the QC-13 form, and then the progress is recorded in easy piping in the weld progress screen. The system validates the Information of welders which are predefined in the project referential. (Refer to Para 3.7) The system will validate the information of WPS recorded against the welder with the welder qualifications which are predefined in the Project referential. (Refer to Para 3.6) The system will alert the user if he records the incorrect information.

QUALITY CONTROL FORM		QC 13		PROJECT: PMP		OWNER: Qatar Liquefied Gas Company								
SPOOL PREFABRICATION		WELDING & DIMENSIONAL CHECK		PROJECT NO: 9833N		QCF Rev No. 0								
SUB CONTRACTOR: DESCON		Date:		Report No: QC13-5-0343										
ISOMETRIC NO: 4002R_LS04920385_01		Spool No: SP01		Rev No: 0F2		Service Class: 1SLX								
Material: CS														
SHOP JOINTS	TRACEABILITY				WELDING				THK	WELD TYPE	DESC ON			
	DESC	IDENT CODE	TRAC NO (1)	SIZE	BEFORE			IN PROGRESS				AFTER		
					WPS NO	CUTTING	BEVELING	RT-UP					WELDER NO (2)	PRE HEAT
002	Fittings	C956PV		16					W-083		18-Jan-2012	9.53	BW	TP QC
	Pipe	C6D29B							W-097					
									W-068					
									W-073					
003	Pipe	C6D29B		16					W-083		18-Jan-2012	9.53	BW	TP QC
	Fittings	C956PV							W-097					
									W-068					
									W-073					

The following are the weld progress columns for which the user needs to fill in this screen

- Cutting Date – User to input the date

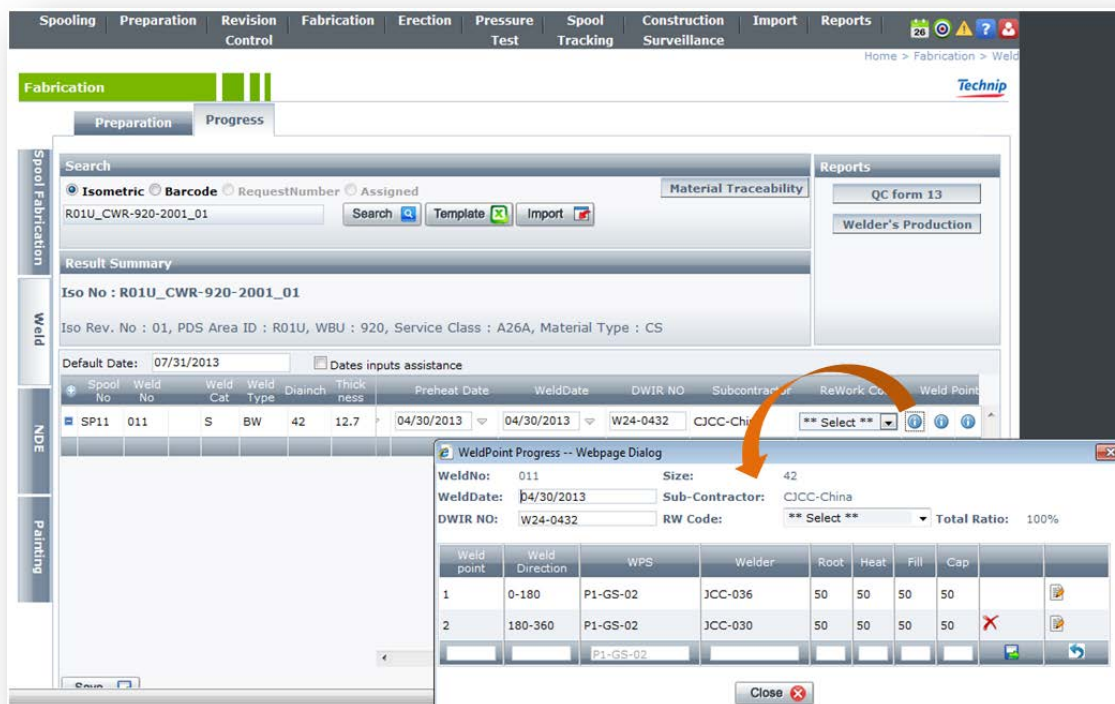
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- Beveling Date – User to input the date
- Fit-up Date – User to input the date
- Preheat Date – User to input the date
- Weld Date – User to input the date
- DWIR No. – User to input the QC Form No
- Subcontractor – Read Only
- Rework Code – User to select from dropdown
- Weld Point No. – User to input the number
- WPS No. – User to input the WPS No.
- Welder Code – User to input the Welder code

If the joint is progressed by only 1 welder (i.e., if there is only 1 weld point), then the user can enter the progress details in the same screen.

If the joint is progressed by more than 1 welder (i.e., if there are more than 1 weld points), then the user has to click the icon  upon which a pop-up is displayed to the user for entering the weld point details.



The screenshot shows the 'Fabrication' tab in the Technip software. A 'WeldPoint Progress -- Webpage Dialog' pop-up window is open, displaying the following information:

WeldNo: 011, Size: 42
WeldDate: 04/30/2013, Sub-Contractor: CICC-China
DWIR NO: W24-0432, RW Code: ** Select **
Total Ratio: 100%

Weld point	Weld Direction	WPS	Welder	Root	Heat	Fill	Cap	
1	0-180	P1-GS-02	JCC-036	50	50	50	50	
2	180-360	P1-GS-02	JCC-030	50	50	50	50	
		P1-GS-02						

Note:

- In the pop-up screen, when the user enters the progress information for the first weld-point and saves, the same WPS No. will be automatically loaded in read-only mode for entering the progress information of the second weld-point.
- In the pop-up screen, when the user enters the progress information for the first weld-point and saves, the user has to enter a different welder code for the second weld-point. The system will validate the welder code.
- In the pop-up screen, Root and Cap are mandatory fields. The total of Root and Cap for a joint should be 100. The total of Heat and Fill for a joint can be 0 or 100.

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- Modification of WPS No, Welder code is not allowed for those weld-points that are selected for examination in any of the batches OR already examined.

7.4 Fabricated

This step will be updated once the dimensional check is finished and all the information is validated and signed by the concerned persons and then spool is considered as “Fabricated”

QUALITY CONTROL FORM		QC 13		PROJECT: PMP		OWNER: Qatar Liquefied Gas Company	
SPOOL FABRICATION		WELDING & DIMENSIONAL CHECK		PROJECT NO: 14002V		QC Rev No: 0	
ISOMETRIC NO: 4002R_L504920385_01		SPOOL NO: SP01		Rev No: 0F2		Service Class: 1SLX	
Material: CS		Welding		Welder No (2)		Welding (VT) DATE	
THK		WELD TYPE		TP QC		DESCON	
SHOP JOINTS	TRACEABILITY			WELDING			
	DESC	IDENT CODE	TRAC NO (1)	SIZE	WPS NO	CUTTING	BEVEILING
002	Pipe	C02095		16			
	Fittings	C02095					
003	Pipe	C02095		16			
	Fittings	C02095					
004	Pipe	C02095		16			
	Fittings	C02095					

1. NO (NDE) with and without before cutting / assembling

2. No row to be filled if 2 welders are used for joint

DIAMENSIONAL CHECK: ☐ ACCEPTABLE ☐ NA

TRACEABILITY CHECK: ☐ ACCEPTABLE ☐ NA

DRIP FLANGE BRIDGING: YES ☐ NO ☐ NA

REINFORCEMENT PAD TESTED: YES ☐ NO ☐ NA

JOINT NO: _____ (CCF W 1 TO 81 ISSUED)

JOINT NO: _____ (CCF W 1 TO 81 ISSUED)

WELDING COMPLETED YES ☐ NO ☐

INSPECTORS	DESCON	Technip Chiyoda Joint Venture	Qatar Liquefied Gas Company
PHASE			
NAME			
SIGNATURE			
DATE			

7.5 QC Release

This step will be updated only when all the joints of the particular spool are NDE completed. The user can be able to know the spool release status from the “Spool wise NDE Status” report

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Spool No.	Material Check	QC Form No.	NDE Status	QC Release Date	W10P Form No.
11147_CW-100-4007_01	NOT OK		Not Released		
11147_CW-100-4008_01	NOT OK		Not Released		
11147_CW-100-4009_01	NOT OK		Not Released		
11147_CW-100-4010_01	NOT OK		Not Released		
11147_CW-100-4011_01	NOT OK		Not Released		
11147_CW-100-4012_01	NOT OK		Not Released		
11147_CW-100-4013_01	NOT OK		Not Released		
11147_CW-100-4014_01	NOT OK		Not Released		
11147_CW-100-4015_01	NOT OK		Not Released		
11147_CW-100-4016_01	NOT OK		Not Released		
11147_CW-100-4017_01	NOT OK		Not Released		
11147_CW-100-4018_01	NOT OK		Not Released		
11147_CW-100-4019_01	NOT OK		Not Released		
11147_CW-100-4020_01	NOT OK		Not Released		
11147_CW-100-4021_01	NOT OK		Not Released		
11147_CW-100-4022_01	NOT OK		Not Released		
11147_CW-100-4023_01	NOT OK		Not Released		
11147_CW-100-4024_01	NOT OK		Not Released		
11147_CW-100-4025_01	NOT OK		Not Released		
11147_CW-100-4026_01	NOT OK		Not Released		
11147_CW-100-4027_01	NOT OK		Not Released		
11147_CW-100-4028_01	NOT OK		Not Released		
11147_CW-100-4029_01	NOT OK		Not Released		
11147_CW-100-4030_01	NOT OK		Not Released		
11147_CW-100-4031_01	NOT OK		Not Released		
11147_CW-100-4032_01	NOT OK		Not Released		
11147_CW-100-4033_01	NOT OK		Not Released		
11147_CW-100-4034_01	NOT OK		Not Released		
11147_CW-100-4035_01	NOT OK		Not Released		
11147_CW-100-4036_01	NOT OK		Not Released		
11147_CW-100-4037_01	NOT OK		Not Released		
11147_CW-100-4038_01	NOT OK		Not Released		
11147_CW-100-4039_01	NOT OK		Not Released		
11147_CW-100-4040_01	NOT OK		Not Released		
11147_CW-100-4041_01	NOT OK		Not Released		
11147_CW-100-4042_01	NOT OK		Not Released		
11147_CW-100-4043_01	NOT OK		Not Released		
11147_CW-100-4044_01	NOT OK		Not Released		
11147_CW-100-4045_01	NOT OK		Not Released		
11147_CW-100-4046_01	NOT OK		Not Released		
11147_CW-100-4047_01	NOT OK		Not Released		
11147_CW-100-4048_01	NOT OK		Not Released		
11147_CW-100-4049_01	NOT OK		Not Released		
11147_CW-100-4050_01	NOT OK		Not Released		

7.6 Sent to Paint

This step will be updated when the QC will declare that the spool is released. The user need to generate the QC form W10P. When the form is signed by the QC. Then the user can update this field.

QC Form-W10P

Generate forms using 'search' method. Generate forms using 'excel import' method.

(PDSArea) Search Priority 4002A_MV04020156_01 (Service class)

(Subcontractor) (Area classification) SP02 (AG-Aboveground)

Note: W10P Form will be generated

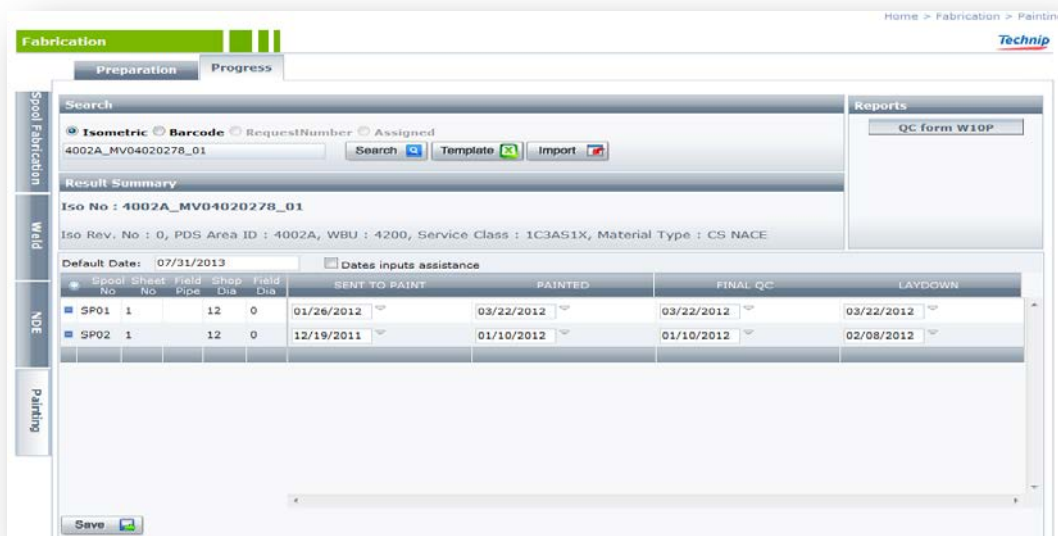
ISO Number 4002A_MV04020156_01

RELEASED FOR PAINTING YES ☐ NO ☐

INSPECTION DEBAI Technip Chloride Joint Venture Qatar Refinery Gas Company

7.7 Painted, Final QC and Laydown

The above fields can be updated once all the painting activities are completed.



Spool No	Field No	Pipe Dia	Shop Dia	SENT TO PAINT	PAINTED	FINAL QC	LAYDOWN
SP01	1	12	0	01/26/2012	03/22/2012	03/22/2012	03/22/2012
SP02	1	12	0	12/19/2011	01/10/2012	01/10/2012	02/08/2012

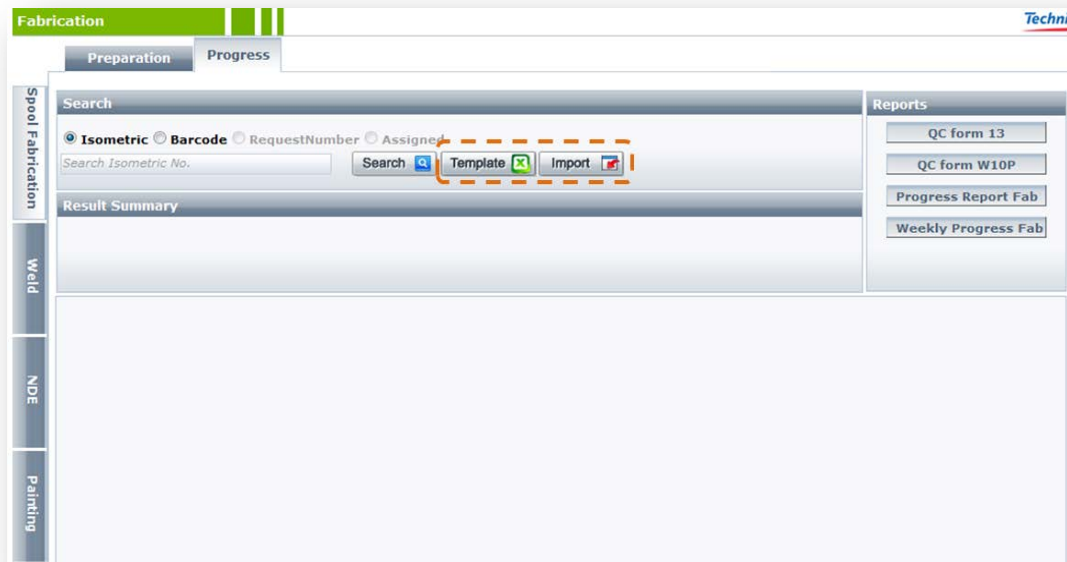
Fabrication Dash Boards

8. IMPORT PROGRESS MODULE

The "Import Progress" screen will allow the user to browse for the excel file containing the progress data, based on the generated template. The user will then specify the type of progress data being imported by selecting one of three possible values from a dropdown list:

- Import prefabrication progress.
- Import erection progress.
- Imports weld progress.
- Import Spool Category Definition

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Note:

- The format of the template should not be modified in any way.
- User should not Delete any existing records or Progress against the spools from Prefabrication and Erection Templates.
- Only date entries are allowed in the progress columns.
- The progress columns should be entered in the date format dd-MMM-yyyy.
- The column names should not be modified.
- New rows should not be added to the template.
- New columns should not be added to the template.

8.1 Import Prefabrication process

- The progress data in the excel sheet will always supersede progress data in the database. Meaning that if there was a conflict between the data in the excel sheet and the actual data in PSMS, that particular cell will be highlighted in yellow. The user will have to confirm the changes and proceed the import if he wishes. The data in the system will be overwritten with the new data on confirmation.
- Import of spools which do not exist will be rejected by the system

The user needs to download the relevant progress template to import the progress. The user can click the option  to download the template.

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Generate Progress Templates

Filter

1801R

Search Line No.

Search Priority

Search Isometric No.

(Serviceclass)

(LB/SB)

(AG)-Aboveground

(Subcontractor)

(Area classification)

(Location)

Cancel

Select the type of Template

☐ Prefabrication Progress
☐ Erection Progress
☐ Weld Progress
☒ Spool Category Definition

Export

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The user can choose the option **Browse...** and select the progress template and click on **Import** to import the progress file.

The system will validate the import file and the errors are highlighted in red color with appropriate error messages. Import will not be performed unless all the errors have been rectified by the user. The user will be prompted with how to view the validated import file: via Excel or Screen.

The Excel file will not be able to highlight the errors and overwritten cells in color. However, the error messages will be shown.

Home > Import > Import Progress

Import Progress

Type:

(Select Type)

Select a file to import:

(Select Type)

Prefabrication Progress

Erection Progress

Weld Progress

Spool Category Definition

Import

Proceed

Cancel

Print

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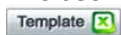
Once the user will click on **Proceed**, the imported information will be saved in the database.

8.2 Import Erection process

The progress data in the excel sheet will always supersede progress data in the database. Meaning that if there was a conflict between the data in the excel sheet and the actual data in PSMS, that particular cell will be highlighted in yellow. The user will have to confirm the changes and proceed the import if he wishes. The data in the system will be overwritten with the new data on confirmation.

Import of spools which do not exist will be rejected by the system

The user needs to download the relevant progress template to import the progress. The user can click the option



to download the template.

8.3 Import Weld Progress

Import Weld progress will update weld progress of Joints associated with the Importing template. During import system shall all the validation like Subcon , WPS and Welder authenticity etc information and coverage of Root, Fill and Capping activities.

If a Joint is already updated as Welded and NDE selection has been done for the joint, then the Joint cannot be editable thro import procedure.

8.4 Import Spool Definition Category

Import Spool Definition Category shall update the User Defined attributes which are created in

Settings / Admin - > Project Definition -> Project Category Definition

The export of Template shall export the list of all spools of selected PDS area with Custom defined attributes as column headings. User can update these attributes at spool level and can import.

Note: All the custom defined attributes are Text fields.

9. FABRICATION REPORTS

In this screen the user can generate the fabrication reports.



9833N_361_Progress_Report_-_FabricationReport.xls

This report will provide the progress information of entire fabrication, the user can use several filters like PDS area, Area classification, subcontractor, material type and each Isometric level etc.



WeeklyProgressFabricationReport.xls

This report will provide the progress information on a weekly and cumulative for the fabrication. The user can use several filters like PDS area, Area classification, subcontractor, material type etc.



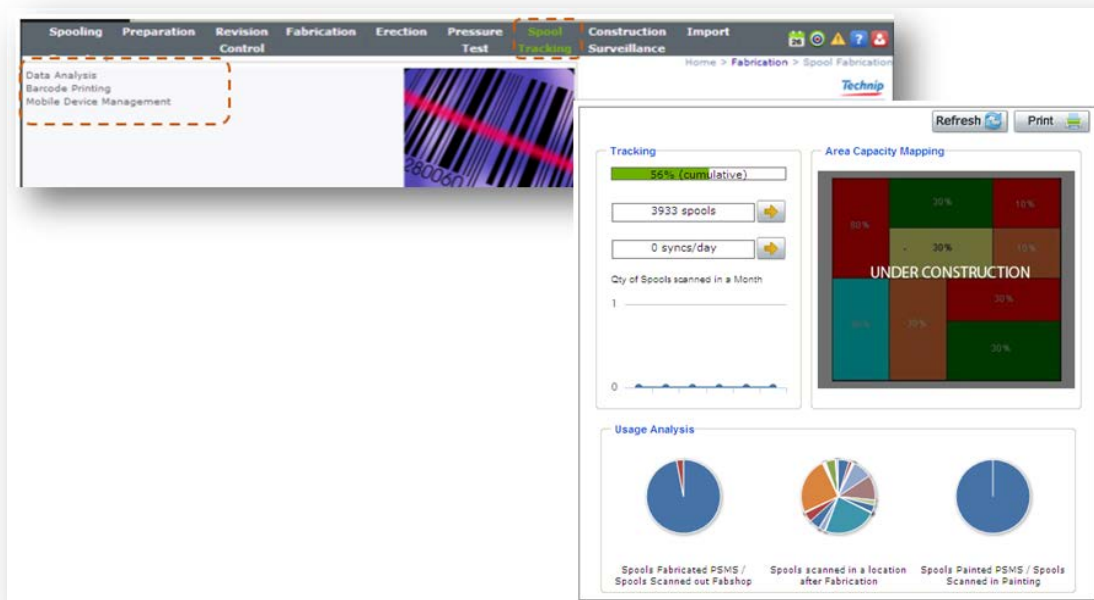
Summary_Report.xls

This report will provide the progress information of each spool status including all activities right from material availability to fabrication, painting, erection and RFT supporting etc.

10. SPOOLTRACKING INTRODUCTION

The spool tracking module allows the user to track the location of the spools on site by using the mobile devices (PDA). It is organized in 3 different sections. It is also having a separate dashboard to know the statistics.

- Data Analysis
- Barcode Printing
- Mobile Device Management



MAIN FUNCTIONALITIES OF SPOOL TRACKING

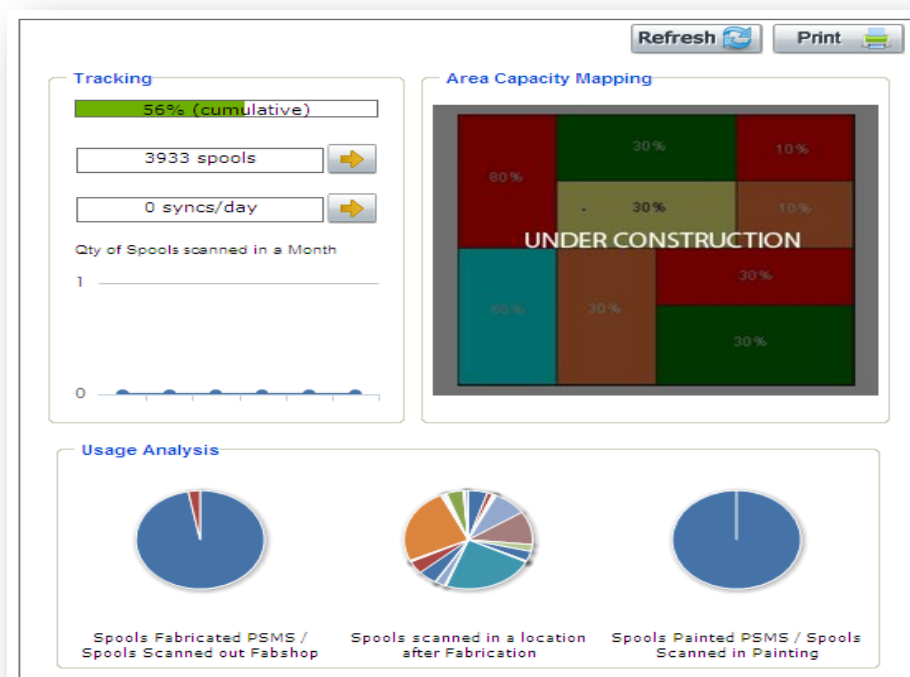
- Locate the spools of the project

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- Track the history location of the spools
- Visualize spool images
- Detect inconsistent scan/records
- Analyze the tracking data
- Manage the storage area capacity
- Print barcodes
- Manage the mobile devices

10.1 Overview of Dashboard



Tracking

- Cumulative percentage of spools scanned (total spools with at least one scan / total of spools of the project)
- Number of spools currently related to the spool tracking functionality (Easy piping status from "Start Fab" to "Erected") + trend
- Average count of synchronization PDA-Computer per day, during the last week + trend
- Curve representing the quantity of spools scanned during the running month

Usage analysis

- Spools scanned out from the fabshop (in red) among all the fabricated spools (according to easy piping status)

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- Repartition of the spools in different areas right after their fabrication
- Spools scanned in the paintshop (in red) among all the painted spools (according to easy piping status)

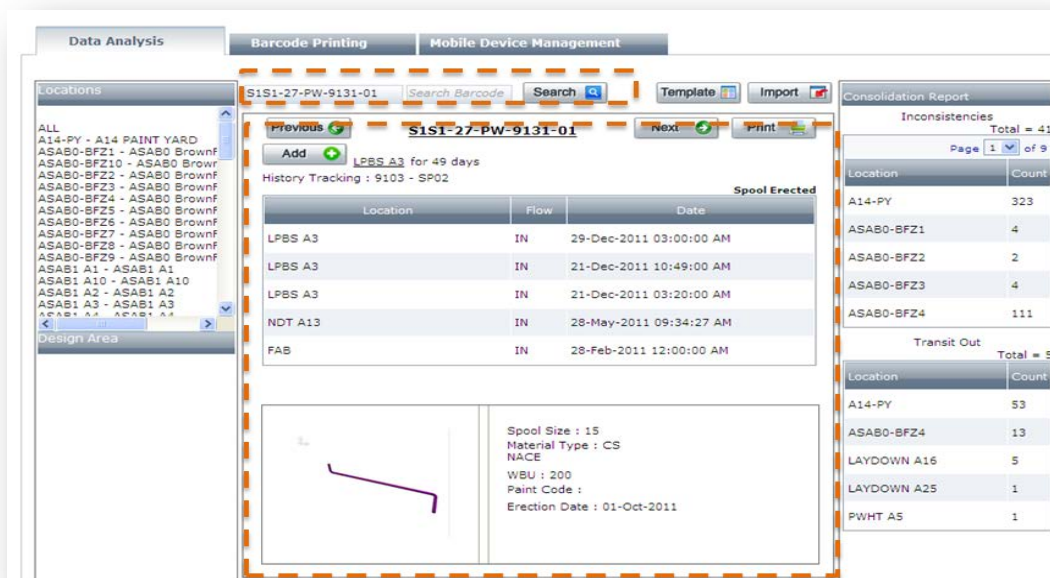
Area capacity mapping

- Map of the locations, filled with the current spool quantities stored versus the location capacity

10.2 Data Analysis

The user can search by isometric or by barcode

- Isometric search allows displaying the first spool of the isometric. The user can then scroll between the other spools of the isometric
- Barcode search allows displaying the corresponding spool. The user can then go to the isometric level by clicking on the isometric number



Data Analysis | Barcode Printing | Mobile Device Management

Locations: ALL, A14-PY - A14 PAINT YARD, ASAB0-BFZ1 - ASAB0 Brownf, ASAB0-BFZ10 - ASAB0 Brownf, ASAB0-BFZ2 - ASAB0 Brownf, ASAB0-BFZ3 - ASAB0 Brownf, ASAB0-BFZ4 - ASAB0 Brownf, ASAB0-BFZ5 - ASAB0 Brownf, ASAB0-BFZ6 - ASAB0 Brownf, ASAB0-BFZ7 - ASAB0 Brownf, ASAB0-BFZ8 - ASAB0 Brownf, ASAB0-BFZ9 - ASAB0 Brownf, ASAB1 A1 - ASAB1 A1, ASAB1 A2 - ASAB1 A2, ASAB1 A3 - ASAB1 A3, ASAB1 A4 - ASAB1 A4

Design Area

Search: S1S1-27-PW-9131-01 | Search Barcode | Search

Template | Import

Previous | Add | S1S1-27-PW-9131-01 | Next | Print

History Tracking : 9103 - SP02

Location	Flow	Date
LPBS A3	IN	29-Dec-2011 03:00:00 AM
LPBS A3	IN	21-Dec-2011 10:49:00 AM
LPBS A3	IN	21-Dec-2011 03:20:00 AM
NDT A13	IN	28-May-2011 09:34:27 AM
FAB	IN	28-Feb-2011 12:00:00 AM

Spool Erected

Spool Size : 15
Material Type : CS
NACE
WBU : 200
Paint Code :
Erection Date : 01-Oct-2011

Consolidation Report

Inconsistencies Total = 41
Page 1 of 9

Location	Count
A14-PY	323
ASAB0-BFZ1	4
ASAB0-BFZ2	2
ASAB0-BFZ3	4
ASAB0-BFZ4	111

Transit Out Total = 5

Location	Count
A14-PY	53
ASAB0-BFZ4	13
LAYDOWN A16	5
LAYDOWN A25	1
PWHT A5	1

- Details of the searched spool
 - Location, duration and tracking history
 - Spool description (material, WBU etc.)
 - Spool image

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The user can modify manually the current location of the spool by clicking on  Icon, It will create a new record

The user can also print the spool details by clicking on  Icon,

The user can use these icons   as a search options.

In the below screen, The location tab allows the user to visualize all the locations defined in the project referential ([Refer to Para 3.23](#))

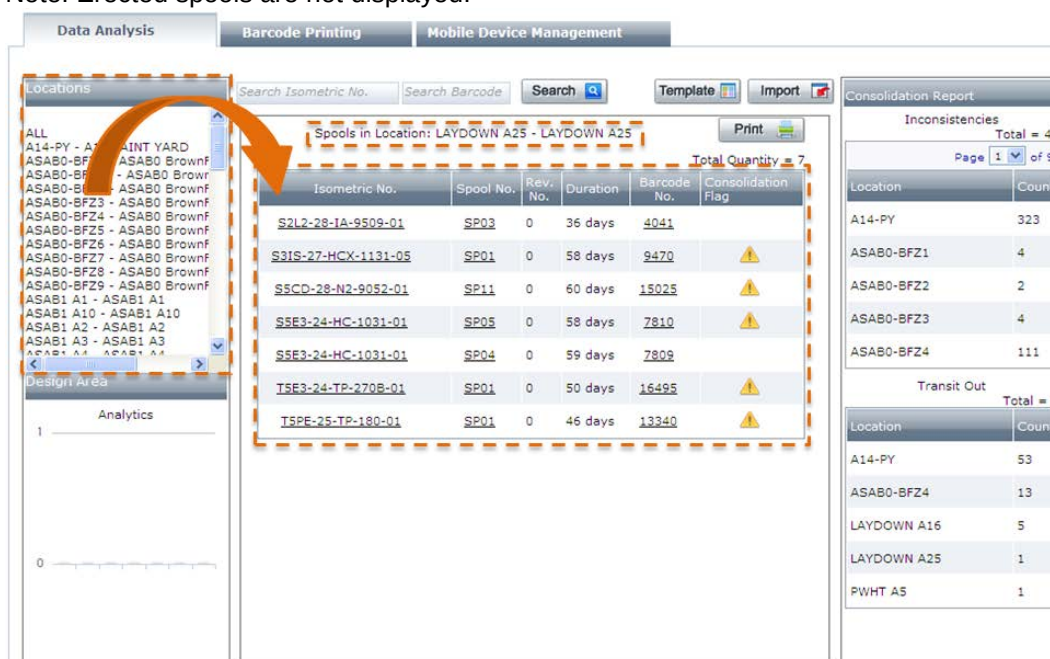
When the user clicks on the any of the location, It will display the list of the spools belongs to the selected location.

- Spool detail (isometric, spool and barcode number)
- Duration in this location
- Flag in case of inconsistency

The user can also click on the tabs Isometric no, Spool no and Barcode to view the details of that particular selection.

The user can also print the spool details by clicking on  Icon,

Note: Erected spools are not displayed.



Locations

Search Isometric No. Search Barcode Search

Template Import

Spools in Location: LAYDOWN A25 - LAYDOWN A25 Total Quantity = 7

Isometric No.	Spool No.	Rev. No.	Duration	Barcode No.	Consolidation Flag
S2L2-28-1A-9509-01	SP03	0	36 days	4041	
S3IS-27-HCX-1131-05	SP01	0	58 days	9470	⚠
S5CD-28-N2-9052-01	SP11	0	60 days	15025	⚠
S5E3-24-HC-1031-01	SP05	0	58 days	7810	⚠
S5E3-24-HC-1031-01	SP04	0	59 days	7809	⚠
T5E3-24-TP-2708-01	SP01	0	50 days	16495	⚠
T5PE-25-TP-180-01	SP01	0	46 days	13340	⚠

Consolidation Report

Inconsistencies Total = 41

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Location	Count
A14-PY	323
ASAB0-BFZ1	4
ASAB0-BFZ2	2
ASAB0-BFZ3	4
ASAB0-BFZ4	111

Transit Out Total = 5

Location	Count
A14-PY	53
ASAB0-BFZ4	13
LAYDOWN A16	5
LAYDOWN A25	1
PWHT A5	1

In the below screen, the design area tab allows the user to view the location of the spools of a selected design area.

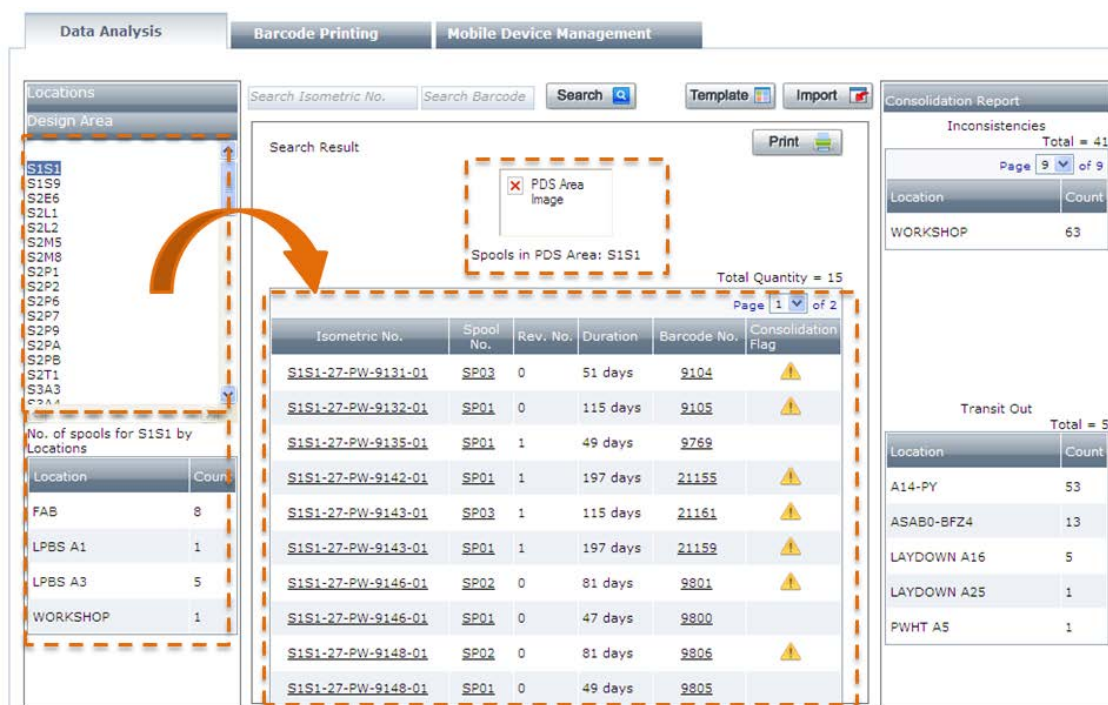
When user selects any design area, It will displays the

- The list of locations
- The list of spools
- An image of the design area

The user can also print the spool details by clicking on  Icon,

Note: Erected spools are not displayed.

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Data Analysis | Barcode Printing | Mobile Device Management

Search Isometric No. | Search Barcode | Search [Icon] | Template [Icon] | Import [Icon]

Locations

Design Area

S1S1
S1S9
S2E6
S2L1
S2L2
S2M5
S2M8
S2P1
S2P2
S2P6
S2P7
S2P9
S2PA
S2PB
S2T1
S3A3
S3A4

No. of spools for S1S1 by Locations

Location	Count
FAB	8
LPBS A1	1
LPBS A3	5
WORKSHOP	1

Search Result

PDS Area Image [Icon]

Spools in PDS Area: S1S1

Total Quantity = 15

Isometric No.	Spool No.	Rev. No.	Duration	Barcode No.	Consolidation Flag
S1S1-27-PW-9131-01	SP03	0	51 days	9104	[Warning Icon]
S1S1-27-PW-9132-01	SP01	0	115 days	9105	[Warning Icon]
S1S1-27-PW-9135-01	SP01	1	49 days	9768	[Warning Icon]
S1S1-27-PW-9142-01	SP01	1	197 days	21155	[Warning Icon]
S1S1-27-PW-9143-01	SP03	1	115 days	21161	[Warning Icon]
S1S1-27-PW-9143-01	SP01	1	197 days	21159	[Warning Icon]
S1S1-27-PW-9146-01	SP02	0	81 days	9801	[Warning Icon]
S1S1-27-PW-9146-01	SP01	0	47 days	9800	[Warning Icon]
S1S1-27-PW-9148-01	SP02	0	81 days	9806	[Warning Icon]
S1S1-27-PW-9148-01	SP01	0	49 days	9805	[Warning Icon]

Page 1 of 2

Consolidation Report

Inconsistencies Total = 41

Page 9 of 9

Location	Count
WORKSHOP	63

Transit Out Total = 5

Location	Count
A14-PY	53
ASAB0-BFZ4	13
LAYDOWN A16	5
LAYDOWN A25	1
PWHT A5	1

CONSOLIDATED REPORTS

Two types of consolidation reports are available: inconsistencies and transit out

- “Inconsistencies” are flagged when the easy piping status of a spool and its location are not matching
Example: a spool is painted, and it is still located in the fabshop

Erected spools, which have a location, scan after their erection date will be considered as inconsistent

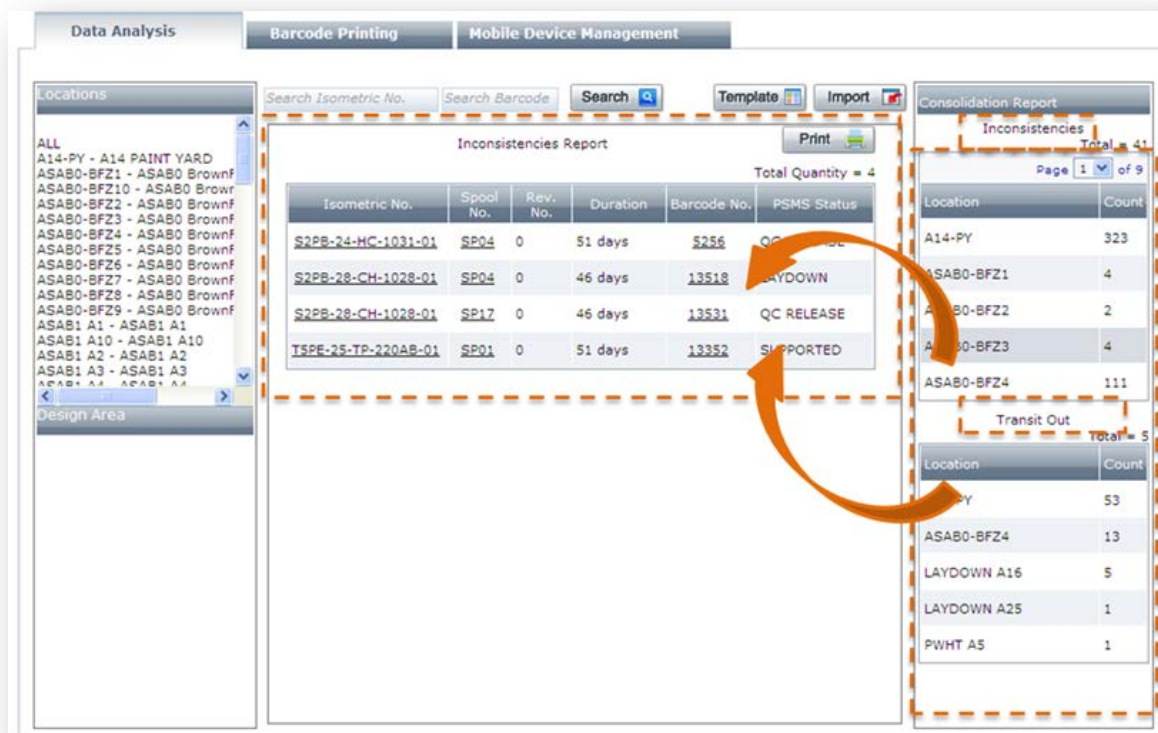
- Transit out are flagged when a spool is scanned OUT a location, but not scanned IN somewhere else

The flag will come only if the spool is out a location during more than two days

The user can also print the spool details by clicking on  Icon,

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Data Analysis | **Barcode Printing** | **Mobile Device Management**

Search Isometric No. Search Barcode Search [Q] Template Import

Inconsistencies Report Print Total Quantity = 4

Isometric No.	Spool No.	Rev. No.	Duration	Barcode No.	PSMS Status
S2PB-24-HC-1031-01	SP04	0	51 days	5256	QC RELEASE
S2PB-28-CH-1028-01	SP04	0	46 days	13518	LAYDOWN
S2PB-28-CH-1028-01	SP17	0	46 days	13531	QC RELEASE
TSPE-25-TP-220AB-01	SP01	0	51 days	13352	SUPPOTED

Consolidation Report Total = 41 Page 1 of 9

Location	Count
A14-PY	323
ASAB0-BFZ1	4
ASAB0-BFZ2	2
ASAB0-BFZ3	4
ASAB0-BFZ4	111

Transit Out Total = 5

Location	Count
A14-PY	53
ASAB0-BFZ4	13
LAYDOWN A16	5
LAYDOWN A25	1
PWHT A5	1

Export 

10.3 Barcode Printing

- The barcode printing functionality allows the user for the creation of spool list (excel format)
- This list will be used to print the barcode itself in another software (Zebra)
- The user can search some spools (section in the left) and add them to a basket (section on the right)
- Search is done by isometric or barcode number
- The user can click on the  to export an excel sheet, filled with the list of spools located on the section on the right

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The screenshot displays the 'Mobile Device Management' section of the CMC software. On the left, a 'Locations' list includes various sites like A14-PY, ASAB0-BFZ1, etc. The central area features a 'Search Barcode' input field and a 'Search' button. A dashed orange box highlights the search area and the 'Export' button. An orange arrow points from the 'Export' button to a Microsoft Excel spreadsheet showing a list of barcode data with columns for Barcode No., ISO Number, Spool No., Spool Rev No., Size, Material, Paint Code, and WBU.

10.4 Mobile Device Management

The mobile device management functionality allows the user to analyze the use of PDA on site. All the PDA Devices and PDA users are defined in the Easy piping referential (Refer Doc EP_Setup) and those will be displayed in this page.

The user can also print the spool details by clicking on  icon.

The screenshot displays the 'Mobile Device Management' section of the CMC software. It shows 'PDA in use: 1' and 'No. of Users: 2'. A 'Device No: 001' is highlighted, showing details like Connectivity (Ethernet), Latest update, Client Application, Most used by (USER), Data Collection Rate (Week), and Most scanned in (LPBS A1). A 'Print' button is visible.

10.5 Offline Synchronization of PDA data

PDA Synchronization can be done in 2 methods.

1. **Online Synchronization** : Online synchronization is, if user resides with in Technip network PDA data can be synchronized directly to easy piping database so that PDA data shall be loaded directly to database and Vice versa.
2. **Offline Synchronization** : Offline synchronization is a method of Synchronising the PDA data in outside Technip network where he has no access to Technip network.

There are 2 steps involved in this method.

2.1 Import PDA data to Easy Piping : This step helps user to import data from PDA to Easy piping.

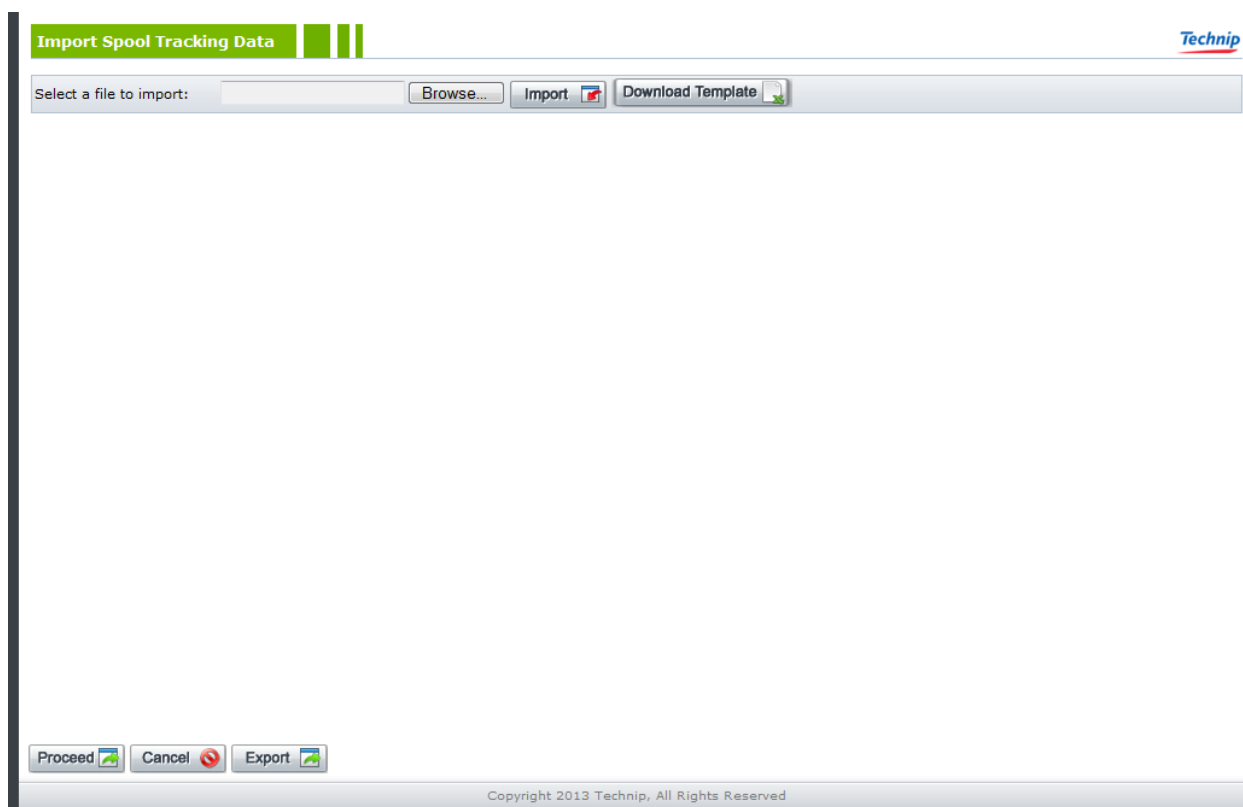
Easy piping has ability to import the file in both Excel and .txt files for this import. To generate excel

template press  button to download the excel template. This process can be useful if there are no PDAs available and spool tracking has been doing manually.

If PDAs are available a text (.txt) shall be generated by PDA with all scanned data. After synchronising the PDA with computer a Text file shall be generated. This text file will be located at (C:\Kalipso Project Updates\TR0001\ToPC) folder. User need to Import this file to Easy piping.

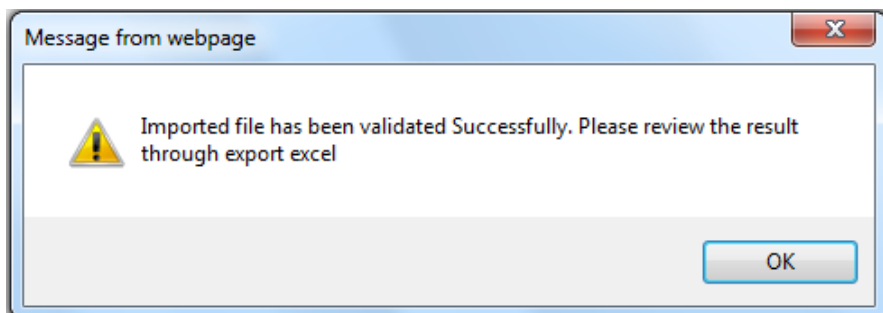
The Import option is provided in easy piping at

Easy piping -> Import -> Import spool Tracking data



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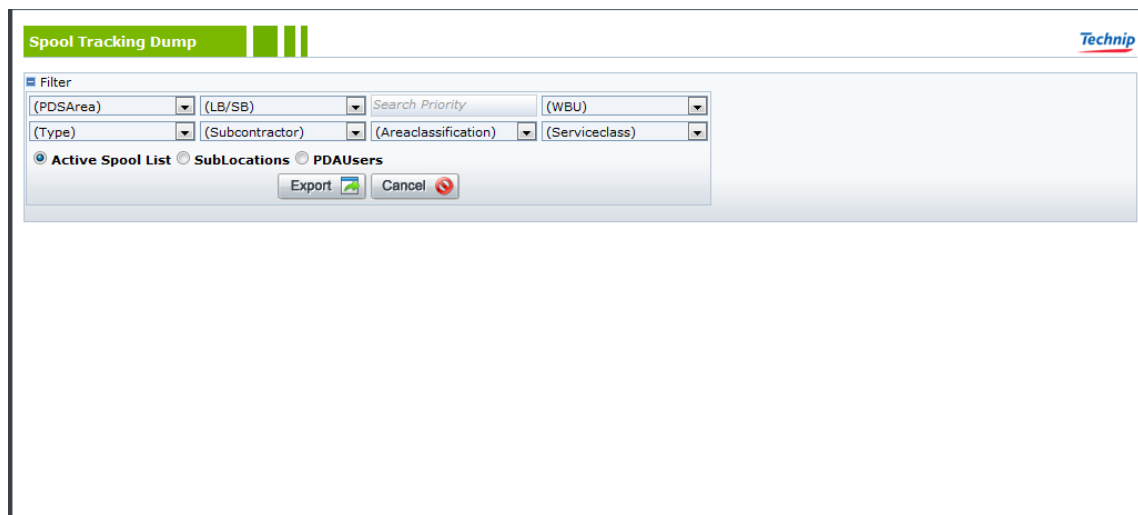
After browsing the file location press import button. System shall import and do the validation check for any inconsistencies. At the end of validation check following message box shall appear to review the validation results.



Click on "Export" button to check if any inconsistencies are there. If any, correct them and reimport the file again. If all data has been validated without any inconsistencies then press "import" button to import the data to easy piping. User can view the data update in Data Analysis tab of Spool tracking.

2.2 Export Easy piping data to PDA : This step export all the files which are required to update PDA from Easy piping. User can export the list required files through Reports module.

Easy piping - > Reports - > Data Dump - > Spool Tracking



Spool Tracking export shall export 3 Different report which are essential to update PDA.

1. **Active Spool List :** Active spool list shall export all the list of spools which are Active. Active spools are defined as spools for which "Start Fab" date is not null and "Erection date" is null. So any spool for which fabrication is not started or erection is completed is not considered as Active spools.
2. **Sub Locations :** List of locations
3. **PDA user :** List of PDA users.

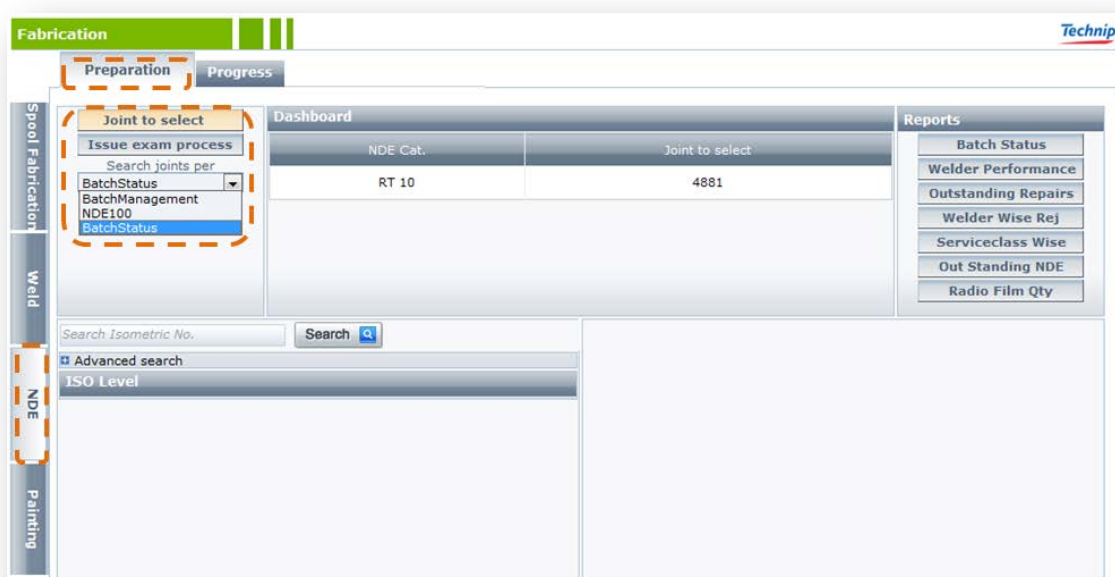
Note : Offline synchronization can be done only by System Admin and Project Admin. Other role user cannot Import or export the PDA data.

[*Easy Piping Part 4: NDE Management*]

11. NDE MANAGEMENT

The NDE (Non Destructive Examination) Management is one of the important modules in Easy Piping. This module helps the user to manage all the NDE activities throughout the operational phase. This module is organized in two different sub modules preparation and progress. This screen will appear in both Fabrication and Erection modules. This modules contains different sections as mentioned below

- Batch Status
- Batch Management
- NDE 100
- Joint to select
- Issue Examination process



What is a Batch?

The batch concept is to manage the NDE activities for “Spot or random examination of 10 or 20% refers respectively to 1 or 2 welds done among 10 weld joints which are entirely examined on the whole circumference for each welder”

A batch of weld is made by grouping the welds executed by one welder belonging to a particular NDE category

- Easy piping will create the batch as soon as the weld progress information is recorded. It will take the required NDE% from the NDE matrix which is defined in the project referential ([Refer to Para 3.9](#)) based on the progressed weld points, weld type, location and piping class. According to the retrieved NDE %, Welder no and location, System will allocate into different batches.
- If the batch already exists. Then system gets the no. of weld points already in the batch. If it is lesser than the size of the batch (10 or 20 depending on NDE %), then the weld point will be added to the batch with status “S” and batch status ‘Joint to Select’. If the batch has the required no. of weld points then a new batch is created with the same characteristics and the weld point will be added to the new batch with status “S” and batch status ‘Joint to Select’.

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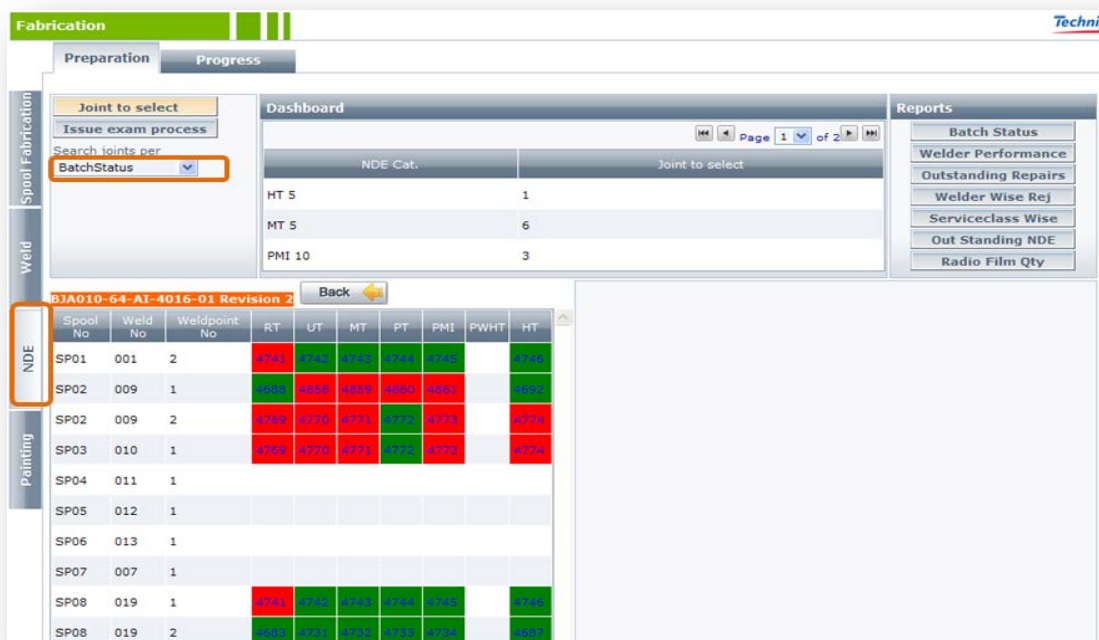
- If a weld point is added to a batch where 4 or more weld-points are rejected, its status will be “SS” because it needs to be tested 100%. Batch status is ‘Awaiting NDE’.

11.1 Batch status

This module helps the user to view the lists of different weld-points of an ISO and displays the batch no. to which it belongs for all examinations. It also helps the user to identify the batch status by color coding defined in the screen.

The batch status of the batch can be identified by viewing the color code.

- Red – Joint to Select
- Orange – Awaiting NDE
- Green – Released



Spool No	Weld No	Weldpoint No	RT	UT	MT	PT	PMI	PWHT	HT
SP01	001	2	4746	4746	4746	4746	4746	4746	4746
SP02	009	1	4746	4746	4746	4746	4746	4746	4746
SP02	009	2	4746	4746	4746	4746	4746	4746	4746
SP03	010	1	4746	4746	4746	4746	4746	4746	4746
SP04	011	1							
SP05	012	1							
SP06	013	1							
SP07	007	1							
SP08	019	1	4746	4746	4746	4746	4746	4746	4746
SP08	019	2	4746	4746	4746	4746	4746	4746	4746

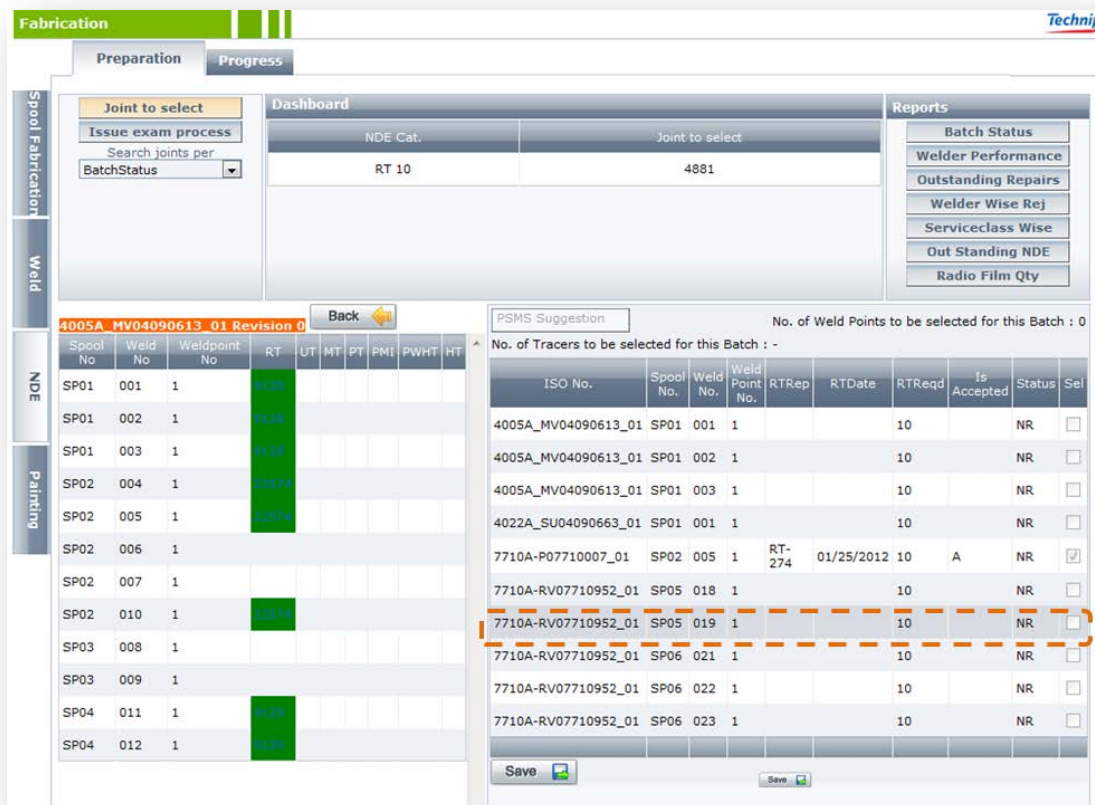
The user can click on the batch to perform all the actions of a batch. As soon as the user clicks on the batch it will show all the joints related to the batch in the same screen. Easy piping will suggest a joint to select for the user to select, the user can select the same joint or he can choose any other joint of his choice. After the selection of joint the status will change from “S” to “SS”

Legend for Joint Status is as follows

- “S” – Joint to be selected
- “SS” – Joint selected and awaiting examination
- “NR” – Joint Examination result updated
- “?” - Selection of joints completed in a Batch

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The user can use the Icon () to save the selected joint.

11.2 Batch Management

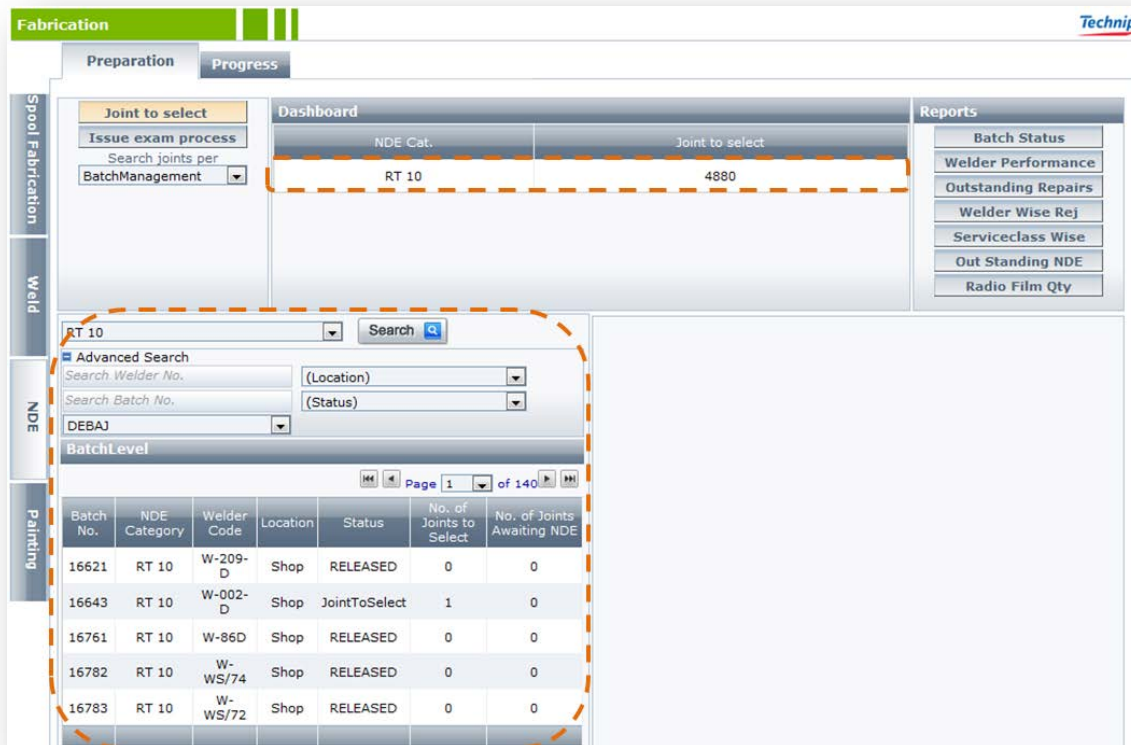
This module helps the user to view the list of batches defined by the Easy Piping. The user can also view the No of joints to select per NDE category in the dashboard section in the same screen. There is an advanced search which helps the user to filter the list of batches based on the following criteria

- NDE Category
- Welder No.
- Location
- Status
- Batch number
- Sub-Contractor

Status could be one of the following:

- Released
- Awaiting NDE
- Joint to Select

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Fabrication

Technip

Preparation **Progress**

Joint to select

Issue exam process

Search joints per

BatchManagement

Dashboard

NDE Cat.	Joint to select
RT 10	4880

Reports

- Batch Status
- Welder Performance
- Outstanding Repairs
- Welder Wise Rej
- Serviceclass Wise
- Out Standing NDE
- Radio Film Qty

RT 10 Search

Advanced Search

Search Welder No. (Location)

Search Batch No. (Status)

DEBAJ

BatchLevel

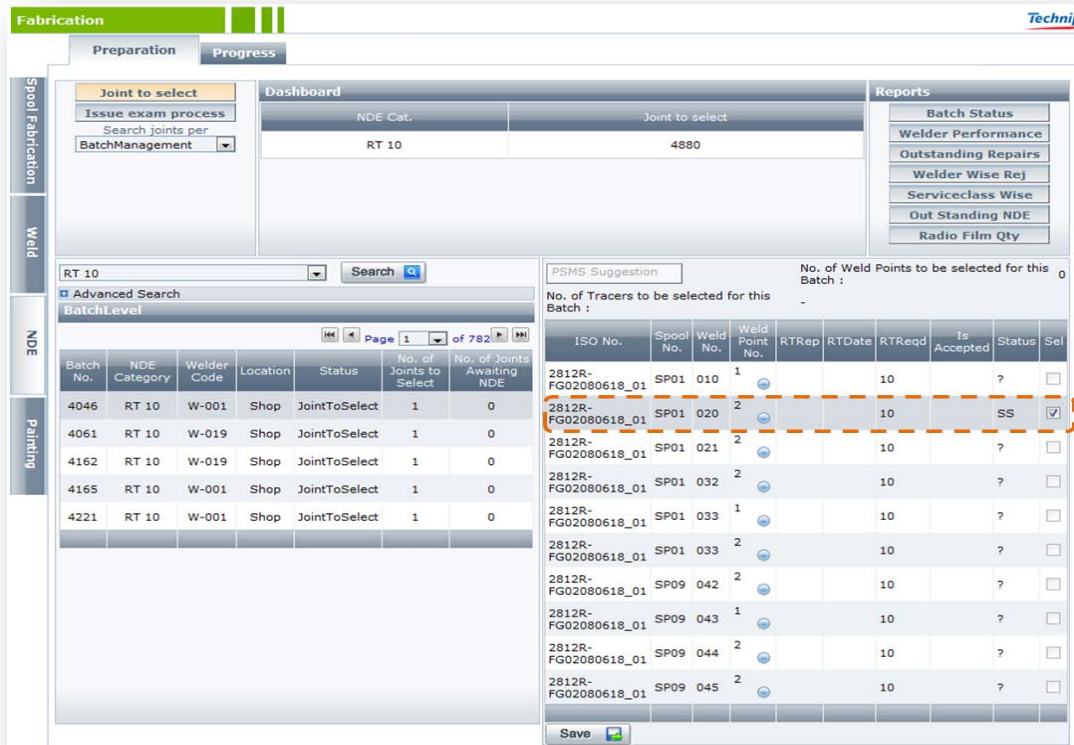
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Batch No.	NDE Category	Welder Code	Location	Status	No. of Joints to Select	No. of Joints Awaiting NDE
16621	RT 10	W-209-D	Shop	RELEASED	0	0
16643	RT 10	W-002-D	Shop	JointToSelect	1	0
16761	RT 10	W-86D	Shop	RELEASED	0	0
16782	RT 10	W-WS/74	Shop	RELEASED	0	0
16783	RT 10	W-WS/72	Shop	RELEASED	0	0

The user can click on the batch to perform all the actions of a batch. As soon as the user clicks on the batch it will show all the joints related to the batch in the same screen.

Easy piping will suggest a joint to select for the user to select, the user can select the same joint or he can choose any other joint of his choice. After the selection of joint the status will change from “S” to “SS” and if the user unselect the joint then the status will change from “SS” to “S”

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Batch No.	NDE Category	Welder Code	Location	Status	No. of Joints to Select	No. of Joints Awaiting NDE
4046	RT 10	W-001	Shop	JointToSelect	1	0
4061	RT 10	W-019	Shop	JointToSelect	1	0
4162	RT 10	W-019	Shop	JointToSelect	1	0
4165	RT 10	W-001	Shop	JointToSelect	1	0
4221	RT 10	W-001	Shop	JointToSelect	1	0

ISO No.	Spool No.	Weld No.	Weld Point No.	RTRep	RTDate	RTReqd	Is Accepted	Status	Sel
2812R-FG02080618_01	SP01	010	1			10		?	☐
2812R-FG02080618_01	SP01	020	2			10		SS	☒
2812R-FG02080618_01	SP01	021	2			10		?	☐
2812R-FG02080618_01	SP01	032	2			10		?	☐
2812R-FG02080618_01	SP01	033	1			10		?	☐
2812R-FG02080618_01	SP01	033	2			10		?	☐
2812R-FG02080618_01	SP09	042	2			10		?	☐
2812R-FG02080618_01	SP09	043	1			10		?	☐
2812R-FG02080618_01	SP09	044	2			10		?	☐
2812R-FG02080618_01	SP09	045	2			10		?	☐

If a selected joint of a batch is accepted. All the other joints of the batch status will be changed to NR and it will disable for selection. The system will show the status of batch as “Released”

If a selected joint of a batch is rejected, the status for the particular joints is changed to NR and the other joints will be changed to “T1”, Easy Piping will suggest 2 tracers for the user to select. The user can select the suggested tracers or any other tracers for the batch.

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The screenshot shows the 'Progress' tab for 'RT 10'. The 'Dashboard' section displays 'NDE Cat: RT 10' and 'Joint to select: 23'. The 'Reports' section includes links for 'Batch Status', 'Welder Performance', 'Outstanding Repairs', 'Welder Wise Rej', 'Serviceclass Wise', 'Out Standing NDE', and 'Radio Film Qty'. The 'Advanced Search' section shows 'BatchLevel' and 'Page 2 of 8'. The 'Main Table' lists joints with columns: Batch No., NDE Category, Welder Code, Location, Status, No. of Joints to Select, and No. of Joints Awaiting NDE. The 'PSMS Suggestion' section shows 'No. of Tracers to be selected for this Batch: 2' and 'No. of Weld Points to be selected for this Batch: 0'. The 'Weld Points Table' lists ISO No., Spool No., Weld No., Weld Point No., RTRep, RTDate, RTReqd, Is Accepted, Status, and Sel. The first row is highlighted with a dashed orange box.

Batch No.	NDE Category	Welder Code	Location	Status	No. of Joints to Select	No. of Joints Awaiting NDE
3861	RT 10	W-001	S	RELEASED	0	0
3862	RT 10	W-001	S	AwaitingNDE	0	2
4181	RT 10	DES-01	S	RELEASED	0	0
4182	RT 10	DES-01	S	JointToSelect	2	0
6161	RT 10	W-004	S	JointToSelect	2	0

ISO No.	Spool No.	Weld No.	Weld Point No.	RTRep	RTDate	RTReqd	Is Accepted	Status	Sel
1801R-LS01920952_01	SP01	001	1	123456	11/22/2012	10	R	NR	☐
1801R-LS01920952_01	SP01	002	1			10		T1	☒
1801R-LS01920952_01	SP01	003	1			10		T1	☐
1801R-LS01920952_01	SP01	004	1			10		T1	☒

Once the user selects the first tracer, the status of that particular joint is changed to T1S and the system shows the status of other joints as T2, after the selection of the second tracer, that particular joint status is changed to T2S and other joints status will be changed to “?”

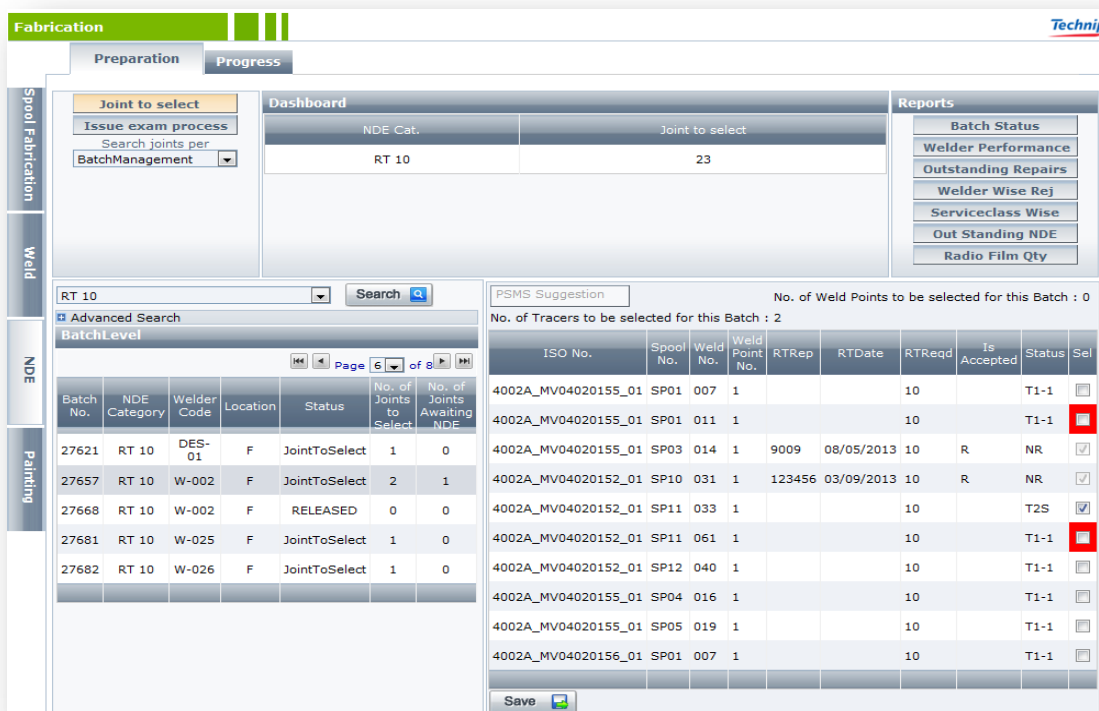
The screenshot shows the 'Progress' tab for 'RT 10' after selecting the first tracer. The 'Dashboard' section displays 'NDE Cat: RT 10' and 'Joint to select: 23'. The 'Reports' section includes links for 'Batch Status', 'Welder Performance', 'Outstanding Repairs', 'Welder Wise Rej', 'Serviceclass Wise', 'Out Standing NDE', and 'Radio Film Qty'. The 'Advanced Search' section shows 'BatchLevel' and 'Page 6 of 8'. The 'Main Table' lists joints with columns: Batch No., NDE Category, Welder Code, Location, Status, No. of Joints to Select, and No. of Joints Awaiting NDE. The 'PSMS Suggestion' section shows 'No. of Tracers to be selected for this Batch: 0' and 'No. of Weld Points to be selected for this Batch: 0'. The 'Weld Points Table' lists ISO No., Spool No., Weld No., Weld Point No., RTRep, RTDate, RTReqd, Is Accepted, Status, and Sel. The first row is highlighted with a dashed orange box.

Batch No.	NDE Category	Welder Code	Location	Status	No. of Joints to Select	No. of Joints Awaiting NDE
27621	RT 10	DES-01	F	JointToSelect	1	0
27657	RT 10	W-002	F	JointToSelect	2	0
27668	RT 10	W-002	F	RELEASED	0	0
27681	RT 10	W-025	F	JointToSelect	1	0
27682	RT 10	W-026	F	JointToSelect	1	0

ISO No.	Spool No.	Weld No.	Weld Point No.	RTRep	RTDate	RTReqd	Is Accepted	Status	Sel
4002A_MV04020155_01	SP01	007	1			10		?	☐
4002A_MV04020155_01	SP01	011	1			10		?	☐
4002A_MV04020155_01	SP03	014	1			10		T1S	☒
4002A_MV04020152_01	SP10	031	1	123456	03/09/2013	10	R	NR	☒
4002A_MV04020152_01	SP11	033	1			10		T2S	☒
4002A_MV04020152_01	SP11	061	1			10		?	☐
4002A_MV04020152_01	SP12	040	1			10		?	☐
4002A_MV04020155_01	SP04	016	1			10		?	☐
4002A_MV04020155_01	SP05	019	1			10		?	☐
4002A_MV04020156_01	SP01	007	1			10		?	☐

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If any of the selected tracers are rejected, The user is required to select another 2 tracer for rejected joint. Easy Piping will suggest 2 more tracers to select, the user can select the same joints or any other joints from the batch.



Batch No.	NDE Category	Welder Code	Location	Status	No. of Joints to Select	No. of Joints Awaiting NDE
27621	RT 10	DES-01	F	JointToSelect	1	0
27657	RT 10	W-002	F	JointToSelect	2	1
27668	RT 10	W-002	F	RELEASED	0	0
27681	RT 10	W-025	F	JointToSelect	1	0
27682	RT 10	W-026	F	JointToSelect	1	0

ISO No.	Spool No.	Weld No.	Weld Point No.	RTRep	RTDate	RTReqd	Is Accepted	Status	Sel
4002A_MV04020155_01	SP01	007	1			10		T1-1	☐
4002A_MV04020155_01	SP01	011	1			10		T1-1	☒
4002A_MV04020155_01	SP03	014	1	9009	08/05/2013	10	R	NR	☒
4002A_MV04020152_01	SP10	031	1	123456	03/09/2013	10	R	NR	☒
4002A_MV04020152_01	SP11	033	1			10		T2S	☒
4002A_MV04020152_01	SP11	061	1			10		T1-1	☒
4002A_MV04020152_01	SP12	040	1			10		T1-1	☐
4002A_MV04020155_01	SP04	016	1			10		T1-1	☐
4002A_MV04020155_01	SP05	019	1			10		T1-1	☐
4002A_MV04020156_01	SP01	007	1			10		T1-1	☐

The user needs to follow the same principal as followed for the 1st lever tracers to complete the batch. The user can use the Icon () to save the selected joint.

Legend for Joint Status is as follows

- “S” – Joint to be selected
- “SS” – Joint selected and awaiting examination
- “NR” – Joint Examination result updated
- “?” - Selection of joints completed in a Batch
- T1, T2, – Tracers in the first level
- T1S, T2S - Selected tracers in the first level
- T1-1, T1-2, T2-1, T2-2, – Tracers in the second level
- T1-1S, T1-2S, T2-1S, T2-2S – Selected tracers in the second level

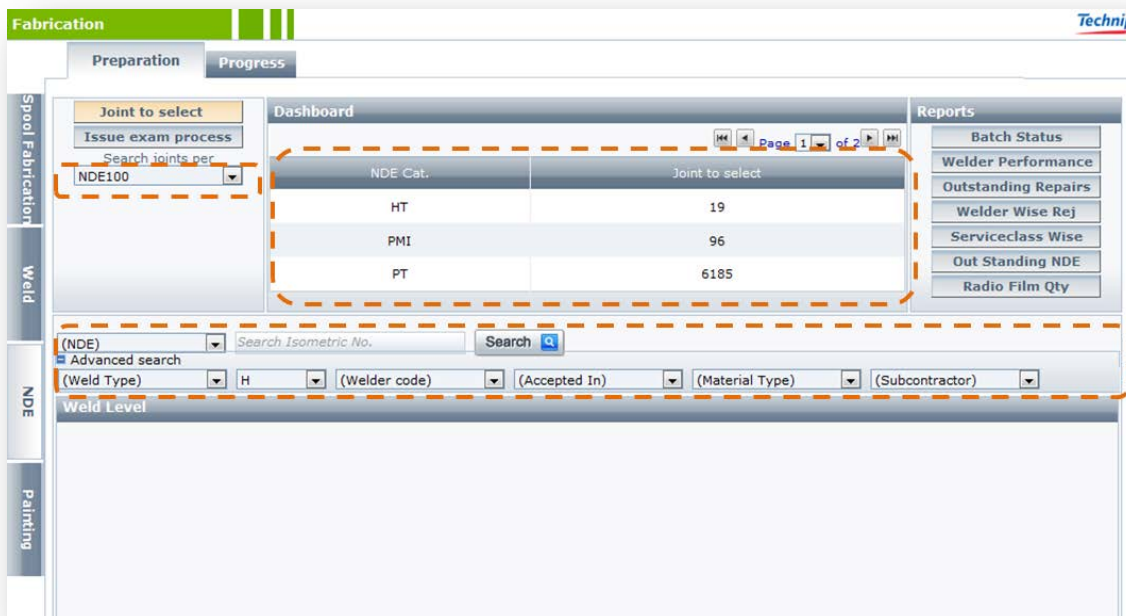
Note:

- If there are any weld-points in a batch that has a sibling, it will be indicated by an icon. 
- If there are any weld-points in the batch that are rejected in any other batch, that particular weld-point will be disabled.
- If there are any weld-points in the batch that have related weld-points in any other batch that are rejected, then that particular weld-point will be disabled.
- If 4 or more weld-points in a batch are rejected, then all the weld-points in the batch should be examined 100%. All the weld-points get automatically selected with status “SS”. Any new weld-points that are added to this batch will automatically get selected and have status “SS”. Status of the batch is “Awaiting NDE”.
- If any 2nd level tracer of a batch is rejected, then all the weld-points in the batch should be examined 100%. All the weld-points get automatically selected with status “SS”. Any new weld-points that are added to this batch will automatically get selected and have status “SS”. Status of the batch is “Awaiting NDE”.

11.3 NDE 100

This module helps the user to select the joints which required 100% NDE for all the categories like (RT, MT, and PWHT etc.) The user can also view the No of joints to select per NDE category in the dashboard section in the same screen.

In this module there is an advanced search which helps the user to unselect the joint and roll back the progress which was recorded mistakenly by the users. This will be for limited users like “System admin” or “Project admin”



NDE Cat.	Joint to select
HT	19
PMI	96
PT	6185

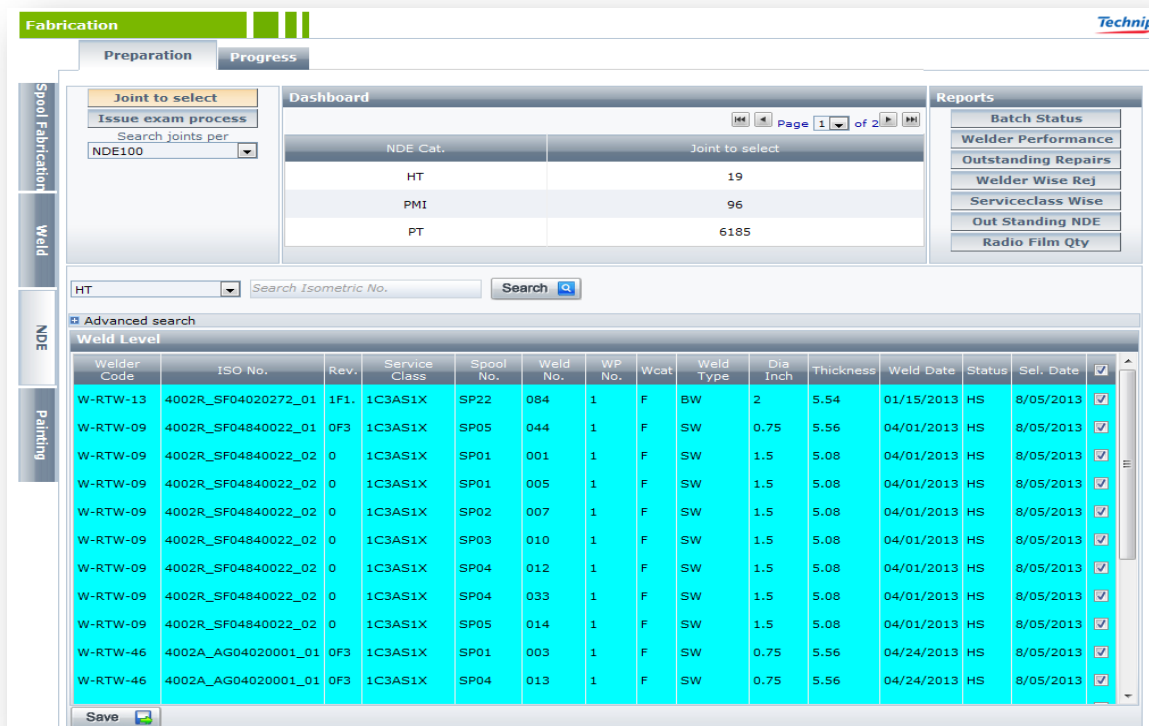
The user needs to select any category of NDE from the dropdown and search for the joints.

The user can use Icon  to view the list of joints to select. The user can select any desired joint or he can select all the joints. The status of the joint will be “H” if it is not selected and the status of the joint will change to “HS” once it is changed.

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In this screen the user can see all the joints which required 100% NDE as per the NDE matrix and also the joints created by EasyPiping R1, R2, R3 as a suffix for the mother joint which is rejected.



Welder Code	ISO No.	Rev.	Service Class	Spool No.	Weld No.	WP No.	Wcat	Weld Type	Dia Inch	Thickness	Weld Date	Status	Sel. Date	✓
W-RTW-13	4002R_SF04020272_01	1F1	1C3AS1X	SP22	084	1	F	BW	2	5.54	01/15/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_01	0F3	1C3AS1X	SP05	044	1	F	SW	0.75	5.56	04/01/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_02	0	1C3AS1X	SP01	001	1	F	SW	1.5	5.08	04/01/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_02	0	1C3AS1X	SP01	005	1	F	SW	1.5	5.08	04/01/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_02	0	1C3AS1X	SP02	007	1	F	SW	1.5	5.08	04/01/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_02	0	1C3AS1X	SP03	010	1	F	SW	1.5	5.08	04/01/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_02	0	1C3AS1X	SP04	012	1	F	SW	1.5	5.08	04/01/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_02	0	1C3AS1X	SP04	033	1	F	SW	1.5	5.08	04/01/2013	HS	8/05/2013	✓
W-RTW-09	4002R_SF04840022_02	0	1C3AS1X	SP05	014	1	F	SW	1.5	5.08	04/01/2013	HS	8/05/2013	✓
W-RTW-46	4002A_AG04020001_01	0F3	1C3AS1X	SP01	003	1	F	SW	0.75	5.56	04/24/2013	HS	8/05/2013	✓
W-RTW-46	4002A_AG04020001_01	0F3	1C3AS1X	SP04	013	1	F	SW	0.75	5.56	04/24/2013	HS	8/05/2013	✓

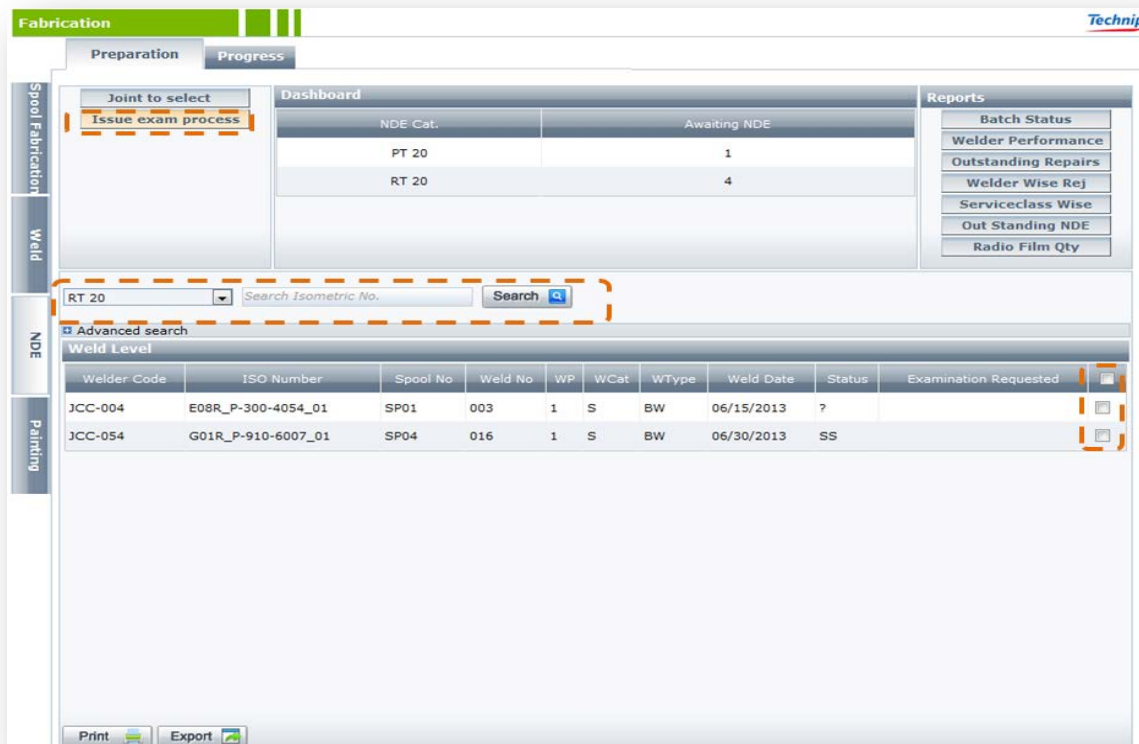
Notes:

- When the joint is “ACCEPTED”, the weld-point will no longer be displayed in this screen.
- When a weld-point is selected, any related weld-points of the selected weld-point also get selected automatically and its status changes to ‘HS’.

11.4 Issue Examination Process

This module helps the user to List out the joints to be examined for a specific NDE category. When the users select all the joints in the “NDE 100” or “Joint to select” all the joints will be viewed in this screen. The user can issue the joints by selecting the desired joints or he can select all the joints by selecting check all.

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Dashboard	
NDE Cat.	Awaiting NDE
PT 20	1
RT 20	4

Advanced search									
Weld Level									
Welder Code	ISO Number	Spool No	Weld No	WP	WCat	WType	Weld Date	Status	Examination Requested
JCC-004	E08R_P-300-4054_01	SP01	003	1	S	BW	06/15/2013	?	
JCC-054	G01R_P-910-6007_01	SP04	016	1	S	BW	06/30/2013	SS	

Notes:

- The user can use Icon  to print all the listed joints to Issue for examination
- The user can use Icon  to export all the listed joints into an excel format to issue for examination

11.5 Examination Progress

This module helps the user to update the examination results for the selected welds to the different categories of NDE .The user can update the progress by “Manually” and “Import procedure” as well.

The user can search the desired spools and welds by the different search options available on the screen.

- Search by Isometric
- Search by Barcode

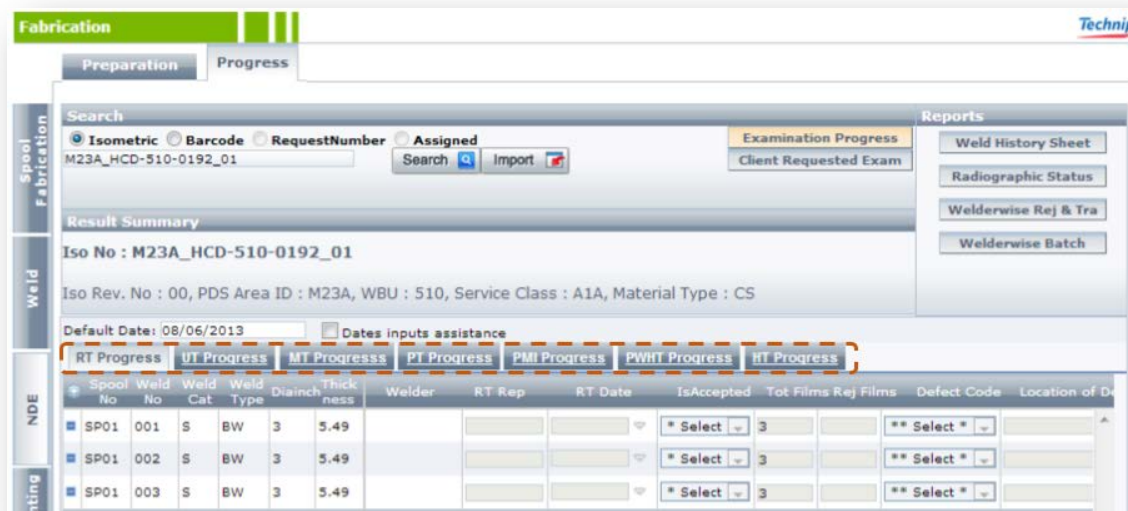
There are different tabs for the different NDE categories to update the examination result.

Note:

- Only the welds that are selected in the ‘Batch Management’ and NDE100 screen for all NDE category examination will be enabled for entry. It will be disabled for all the other welds
- If a weld has more than 1 weld-point, user has to click on  which will open a popup where the user can enter the examination results of all weld-points.
- Once the progress is recorded for weld-points in the examination progress screen. results will be disabled and modification will not be possible except for the following fields:
RT Rep. RT Date, Rejected Films, Defect Code, Location of Defect.

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- If any of the weld-points are rejected, new revision of the rejected weld-point is created with suffix R1. For further revision R2, R3, etc. These newly created welds because of revision will be tested 100% for RT examination. It will appear in the NDE 100 module.
- If any of the weld-points of a weld is rejected, then the entire weld will be displayed as rejected.

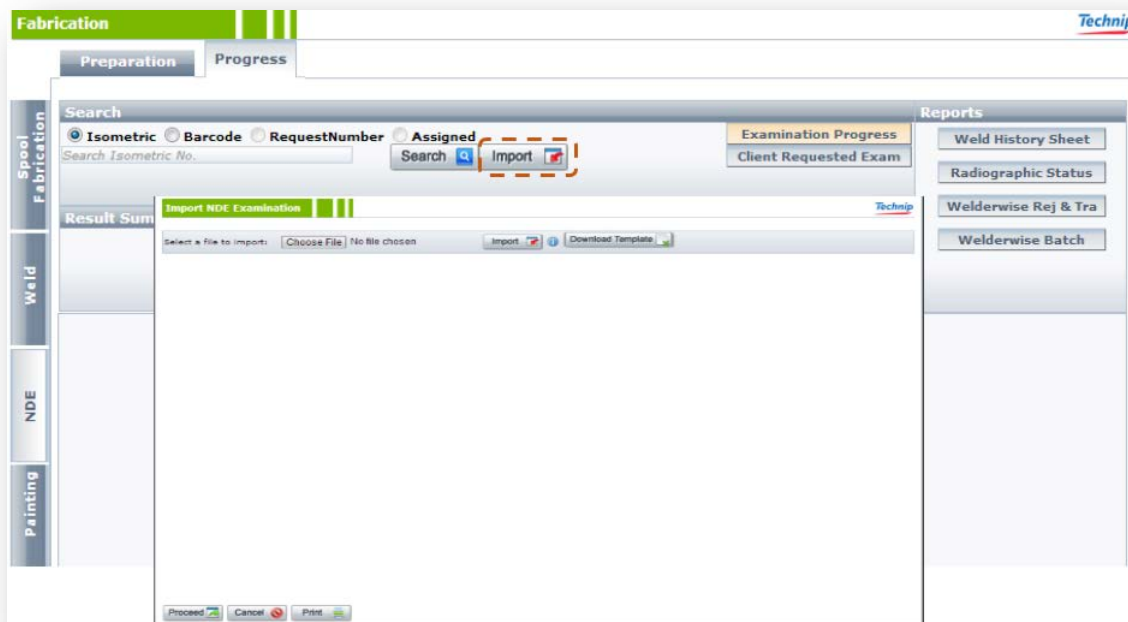


Spool No	Weld No	Weld Cat	Weld Type	Dia inch	Thick ness	Welder	RT Rep	RT Date	IsAccepted	Tot Films	Rej Films	Defect Code	Location of D
SP01	001	S	BW	3	5.49				* Select	3		** Select *	
SP01	002	S	BW	3	5.49				* Select	3		** Select *	
SP01	003	S	BW	3	5.49				* Select	3		** Select *	

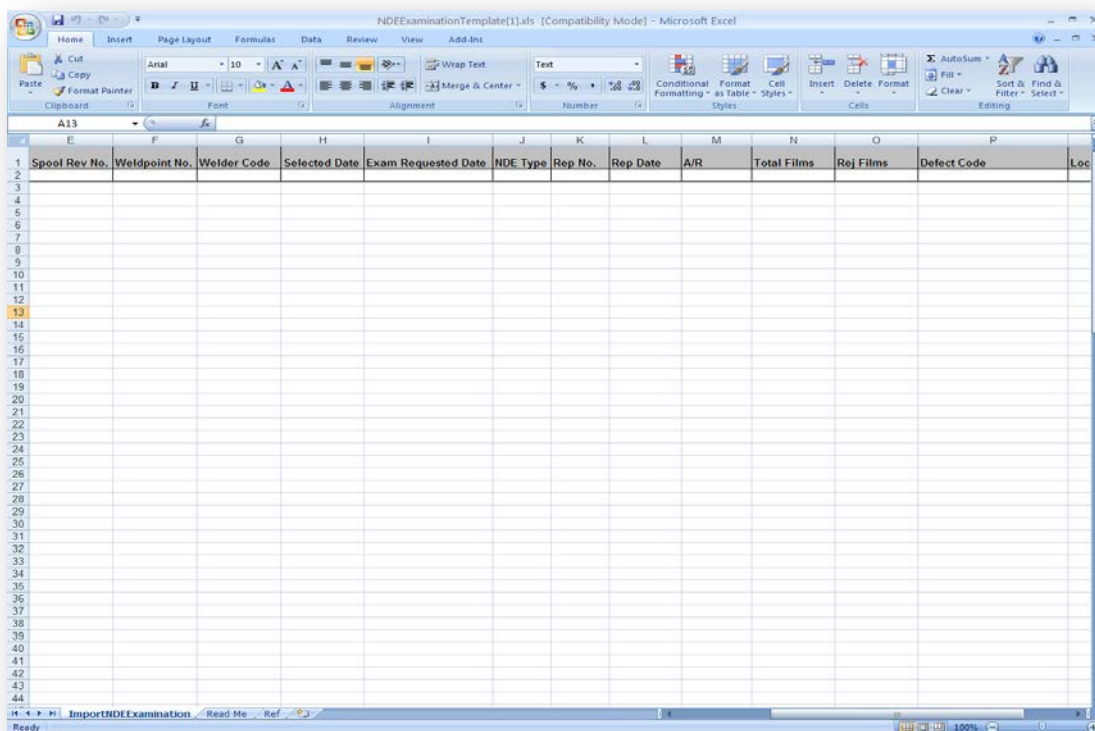
Import procedure for NDE Examination Progress

This module helps the user to import all the Examination progress into the system. In this module the system allows the user to generate a template, this template is duly filled by the user and import the same template into the system.

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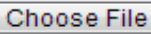
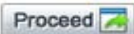
The user can use the icon  to download the template and fill the template with the relevant information



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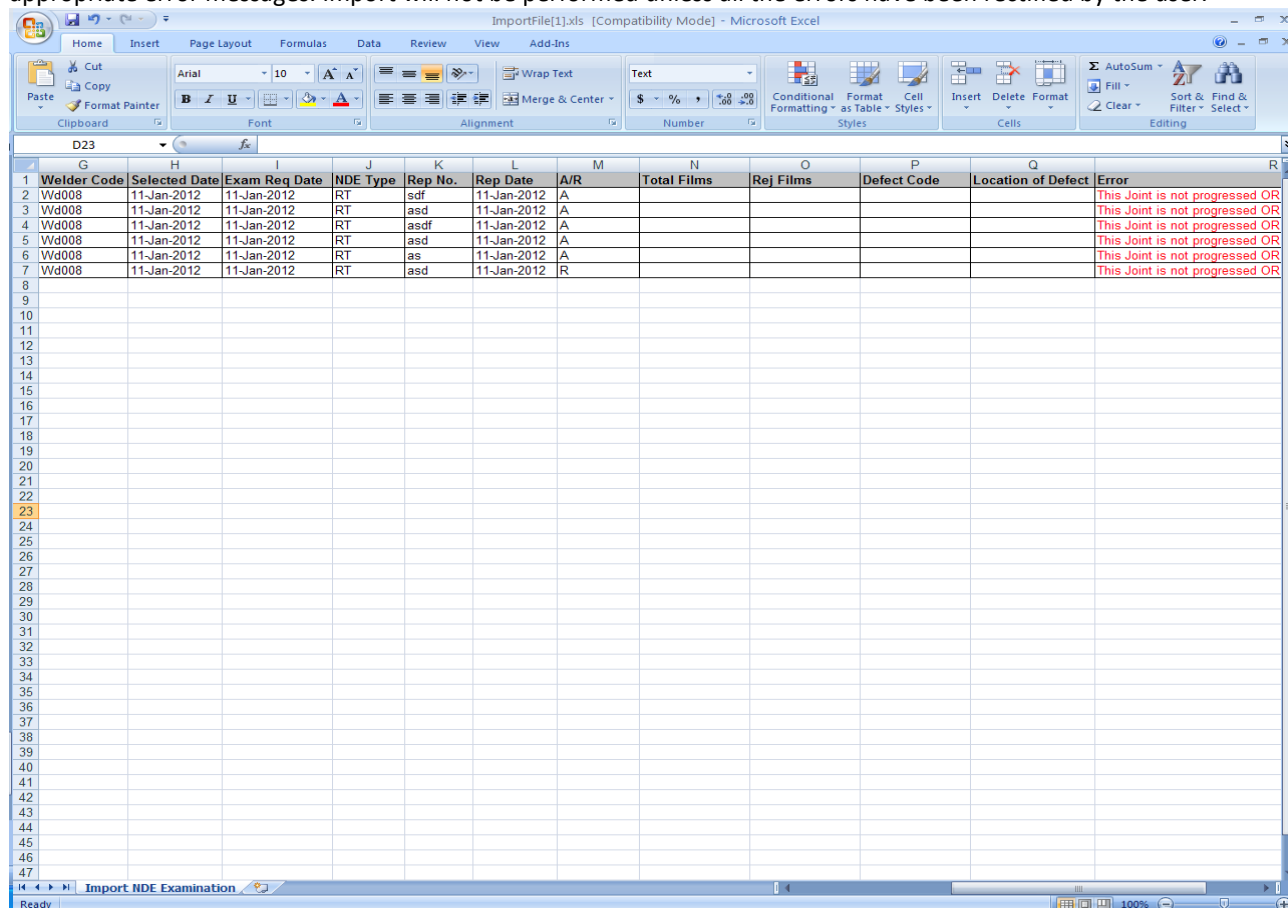
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The system will allow the user to browse the file, the user can use the Icon  and select the Import file and  all the information is imported into the system.

Note

- System will allow .xls or .xlsx files only
- Empty file is not allowed
- ISO number, spool number, weld number, ISO Revision number, spool revision number which do not exist will be rejected by the system.
- Import of welds should have weld progress date otherwise rejected by the system.
- Import of welds and NDE Type which do not exist will be rejected by the system, along with the error message - Invalid NDE Type
- Duplicates values are not allowed

The system will validate the information provided by the user and the system will highlight the errors in red with the appropriate error messages. Import will not be performed unless all the errors have been rectified by the user.



	G	H	I	J	K	L	M	N	O	P	Q	R
	Welder Code	Selected Date	Exam Reg Date	NDE Type	Rep No.	Rep Date	A/R	Total Films	Rej Films	Defect Code	Location of Defect	Error
2	Wd008	11-Jan-2012	11-Jan-2012	RT	sdf	11-Jan-2012	A					This Joint is not progressed OR
3	Wd008	11-Jan-2012	11-Jan-2012	RT	asd	11-Jan-2012	A					This Joint is not progressed OR
4	Wd008	11-Jan-2012	11-Jan-2012	RT	asdf	11-Jan-2012	A					This Joint is not progressed OR
5	Wd008	11-Jan-2012	11-Jan-2012	RT	asd	11-Jan-2012	A					This Joint is not progressed OR
6	Wd008	11-Jan-2012	11-Jan-2012	RT	as	11-Jan-2012	A					This Joint is not progressed OR
7	Wd008	11-Jan-2012	11-Jan-2012	RT	asd	11-Jan-2012	R					This Joint is not progressed OR
8												
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
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47												

The user will be prompted with how to view the validated import file: via Excel or Screen.

The Excel file will not be able to highlight the errors and overwritten cells in color. However, the error messages will be shown. If the user selects to view the validations in the screen.

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Select a file to import:

Isometric No	Rev No	Spool No	Weld No	Spool Rev No	Weldpoint No	Welder Code	Sel Date	Exam Req Date	NDE Type	Rep No	Rep Date	A/R	Total Films	Rej Films	Defect Code	Location of Defect	Error
4004A_AG04090512_01	01	SP01	001		1	Wd008	11-Jan-2012	11-Jan-2012	RT	asf	11-Jan-2012	A					This Joint is not progressed OR is invalid
4004A_AG04090512_01	01	SP02	002		1	Wd008	11-Jan-2012	11-Jan-2012	RT	asd	11-Jan-2012	A					This Joint is not progressed OR is invalid
4004A_AG04090512_01	01	SP03	003		1	Wd008	11-Jan-2012	11-Jan-2012	RT	asdf	11-Jan-2012	A					This Joint is not progressed OR is invalid
4004A_AG04090512_01	01	SP04	004		1	Wd008	11-Jan-2012	11-Jan-2012	RT	asd	11-Jan-2012	A					This Joint is not progressed OR is invalid
B06A-137-GLP-01671-01	2	SP01	001		1	Wd008	11-Jan-2012	11-Jan-2012	RT	as	11-Jan-2012	A					This Joint is not progressed OR is invalid
B06A-137-GLP-01671-01	2	SP02	002		1	Wd008	11-Jan-2012	11-Jan-2012	RT	asd	11-Jan-2012	R					This Joint is not progressed OR is invalid

11.6 Client Examination Progress

This module helps the user to update the NDE examination progress for the welds from the two categories

- Batch is cleared and an additional joint need to be tested in that batch.
- Not defined in the NDE Matrix, but the client/QC inspector calls for NDE examination

The user can search the desired spools and welds by the different search options available on the screen.

- Search by Isometric
- Search by Barcode

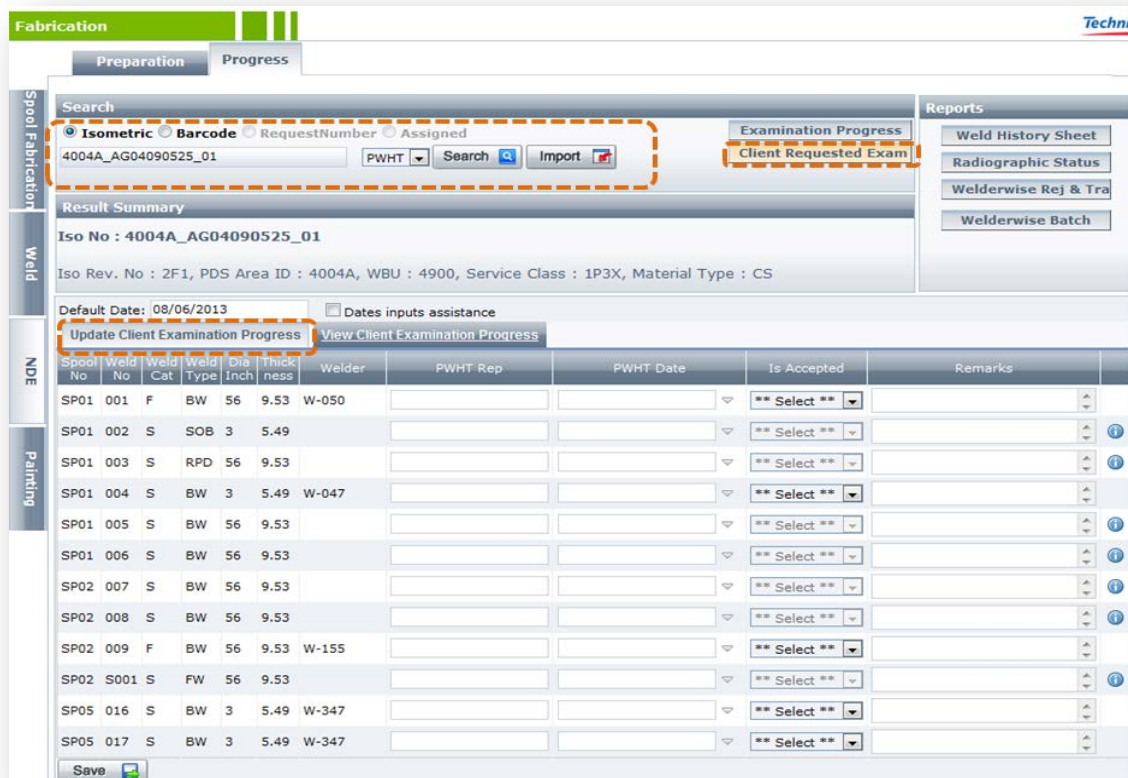
The user can view two tabs available in this screen as follows

- Update Client Examination Progress
- View Client Examination Progress

In the “Update Client Examination progress” tab the user can update the progress for any joint that has been selected.

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Note:

- If a weld has more than 1 weld-point, user has to click on  which will open a popup where the user can enter the examination results of all weld-points.
- Once the progress is recorded for weld-points in the examination progress screen. results will be disabled and modification will not be possible except for the following fields:
RT Rep. RT Date, Rejected Films, Defect Code, Location of Defect.
- If any of the weld-points are rejected, new revision of the rejected weld-point is created with suffix R1. For further revision R2, R3, etc. These newly created welds because of revision will be tested 100% for RT examination. It will appear in the NDE 100 module.
- If any of the weld-points of a weld is rejected, then the entire weld will be displayed as rejected.
- The client requested joints will be highlighted in orange color in the Batch management screen for easy identification.
- Tracer management is not handled

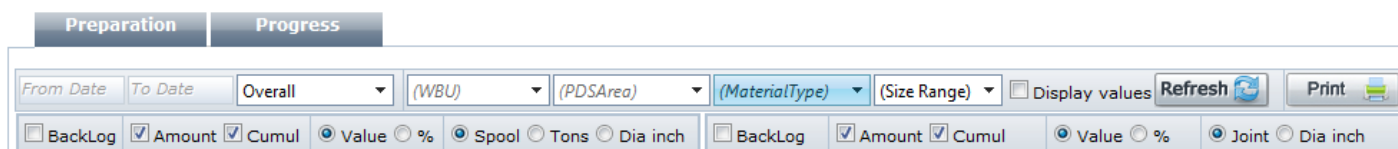
In the “View client examination progress” tab, user can view all the updated progress in a read only mode

11.7 Fabrication Dash Board

Fabrication Dashboard provides graphical representation of progressed quantities and backlogs for various Fabrication activities at Spool level and Joints level. Various Filter option are provided to customize the filters for user convenience.



11.7.1 Filters

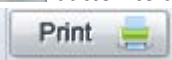


Filters provided to customise the search criteria for user convenience. This option shall enable user to view the progress and backlog quantities in terms of Dia- inch, No of spools or Tons Etc. The Main option in the Filters are

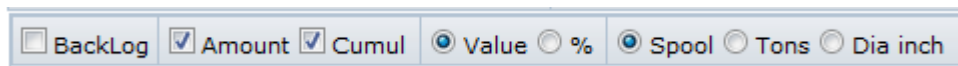
Global Filters :

1. Time scale : Facilitates user to select specified time period or predefined time.
2. WBU
3. Design Area (PDS Area)
4. Material Type
5. Size Range

After selection of Global filters user need to Click on  button to apply filters on Graphs.

Print option is provided to print the graph with selected filters. 

Spool Fabrication Progress Filters:



1. Backlog : Provide the information Backlog for each and every activity of Spool Fabrication progress at Spool Level.
2. Amount :Display the periodic values (Bar chart)
3. Cumul : Display the overall values (Line chart)
4. Values or % : Display the values of progress % of Total scope.
5. Spools, Tons or Dia Inch : Display the values of in No of spools or Tons or Diainch (Spool level Shop Scope) based on the selection.

Joint Fabrication Progress Filters:

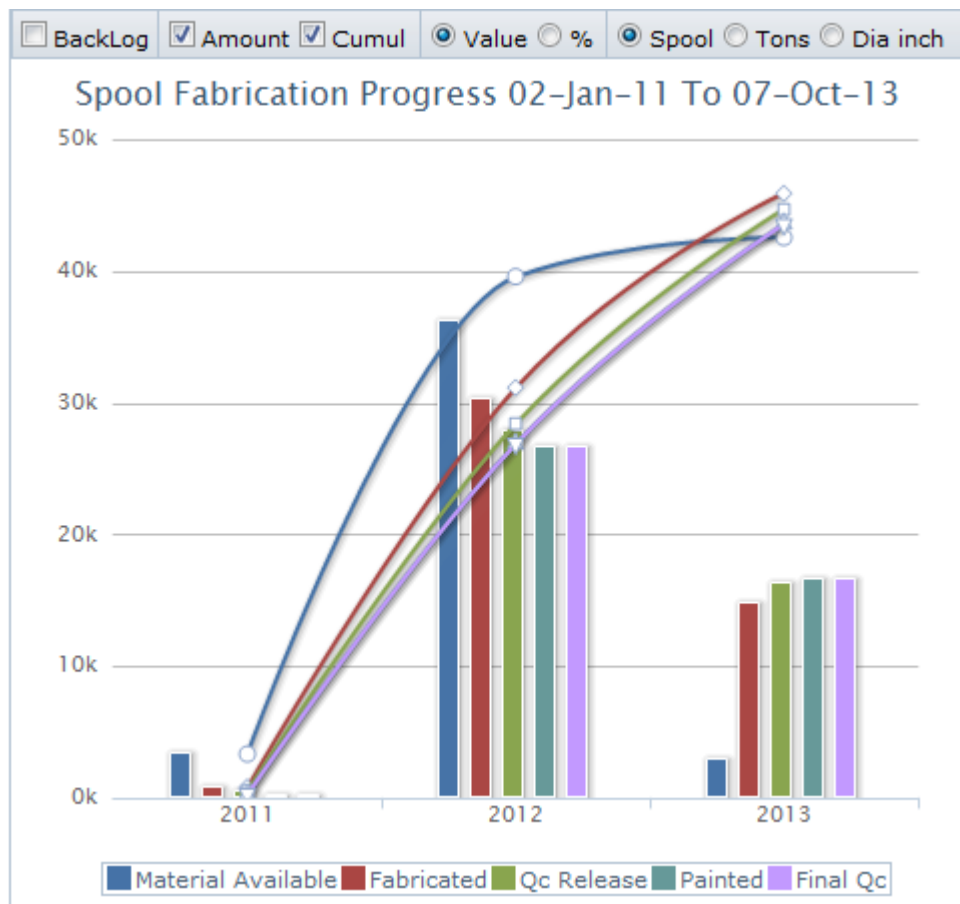


1. Backlog : Provides the information of Backlog for all activities of Joint fabrication and NDE process. (Note: Backlog for NDE progress shall not be displayed if other filters are on except Time filters)
2. Amount :Display the periodic values (Bar chart)
3. Cumul : Display the overall values (Line chart)
4. Values or % : Display the values of progress % of Total scope.
5. Joint or Dia inch : Displays the values in No of Joints wise of Dia- inch wise

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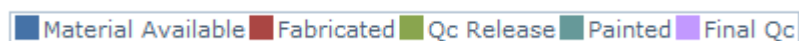
11.7.2 Spool Fabrication Chart :



Spool Fabrication chart provides progress and backlog information for following Activities at Spool level.

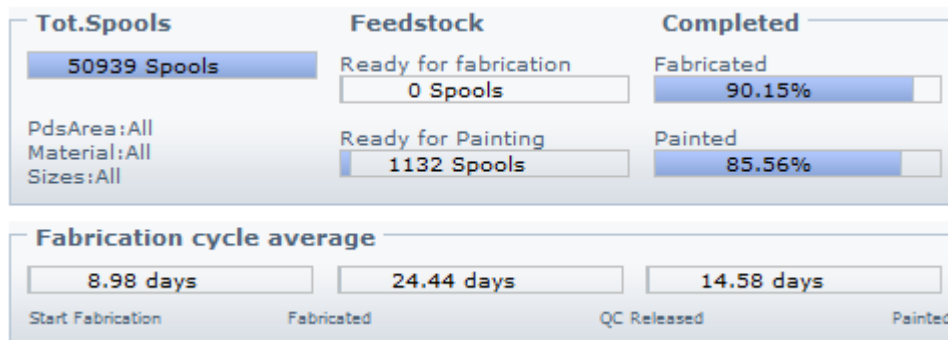
1. Material Availability
2. Fabricated
3. QC Release
4. Painted
5. Final QC

User can switch the visibility of Activities.



By Clicking the above icons user can show or hide the activity.

11.7.1 Spool Fabrication Feed Stock Quantities and KPI :



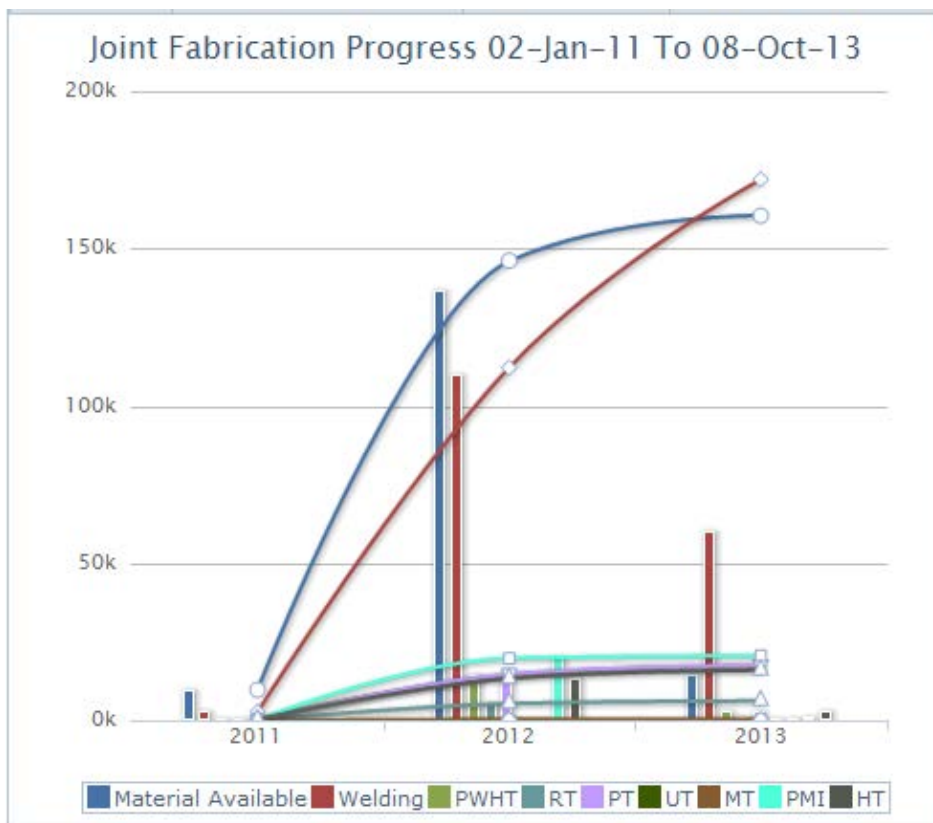
Spool Fabrication Feed stock Quantities represent the overall Scope and Feed stock availability data.

1. Feed stock quantities always display the overall quantities, So Time Filters in global filters are not applicable.
2. Total Spools always represent the overall scope of the project.
3. All the values can be displayed in Spool , Tons or Dia Inch based on the selection in the Spool Fabrication filter

☒ Spool ☐ Tons ☐ Dia inch

KPI For Spool Fabrication : KPI represents avg time taken for each activities based on historical data of the project.

11.7.2 Joint Fabrication Chart :

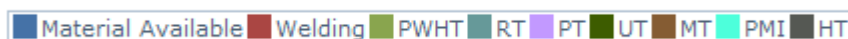


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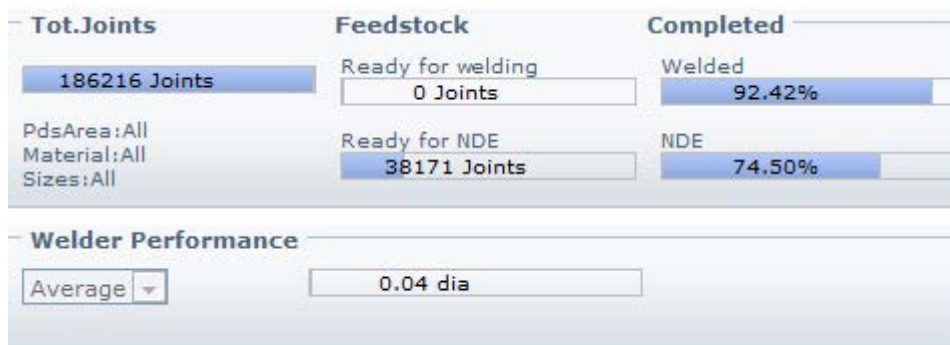
Joint Fabrication chart provides progress and backlog information for following Activities at Joint level. These values are represents only Shop Joints for which the Dia-Inch Factor is “Yes”.

1. Material Availability
2. Welding
3. PWHT
4. RT
5. PR
6. MT
7. PMI
8. HT

User can switch the visibility of progress by clicking the Following button on the Graph :

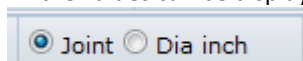


11.7.3 Joint Fabrication Feed Stock Quantities and KPI:



Joint Fabrication Feed stock Quantities represent the overall Scope and Feed stock availability data.

1. Feed stock quantities always display the overall quantities, So Time Filters in global filters are not applicable.
2. Total joints always represent the overall scope of the project.
3. All the values can be displayed in No of Joint or Dia-Inch based on the selection in the Spool Fabrication filter

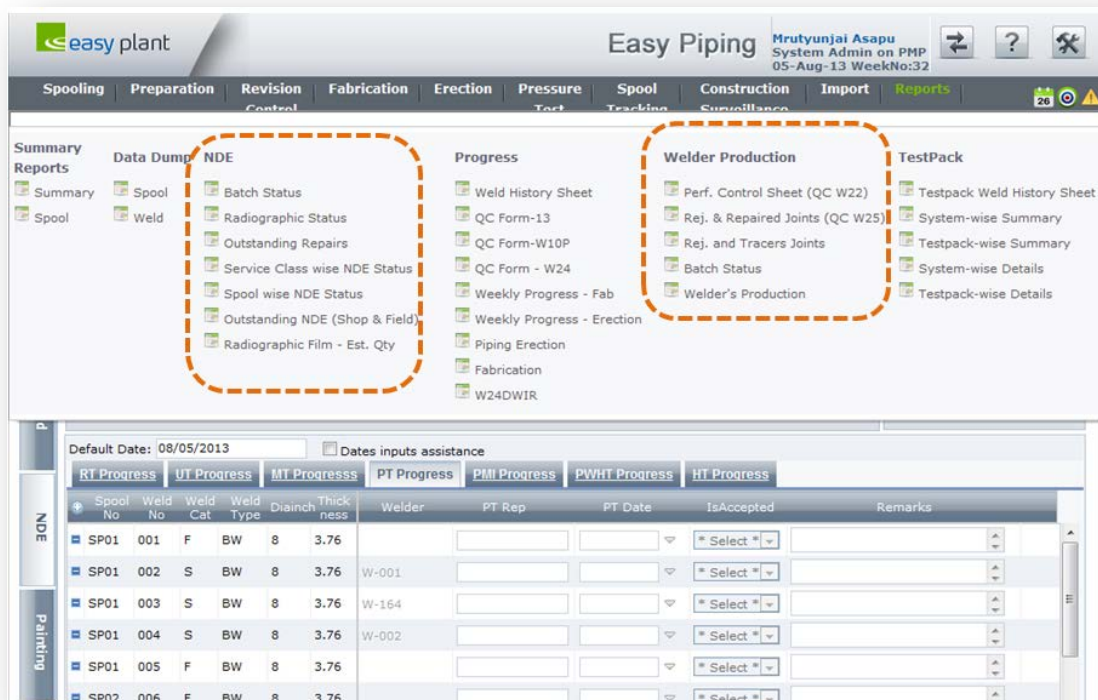


KPI For Welder Performance : KPI represents the overall Avg Dia Welded by a welder per day.

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11.8 NDE Reports

Easy Piping will allow the user to see certain reports related to the NDE which explained below in detail. The user can access the reports from the same screen or a separate tab which is only for all the reports.



Welders_Perf_Contr
ol_Sheet.xls

Welders_Perf_Control Sheet: gives for each welder weekly and cumulative statistics on percentage / number of joints welded accepted /rejected and type of defects



WelderwiseRejected
andTracersJoints.xls

Welder wise rejected and tracer joints: It will list all the rejected joints with the corresponding Tracers. Used to track backlog in penalty shoot.



WelderwiseRejected
andRepairedJoints.xls

Welder wise rejected and repaired joints: It will list for each welder the joints to be repaired - repaired. Used to track backlog in repairs.



WelderwiseBatchStat
us.xls

Welder wise batch status: It will gives examination status for a welder

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Welders_Production.
xls

Welders Production: It will give the production report for the certain selected period for a selected welder or all welders during that period.



9833N_4941_Batch_
Status_Report.xls

Batch status: It will gives released status of each welds regarding NDE



Radiographic_Film_St
atus_Report.xls

Radiographic status: It will give statistics on percentage / number of joints/films accepted /rejected and type of defects - weekly and cumulative



Outstanding_Repairs
_Report.xls

Outstanding Repairs: list all the joints to be repaired with all necessary details, including pending days since NDE done



Service_Class_wise_
NDE_Status_Report.xls

Service class wise NDE status: gives NDE status for each piping class. Used to check examination percentage compliance



Spool_wise_NDE_Sta
tus.xls

Spool wise NDE status: It will list all spools status regarding NDE, spool is Qc released or not and display the corresponding QC and W10 report number



Outstanding_NDE.xls

Outstanding NDE: lit all the joints awaiting NDE reports, including pending days since requested. Used for NDE backlog tracking



Radiographic_Film_E
stimated.xls

Radiographic film – Est. qty: It will give an estimated quantity of radiographic films to be shoot.

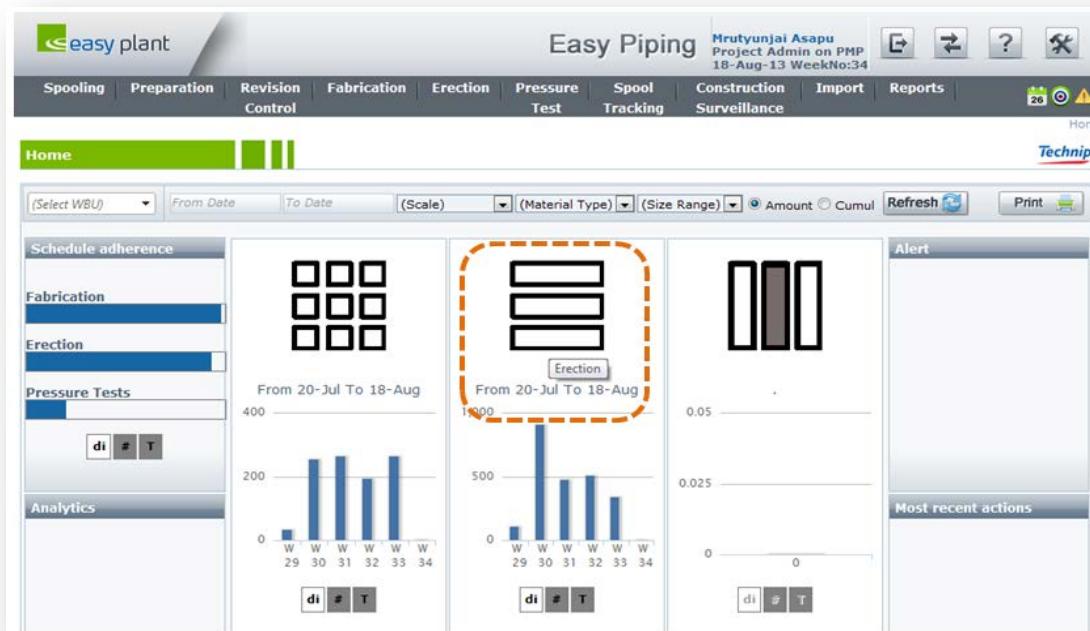
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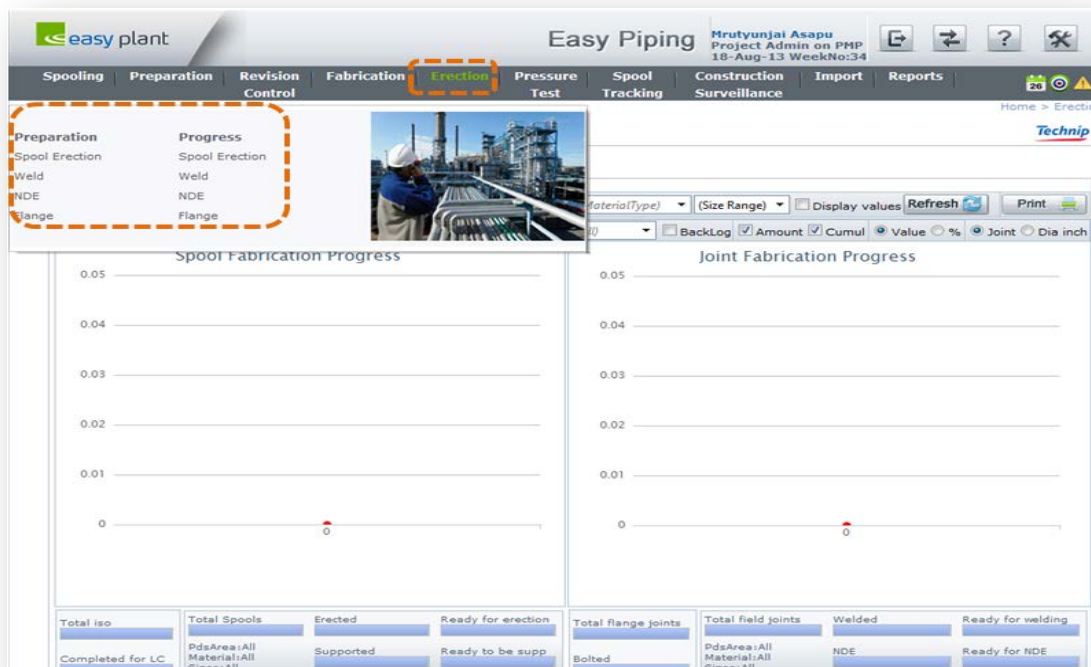
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[*Easy Piping Part 5: Erection Module*]

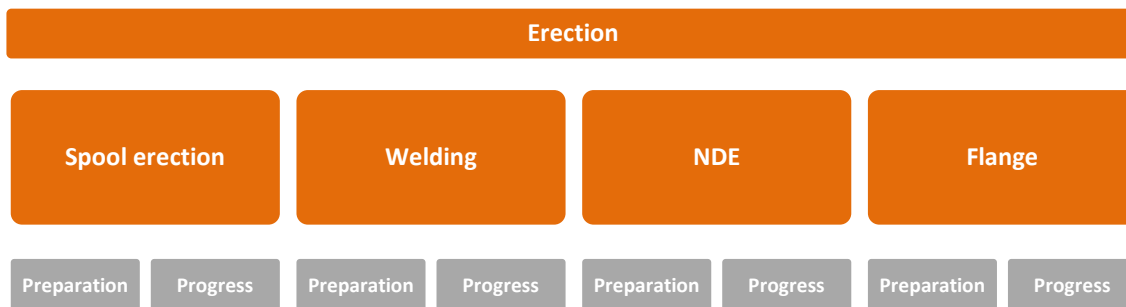
12. ERECTION MODULE

The Erection module is organized in 4 different sections corresponding to the different site activities. These sections allow the user to plan & prepare the workload and record the Progress for each activity. The user can also generate the Erection reports from this screen. All the sections and its uses are defined clearly in the below details.





ARTICLE I. DIFFERENT SECTIONS DURING ERECTION PHASE



12.1 Spool Erection

This screen contains two different sections which is preparation and progress. The preparation screen allows the user to plan and prepare the workload. The progress screen consists of several steps that user needs to follow. There can be some additional steps which shall be added if it is required for a project ([Refer to Para 1.1](#)). Definition of each step has to be agreed in the initial stage of the project

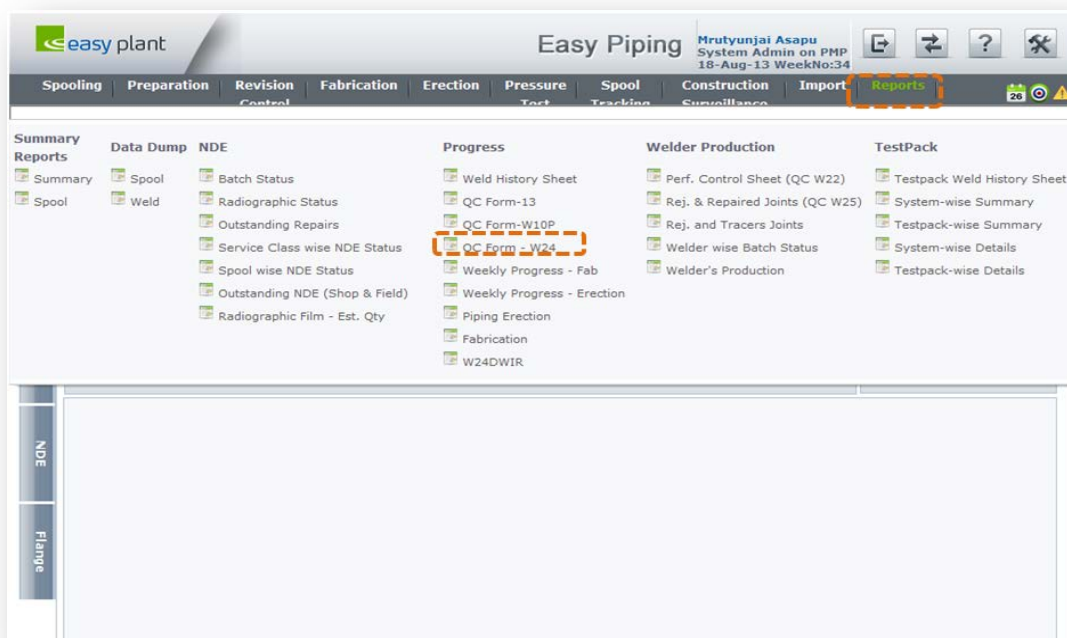
- To Site
- Erected
- Welded/Bolted
- Supported
- RFT

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In order to update progress for the above steps in Easy piping, the users have to follow the QC form (W-24 Form) The user can generate the QC forms in two methods.

3. Using search method – User can search for single Iso and spool
4. Using Excel Import method – User can List upto Max 50 spools in excel and generate the forms

- W-24 form is the daily progress report form, at spool level.
- The QC Form should be duly filled by workers about the details of the Field joints for the selected spool
 - Material traceability
 - Welding



12.2 Material Check

In the weld progress screen, the user can access the “material traceability” pop-up window. The user will enter the heat numbers filled in the QC – 13 forms by foreman. Easy piping will validate the entered heat numbers which are predefined in the project referential. ([Refer to Para 3.12](#))

When heat numbers are correctly recorded, the “material check” status of the corresponding spools will be automatically updated.

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Fabrication

Preparation Progress

Search: 4002A_MV04020151_01

Result Summary

Iso No : 4002A_MV04020151_01

Iso Rev. No : 2F3, PDS Area ID : 4002A, WBU : 4200, Service Class : 1C3AS1X, Material Type : CS NACE

Default Date: 08/18/2013 ☐ Dates inputs assistance

Spool No Weld No Weld Cat Weld Type Dia/Inch Thickness Cutting Date Beveling Date Fitup Date Preheat Date Weld Date DWIR

PSMS_ND00_06:Material Traceability - Microsoft Internet Explorer provided by TECHNIP Abu Dhabi

Material Traceability

Filter: (PDSArea) Search Line No. (AG)-Aboveground 4002A_SU04090679_01 (Serviceclass)

(Subcontractor) Search Priority (Areaclassification) Y

Fill Option: ☐ Default Trac. Number: Note: Please Save, Before going to next page !

ISO Number	Spool No	Sheet No	Weld No	Weld Cat	Weld Type	Size	Group 1	Ident Code 1	Trac. Number 1	Group 2	Ident Code 2	Trac. Number 2
4002A_SU04090679_01	SP01	1	001	S	BW	4	Flanges	CDJ11C		Fittings	COF8YX	
4002A_SU04090679_01	SP01	1	002	S	BW	4	Fittings	COF8YX		Fittings	CNAPK7	
4002A_SU04090679_01	SP01	1	003	S	BW	2	Fittings	CNAPK7		Flanges	CN41CS	
4002A_SU04090679_01	SP01	1	003RC	F	BW	2	(Select)			(Select)		

12.3 Weld Progress

In the Erection module, the weld progress screen will show only the field joints. Welding progress is filled in the W-24 form, and then the progress is recorded in easy piping in the weld progress screen. The system validates the Information of welders which are predefined in the project referential. (Refer to Para 3.7)

The system will validate the information of WPS recorded against the welder with the welder qualifications which are predefined in the Project referential. (Refer to Para 3.6) The system will alert the user if he records the incorrect information.

				PROJECT: PMP		PROJECT NO: 9833N		SUBCONTRACTOR: ELETECH							
OWNER Jatar Liquefied Gas Company				Date:											
QUALITY CONTROL FORM		W24		SPOOL ERECTION		DAILY WELDING PROGRESS		QCF Rev No.0							
ISO NO:		4002A_AV04877001_01		Rev:		0F1		Service Class: WIAIX		Matn: CS		Report No: W24-F-0824			
SPOOL No	DATE SITE	SIZE	Field Joints	Weld Type	Thk	TRACEABILITY			PROGRESS	WPS NO	WELDING		PRE HEAT	WELDING VT DATE	ELETECH TP
						DESC	IDENT CODE	TRAC No			WELDER NO (*)	Root			
SP01	13-Feb-2013	2	004	BW	5.54	Fittings	C8P32K			WPS-PMP-05	W-RTW-01	W-RTW-01		21-Mar-2013	
						Pipe	C6CNMS								
SP01	13-Feb-2013	2	012	BW	5.54					WPS-PMP-05	W-RTW-36	W-RTW-36		14-Mar-2013	
SP02	13-Feb-2013	2	007	BW	5.54	Pipe	C6CNMS			WPS-PMP-05	W-RTW-01	W-RTW-01		02-Apr-2013	
						Fittings	C8PATA								
SP03	13-Feb-2013	2	009	BW	5.54	Pipe	C6CNMS			WPS-PMP-05	W-RTW-01	W-RTW-01		02-Apr-2013	
						Pipe	C6CNMS								
SP04	13-Feb-2013	1.5	S001	TRN	7.14	Pipe	C6CNMS			WPS-PMP-05	W-RTW-01	W-RTW-01		03-Apr-2013	
						Supports	..								

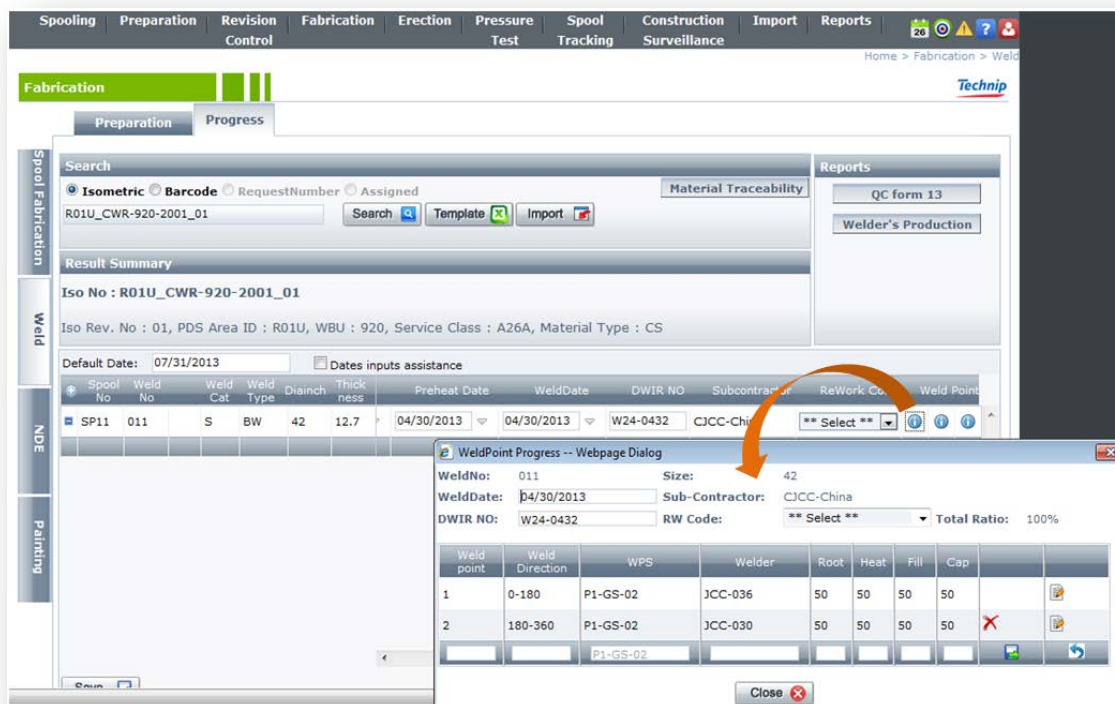
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The following are the weld progress columns for which the user needs to fill in this screen

- Cutting Date – User to input the date
- Beveling Date – User to input the date
- Fit-up Date – User to input the date
- Preheat Date – User to input the date
- Weld Date – User to input the date
- DWIR No. – User to input the QC Form No
- Subcontractor – Read Only
- Rework Code – User to select from dropdown
- Weld Point No. – User to input the number
- WPS No. – User to input the WPS No.
- Welder Code – User to input the Welder code

If the joint is progressed by only 1 welder (i.e., if there is only 1 weld point), then the user can enter the progress details in the same screen.

If the joint is progressed by more than 1 welder (i.e., if there are more than 1 weld points), then the user has to click the icon  upon which a pop-up is displayed to the user for entering the weld point details.



The screenshot shows the Technip software interface with the 'Fabrication' tab selected. The 'Weld Point Progress' dialog box is open, displaying the following information:

WeldNo: 011, Size: 42
WeldDate: 04/30/2013, Sub-Contractor: CICC-China
DWIR NO: W24-0432, RW Code: ** Select **
Total Ratio: 100%

Weld point	Weld Direction	WPS	Welder	Root	Heat	Fill	Cap	
1	0-180	P1-GS-02	JCC-036	50	50	50	50	
2	180-360	P1-GS-02	JCC-030	50	50	50	50	
		P1-GS-02						

The dialog box also includes a 'Close' button at the bottom right.

Note:

- In the pop-up screen, when the user enters the progress information for the first weld-point and saves, the same WPS No. will be automatically loaded in read-only mode for entering the progress information of the second weld-point.
- In the pop-up screen, when the user enters the progress information for the first weld-point and saves, the user has to enter a different welder code for the second weld-point. The system will validate the welder code.

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- In the pop-up screen, Root and Cap are mandatory fields. The total of Root and Cap for a joint should be 100. The total of Heat and Fill for a joint can be 0 or 100.
- Modification of WPS No, Welder code is not allowed for those weld-points that are selected for examination in any of the batches OR already examined.

12.4 To site

This step will be updated by the user from the W-24 QC form. Once the spools are received at site and confirmed by the area supervisor.

QUALITY CONTROL FORM		W24		SPOOL ERECTION		DAILY WELDING PROGRESS		QCF Rev No.0		Date :				
PROJECT :		PMP		PROJECT NO :		3633N		SUBCONTRACTOR :		ELETECH				
OWNER		Jatar Liquefied Gas Company		Rev :		0F1		Service Class :		W1A1X				
ISO NO :		4002A_AV04877001_01		Material :		CS		Report No :		W24-F-0824				
SPOOL No	DATE SITE	SIZE	Field Joints	Weld Type	Thk	TRACEABILITY		PROGRESS	WPS NO	WELDER NO (*)		PRE HEAT	WELDING VT DATE	ELETECH TP
						DESC	IDENT CODE			TRAC No	Root			
SP01	13-Feb-2013	2	004	B/W	5.54	Fittings	C8P3CK		WPS-PMP-05	W-RTW-01	W-RTW-01		21-Mar-2013	
						Pipe	C8CNM5							
SP01	13-Feb-2013	2	012	B/W	5.54				WPS-PMP-05	W-RTW-36	W-RTW-36		14-Mar-2013	
SP02	13-Feb-2013	2	007	B/W	5.54	Pipe	C8CNM5		WPS-PMP-05	W-RTW-01	W-RTW-01		02-Apr-2013	
						Fittings	C8PA7A							
SP03	13-Feb-2013	2	009	B/W	5.54	Pipe	C8CNM5		WPS-PMP-05	W-RTW-01	W-RTW-01		02-Apr-2013	
						Pipe	C8CNM5							
SP04	13-Feb-2013	1.5	S001	TRN	7.14	Pipe	C8CNM5		WPS-PMP-05	W-RTW-01	W-RTW-01		03-Apr-2013	
						Supports	..							

12.5 Erected

This step will be updated by the user from the W-24 QC form. Once the spools are erected at site and confirmed by the area supervisor.

Erection **Progress**

Search: **4002A_MV04020156_01**

Result Summary
 Iso No : 4002A_MV04020156_01
 Iso Rev. No : 0, PDS Area ID : 4002A, WBU : 4200, Service Class : 1C3AS1X, Material Type : CS NACE

Default Date: 08/19/2013 ☐ Dates inputs assistance

Spool No	Sheet No	Field No	Shop Dia	Field Dia	TO SITE	ERECTED	WELDED BOLT	SUPPORTED	RFT
SP01	1	60	26		10/16/2012	10/17/2012	10/28/2012	10/28/2012	02/02/2013
SP02	1	6	0		10/16/2012	10/28/2012	10/28/2012	10/28/2012	10/28/2012
SP03	2	48	24			01/14/2013	01/30/2013	01/30/2013	02/02/2013
SP04	2								
SP05	2								
SP06	2								
SP07	3								
SP08	3								

WELDING

WPS NO	WELDER NO	WELDED BOLT	PRE HEAT	WELDING DATE	TP
WPS-PMP-05	W-RTW-01	W-RTW-01		21-Mar-2013	
WPS-PMP-05	W-RTW-36	W-RTW-36		14-Mar-2013	
WPS-PMP-05	W-RTW-01	W-RTW-01		02-Apr-2013	
WPS-PMP-05	W-RTW-01	W-RTW-01		02-Apr-2013	
WPS-PMP-05	W-RTW-01	W-RTW-01		03-Apr-2013	

12.6 Welded/Bolted

This step will be updated by the user from the W-24 QC form. Once the spools are erected and all the joints of the spools are welded or bolted.

12.7 Supported

This step will be updated by the user from the W-23 QC form. Once all the supported are erected and welded.

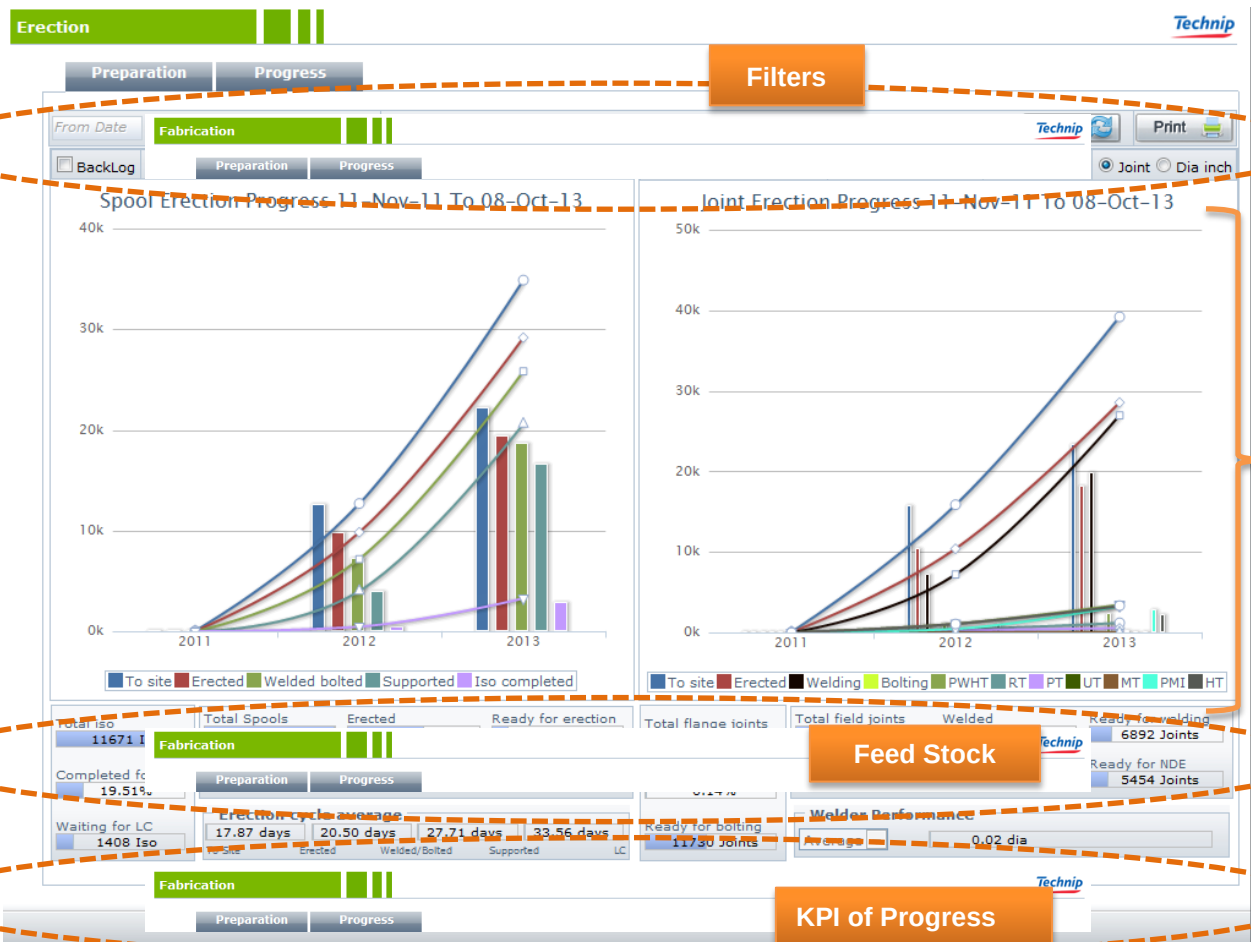
12.8 RFT

This step is an automatic update by the Easy Piping system, if all the predecessor steps are completed.

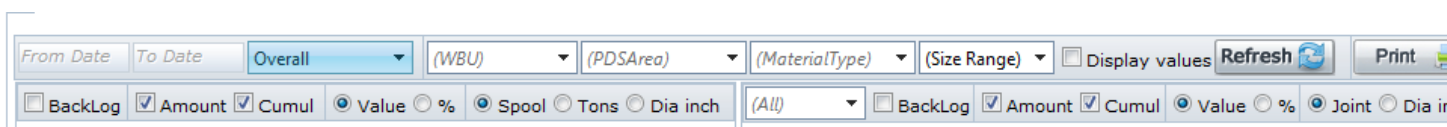
12.9 ERECTION DASH BOARD

Erection Dashboard provides graphical representation of progressed quantities and backlogs for various Erection activities at Spool level, Joints level and Flange Joint . Various Filter option are provided to customize the filters for user convenience.

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12.9.1 Filters



Filters provided to customise the search criteria for user convenience. This option shall enable user to view the progress and backlog quantities in terms of Dia- inch, No of spools, No of Flange Joints or Tons Etc. The Main option in the Filters are

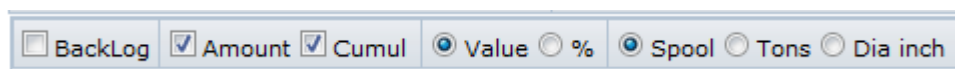
Global Filters :

1. Time scale : Facilitates user to select specified time period or predefined time.
2. WBU
3. Design Area (PDS Area)
4. Material Type
5. Size Range

After selection of Global filters user need to Click on  button to apply filters on Graphs.

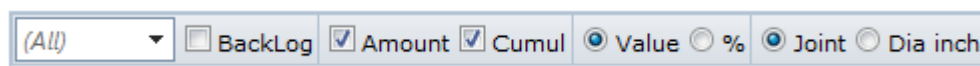
Print option is provided to print the graph with selected filters. 

Spool Erection Progress Filters:



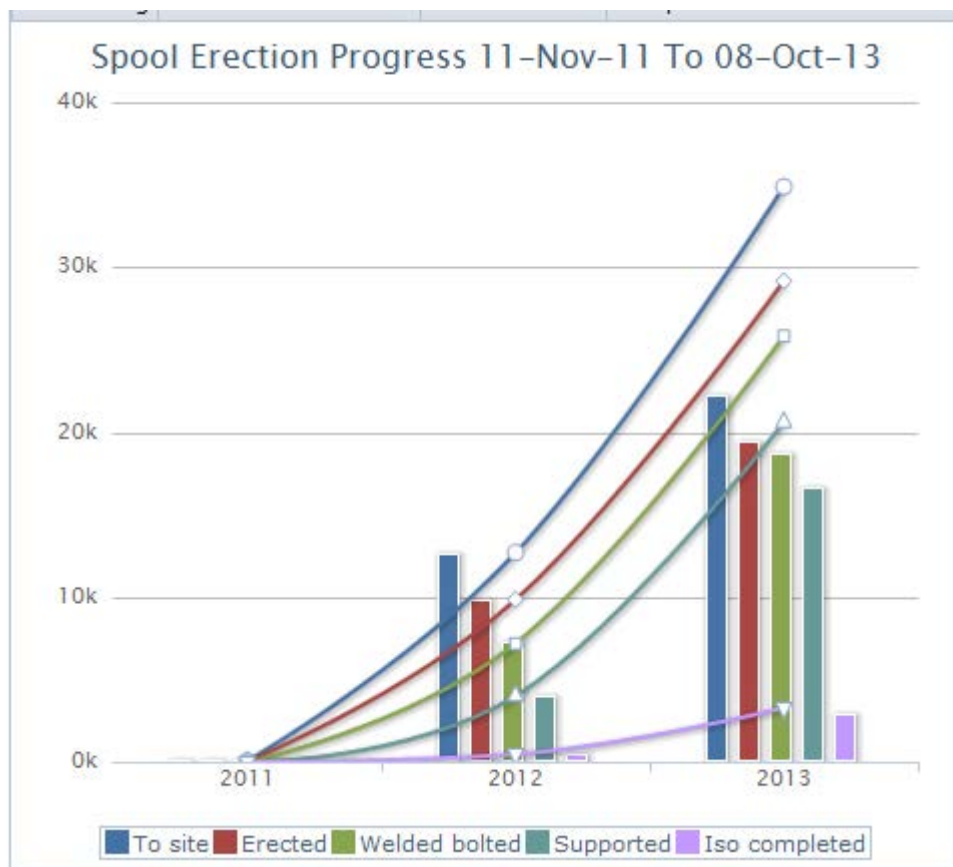
1. Backlog : Provide the information Backlog for each and every activity of Spool Fabrication progress at Spool Level.
2. Amount :Display the periodic values (Bar chart)
3. Cumul : Display the overall values (Line chart)
4. Values or % : Display the values of progress % of Total scope.
5. Spools, Tons or Dia Inch : Display the values of in No of spools or Tons or Diainch (Spool level Field Scope) based on the selection.

Joint Filed Progress Filters:



1. All: If user wishes to view the Flange joint progress then he need to select the “All” option in the Dash Board.
2. Backlog : Provides the information of Backlog for all activities of Joint fabrication and NDE process. (Note: Backlog for NDE progress shall not be displayed if other filters are on except Time filters)
3. Amount :Display the periodic values (Bar chart)
4. Cumul : Display the overall values (Line chart)
5. Values or % : Display the values of progress % of Total scope.
6. Joint or Dia inch : Displays the values in No of Joints wise of Dia- inch wise

12.9.2 Spool Erection Chart :



Spool Erection chart provides progress and backlog information for following Activities at Spool level.

1. Sent to Site
2. Erected
3. Welded & Bolted
4. Supported
5. Iso Completed

User can switch the visibility of Activities.



By Clicking the above icons user can show or hide the activity.

12.9.3 Spool Erection Feed Stock Quantities and KPI

Total iso 11671 Iso	Total Spools 50939 Spools	Erected 57.30%	Ready for erection 5723 Spools												
Completed for LC 19.51%	PdsArea:All Material:All Sizes:All	Supported 40.48%	Ready to be supp 5224 Spools												
Waiting for LC 1408 Iso	Erection cycle average <table border="1"> <tbody> <tr> <td>17.87 days</td> <td>20.50 days</td> <td>27.71 days</td> <td>33.56 days</td> </tr> <tr> <td>To Site</td> <td>Erected</td> <td>Welded/Bolted</td> <td>Supported</td> </tr> <tr> <td></td> <td></td> <td></td> <td>LC</td> </tr> </tbody> </table>			17.87 days	20.50 days	27.71 days	33.56 days	To Site	Erected	Welded/Bolted	Supported				LC
17.87 days	20.50 days	27.71 days	33.56 days												
To Site	Erected	Welded/Bolted	Supported												
			LC												

Spool Erection Feed stock Quantities represent the overall Scope and Feed stock availability data.

1. Feed stock quantities always display the overall quantities, So Time Filters in global filters are not applicable.
2. Total Spools always represent the overall scope of the project.
3. All the values can be displayed in Spool , Tons or Dia Inch based on the selection in the Spool Fabrication filter

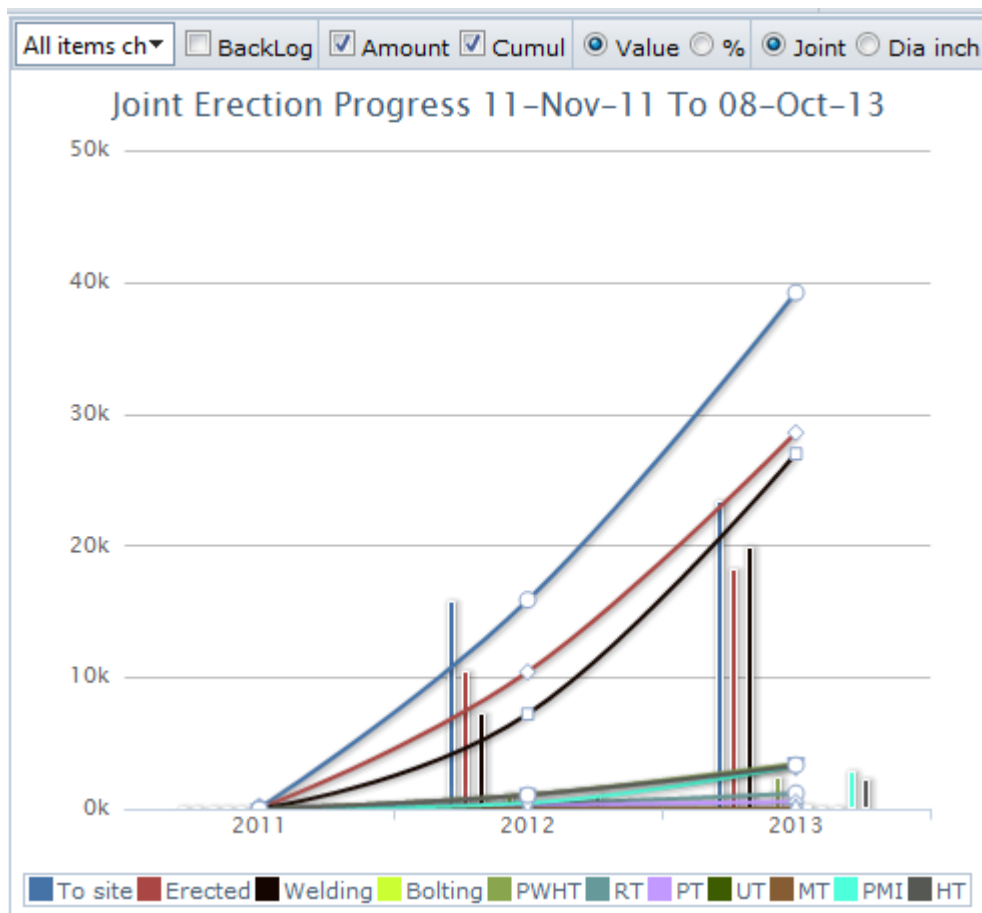
☒ Spool
 ☐ Tons
 ☐ Dia inch

4. Total Iso : Total No Iso in the Project
5. Completed for LC : % Iso completed Line check
6. Waiting for LC : No Iso awaiting for Line check

KPI For Spool Erection: KPI represents avg time taken for each erection activity based on historical data of the project.

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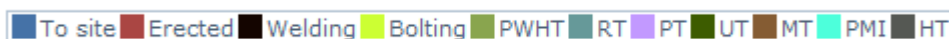
12.9.4 Joint Erection Chart



Joint Erection chart provides progress and backlog information for following Activities at Joint level. These values are represents only Shop Joints for which the Dia-Inch Factor is "Yes".

1. To Site
2. Erected
3. Welding
4. Bolting
5. PWHT
6. RT
7. PR
8. MT
9. PMI
10. HT

User can switch the visibility of progress by clicking the Following button on the Graph :



12.9.5 Joint Erection Feed Stock and KPI :

Total flange joints 25517 Joints	Total field joints 43299 Joints	Welded 50.92%	Ready for welding 6892 Joints
Bolted 0.14%	PdsArea:All Material:All Sizes:All	NDE 38.65%	Ready for NDE 5454 Joints
Ready for bolting 11730 Joints	Welder Performance Average 0.02 dia		

Joint Erection Feed stock Quantities represent the overall Scope and Feed stock availability data.

1. Flange Joints scope and Progress
2. Feed stock quantities always display the overall quantities, So Time Filters in global filters are not applicable.
3. Total joints always represent the overall scope of the project.
4. All the values can be displayed in No of Joint or Dia-Inch based on the selection in the Spool Fabrication filter

☒ Joint ☐ Dia inch

KPI For Welder Performance : KPI represents the overall Avg Dia Welded (Erection Joints) by a welder per day.

13. ERECTION REPORTS

In this module the user can generate the erection reports from the same screen.



ProgressPipingErection.xls

This report will provide the erection status for entire project. The user can use several filters like PDS Area, Area Classification, Material type, Subcontractor and LB/SB etc.



WeeklyProgressPipingErection.xls

This report will provide the erection status on a weekly basis. The user can search for particular period of time. The user can also search by filters like PDS Area, Area Classification, Material type, Subcontractor and LB/SB etc.

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The screenshot shows the Technip software interface for the Erection section. The main window has a green header bar with the word 'Erection' and the Technip logo. Below the header, there are two tabs: 'Preparation' and 'Progress'. The 'Progress' tab is currently selected. On the left side, there is a vertical navigation menu with options: 'Spool Erection', 'Weird', 'NDE', and 'Flange'. The main content area is divided into two sections. The top section is titled 'Search' and contains radio buttons for 'Isometric' (selected), 'Barcode', 'RequestNumber', and 'Assigned'. Below these is a text input field labeled 'Search Isometric No.' and three buttons: 'Search', 'Template', and 'Import'. The bottom section is titled 'Result Summary' and is currently empty. On the right side, there is a 'Reports' section with two buttons: 'Progress Erection' and 'Weekly Progress Erection'. The 'Progress Erection' button is highlighted with a dashed orange border.

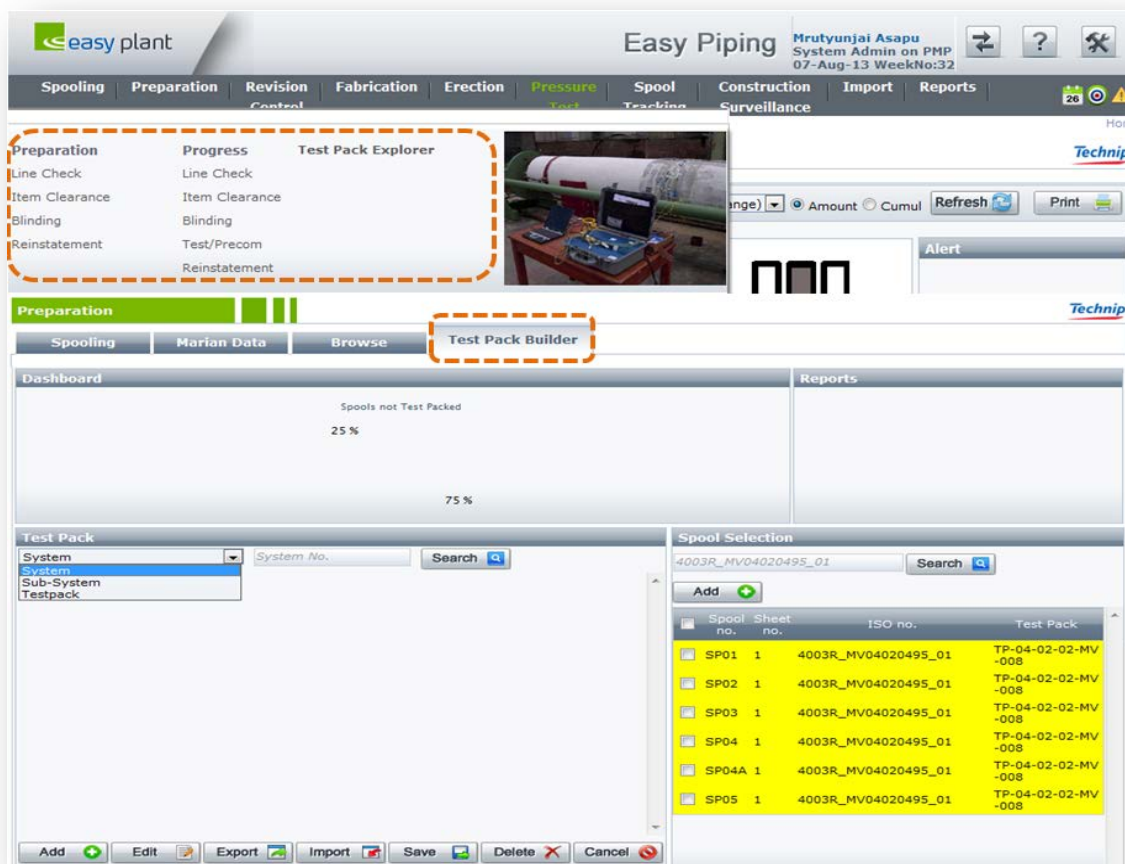
[*Easy Piping Part 6: Testpack & Flange Management*

14. TESTPACK MANAGEMENT

The Test pack module helps the user to manage all the test packs and the process flow of test packages during the project phase. The system allows the user to create and edit the test packages at any stage of the project. This module also helps the user to manage the Flange management (Bolt Torquing). This list of main functionalities are as follows.

- Test packs Preparation.
- Test pack review – Edit and revision management
- Line check Preparation/Progress
- Item Clearance preparation/Progress(Punch item)
- Blinding preparation/Progress
- Reinstatement Preparation/Progress
- Test pack Explorer – Navigation of status

- Tracking reports for construction, Bolting and NDE



15. TESTPACK PREPARATION

In the Easy piping test packs can be created in two methods manual edit method and import procedure. The user can see this screen under preparation module as “Test pack Builder”.

15.1 Testpack builder

This module helps the user to create a test pack and browse the test packs for review and edit the content if necessary.

The user can browse in a hierarchical level in this screen

- System level – The system will show all the test packs related to the searched system
- Sub system level – The system will show all the test packs related to the searched subsystem
- Test pack level – The system will show the particular test pack related to the searched test pack

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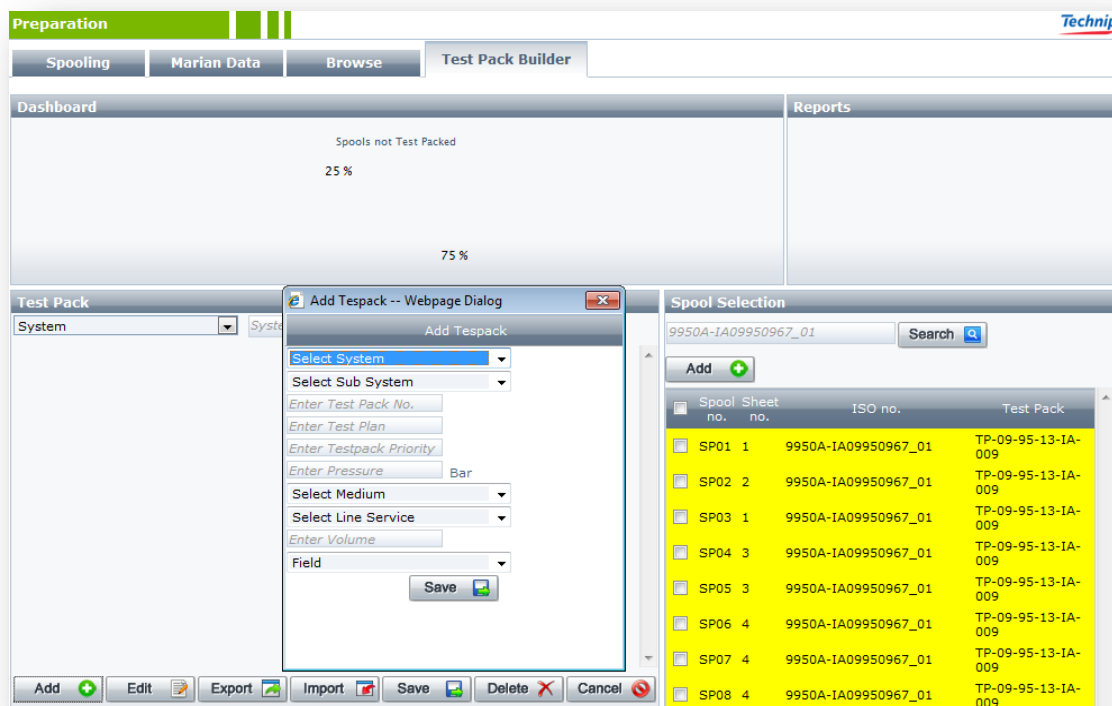
Note: After the search results, if the user click on the test pack it will show the details of test pack like isometric and spools.

The user can start creating the test packs manually by start clicking  in the screen. When the user clicks on  the system will show a pop-up where the below fields are shown.

- System
- Sub-system
- Test pack No
- Test plan : date from - to
- Test pack priority
- Test pressure
- Test medium
- Service class
- Test pack volume
- Test pack location

Once the test pack is added from the pop up the user can search for desired Isometrics and spools and add to the test pack in the same screen.

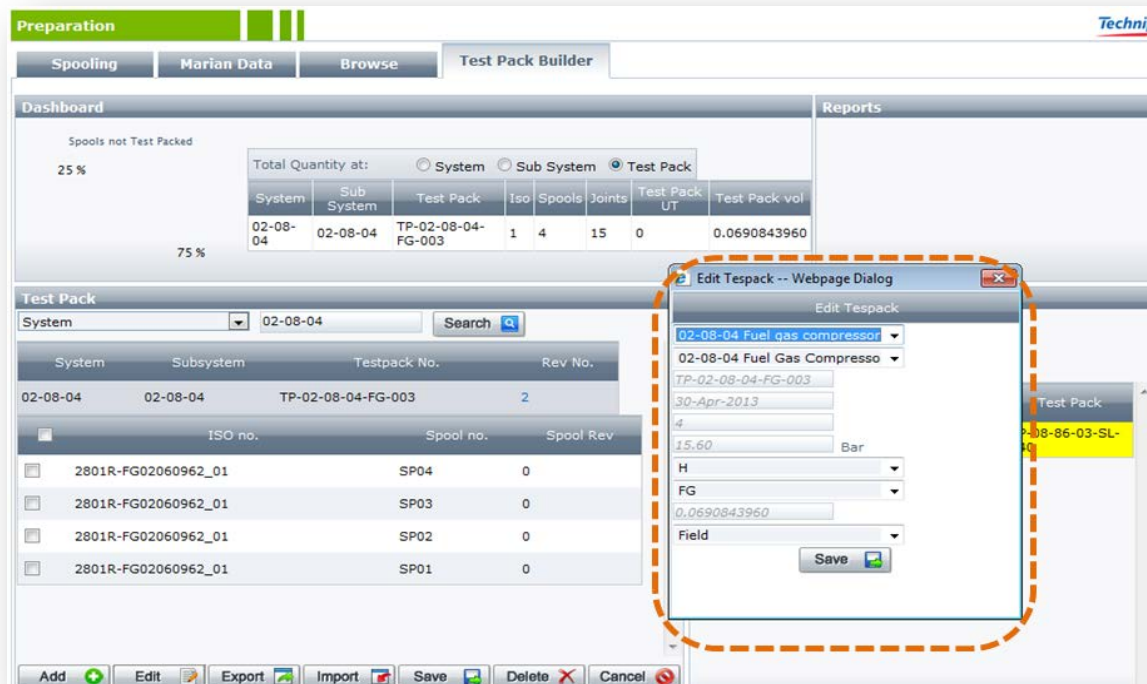
In the aforementioned fields, System, subsystem and service class are predefined referential ([Refer to Para 3.19](#), [Para 3.20](#), [Para 3.25](#))



Spool no.	Sheet no.	ISO no.	Test Pack
SP01	1	9950A-IA09950967_01	TP-09-95-13-IA-009
SP02	2	9950A-IA09950967_01	TP-09-95-13-IA-009
SP03	1	9950A-IA09950967_01	TP-09-95-13-IA-009
SP04	3	9950A-IA09950967_01	TP-09-95-13-IA-009
SP05	3	9950A-IA09950967_01	TP-09-95-13-IA-009
SP06	4	9950A-IA09950967_01	TP-09-95-13-IA-009
SP07	4	9950A-IA09950967_01	TP-09-95-13-IA-009
SP08	4	9950A-IA09950967_01	TP-09-95-13-IA-009

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The user can use the option  to edit any parameters for the existing test pack. The system will display a pop up as “Edit Test package” with existing parameters and user can edit the parameters and add the new spools if any,



The user can use the options  after the editing is completed. The user can also delete the inappropriate data if any by using the option .

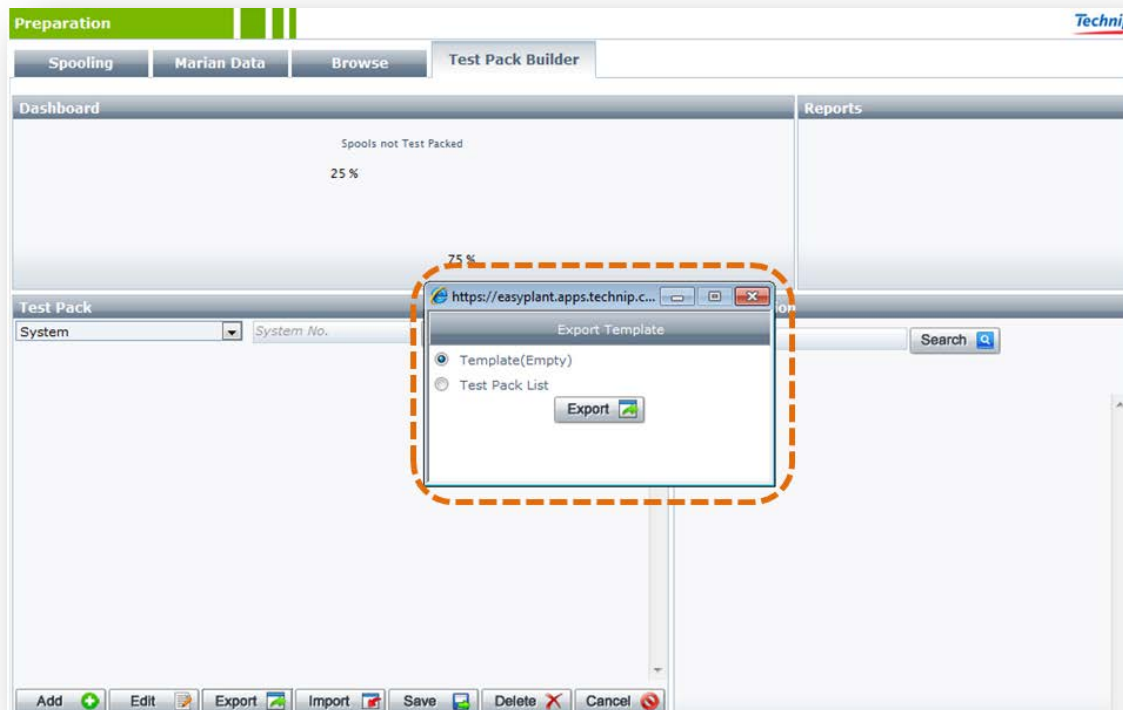
15.2 Testpack import

This module also helps the user to Import the test pack details through import procedure. System allows the user to generate the test pack template where users have to fill the required data and import into the system. When the user clicks on  system will display a popup with two types of templates. The user can choose the type of test pack he wants to use

Empty Template – It will give the empty template with no data related to test packs.

Test pack list – It will give the total list of test packs which are previously assigned.

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The user needs to fill the following details in the template before import.

- System
- Subsystem
- Test Pack
- Test pack rev
- Test Medium : Dropdown list H, P or V
- Test Pressure (Editable)
- Test plan date (Editable)
- Iso No (Read only field)
- Rev (Read only field)
- Spool number (Read only field)
- Spool revision (Read only field)
- Test pack Location

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System	Sub System	Test Pack No	Test Medium	Test Pressure	Test Plan	Testpack Priority	volume	Isometric	Iso rev no	spoolno	Spool rev	Testpack Location
00-02-10	00-02-10	TP-00-02-10-AV-001	V		31-May-2013	6	0.0672	6003A_AV00020002_01	1	SP01	1	Field
00-02-10	00-02-10	TP-00-02-10-AV-001	V		31-May-2013	6	0.0672	6003A_AV00020002_01	1	SP02	1	Field
00-02-10	00-02-10	TP-00-02-10-AV-001	V		31-May-2013	6	0.0672	6003A_AV00020002_01	1	SP03	1	Field
00-02-10	00-02-10	TP-00-02-10-AV-001	V		31-May-2013	6	0.0672	6003A_AV00020002_01	1	SP04	1	Field
00-02-10	00-02-10	TP-00-02-10-AV-001	V		31-May-2013	6	0.0672	6003A_AV00020353_01	1	SP02	1	Field
00-92-24	00-92-24	TP-00-92-24-SC-007	P	10.5	30-Jun-2013	7	0.0318664	6003A_SC00020001_01	0	SP01	0	Field
00-92-24	00-92-24	TP-00-92-24-SC-007	P	10.5	30-Jun-2013	7	0.0318664	6003A_SC00020001_01	0	SP02	0	Field
00-92-24	00-92-24	TP-00-92-24-SC-007	P	10.5	30-Jun-2013	7	0.0318664	6003A_SC00020001_01	0	SP03	0	Field
00-92-24	00-92-24	TP-00-92-24-SC-007	P	10.5	30-Jun-2013	7	0.0318664	6003A_SC00020001_01	0	SP04	0	Field
00-92-24	00-92-24	TP-00-92-24-SC-007	P	10.5	30-Jun-2013	7	0.0318664	6003A_SC00020001_01	0	SP05	0	Field
00-92-25	00-92-25	TP-00-92-25-SCC-001	H	5.25	31-May-2013	6	0.0022146	6003A_SCC00020351_01	0	SP01	0	Field
00-92-25	00-92-25	TP-00-92-25-SCC-002	H	25.06	31-May-2013	6	1.8218361	6001R_SCC00920352_01	1	SP01	1	Field
00-92-25	00-92-25	TP-00-92-25-SCC-002	H	25.06	31-May-2013	6	1.8218361	6001R_SCC00920352_01	1	SP02	1	Field
00-92-25	00-92-25	TP-00-92-25-SCC-002	H	25.06	31-May-2013	6	1.8218361	6001R_SCC00920352_01	1	SP03	1	Field
00-92-25	00-92-25	TP-00-92-25-SCC-002	H	25.06	31-May-2013	6	1.8218361	6001R_SCC00920352_01	1	SP04	1	Field
00-92-25	00-92-25	TP-00-92-25-SCC-002	H	25.06	31-May-2013	6	1.8218361	6001R_SCC00920352_01	1	SP05	1	Field
00-92-25	00-92-25	TP-00-92-25-SCC-002	H	25.06	31-May-2013	6	1.8218361	6001R_SCC00920352_01	1	SP06	1	Field

The user can choose the option  to import the test pack template once it is filled.

Notes:

- Import of iso / spools which are not defined in PSMS will be rejected by the system
- Date column should have the following date format (dd-MM-yyyy).
- To add an additional spools to an existing test pack it should be done manually the system will display the message “To add additional spools manually”
- The data in the excel sheet will always supersedes data in the database. Meaning that if there was a conflict between the data in the excel sheet and the actual data in Easy Piping that particular cell will be displayed with the errors. The user will have to confirm the changes and proceed the import if he wishes. The data in the system will be overwritten with the new data on confirmation.

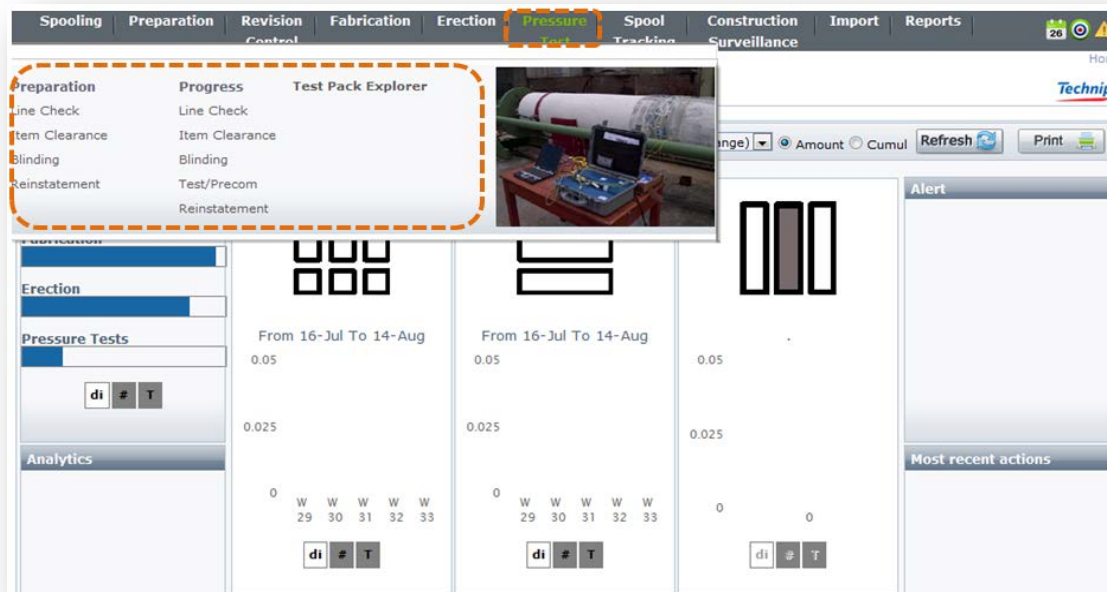
Failure of any of these rules will be validated by the system and the errors will be displayed with the appropriate error messages. Import will not be performed unless all the errors have been rectified by the user.

16. PRESSURE TEST MODULE

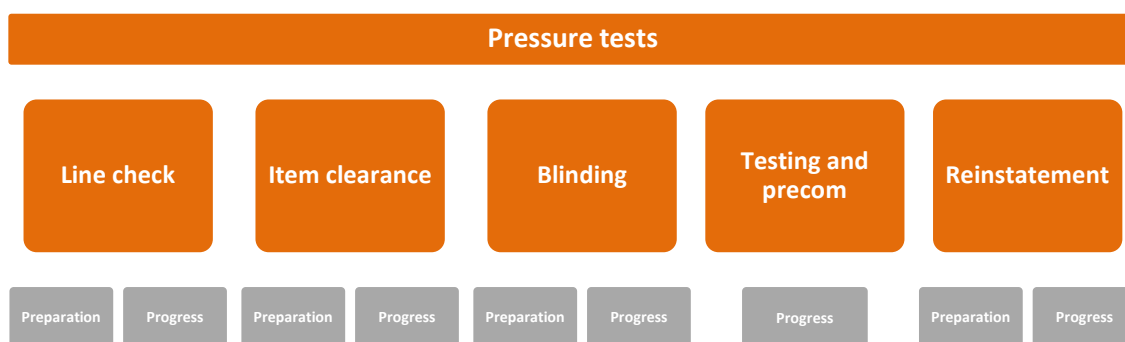
This module helps the user to prepare the workload and update the progress for the same. This section is organized into 5 different sections corresponding to different testing activities like

- Line check
- Item Clearance
- Blinding
- Testing and ProComm
- Reinstatement

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- Each activities is divided in two sub-modules
 - Preparation – This module allows the user to prepare the workload
 - Progress – This module allows the user to enter the progress related to the corresponding activity



16.1 Line Check Preparation

This module allows the user to assign the line checking activity for the line checkers. The line checkers are predefined in the project referential ([Refer to Para 3.21](#)). The user can assign the workload in two different levels

- Isometric level
- Test pack level

The user can search for the Iso or test pack. The system will display the list of Iso's or test packs which are ready for the line checking based on the criteria

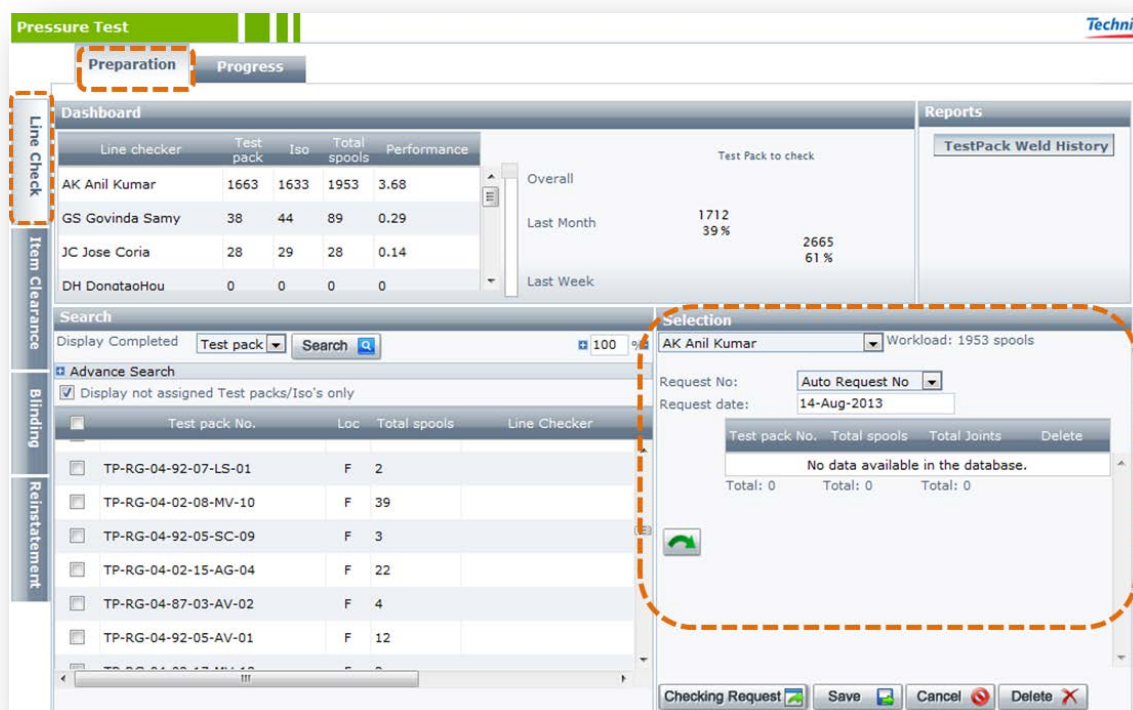
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- Spools are supported
- Weld joints are welded

During the search process the user can use the “Advance search” options which will filters based on the selection. The below mentioned are the following categories

- Test pack location
- System and subsystem
- PDS Area
- Area Classification

The user will then select a line checker and assign one or several Iso or test pack to be line checked
The user can generate a checking request, in which these Iso or test pack will be listed.



Pressure Test

Preparation Progress

Line Check

Dashboard

Line checker	Test pack	Iso	Total spools	Performance
AK Anil Kumar	1663	1633	1953	3.68
GS Govinda Samy	38	44	89	0.29
JC Jose Coria	28	29	28	0.14
DH DonotaoHou	0	0	0	0

Search

Display Completed Test pack Search 100 %

Advance Search

☒ Display not assigned Test packs/Iso's only

Test pack No.	Loc	Total spools	Line Checker
TP-RG-04-92-07-LS-01	F	2	
TP-RG-04-02-08-MV-10	F	39	
TP-RG-04-92-05-SC-09	F	3	
TP-RG-04-02-15-AG-04	F	22	
TP-RG-04-87-03-AV-02	F	4	
TP-RG-04-92-05-AV-01	F	12	

Selection

AK Anil Kumar Workload: 1953 spools

Request No: Auto Request No

Request date: 14-Aug-2013

Test pack No.	Total spools	Total Joints	Delete
No data available in the database.			
Total: 0	Total: 0	Total: 0	

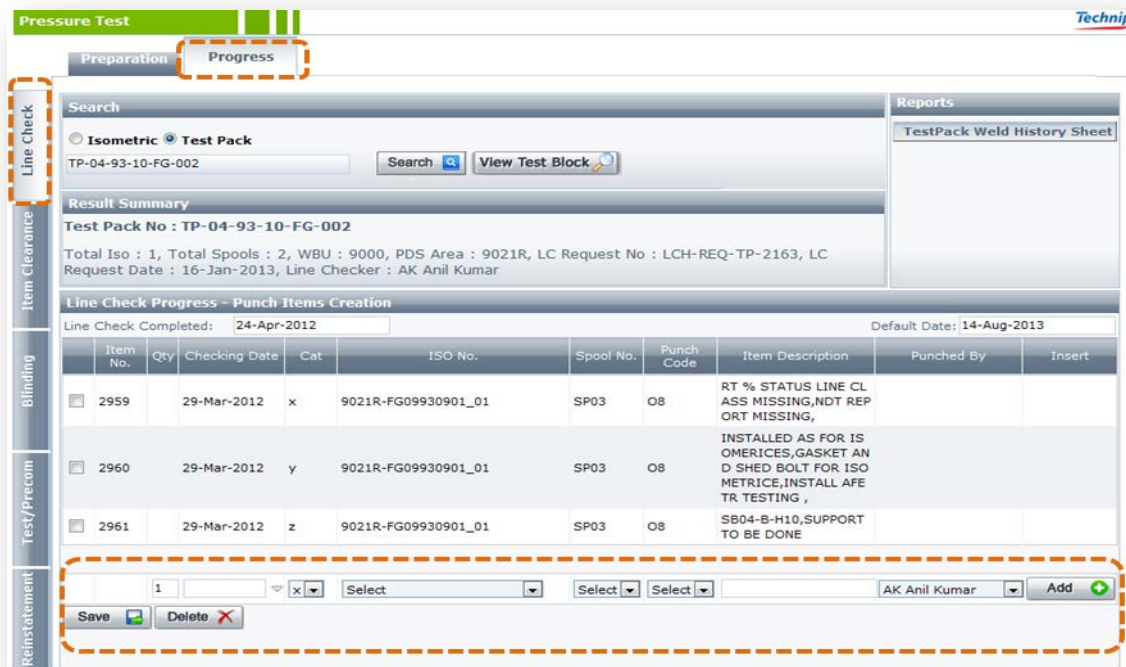
Checking Request Save Cancel Delete

16.2 Line Check Progress

This module allows the user to update the progress for the line checking activity. Once the assigned test pack is line checking done the user can record the progress.

The user can search for the test pack or Isometric and fill the following details in this screen.

- The item numbering is automatic
- Checking date
- Category (X, Y or Z)
- Localization of the item : isometric and spool
- Punch code
- Description (will come automatically with the punch code, but can be modified)
- Punch done by: Punch originator



Pressure Test

Preparation **Progress**

Line Check

Search

☐ Isometric ☒ Test Pack

TP-04-93-10-FG-002

Result Summary

Test Pack No : TP-04-93-10-FG-002

Total Iso : 1, Total Spools : 2, WBU : 9000, PDS Area : 9021R, LC Request No : LCH-REQ-TP-2163, LC Request Date : 16-Jan-2013, Line Checker : AK Anil Kumar

Line Check Progress - Punch Items Creation

Line Check Completed: 24-Apr-2012 Default Date: 14-Aug-2013

Item No.	Qty	Checking Date	Cat	ISO No.	Spool No.	Punch Code	Item Description	Punched By	Insert
2959		29-Mar-2012	x	9021R-FG09930901_01	SP03	O8	RT % STATUS LINE CLASS MISSING, NDT REPORT MISSING,		
2960		29-Mar-2012	y	9021R-FG09930901_01	SP03	O8	INSTALLED AS FOR ISOMERICS, GASKET AND SHED BOLT FOR ISO METRICE, INSTALL AFTER TESTING,		
2961		29-Mar-2012	z	9021R-FG09930901_01	SP03	O8	SB04-B-H10, SUPPORT TO BE DONE		

1 AK Anil Kumar

Note: Line check completed date is mandatory to proceed for the next step.

16.3 Item Clearance Preparation

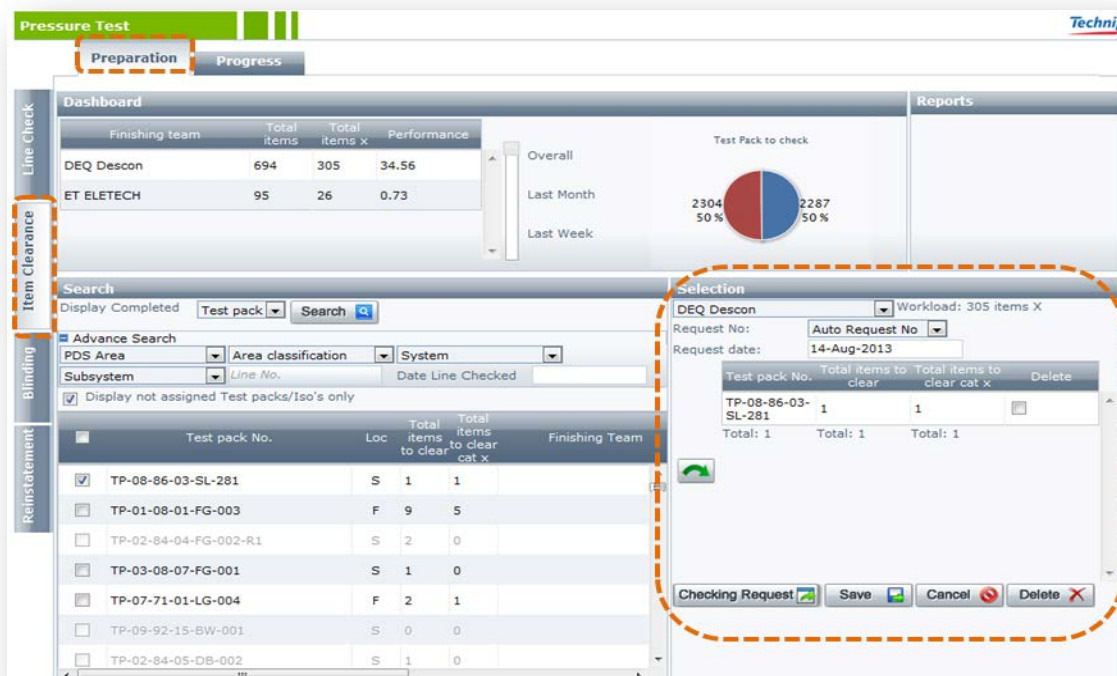
This module allows the user to assign the job for Item clearance to the finishing team. The finishing teams are predefined in the project referential ([Refer to Para 3.17](#)). The user can assign the workload in two different levels

- Isometric level
- Test pack level

During the search process the user can use the “Advance search” options which will filters based on the selection. The below mentioned are the following categories

- Test pack location
- System and subsystem
- PDS Area
- Area Classification

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Pressure Test

Preparation | **Progress**

Line Check
Item Clearance
Blinding
Reinstatement

Dashboard

Finishing team	Total items	Total items x	Performance
DEQ Descon	694	305	34.56
ET ELETECH	95	26	0.73

Overall
Last Month
Last Week

Test Pack to check

2304 50% | 2287 50%

Search

Display Completed Test pack Search

Advance Search

PDS Area Area classification System
Subsystem Line No. Date Line Checked

☒ Display not assigned Test packs/Iso's only

Test pack No.	Loc	Total items to clear	Total items to clear cat x	Finishing Team
☒ TP-08-86-03-SL-281	S	1	1	
☐ TP-01-08-01-FG-003	F	9	5	
☐ TP-02-84-04-FG-002-R1	S	2	0	
☐ TP-03-08-07-FG-001	S	1	0	
☐ TP-07-71-01-LG-004	F	2	1	
☐ TP-09-92-15-BW-001	S	0	0	
☐ TP-02-84-05-DB-002	S	1	0	

Selection

DEQ Descon Workload: 305 items X

Request No: Auto Request No
Request date: 14-Aug-2013

Test pack No.	Total items to clear	Total items to clear cat x	Delete
TP-08-86-03-SL-281	1	1	☐
Total: 1	Total: 1	Total: 1	

Checking Request Save Cancel Delete

The user will then select a Finishing team and assign one or several Iso or test pack to complete item clearance. The user can generate a Item clearance request, in which these Iso or test pack will be listed.

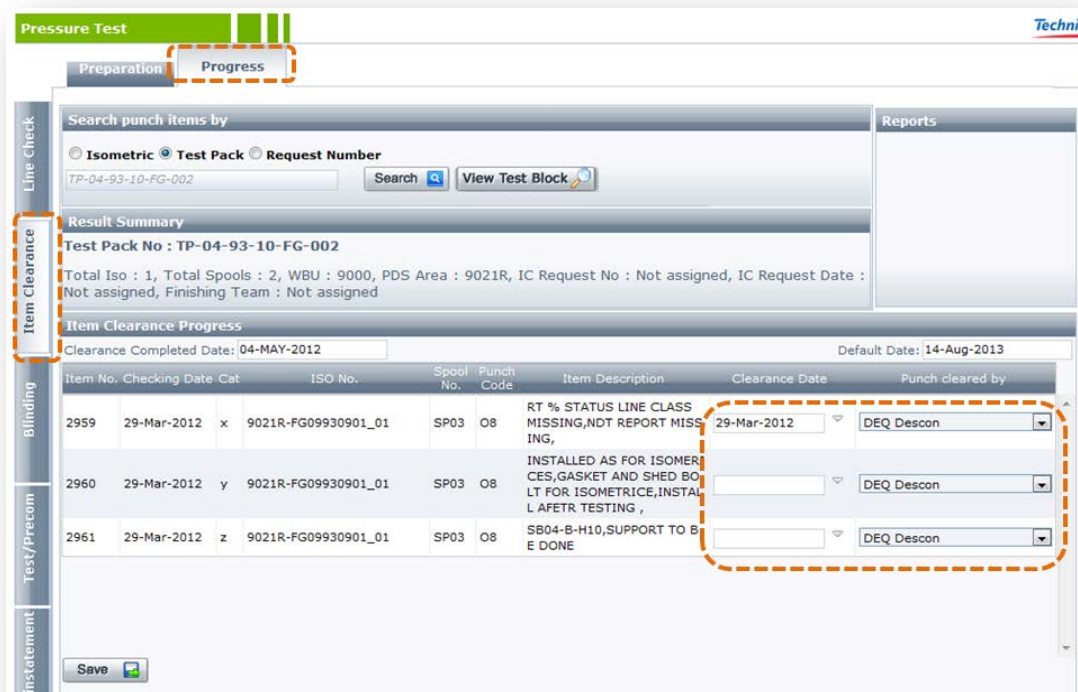
16.4 Item Clearance Progress

This module allows the user to update the progress for the Item clearance activity. Once the assigned test pack is completed item clearance. The user can record the progress.

The user can search for the test pack or Isometric and fill the following details in this screen.

- Punch item clearance date
- Punch cleared by (finishing team)

Note: Line check completed date is mandatory to proceed for the next step.



Pressure Test

Preparation **Progress**

Search punch items by

☐ Isometric ☒ Test Pack ☐ Request Number

TP-04-93-10-FG-002 Search View Test Block

Result Summary

Test Pack No : TP-04-93-10-FG-002

Total Iso : 1, Total Spools : 2, WBU : 9000, PDS Area : 9021R, IC Request No : Not assigned, IC Request Date : Not assigned, Finishing Team : Not assigned

Item Clearance Progress

Clearance Completed Date: 04-MAY-2012 Default Date: 14-Aug-2013

Item No.	Checking Date	Cat	ISO No.	Spool No.	Punch Code	Item Description	Clearance Date	Punch cleared by
2959	29-Mar-2012	x	9021R-FG09930901_01	SP03	O8	RT % STATUS LINE CLASS MISSING, NDT REPORT MISSING,	29-Mar-2012	DEQ Descon
2960	29-Mar-2012	y	9021R-FG09930901_01	SP03	O8	INSTALLED AS FOR ISOMETRIC, GASKET AND SHED BOLT FOR ISOMETRIC, INSTALL AFTER TESTING,		DEQ Descon
2961	29-Mar-2012	z	9021R-FG09930901_01	SP03	O8	SB04-B-H10, SUPPORT TO BE DONE		DEQ Descon

Save

16.5 Blinding Preparation

This module helps the user to assign the blinding activity to the blinding team. The blinding teams are predefined in the project referential ([Refer to Para 3.16](#)).

During the search process the user can use the "Advance search" options which will filter based on the selection. The below mentioned are the following categories

- Test pack location
- System and subsystem
- PDS Area
- Area Classification
- Date RFT

The user can generate a blinding request, in which these test packs will be listed.

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Pressure Test

Preparation **Progress**

Dashboard

Blinding team	Test packs	Joints	Blinding UT	Performance
DEQ Descon	547	0	0	
DEQ Descon	564	2151	0	
ET ELETECH	50	0	0	
ET ELETECH	104	417	0	

Test Pack to blind

Overall: 1552 (34%), 3039 (66%)

Selection

DEQ Descon Workload: 0 UT

Request No: Auto Request No

Request date: 14-Aug-2013

Test pack No.	Blinding UT	Quantity of joints	Delete
TP-02-08-06-N-001	0	2	
Total: 1	Total: 0	Total: 2	

Checking Request Save Cancel Delete

16.6 Blinding Progress

This module allows the user to update the progress for the Blinding activity. Once the assigned test packs are blinding done. The user can record the progress.

The user can search for the test pack and fill the following details in this screen.

- Blinding date

Pressure Test

Preparation **Progress**

Search test pack by

System Sub System Test Pack Request Number

TP-04-93-10-FG-002 Search View Test Block

Result Summary

Test Pack No : TP-04-93-10-FG-002

System : 04-93-10, Sub System : 04-93-10, Total Iso : 1, Total Spools : 2, WBU : 9000, PDS Area : 9021R, Blinding Request No : Not assigned

Blinding Progress

Default Date: 14-Aug-2013

Test pack No.	RFT Date	Blinding Request Date	Blinding team	Blinding date
TP-04-93-10-FG-002		Not assigned	Not assigned	05-Apr-2012

Save

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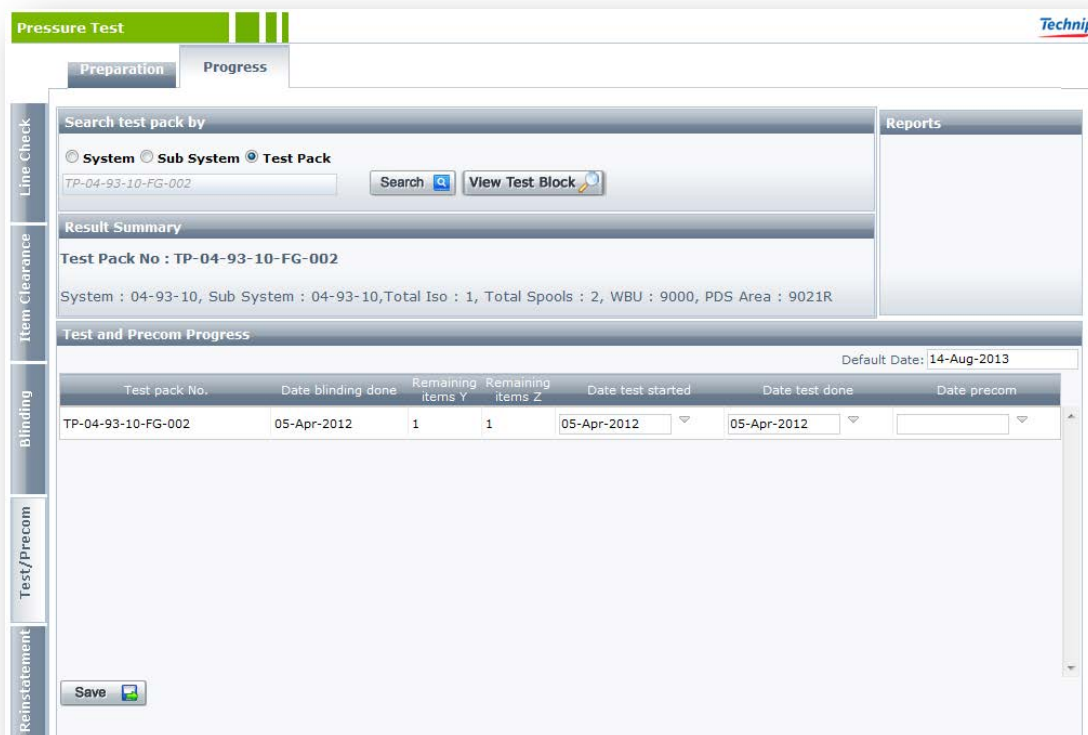
16.7 Testing and Pre commissioning progress

- Easy piping does not manage the preparation of the testing and pre-commissioning. However, the user can enter the dates related to these activities
- Testing start date
- Testing done date
- Pre-commissioning date

The user will search for a test pack using the appropriate search field

These dates are useful to be entered, since it allow to manage the reinstatement activity for the flange joints category Y and Z

- Category Y : reinstatement to be done after testing and before pre-commissioning
- Category Z : reinstatement to be done after pre-commissioning



Pressure Test

Preparation Progress

Search test pack by

☐ System ☐ Sub System ☒ Test Pack

TP-04-93-10-FG-002

Search View Test Block

Result Summary

Test Pack No : TP-04-93-10-FG-002

System : 04-93-10, Sub System : 04-93-10, Total Iso : 1, Total Spools : 2, WBU : 9000, PDS Area : 9021R

Test and Precom Progress

Default Date: 14-Aug-2013

Test pack No.	Date blinding done	Remaining items Y	Remaining items Z	Date test started	Date test done	Date precom
TP-04-93-10-FG-002	05-Apr-2012	1	1	05-Apr-2012	05-Apr-2012	

Save

16.8 Reinstatement Preparation

This module allows the user to assign some test packs for the reinstatement to the reinstatement team. The reinstatement teams are predefined in the project referential ([Refer to Para 3.18](#)).

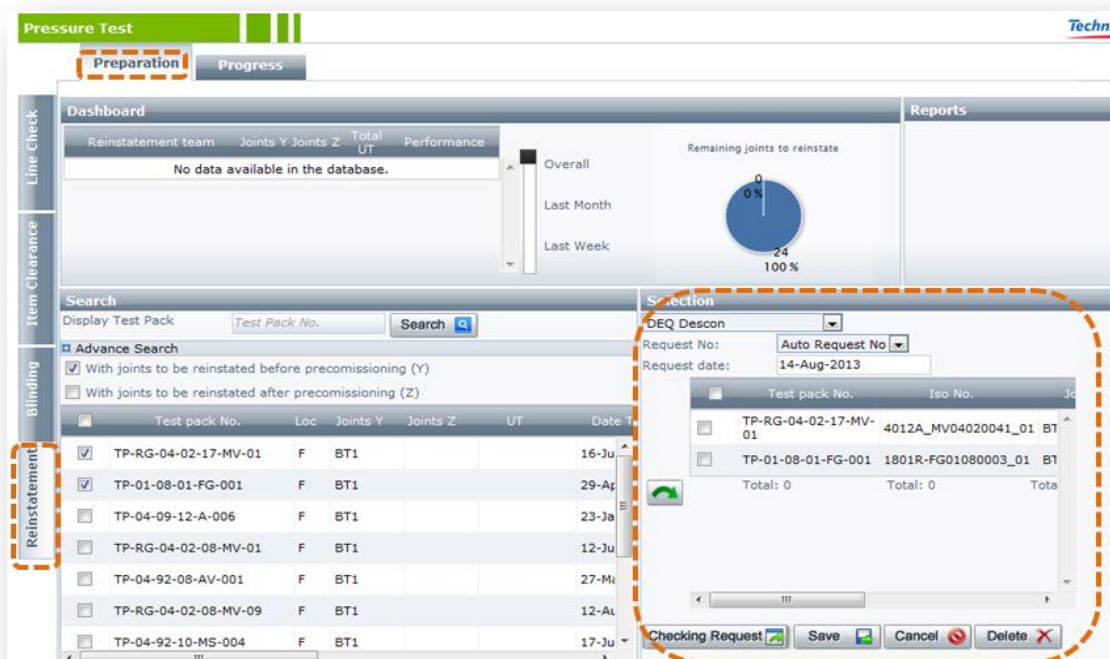
The user will search flange joints (test pack wise) in the search section

- Only the flange joints Y belonging to a test pack which has been tested will appear

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- Only the flange joints Z belonging to a test pack which has been pre-commissioned will appear

The user will then select reinstatement team and assign one or several flange joint to be reinstated
The user can generate a reinstatement request, in which these flange joints will be listed



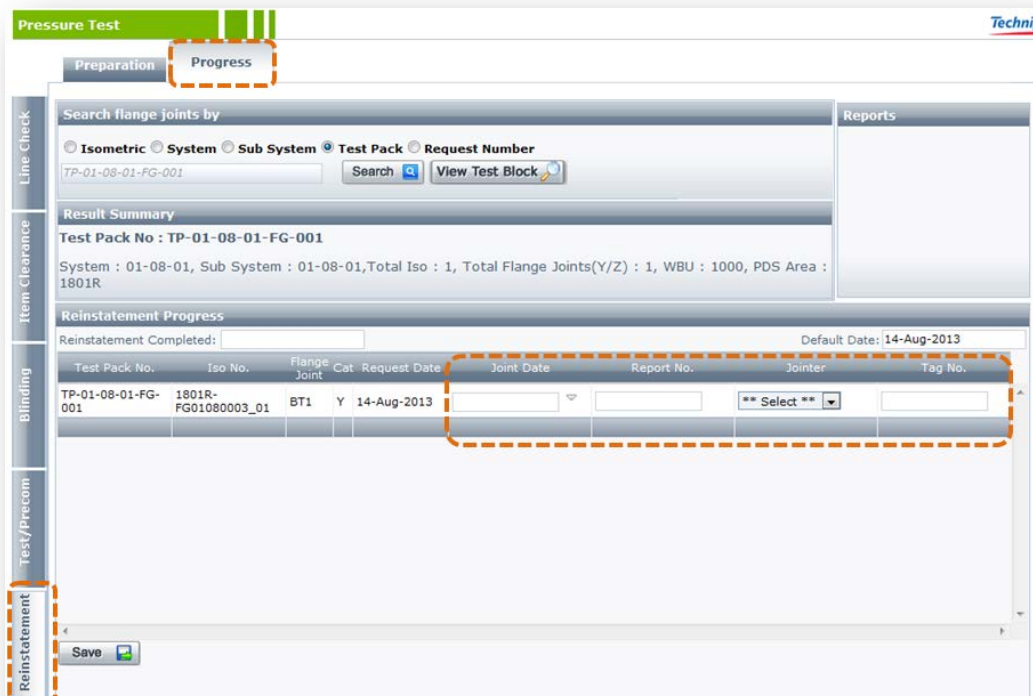
16.9 Reinstatement Progress

This module allows the user to update the progress for the reinstatement activity. Once the assigned test packs are reinstatement done. The user can record the progress.

The user will search flange joints (test pack wise) in the search section

The user can search for the test pack and fill the following details in this screen.

- Joint date
- Report number
- Jointer No (The jointer numbers are predefined in the project referential Refer to [Para 3.15](#))
- Tag number



Pressure Test

Preparation Progress

Search flange joints by

Isometric System Sub System Test Pack Request Number

TP-01-08-01-FG-001 Search View Test Block

Result Summary

Test Pack No : TP-01-08-01-FG-001

System : 01-08-01, Sub System : 01-08-01, Total Iso : 1, Total Flange Joints(Y/Z) : 1, WBU : 1000, PDS Area : 1801R

Reinstatement Progress

Reinstatement Completed: Default Date: 14-Aug-2013

Test Pack No.	Iso No.	Flange Joint	Cat	Request Date	Joint Date	Report No.	Joints	Tag No.
TP-01-08-01-FG-001	1801R-FG01080003_01	BT1	Y	14-Aug-2013			** Select **	

Save

17. TESTPACK HOMEPAGE

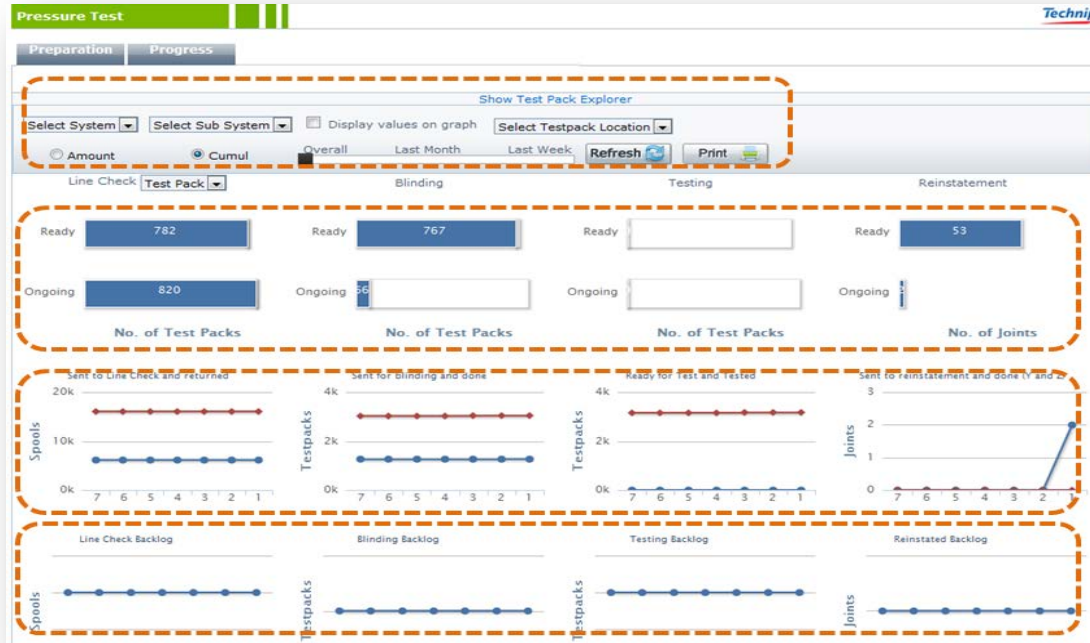
The pressure test homepage displays global information regarding the testing activities like Line checking, item clearance, testing and reinstatement

The bargraph shows the quantities for each activities

- What is ready to be done: “ready”
- What has been sent to be done: “ongoing”
- At test pack, isometric or flange joint level

On top of the page, some filters and options are available

- Filters are applied and curves and backlogs



18. TESTPACK EXPLORER

This screen allows the user to access all the information regarding the testing activities in hierarchical levels as follows.

Clicking on a test pack will make appearing some information at the isometric level. Then, clicking on a isometric will make appearing information at spool level

- Test pack level
- Isometric Level
- Spool Level

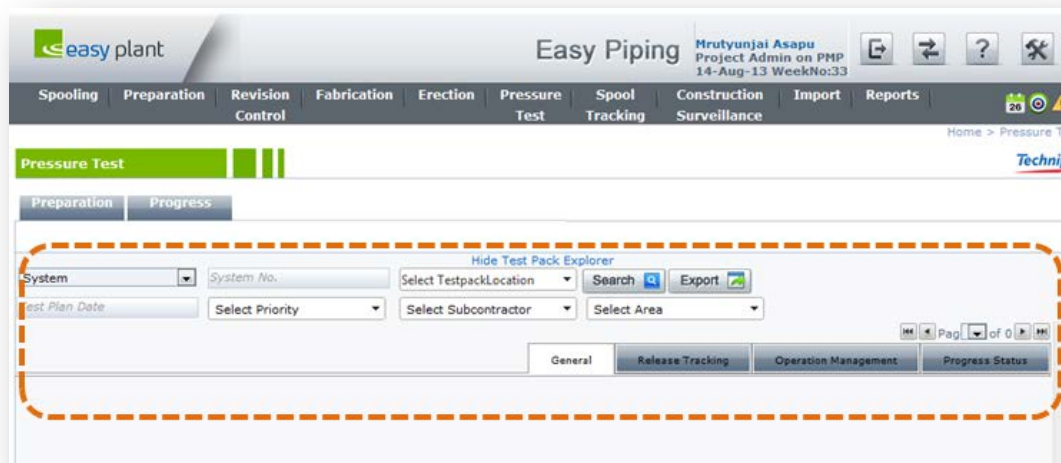
This screen is organized in 4 different sections as per the test pack follow up and the details are mentioned below

- General
- Release tracking
- Operation Management
- Progress Status

The system will allows the user to see the information based on the below filter also.

- System No
- Test pack Location
- Area classification
- Subcontractor
- Priority

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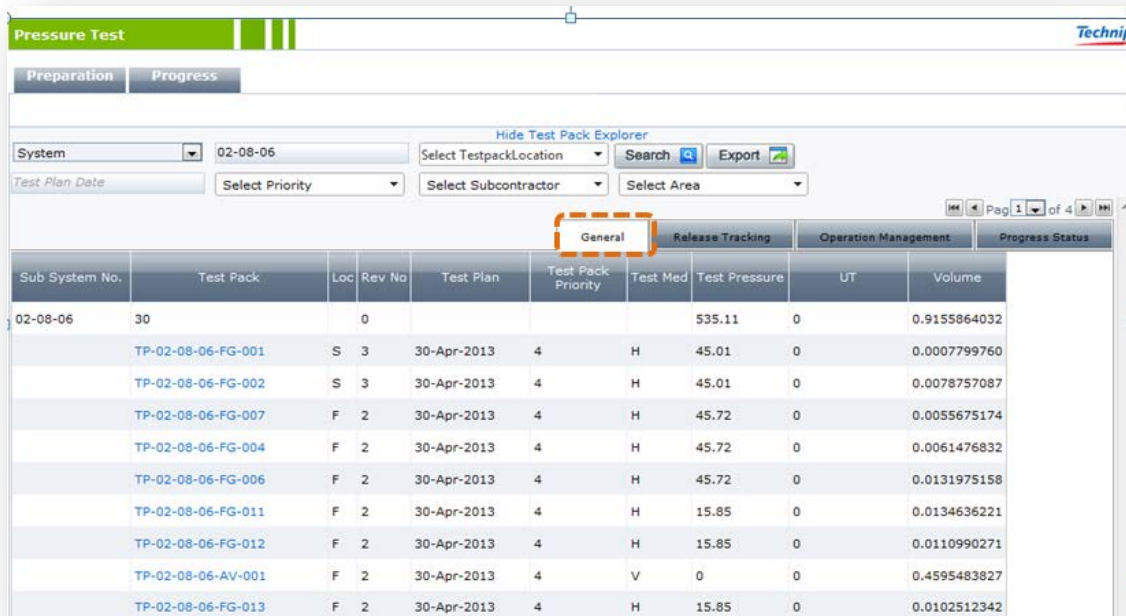


18.1 General - Testpack Level

This screen allows the user to visualize the general information related to the test pack definition. The below mentioned are the details shown in this screen.

- Subsystem No
- Test pack No
- Test pack Location
- Rev Number (auto generated by the system)
- Test planned date
- Test pack Priority
- Test medium
- Test pressure
- Unit of time
- Volume

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Sub System No.	Test Pack	Loc	Rev No.	Test Plan	Test Pack Priority	Test Med	Test Pressure	UT	Volume
02-08-06	30		0				535.11	0	0.9155864032
	TP-02-08-06-FG-001	S	3	30-Apr-2013	4	H	45.01	0	0.0007799760
	TP-02-08-06-FG-002	S	3	30-Apr-2013	4	H	45.01	0	0.0078757087
	TP-02-08-06-FG-007	F	2	30-Apr-2013	4	H	45.72	0	0.0055675174
	TP-02-08-06-FG-004	F	2	30-Apr-2013	4	H	45.72	0	0.0061476832
	TP-02-08-06-FG-006	F	2	30-Apr-2013	4	H	45.72	0	0.0131975158
	TP-02-08-06-FG-011	F	2	30-Apr-2013	4	H	15.85	0	0.0134636221
	TP-02-08-06-FG-012	F	2	30-Apr-2013	4	H	15.85	0	0.0110990271
	TP-02-08-06-AV-001	F	2	30-Apr-2013	4	V	0	0	0.4595483827
	TP-02-08-06-FG-013	F	2	30-Apr-2013	4	H	15.85	0	0.0102512342

18.2 Release Tracking - Testpack Level

This screen displays all the information useful to know whether a test pack is ready for test or not

- Quantity of welded joints which have still to be welded
- Quantity of flange joints which have still to be bolted
- Quantity of welded joints which have still to be NDE tested
- Quantity of isometrics to complete, in order to make the test pack ready for line checking (all spools are supported and welded bolted)
- Quantity of isometrics to be returned from line checking after being assigned in the preparation module
- Quantity of item category X (to be cleared before testing) from line checking still to be cleared
- Quantity of isometrics to be QC released for test (all welded joints are NDE/PWHT released)
- Quantity of isometrics to be Ready For Test (QC released + complete + line check done + all item X cleared)

All the quantity icons are able to click, when the user clicks on the number the system will display a pop up of the particular screen. User can export the list of work to be done.

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Joints To NDE												
SYSTEM	SUBSYSTEM	TESTPACKNO	ISONUMBER	SPOOLNO	WELDNO	RT	UT	MT	PT	PMI	PWHT	HT
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP04	013	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP04	014	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP02	007	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP03	008	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP03	009	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP03	010	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP03	011	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP01	001	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP01	002	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
02-08-06	02-08-06	TP-02-08-06-AV-001	2802A-AV02080310_01	SP01	003	NOT RELEASED	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE

Some clickable links are available, in order to reach other easy piping functionality

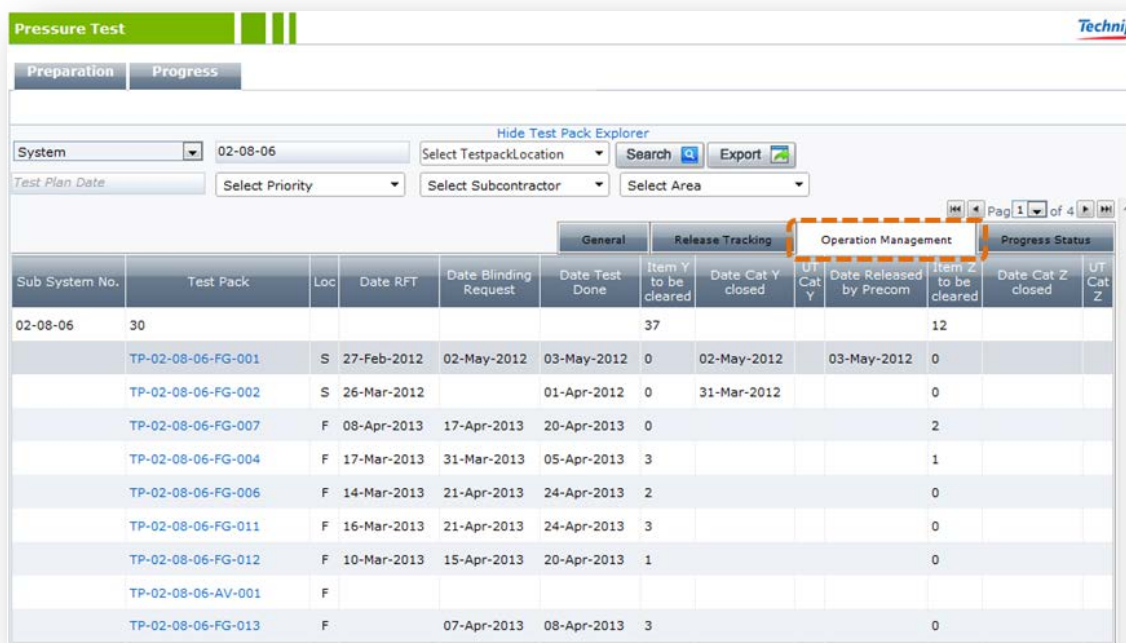
- Isometric to send for checking : to line check - preparation
- Isometric to return form checking : to line check - progress
- Item x to be cleared : to item clearance - preparation

Pressure Test												
Preparation Progress												
System: 02-08-06 Select TestpackLocation Search Export Test Plan Date Select Priority Select Subcontractor Select Area												
General Release Tracking Operation Management Progress Status												
Sub System No.	Test Pack	Loc	Joints to be welded	Joints to be bolted	Joints to NDE	Iso to Complete	Iso to Send for checking	Iso to Return from checking	Item x to be cleared	Iso to be QC Released	Iso to be Ready for Test	
02-08-06	30		15	0	88	6	20	20	51	16	34	
	TP-02-08-06-FG-001	S	0	0	0							
	TP-02-08-06-FG-002	S	0	0	0							
	TP-02-08-06-FG-007	F	0	0	0							
	TP-02-08-06-FG-004	F	0	0	0							
	TP-02-08-06-FG-006	F	0	0	0							
	TP-02-08-06-FG-011	F	0	0	0							
	TP-02-08-06-FG-012	F	0	0	0							
	TP-02-08-06-AV-001	F	14	0	69							
	TP-02-08-06-FG-013	F	0	0	0							

18.3 Operation Management - Testpack Level

This screen displays all the dates and information regarding the blinding, testing and reinstatement activities

- Date the test pack is ready for test
- Date the test pack is assigned to be blinded
- Date the test is done
- Quantity of items (categories Y and Z) to be cleared, the date when all items are cleared, and the corresponding units of time
- Date pre-commissioning



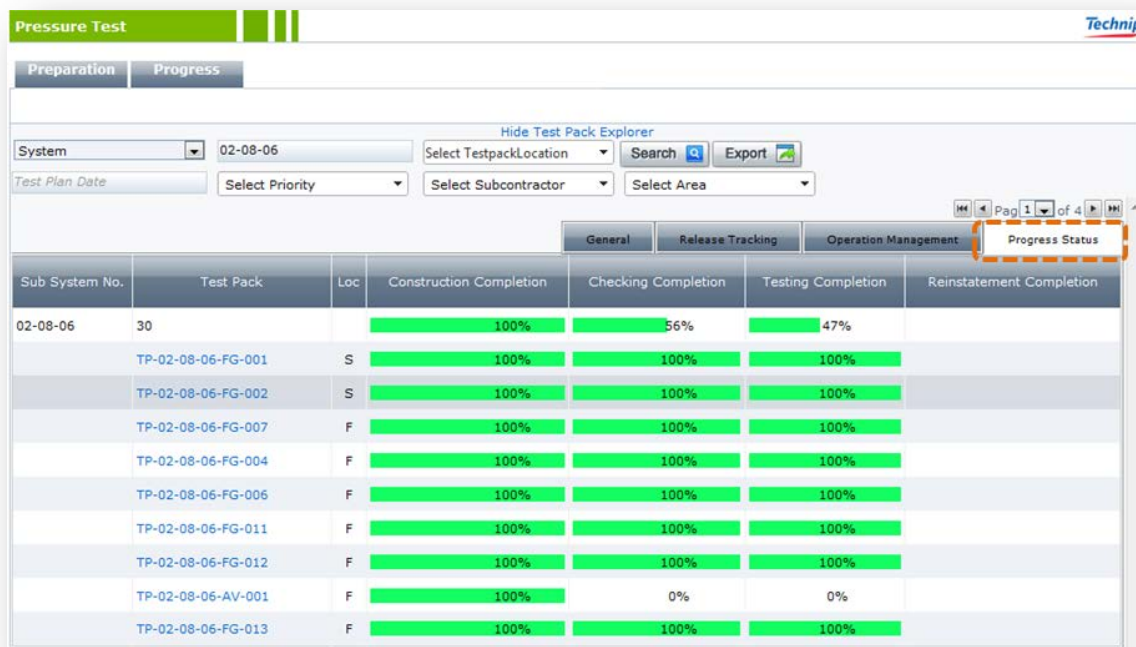
Sub System No.	Test Pack	Loc	Date RFT	Date Blinding Request	Date Test Done	Item Y to be cleared	Date Cat Y closed	UT Cat Y	Date Released by Precom	Item Z to be cleared	Date Cat Z closed	UT Cat Z
02-08-06	30					37				12		
	TP-02-08-06-FG-001	S	27-Feb-2012	02-May-2012	03-May-2012	0	02-May-2012		03-May-2012	0		
	TP-02-08-06-FG-002	S	26-Mar-2012		01-Apr-2012	0	31-Mar-2012			0		
	TP-02-08-06-FG-007	F	08-Apr-2013	17-Apr-2013	20-Apr-2013	0				2		
	TP-02-08-06-FG-004	F	17-Mar-2013	31-Mar-2013	05-Apr-2013	3				1		
	TP-02-08-06-FG-006	F	14-Mar-2013	21-Apr-2013	24-Apr-2013	2				0		
	TP-02-08-06-FG-011	F	16-Mar-2013	21-Apr-2013	24-Apr-2013	3				0		
	TP-02-08-06-FG-012	F	10-Mar-2013	15-Apr-2013	20-Apr-2013	1				0		
	TP-02-08-06-AV-001	F										
	TP-02-08-06-FG-013	F		07-Apr-2013	08-Apr-2013	3				0		

18.4 Progress Status - Testpack Level

This screen displays the percentage of completion of the test pack

- Construction
- Line checking
- Testing
- Reinstatement

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18.5 Spool Status - Isometric Level

- The user can click on a test pack number, in order to access the isometric level
- Two tabs are available at this level : “spool status” and “isometric status”
- To go back to the Higher level, the user has to click on the system – subsystem number
- The user can also scroll between the different test pack of the sub-system using the arrows
- The “spool status” tab displays the current status of the spools belonging to the test pack, at the isometric level
 - A figure identifies the status (example: 12 = Ready For Test)
 - A tool tip is also provided to indicate the status
 - A color coding is done, to quickly identify the level of completion (red, orange and green)

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Hide Test Pack Explorer

System 04-09-08 Select TestpackLocation Search Export

Test Plan Date Select Priority Select Subcontractor Select Area

TP-04-09-08-CH-001 Manual QC Release Date System 04-09-08 - Sub Sytem 04-09-08

ISO No.	Rev	Diam	Loc	Joints to be welded	Joints to be bolted	Joints to NDE	Date ISO Completed	Date ISO sent for checking	Date is returned from checking	Item x to be cleared	Date ISO is QC Released for test	Date ISO is Ready for Test
4001R_CH04090511_01	1F1	2	F	0	0	34			21-Jan-2013	7		
4005R_CH04090504_01	0F1	2	F	1	0	7			21-Jan-2013	5		
4007A_CH04090504_01	0	2	F	0	0	12	23-Jun-2013		21-Jan-2013	9		
4011R_CH04090504_01	0F3	2	F	1	0	2		06-Jan-2013	21-Jan-2013	6		
4011R_CH04090511_01	0F1	2	F	0	0	1	09-Dec-2012	21-Jan-2013	21-Jan-2013	4		
4021R_CH04090504_01	0F1	2	F	2	0	6			21-Jan-2013	13		
4025A_CH04090504_01	0F3	2	F	7	0	57			21-Jan-2013	4		
4025A_CH04090506_01	0	1.50	F	3	0	11			21-Jan-2013	2		

18.7 Spool Status – Spool Level

- The user can also click on a isometric number, in order to access the spool level
- To go back to the upper level, the user has to click on the system – subsystem – test pack number
- The user can also scroll between the different isometrics of the test pack using the arrows
- The “spool status detailed” tab displays the detailed current status of the spools belonging to the test pack, and the selected isometric
- The status of each spool is detailed
- A color coding is done, to quickly identify the level of completion (red, orange and green)

System 22-P-001 - Sub Sytem 27-US-010 - TestPack TPS-A-27-US-010-CH-138

S2P2-27-CH-1005-01 Spool Status Detailed

Spool No.	Send to Plan	Start Fab	Fabricated	QC Released	Sent To Paint	Painted	Final QC	Laydown	To site	Erected	Welded/Bolted	Supported	RFT
SP01													
SP02													
SP03													
SP04													
SP05													
SP06													
SP07													
SP08													
SP09													
SP10													
SP11													
SP12													
SP13													
SP14													

19. FLANGE MANAGEMENT

This module helps the user to manage the bolt torquing and tensioning activities related to the flange connections. The flange management is interrelated to the test packages.

Main functionalities of flange management are as follows:

- Numbering and quantifying works.
- Identify joints to be performed before test or after test and so the requirements of permanent or temporary gasket for the jointing.
- Identify method of tightening required.
- Define torquing values.
- Records execution progress and traceability if required.
- Calculating estimations of number of jointing to forecast and anticipate the required resources
- Editing the torquing program and all reports necessary for backlog tracking
- Compiling records of jointing, jointers, in order to edit history sheet for test packs (if required)

The flange management functions in different stages as follows

- Import bolting report data
- Revision management and flange browse
- Progress import of bolting

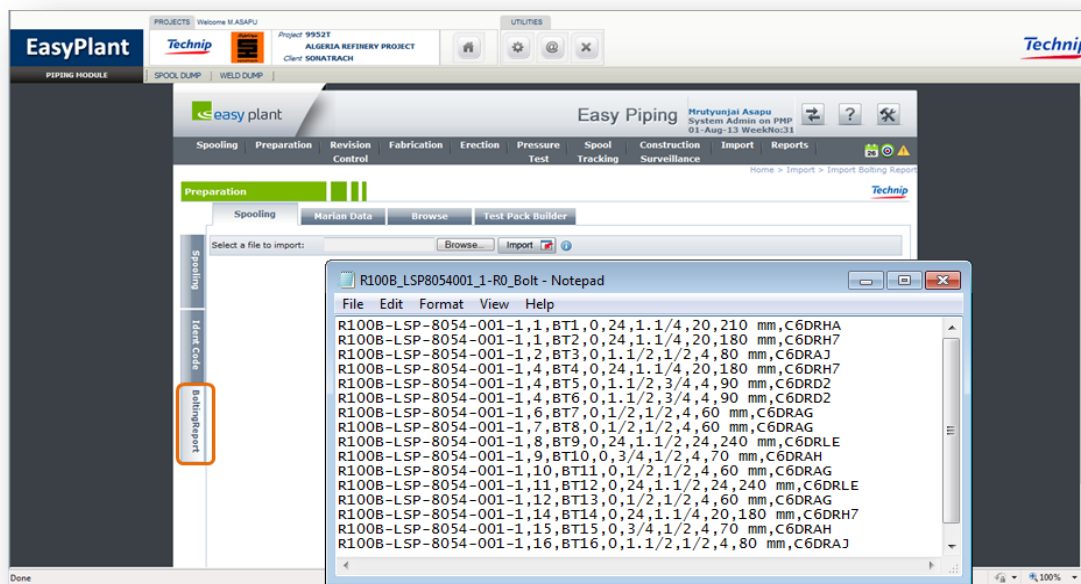
19.1 Import Bolting Report Data

A dedicated screen in Easy piping is handling the import process, in which the user will browse to select the file to be imported, the file will strictly following the agreed template, otherwise, Easy piping will not be able to interpret the file and import the data appropriately

Easy piping will show the imported data in a preview pane before doing the actual import to the database. At this point, Easy Piping will validate the data, and alert the user if inconsistencies are reported. The user will check the consistency of the data and will chose to proceed.

Importing rules are:

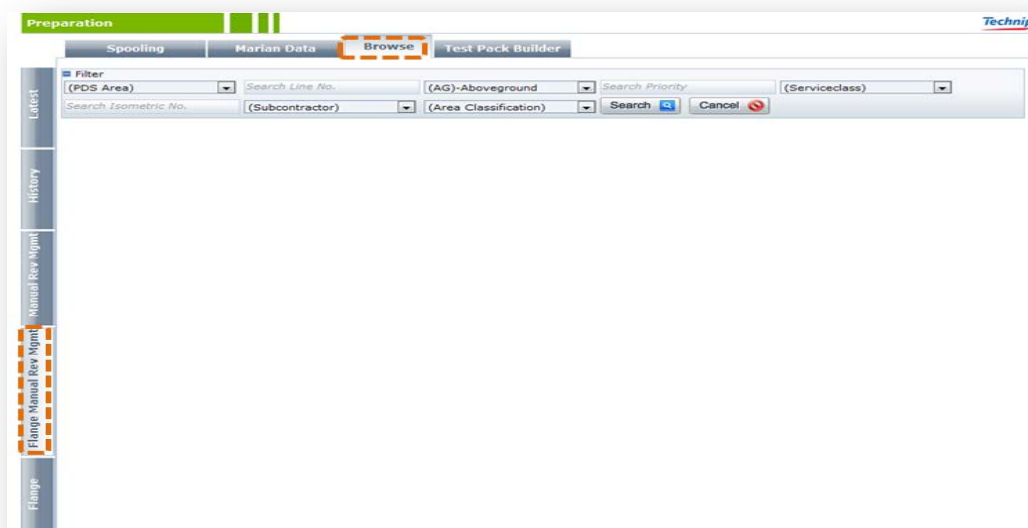
- Easy piping will validate whether the Spool data (including revision) that is being imported already exists in the database. If not, the import process will notify the user that corresponding weld sum data does not exist and invite the user to import it.
- Diameter size that will be provided in fractional format will be converted to decimal format by the system. For ex: The acceptable format in the import file for fractional numbers is 2.1/2. This will be read by the system as 2.5.
- The value of the "BT No." should be always starting with BT characters. Example: BT1 or BT21.
- All the columns in the imported text file will be stored as 'text' in the database, except for the "Diam" and "Qty" column; the value will be stored as a 'number'.
- Easy piping will validate whether the flange rating for the imported joint exist in the rating referential by checking the rating and diam, otherwise an error message will appear.



19.2 Flange Manual Revision Management

The “Flange Manual Revision Management” module allows the user to do Flange Revision management manually. This Flange Revision management will happen without getting revised ‘Bolting Report’ (txt) file from SPOOLGEN.

A separate screen is available for this module, which comes under ‘Preparation’ menu, and ‘Browse’ tab and sub-tab named as ‘Flange Manual Revision Management’.



When the user click on the search button, a list of Isometrics will be displayed in a grid that satisfies the search criteria.

From Search list user can select an Isometric for manual revision management. By selecting any one of the Isometric, system will redirect to Flange manual revision management page.

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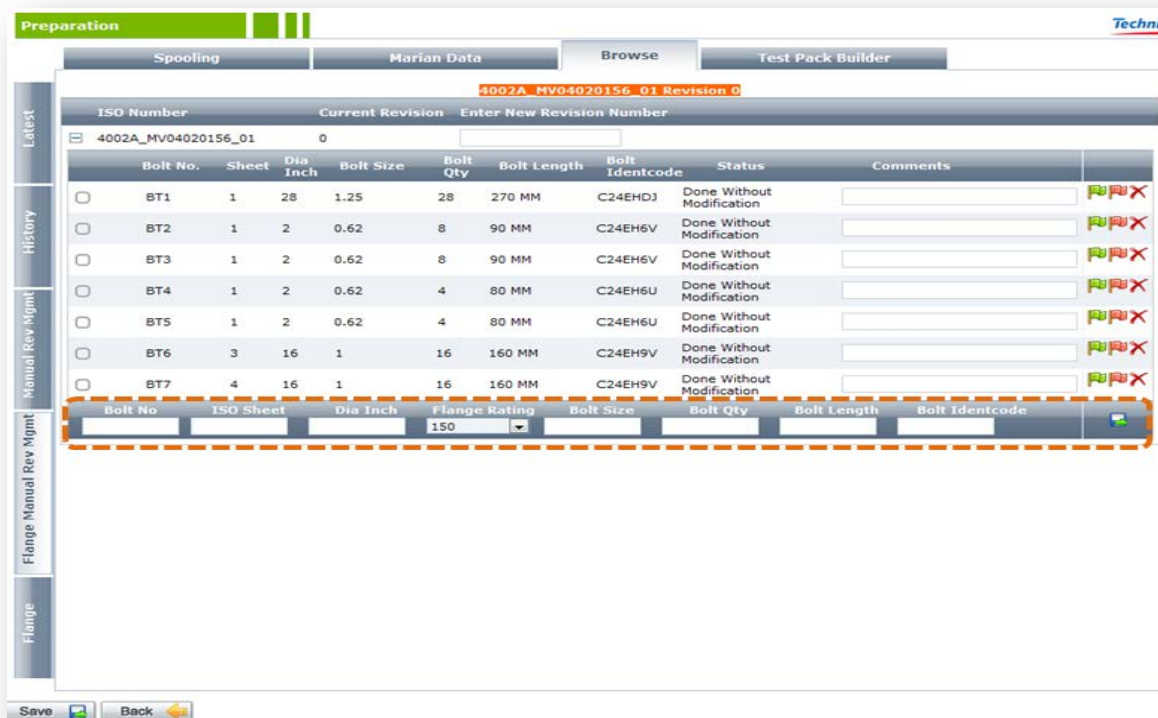
Bolt No.	Sheet	Dia Inch	Bolt Size	Bolt Qty	Bolt Length	Bolt Identcode	Status	Comments
BT1	1	28	1.25	28	270 MM	C24EHDJ	Done Without Modification	
BT2	1	2	0.62	8	90 MM	C24EH6V	Done Without Modification	
BT3	1	2	0.62	8	90 MM	C24EH6V	Done Without Modification	
BT4	1	2	0.62	4	80 MM	C24EH6U	Done Without Modification	
BT5	1	2	0.62	4	80 MM	C24EH6U	Done Without Modification	
BT6	3	16	1	16	160 MM	C24EH9V	Done Without Modification	
BT7	4	16	1	16	160 MM	C24EH9V	Done Without Modification	

By default all the Flange joint status is 'Done without Modification'. The progress information will either be copied to the newer revision of the Flange or not, as explained in the table below:

	Status	Action
	Not Done	The user will select this value if there is no progress information. No action to be taken in this case, the newer version of the spool will remain empty, same applies to the weld
	Cancelled	The user will select this value if the spool has been cancelled. No action to be taken in this case, the spool will not appear in the newer revision.
	Done, but revised w/o modification	This value refers to Spools which are fabricated in the older revision, but the newer version of the spool hasn't changed, so the progress information of the older revision will be copied to the newer revision. Weld progress information of the older revision will also be copied to welds in the newer revision. The 'Spool Revision No.' will remain the same.

Note:

- In the below screen, all the fields are mandatory, the user have to fill all the details.
- The value of the "Bolt No." should have minimum three characters. Example: BT01
- Bolt No should start with 'BT'.
- PSMS will check whether there are duplicate Flange joints (same Bolt no.). If duplicates exist, an error message will appear against the particular Flange joint.
- Dia inch, Bolt Size and BoltQty accept only numeric values, otherwise system will validate.



ISO Number	Current Revision	Enter New Revision Number
4002A_MV04020156_01	0	

Bolt No.	Sheet	Dia Inch	Bolt Size	Bolt Qty	Bolt Length	Bolt Identcode	Status	Comments
BT1	1	28	1.25	28	270 MM	C24EHDJ	Done Without Modification	
BT2	1	2	0.62	8	90 MM	C24EH6V	Done Without Modification	
BT3	1	2	0.62	8	90 MM	C24EH6V	Done Without Modification	
BT4	1	2	0.62	4	80 MM	C24EH6U	Done Without Modification	
BT5	1	2	0.62	4	80 MM	C24EH6U	Done Without Modification	
BT6	3	16	1	16	160 MM	C24EH9V	Done Without Modification	
BT7	4	16	1	16	160 MM	C24EH9V	Done Without Modification	

Bolt No.	ISO Sheet	Dia Inch	Flange Rating	Bolt Size	Bolt Qty	Bolt Length	Bolt Identcode
			150				

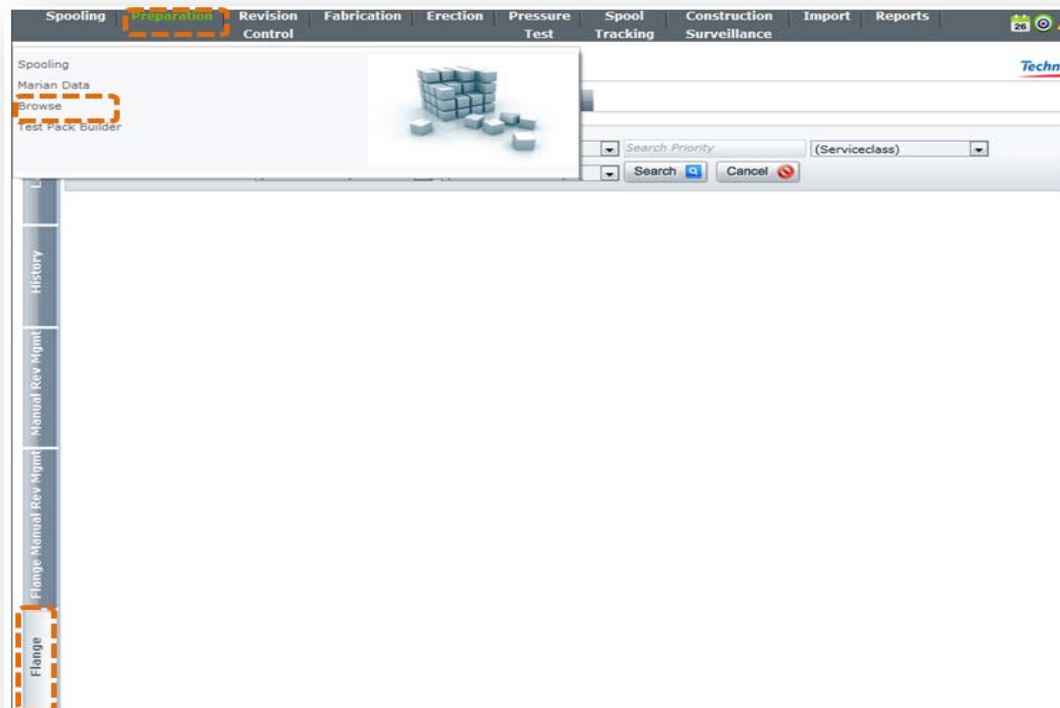
Save Flange Manual Rev Management:

Before save, user has to enter the new revision number the specific ISO. If the entered ISO Revision number is already exists system will discard the revision number then user has to enter new revision number accordingly.

19.2.1 Browse Flange

This module allows the user to navigate in a hierarchical manner between the Isometrics and flange joints and the user can also edit some information in the same screen. This sub module comes under “Preparation Module”

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The user can search for the different search criteria options as mentioned below.

- PDS Area
- Line No.
- ISO No.
- Priority
- Type
- Service Class
- Subcontractor
- Area classification

The search results will be displayed in a hierarchical format starting from the ISO, the user can select to expand and explore a specific ISO, in which the child Flange joints will appear.

In this screen the user can add more than one Flange joint to the ISO. When the user is adding the Flange joint has to follow some validation rules as follows.

- In the below screen, all the fields are mandatory, the user have to fill all the details.
- The value of the "Bolt No." should have minimum three characters. Example: BT01
- Bolt No should start with 'BT'.
- PSMS will check whether there are duplicate Flange joints (same Bolt no.). If duplicates exist, an error message will appear against the particular Flange joint.
- Dia inch, Bolt Size and BoltQty accept only numeric values, otherwise system will validate.

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Preparation

Spooling Marian Data Browse Test Pack Builder

Filter (PDS Area) Search Line No. (AG)-Aboveground Search Priority (Serviceclass) Search Isometric No. (Subcontractor) (Area Classification) Search Cancel

ISO Number	Rev	Service Class	Type	PDS Area	Line No.	Shop Sub	Field Sub	Mat Type
4005A_MV04090613_01	0	3C0KLX	AG	4005A	MV04090613-2"(3C0KLX)-N	DESCON	DESCON	SS
Bolt No.	Sheet	Dia Inch	Bolt Size	Bolt Qty	Bolt Length	Bolt Identcode	Flange Rating	
BT1	1	2	0.62	8	90 MM	C6DRBS	300	
BT2	1	2	0.62	8	90 MM	C6DRBS	300	
BT3	1	2	0.62	8	90 MM	C6DRBS	300	
BT4	1	2	0.62	8	150 MM	C6DRAS	300	
BT5	1	2	0.62	8	90 MM	C6DRBS	300	
4005A_SCC04090502_01	0F1	3C0KLX	AG	4005A	SCC04090502-2"(3C0KLX)-N	DESCON	DESCON	SS
4003A_FG04020001_01	0F1	3C0KLX	AG	4003A	FG04020001-4"(3C0KLX)-N	DESCON	CCT	SS
4004A_BD04090545_01	0	1S1X	AG	4004A	BD04090545-3"(1S1X)-1	DESCON	DESCON	CS
4004A_LS04090524_01	1	1S1X	AG	4004A	LS04090524-6"(1S1X)-1	DESCON	DESCON	CS
4002A_MV04020079_02	1	3C0KLX	AG	4002A	MV04020079-14"(3C0KLX)-2	DEBAJ	ELETECH	SS
4005A_MV04090502_01	2F2	1C3AS1X	AG	4005A	MV04090502-16"(1C3AS1X)-4A	DESCON	DESCON	CS NACE
4025A_AG04090520_01	2F2	1C0KLNK	AG	4025A	AG04090520-10"(1C0KLNK)-2	DESCON	DESCON	SS NACE
1812R-WB01080304_01	0	3P1X	AG	1812R	WB01080304-2"(3P1X)-N	DESCON	DESCON	CS
4004A_BD04090547_01	0	1S1X	AG	4004A	BD04090547-3"(1S1X)-1	DESCON	DESCON	CS

19.2.2 Flange Joint Progress Template

The flange joint progress can be updated in the easy piping with two methods.

- Flange joint progress Import Method
- Flange joint progress input method.

The flange joint progress template helps the user to import the Flange joint progress. This template is also used to link the test packs to the flange joints for the relevant Isometrics.

The user can download the template from the import module.

easy plant Easy Piping Mrutyunjai Asapu System Admin on PMP 18-Aug-13 WeekNo:34

Spooling Preparation Revision Control Fabrication Erection Pressure Test Spool Tracking Construction Surveillance Reports

Generate Progress Templates

Import Progress

Import Ident Code

Import NDE Examination

Import Bolting Report

Generate Flange Joint Progress Template

Import Flange Joint Progress

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PMP - Flange Joint Progress Template																			
PDS Area: ALL				Line No: Search Line No.				Area Classification: ALL											
ISO No.: Search Isometric No				Contract: ALL															
Type: (AG) Aboveground				Class: ALL															
Test Pack	Iso Number	Rev No.	Sheet No./BT No.	Diam	Rating	Bolt Size	Bolt Qty	Bolt Length	Bolt Ident	UT	Joint Method	Jointing Value	Joint Timing	Cat.	Reason	Joint Date	Report Nbr	Jointer	Tag Nbr
TP-08-02-01-CV-001	1801R-CV01060001_01	3	1	BT1	2	150	0.62	4	90 MM	C31JKDD	Manual								
TP-02-09-03H-G-001	1801R-FG01090003_01	0F2	1	BT1	24	150	1.25	20	180 MM	C31JKGV	12.35	Torquing	Before Precom	Y	Test blind				
TP-03-08-03FG-001	1801R-FG01080003_01	0F2	1	BT1	24	150	1.25	20	180 MM	C31JKGV	12.35	Torquing	Before Precom	Y	Test blind				
TP-01-09-10-LO-006	1802A-LO01080801_01	1F1	1	BT1	3	150	0.62	4	100 MM	CBUC452		Manual				16-Feb-2010	TP-001	BT-40	T-001
TP-01-09-10-LO-006	1802A-LO01080802_01	1	1	BT1	6	150	0.75	10	110 MM	CBUC469		Manual				18-Feb-2010	TP-001	UT-40	T-001
TP-02-05-11-SV-007	1810A-SV01020950_01	1	1	BT1	20	500	2	20	430 MM	CBUC4TP	15.675	Torquing	Before Precom	Y	Test blind				
TP-08-03-01-FG-002	1812R-FG01080651_01	1	1	BT1	24	150	1.25	20	450 MM	C31JKLN	3.25	Torquing	After precom	Z	INA				
TP-03-09-11-SV-001	2810A-SV02020950_01	1	1	BT1	20	500	2	20	430 MM	CBUC4TP	4.125	Torquing	After precom	Z	INA				
TP-01-05-02FG-002	2812R-FG02080651_01	1	1	BT1	24	150	1.25	20	450 MM	C31JKLN	3.25	Torquing	After precom	Z	INA				
TP-11-08-21-AG-009	4005R-AG04090544_01	2F1	1	BT1	40	150	1	44	180 MM	C24EH9X	5.25	Torquing	360	Before pressure Test	X	INA			
TP-11-08-21-AG-009	4005R-AG04090545_01	1	1	BT1	34	150	0.88	10	200 MM	C24EH9G	4.5	Torquing	255	Before pressure Test	X	INA			
TP-11-08-21-AG-009	4006A-AG04090520_01	1	1	BT1	40	150	1	44	240 MM	C08REX	5.25	Torquing	450	Before pressure Test	X	INA			

The above template consists several parameters as mentioned below.

- Test Pack
- Iso No (Read only field)
- Rev (Read only field)
- Sheet No. (Read only field)
- BT No (Read only field)
- Diam : (Read only field)
- Rating : (Editable)
- Bolt size : (Read only field)
- Bolt ident (Read only field)
- Bolt qty : (Read only field)
- Bolt Length (Read only field)
- UT: (Read only field)(UT calculation will do at the time of export (Dynamic). Using this formula: **UT = Reference point quantity x coef Diam x Coef Rating x Coef Punch**)
- **Note:** If coef punch is null (That means it is not defined) then the UT value is Null.
- Jointing method: can be only as per project referential.
- Jointing value : editable
- Joint period: can be only as per project definition table: for example Before Test / Before PMC / After PMC
- Joint Category: can be only as per project definition table : for example X, Y, Z
- Reason can be only as per project definition table.

The following are the joint progress along with their data types:

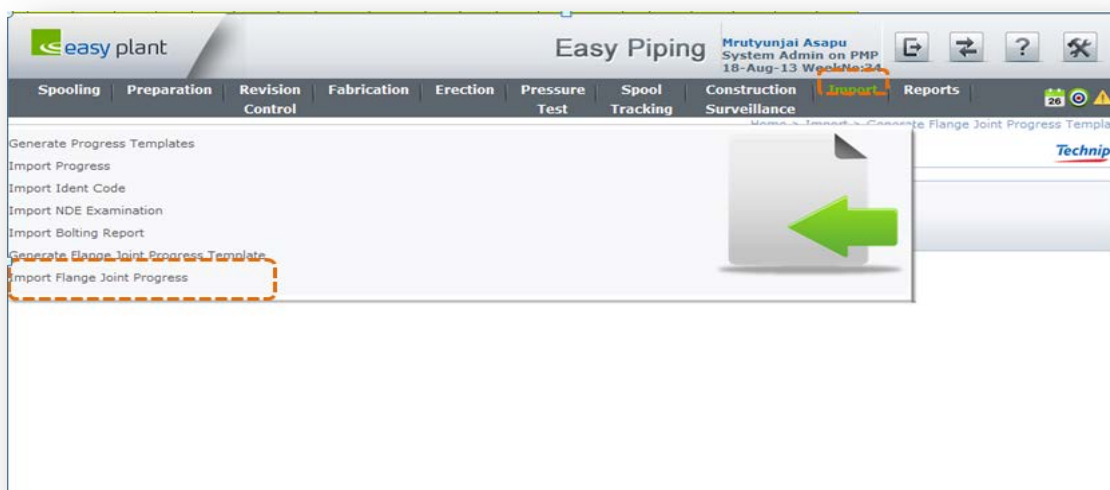
- Joint date : date format (dd-MMM-yyyy)

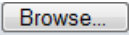
- Report Nbr
- Jointer (Drop down)
- Tag nbr

In case a joint has 2 or more progress records, records will be displayed by one or more additional line.

19.2.3 Flange joint progress import method

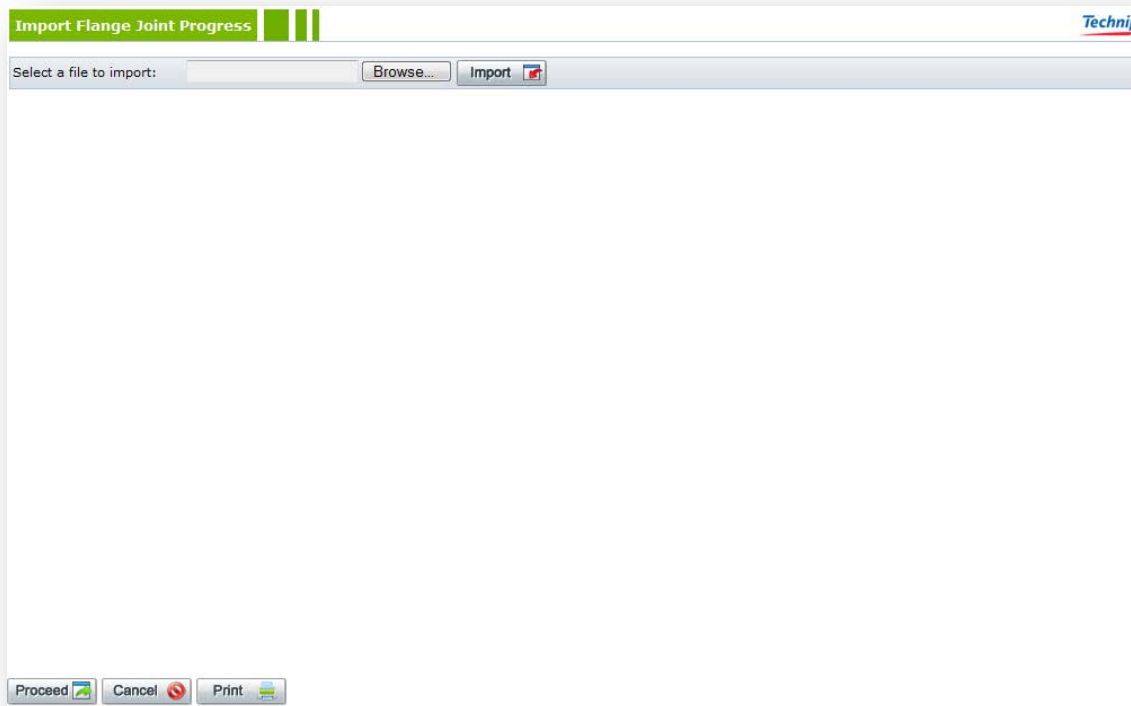
The user can import the progress from the “Import module” The user can update the template with the relevant information and import from this screen



The user can use the Icon  and select the file and use the icon  to import all the progress information into Easy Piping.

Note:

- The progress data in the excel sheet will always supersede progress data in the database. Meaning that if there was a conflict between the data in the excel sheet and the actual data in PSMS, that particular cell will be highlighted in yellow. The user will have to confirm the changes and proceed the import if he wishes. The data in the system will be overwritten with the new data on confirmation.
- Import of flange joints which are not defined will be rejected by the system
- Jointing method column and Jointing value column should have values as per Torquing requirements project referential.
- Joint period column should have values as per project definition table: for example Before Test / Before PMC / After PMC
- Punch column should have values as per project definition table: for example X, Y, Z.
- Reason column can have alphanumeric values. should have values as per project definition table For example NA, Test blind, vent point, drain point, blowing point.
- Joint date column should have the following date format (dd-MMM-yyyy).
- Report Nbr can have alphanumeric values. No validations.
- Jointer column can have values that are defined in the Jointer referential.
- Tag nbr can have alphanumeric values. No validations.



19.2.4 Flange joint progress input method

The flange joint progress screen comes under the spool erection module. In this screen the user can search for a particular Isometric and the system will display all the flange joints related to the isometric. If the user has defined some parameters like Jointing method, category and reasons for a particular flange joint initially, he can edit the information in this screen. The Flange joint information details to be shown are:

- Jointing method: can be only as per project referential.
- Joint period: can be only as per project definition table: for example Before Test / Before PMC / After PMC
- Joint Category: can be only as per project definition table: for example X, Y, Z
- Reason can be only as per project definition table.
- Joint date : date format (dd-MMM-yyyy)
- Report Nbr
- Joiner (Drop down)
- Tag nbr

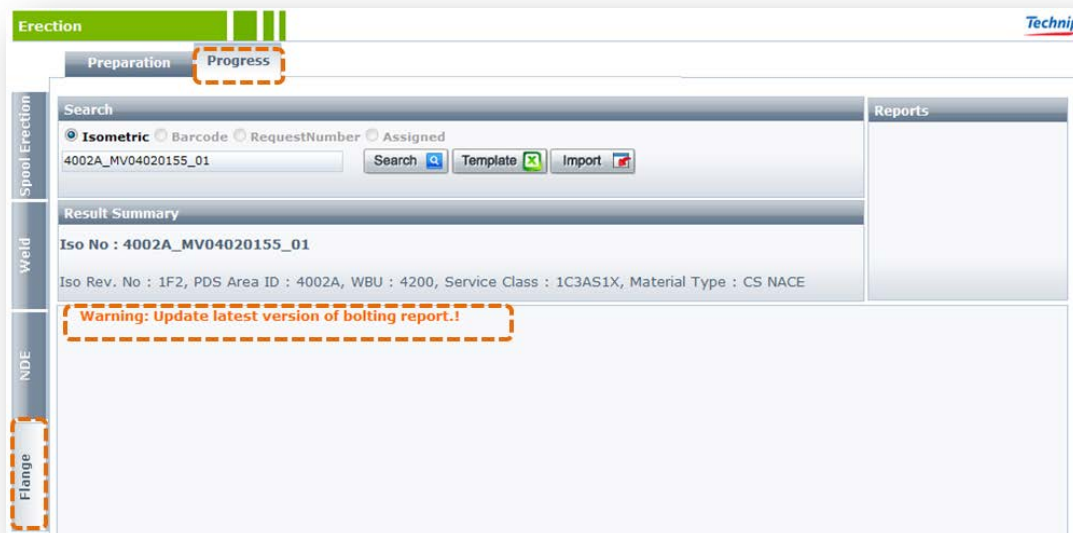
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If the Flange joint is progressed by only 1 Jointer, then the user can enter the progress details in the same screen.

If the flange joint is progressed by 2 jointers, the system will display an icon () in each and every row. The user can click on this icon, the system will display a pop up where user can update the progress.

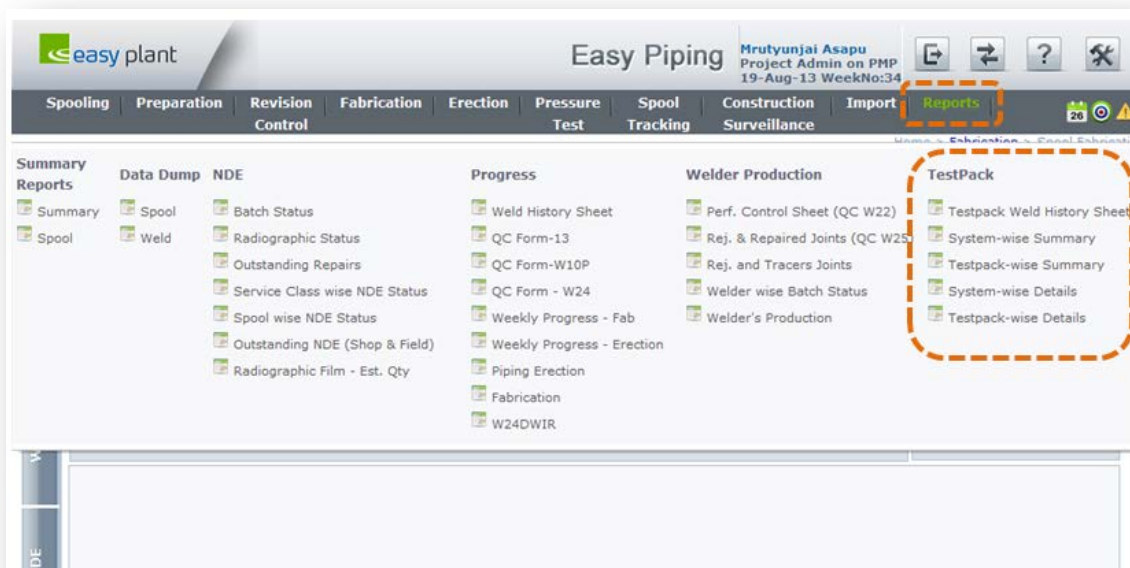
Note:

When the user opens the erection progress screen, the system will check if the rev number of the corresponding spool is matching with the rev number of flange data related. Otherwise, the system will disable the “flange joint progress” grid and will display a warning message to the user.



20. TESTPACK MANGEMENT REPORTS

In Easy piping the user can generate the reports for the test pack status. The below Image shows the location of the reports



9833N_361_Test_Pack_Weld_History_Sheet

This report will provide the weld history of all the spools related to the selected test pack.

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System-wise_Summary_Report.xls

This report will provide the status of the test packs on a system level for the entire project. The user can select either system or sub system.



Testpack-wise_Summary_Report.xls

This report will provide the status of each test pack with the construction and testing information.



System-wise_Details_Report.xls

This report will provide the status of system completion including the construction and test pack status.



Testpack-wise_Details_Report.xls

This report will provide the information of each test pack and punch items also.